Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**CALCULATION** **3110 PRIMARY CRUSHING** **FOUNDATIONS & SUPERSTRUCTURE**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
Employer	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is prepared for design of Primary Crusher House area foundations & superstructure under area code WBS 3110. The analysis & design is carried out in accordance with project specifications,codes & standards. The design calculation ensures the safety and serviceability of the framing system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

In the specified area, there are Primary Crusher supporting structure's foundation & superstructure, Screen supporting structure's foundation & superstructure, Hopper supporting structure foundation and HGTU Tower foundation. The analysis has been carried out in STAAD PRO software, considering the load data attached here with in Attachment 1. Design of slab, beam, column & foundation are done in MS Excel .The load combinations for serviceability check and member's strength calculation are as per SBC.

4. DESIGN REQUIREMENTS

4.1. Documents & Codes

Documents (Refered):

i) Doc. No. MD-216-1000-EG-ST-CRT-1001 [Structural Engineering Design Criteria]
ii) Doc. No. MD-101-SMEM-EG-ST-CRT-1001 [Structural Design Criteria - Process Building & Structures]
iii) Doc. No. MD-101-SMEM-EG-CV-CRT-1001 [Civil Design Criteria - General]

Codes (Refered) :

i) SBC 301 (latest version)
ii) SBC 304-CR-18
iii) ACI 318(M)
iv) ACI SP-17M(09)

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4.2. Material Property

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3. Grouting

- Grout shall be cementitious under steel column bases of Strength 40 MPa

4.4. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.5. Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to Sea Water or Soil	Foundation	75
Exposed to weather	Column & Floor Beam	50
	Wall and Slab	50

4.6. Allowable Soil Bearing Capacity

Recommended net allowable bearing capacity as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A02, bore hole BH-19.

BH-no	Depth of foundation below FGL(m)	Recommended net allowable bearing capacity(kN/m^2)	Vertical modulus of subgrage reaction($kN/m^2/m$)
BH-19	1.5	600	72000

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5. DESIGN CRITERIA

5.1. Design Loads for Crusher Supporting Structure

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3110-EG-ME-LPX-1001/1002 & MD-216-3110-EG-ME-ELV-1007/1008/1009) attached in 'Attachment 2'. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5.1.1. Load Calculations

Structure Self Weight (D1) :

a. Selfweight of structural frame

b. 200mm thick RCC floor. Floor load = 0.20*24 = 4.8 kN/m^2

c. Dead weight of 1000mm wide STEEL stair -

Dead weight of ISMC 200 = 22.3 Kg/m

Dead load per m run = 22.3/cos(38°)= 28.30 Kg/m

Dead weight of 50 thick gratings = 50 Kg/sqm

Dead load per m run = 50*0.5 = 25.00 Kg/m

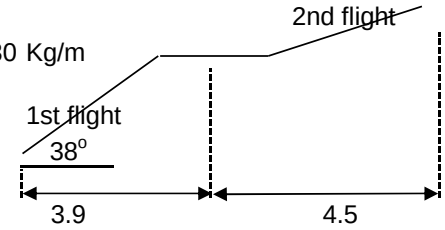
Total dead load =28.3+25 = 53.30 Kg/m

Point load on RCC beam at 1st and 3rd support

= 53.3*2.25 = 120 Kg i.e 1.2 kN

Point load on RCC beam at intermediate support

= 53.3*(4.5+3.9)/2 = 224 Kg i.e 2.24 kN



d. Load form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
17	10	-0.028	-7.934	0.005	0.008	0	0
18	10	-0.027	-7.973	0.01	0.016	0	0
19	10	0.048	-10.994	-0.011	-0.019	0	0
20	10	0.007	-10.243	-0.004	-0.007	0	0

e. Weight of soil over foudation

Unit weight of soil = 20 kN/m^3

Height of soil = 2.4 m

Pressure = 48 kN/m^2

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Equipment Dead Weight (D2) :

a. Vertical load per anchor point = 200 kN (Table 5, MD-216-3110-EG-ME-ELV-1008-A01)
Self weight of crusher = 3/8*200 = 75 kN at load point PC/1
Hopper weight/Material weight = 3/5 As per load data attached in Attachment 1
b. load at load point PC/2 = 45 kN
c. load at load point JB/1 = 155 kN (for JIB crane)
b. Load form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
17	11	0	-2.466	-0.003	0.013	0	0
18	11	-0.003	-8.243	0.005	0.025	0	0
19	11	0.002	-7.931	-0.007	0.015	0	0
20	11	0.001	-16.36	0.005	0.036	0	0

Live Load(L1)

a. Live load intensity on crusher floor = 5 kN/m^2
b. Live load on 1000mm wide STEEL stair -
Live load intensity = 4 KN/m^2
Dead load per m run = 4*0.5 = 2.00 kN/m
Point load on RCC beam at 1st and 3rd support
= 2.0*2.25 = 4.5 kN
Point load on RCC beam at intermediate support
= 2.0*(4.5+3.9)/2 = 8.4 kN

c. Load form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
17	12	-0.096	-8.153	0.025	0.036	0	0
18	12	-0.09	-6.676	0.033	0.05	0	0
19	12	0.172	-16.625	-0.037	-0.071	0	0
20	12	0.014	-11.885	-0.021	-0.046	0	0

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Equipment Live Load(L2) :

a. Vertical load per anchor point = 200 kN (Table 5, MD-216-3110-EG-ME-ELV-1008-A01)
Material load in crusher = 5/8*200 = 125 kN at load point PC/1
Hopper weight/Material weight = 3/5 As per load data attached in Attachment 1

b. Load form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
17	14	0.461	44.513	11.814	20.29	0	0
18	14	-0.46	-44.513	11.836	20.31	0	0
19	14	0.461	62.727	25.696	43.81	0	0
20	14	-0.462	-62.727	25.654	43.75	0	0

Equipment Load Reversible(ELL_REV)

a. at load point PC/1

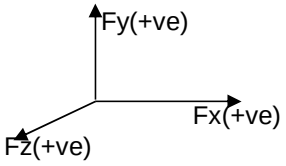
Vertical load	Horizontal load	
	Longitudinal	Transversal
Fy	Fx	Fz
kN	kN	kN
31	12	3

load at load point PC/2

Vertical load	Horizontal load	
	Longitudinal	Transversal
Fy	Fx	Fz
kN	kN	kN
65	4.5	2.5

at load point JB/1

Moment about center
Mx
kN-m
370



STAAD_pro axis

b. Load form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
17	15	0.017	31.465	0.003	0.202	0	0
18	15	-0.018	-31.465	0.013	0.212	0	0
19	15	0.002	46.606	-0.004	0.255	0	0
20	15	-0.002	-46.606	-0.011	0.249	0	0

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Wind Load Calculation (as per SBC 301):

A03

Velocity Pressure at height z, $q_z=0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 Eq. 7.2-12

Design Parameters :

Basic wind speed,V :	155 Km/h	cl 4.2.6.a (MD-216-1000-EG-ST-CRT-1001)	
Structure Type :	Main wind force resisting system		
Wind Directionality Factor, K_d :	0.85		Table 6.4-1
Exposure category :	C	cl 4.2.6.a (MD-216-1000-EG-ST-CRT-1001)	
Topographic Factor, K_{zt} :	1		Eq. 6.4-1
The building and structure classification category :		II	Table 1.6-1
Importance factor, I :	1		Table 6.5-1
Enclosure classification :	Open Buildings		

Design parameters for Gust factor calculation :

Mean roof height,h =	4.980 m		
Type of structure :	CONCRETE		
Damping ratio, β =	0.05		
Fundamental frequency of building :	6.930 Hz	A03	> 1Hz, Rigid structure
Minimum height, z_{min} =	0.2 m		Table 7.2-1
Eqv. Height, $\hat{z}$ =	2.99 m		Table 7.2-1
Factor c =	0.2		Table 7.2-1
Factor $\bar{\varepsilon}$=	0		Table 7.2-1
Factor l(m) =	150		Table 7.2-1
$g_Q = g_v$ =	3.4		Eq 7.2-6
$I_{\hat{z}} = c(10/\hat{z})^{1/6}$ =	0.24		Eq 7.2-2
$L_{\hat{z}} = l(\hat{z}/10)^{\varepsilon}$ =	150 m		Eq 7.2-4

For wind in X direction

$Q_x = \sqrt{(1/(1+0.63((B+h)/L_{\hat{z}})^{0.63})}$ 0.933 Eq 7.2-3

For wind in Z direction

$Q_Z = \sqrt{(1/(1+0.63((B+h)/L_{\hat{z}})^{0.63})}$ 0.938 Eq 7.2-3

Calculation of Gust effect factor, G :

Wind in X direction :

Horz. Dimension of building measured parallel to wind direction, L =			8.4 m
Horz. Dimension of building measured normal to wind direction, B =			10 m
$G_x = 0.925(1+1.7g_Q I_{\hat{z}} Q_x)/(1+1.7g_v I_{\hat{z}})$ =	0.889	Eq 7.2-1	

Wind in Z direction :

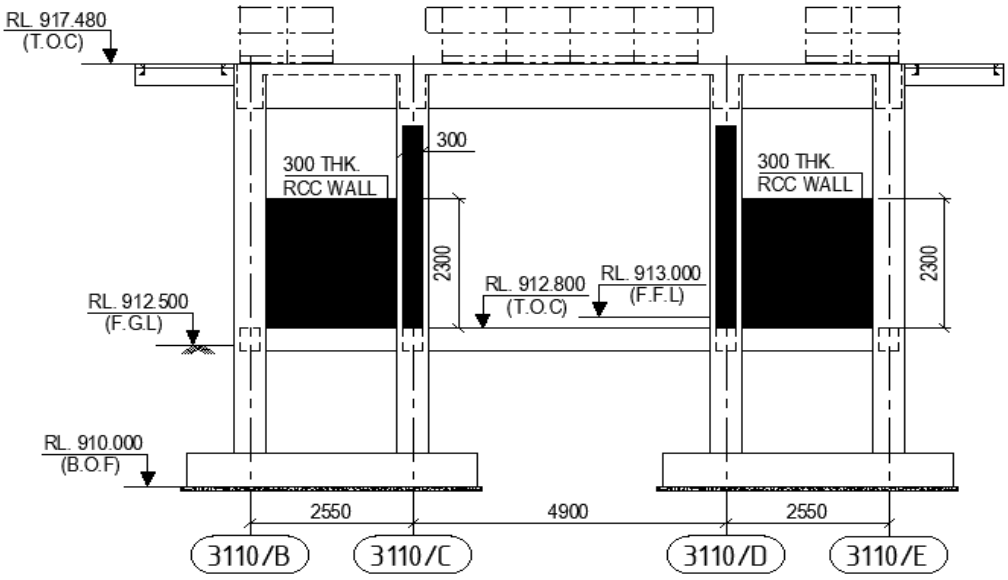
Horz. Dimension of building measured parallel to wind direction, L =			10 m
Horz. Dimension of building measured normal to wind direction, B =			8.4 m
$G_Z = 0.925(1+1.7g_Q I_{\hat{z}} Q_z)/(1+1.7g_v I_{\hat{z}})$ =	0.891	Eq 7.2-1	

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Calculation of Force Coefficient, C_f : A03

Wind in X direction



ELEVATION AT GRID - 3110/3

As the structure is symetrical, consider half of the frame for solidity calculation

Gross area, A_g = 4.98 x (2.55+4.9/2) = 24.9 sqm

Solid area for,

3110/B & 3110/E column = 0.5*4.98+ 0.8*2.55/2+2.05/2*2.3 = 5.87 sqm

A03 3110/C & 3110/D column = 0.5*4.98+ 0.8*(2.55/2 + 4.9/2)+2.05/2*2.3 = 7.83 sqm

Solid area, A_f = 13.7 sqm ε = 13.7/24.9 = 0.55

Force coefficient, C_f = 1.6 fig 7.2-17

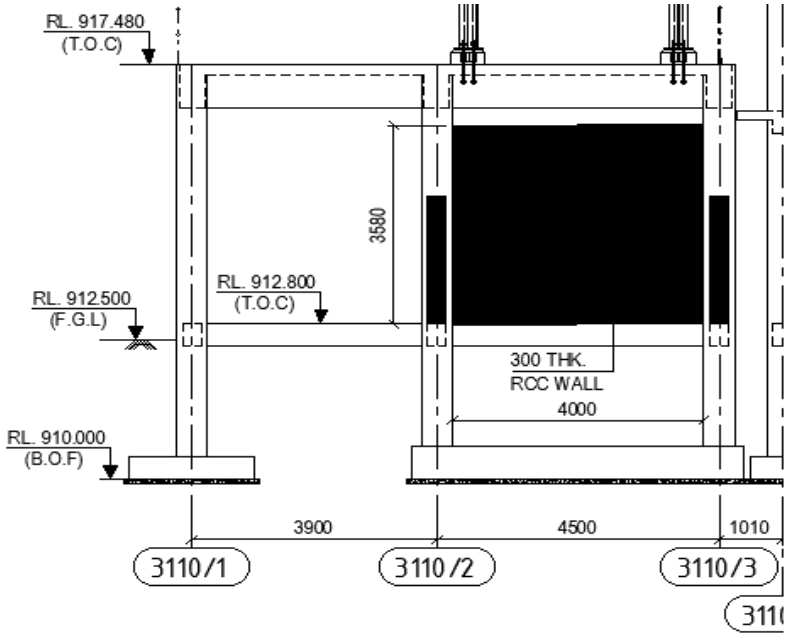
Wind Load(W+X)						Force on column			Udl/column		
						B&E	C&D		B&E	C&D	
z	K_z	q_z	Gx	Cf	p	KN	KN		KN/m	KN/m	
m		KN/m^2			KN/m^2						
0-5	0.86	0.83	0.889	1.6	1.18	6.9329	9.25		1.39	1.86	A03

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Wind in Z direction :

A03



ELEVATION AT GRID - 3110/D

Gross area,A_g = 4.98 x (3.9+4.5) = 41.84 sqm

Solid area for,

3110/1 column = 0.5*4.98+ 0.8*3.9/2 = 4.05 sqm

3110/2 column = 0.5*4.98+ 0.8*(4.5/2 +3.9/2)+3.58*4/2 = 13.01 sqm

A03

3110/3 column = 0.5*4.98+ 0.8*4.5/2+3.58*4/2 = 11.45 sqm

Solid area,A_f = 28.5 sqm ε = 28.51/41.84 = 0.6814

Force coefficient, C_f = 1.6 fig 7.2-17

Wind Load(W+Z)						Force on column			Udl/column		
						1	2	3	1	2	3
z	K_z	q_z	Gz	Cf	p	KN	KN	KN	KN/m	KN/m	KN/m
m		KN/m^2			KN/m^2						
0-5	0.86	0.83	0.891	1.6	1.18	4.7973	15.411	13.56	0.96	3.09	2.72

A03

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Wind Load(W+x) form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		KN			kN-m		
17	3	6.721	11.628	0.002	0.004	0	0
18	3	6.72	11.698	-0.002	-0.005	0	0
19	3	6.721	-11.629	-0.003	-0.003	0	0
20	3	6.72	-11.697	0.002	0.006	0	0

Wind Load(W+z) form rock breaker supporting structure

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		KN			kN-m		
17	4	0.001	9.003	7.176	11.01	0	0
18	4	-0.001	-9.003	7.175	11.01	0	0
19	4	-0.001	9.27	7.177	11.01	0	0
20	4	0.001	-9.27	7.174	11	0	0

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Seismic parameters:

For Ordinary reinforced concrete moment frames

Site Class :	B		Table 14.1.1
Occupancy Category :	II		Table 1.6-1
Spectral Acceleration, S_s :	0.1	g	
Spectral Acceleration, S_1 :	0.02	g	
Importance Factor ,I :	1		Table 9.5
Response Modification Coefficient ,R :	2.5		Table 10.2

Site Coefficient :

F_a :	1	Table 9.4.3a
F_v :	1	Table 9.4.3b

Maximum Spectral Response Accelerations for Short and 1-second Periods :

$S_{MS} = F_a S_s$:	0.1	Eq 9.4.3-1
$S_{M1} = F_v S_1$:	0.02	Eq 9.4.3-2

Design Spectral Response Accelerations for Short and 1-second Periods :

$S_{DS} = 2/3 S_{MS}$:	0.067	Eq 9.4.4-1
$S_{D1} = 2/3 S_{M1}$:	0.013	Eq 9.4.4-2

Period Coefficient, Ct : 0.044 Table 10.9.3.2

5.1.2 Vibration Check : A03

For primary crusher
Motor power rating : 200 kW (as per dwg no MD - 216 -3110 - EG- ME -GAD - 1007)
Motor frequency : 60 Hz
Total weight of crusher : 52.1 Mton
Operating frequency of crusher : 3.3 Hz

Analysis method : static (as per doc no MD-101-SMEM-EG-ST-CRT-0001)
Additional requirement : weight ratio > 3
Weight of crusher supporting structure : 275 Mton (self-weight of structure)
Weight ratio : 275 / 52.1 = 5.2783 ok

For Natural frequency of structure : 7.49 HZ as per Modal analysis done by ANSYS.Refer document no MD-216-3110-EG-CV-CAL-1006 (Calculation_Primary Crushing_Modification of Crusher Supporting Structure).

Remarks : The natural frequency values are found to be high-tuned to 2 times of working frequency (3.3 Hz).

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5.1.3. Load Combinations

Primary Load Cases are –

1 Seismic load (E +X)
2 Seismic load (E +Z)
3 Wind Load(W+X)
4 Wind Load(W+Z)
10 Structure Self Weight (D1)
11 Equipment Dead Weight (D2)
12 Live Load(L1)
13 Equipment Live Load(L2)
14 Equipment Load Reversible(ELL_REV)

Load Combinations for Allowable Stress Design
SBC 301 clause 2.4.1

Load Case	Load Combination
101	D1 + D2
102	D1 + D2 + L1
103	D1 + D2 + L1 + L2
104	D1 + D2 + L1 + L2 + ELL_REV)
105	D1 + D2 + L1 + L2 - ELL_REV)
106	D1 + D2 + W(+X)
107	D1 + D2 + W(-X)
108	D1 + D2 + W(+Z)
109	D1 + D2 + W(-Z)
110	D1 + D2 + 0.7E(+X)
111	D1 + D2 + 0.7E(-X)
112	D1 + D2 + 0.7E(+Z)
113	D1 + D2 + 0.7E(-Z)
114	D1 + D2 + 0.75W(+X) + 0.75L1
115	D1 + D2 + 0.75W(-X) + 0.75L1
116	D1 + D2 + 0.75W(+Z) + 0.75L1
117	D1 + D2 + 0.75W(-Z) + 0.75L1
118	D1 + D2 + 0.75W(+X) + 0.75(L1+L2)
119	D1 + D2 + 0.75W(-X) + 0.75(L1+L2)
120	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2)
121	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2)
122	D1 + D2 + 0.75W(+X) + 0.75(L1+L2+ELL_REV)
123	D1 + D2 + 0.75W(-X) + 0.75(L1+L2+ELL_REV)
124	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2+ELL_REV)
125	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2+ELL_REV)

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126	D1 + D2 + 0.75W(+X) + 0.75(L1+L2-ELL_REV)
127	D1 + D2 + 0.75W(-X) + 0.75(L1+L2-ELL_REV)
128	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2-ELL_REV)
129	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2-ELL_REV)
130	D1 + D2 + 0.75*0.7E(+X) + 0.75(L1+L2)
131	D1 + D2 + 0.75*0.7E(-X) + 0.75(L1+L2)
132	D1 + D2 + 0.75*0.7E(+Z) + 0.75(L1+L2)
133	D1 + D2 + 0.75*0.7E(-Z) + 0.75(L1+L2)
134	0.6(D1 + D2) + W(+X)
135	0.6(D1 + D2) + W(-X)
136	0.6(D1 + D2) + W(+Z)
137	0.6(D1 + D2) + W(-Z)
138	0.6(D1 + D2) + 0.7E(+X)
139	0.6(D1 + D2) + 0.7E(-X)
140	0.6(D1 + D2) + 0.7E(+Z)
141	0.6(D1 + D2) + 0.7E(-Z)

Load Combinations for Strength Design

SBC 301 clause 2.3.2

Load Case	Load Combination
201	1.4(D1 + D2)
202	1.4(D1 + D2) +1.7 L1
203	1.4(D1 + D2) +1.7 (L1 + L2)
204	1.4(D1 + D2) +1.7 (L1 + L2 +ELL_REV)
205	1.4(D1 + D2) +1.7 (L1 + L2 - ELL_REV)
206	1.2(D1 + D2) + 0.8W(+X)
207	1.2(D1 + D2) + 0.8W(-X)
208	1.2(D1 + D2) + 0.8W(+Z)
209	1.2(D1 + D2) + 0.8W(-Z)
210	1.2(D1 + D2) + 1.6W(+X) + L1
211	1.2(D1 + D2) + 1.6W(-X) + L1
212	1.2(D1 + D2) + 1.6W(+Z) + L1
213	1.2(D1 + D2) + 1.6W(-Z) + L1
214	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2)
215	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2)
216	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2)
217	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2)
218	1.2(D1 + D2) + 1.6W(+X) + (L1+L2+ELL_REV)
219	1.2(D1 + D2) + 1.6W(-X) + (L1+L2+ELL_REV)
220	1.2(D1 + D2) + 1.6W(+Z) + (L1+L2+ELL_REV)
221	1.2(D1 + D2) + 1.6W(-Z) + (L1+L2+ELL_REV)

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222	1.2(D1 + D2) + 1.6W(+X) + (L1+L2-ELL_REV)
223	1.2(D1 + D2) + 1.6W(-X) + (L1+L2-ELL_REV)
224	1.2(D1 + D2) + 1.6W(+Z) + (L1+L2-ELL_REV)
225	1.2(D1 + D2) + 1.6W(-Z) + (L1+L2-ELL_REV)
226	1.2(D1 + D2) + 1.6E(+X) + (L1 + L2)
227	1.2(D1 + D2) + 1.6E(-X) + (L1 + L2)
228	1.2(D1 + D2) + 1.6E(+Z) + (L1 + L2)
229	1.2(D1 + D2) + 1.6E(-Z) + (L1 + L2)
230	0.9(D1 + D2) + 1.6W(+X)
231	0.9(D1 + D2) + 1.6W(-X)
232	0.9(D1 + D2) + 1.6W(+Z)
233	0.9(D1 + D2) + 1.6W(-Z)
234	0.9(D1 + D2) + 1.6E(+X)
235	0.9(D1 + D2) + 1.6E(-X)
236	0.9(D1 + D2) + 1.6E(+Z)
237	0.9(D1 + D2) + 1.6E(-Z)

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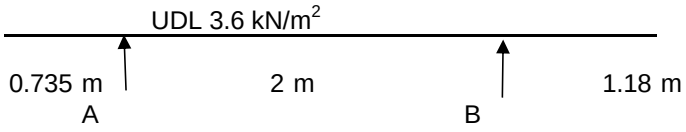
5.2. Design Loads for Screen Supporting Structure

All Design Loads are indicated in Foundation Load Plan (Doc. No. MD-216-3110-EG-ME-LPX-1001/1002 & MD-216-3110-EG-ME-ELV-1007/1008/1009) attached in 'Attachment 2'. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5.2.1. Load Calculations

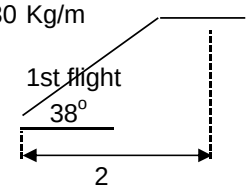
Structure Self Weight (D1) :

a. Selfweight of structural frame
b. 150mm thick RCC floor. Floor load = 0.15*24 = 3.6 kN/m^2 at RL 919.89
c. 1m wide 150mm thick RCC floor. at RL 916.63



Reaction at A = 9.85 kN/m Reaction at B = 11.45 kN/m
Moment at A = 0.97 kN-m/m Moment at B = 2.506 kN-m/m

d. Dead weight of 1000mm wide STEEL stair -
Dead weight of ISMC 200 = 22.3 Kg/m
Dead load per m run = 22.3/cos(38°)= 28.30 Kg/m
Dead weight of 50 thick gratings = 50 Kg/sqm
Dead load per m run = 50*0.5 = 25.00 Kg/m
Total dead load =28.3+25 = 53.30 Kg/m
Point load on RCC beam support
= 53.3*1 = 53.3 Kg i.e. 0.53 kN



e. Weight of soil over foudation
Density of soil = 20 kN/m^3
Height of soil = 2.5 m
Pressure = 50 kN/m^2

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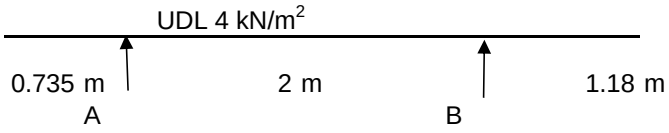
Equipment Dead Weight (D2) :

Total Load for Vibrating Grizzy screen at laod point VG/A = 43 kN — Reference drawing
at laod point VG/B = 43 kN — attached in attachment2
at laod point VG/C = 15 kN

Dead load as per drawing no . MD-216-3110-EG-ME-GAD-1001 — attached in attachment2
Load for Vibrating Grizzy screen at laod point VG/A = 20 kN
at laod point VG/B = 20 kN
Remaining 23 kN load considered as equipment live load.

Live Load(L1)

a. Live load intensity on screen floor = 5 kN/m^2
b. Live load intensity on platform = 4 kN/m^2



Reaction at A = 10.9 kN/m — Reaction at B = 12.72 kN/m
Moment at A = 1.08 kN-m/m — Moment at B = 2.78 kN-m/m

c. Live load on 1000mm wide STEEL stair -
Live load intensity = 4 kN/m^2
Dead load per m run = 4*0.5 = 2.00 kN/m
Point load on RCC beam support
= 2.0*1 = 2 kN

Equipment Live Load(L2) :

Load for Vibrating Grizzy screen at laod point VG/A = 23 kN — Reference drawing
at laod point VG/B = 23 kN — attached in attachment1

Equipment Reversible Load(L3)

at load point VG/A

Vertical load	Horizontal Longitudinal
Fy	Fx
kN	kN
33	29

at load point VG/B

Vertical load	Horizontal load Longitudinal
Fy	Fx
kN	kN
33	29

at load point VG/C

Vertical load	Horizontal Longitudinal
Fy	Fx
kN	kN
23	23

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Equipment Start & Stop_Condition [L4]

at load point VG/A

Vertical load	Horizontal Longitudinal
Fy	Fx
kN	kN
45	35

at load point VG/B

Vertical load	Horizontal load Longitudinal
Fy	Fx
kN	kN
45	35

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Wind Load Calculation (as per SBC 301):

Velocity Pressure at height z, $q_z=0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 Eq. 7.2-12

Design Parameters :

Basic wind speed,V :	155 Km/h	cl 4.2.6.a (MD-216-1000-EG-ST-CRT-1001)	
Structure Type :	Main wind force resisting system		
Wind Directionality Factor, K_d :	0.85		Table 6.4-1
Exposure category :	C	cl 4.2.6.a (MD-216-1000-EG-ST-CRT-1001)	
Topographic Factor, K_{zt} :	1		Eq. 6.4-1
The building and structure classification category :		II	Table 1.6-1
Importance factor, I :	1		Table 6.5-1
Enclosure classification :	Open Buildings		

Design parameters for Gust factor calculation :

Mean roof height,h =	7.090 m		
Type of structure :	CONCRETE		
Damping ratio, β =	0.05		
Fundamental frequency of building :	1.65 Hz	> 1Hz, Rigid structure	
Minimum height, z_{min} =	0.2 m		Table 7.2-1
Eqv. Height, $\hat{z}$ =	4.22 m		Table 7.2-1
Factor c =	0.2		Table 7.2-1
Factor $\bar{\varepsilon}$=	0		Table 7.2-1
Factor l(m) =	150		Table 7.2-1
$g_Q = g_v$ =	3.4		Eq 7.2-6
$I_{\hat{z}} = c(10/\hat{z})^{1/6}$ =	0.23		Eq 7.2-2
$L_{\hat{z}} = l(\hat{z}/10)^{\varepsilon}$ =	150 m		Eq 7.2-4

For wind in X direction

$Q_x = \sqrt{(1/(1+0.63((B+h)/L_{\hat{z}})^{0.63})}$ 0.928 Eq 7.2-3

For wind in Z direction

$Q_Z = \sqrt{(1/(1+0.63((B+h)/L_{\hat{z}})^{0.63})}$ 0.950 Eq 7.2-3

Calculation of Gust effect factor, G :

Wind in X direction :

Horz. Dimension of building measured parallel to wind direction, L =			2 m
Horz. Dimension of building measured normal to wind direction, B =			10 m
$G_x = 0.925(1+1.7g_Q I_{\hat{z}} Q_x)/(1+1.7g_v I_{\hat{z}})$ =	0.887	Eq 7.2-1	

Wind in Z direction :

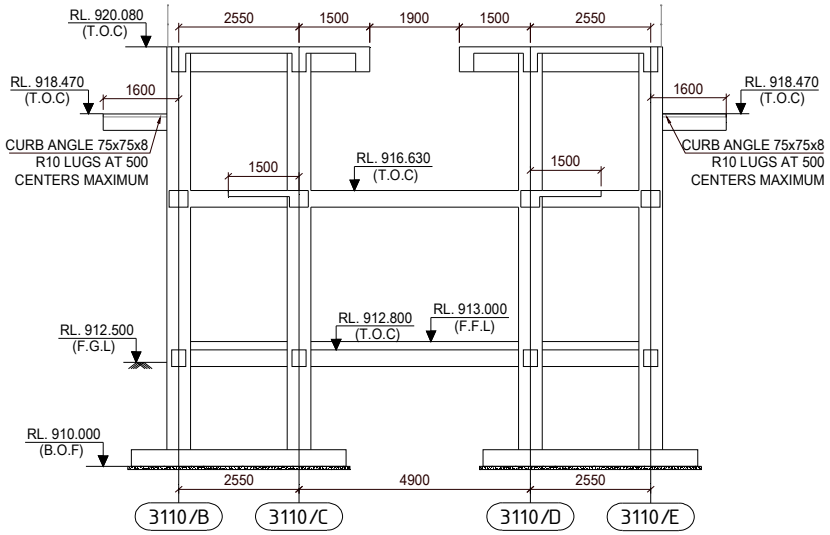
Horz. Dimension of building measured parallel to wind direction, L =			10 m
Horz. Dimension of building measured normal to wind direction, B =			2 m
$G_Z = 0.925(1+1.7g_Q I_{\hat{z}} Q_z)/(1+1.7g_v I_{\hat{z}})$ =	0.899	Eq 7.2-1	

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Calculation of Force Coefficient, C_f :

Wind in X direction



ELEVATION AT GRID - 3110/4

(SCALE 1:100)

along X direction

As the structure is symetrical, consider half of the frame for solidity calculation

Gross area,A_g = 7.08 x (2.55+4.9/2) = 35.4 sqm

Solid area for,

3110/B & 3110/E column = 0.5*7.08+ 0.6*2.55/2 + 0.4*2.55/2 = 4.82 sqm

3110/C & 3110/D column = 0.5*7.08+ 0.6*(2.55/2 + 4.9/2)+ 0.4*(2.55/2 + 4.9/2) 7.27 sqm

Solid area,A_f = 12.1 sqm ε = 12.08/35.4 = 0.34

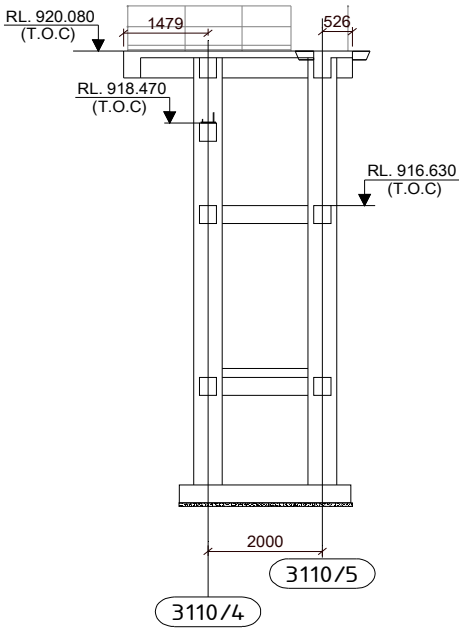
Force coefficient, C_f = 1.6 fig 7.2-17

Wind Load (W+x)						Force on column			Udl/column		
						B & E	C & D		B & E	C & D	
z	K_z	q_z	Gx	Cf	p	kN	kN		kN/m	kN/m	
m		kN/m^2			kN/m^2						
0-5	0.86	0.83	0.89	1.6	1.18	5.68	8.568		0.8021	1.21	
6	0.90	0.87			1.23	5.95	8.9701		0.8397	1.27	
8	0.95	0.92			1.3	6.27	9.465		0.886	1.34	

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Wind in Z direction :



ELEVATION AT GRID - 3110/B
(SCALE 1:100)

Row along Z direction

Gross area,A_g = 7.08 x 2 = 14.16 sqm

Solid area for,

3110/4 column = 0.5*7.08+ 0.6*1 + 0.6*1.479 + 0.4*1 = 5.427 sqm

3110/5 column = 0.5*7.08+ 0.6*1 + 0.6*0.526 + 0.4*1 = 4.856 sqm

Solid area,A_f = 10.3 sqm ε = 10.29/14.16 = 0.73

Force coefficient, C_f = 1.6 fig 7.2-17

Wind Load (W+z)						Force on column			Udl/column for frame		
						4	5		4	5	
z	K_z	q_z	Gz	Cf	p	kN	kN		kN/m	kN/m	
m		kN/m^2			kN/m^2						
0-5	0.86	0.83	0.90	1.6	1.2	6.49	5.8039		0.916	0.8198	
6	0.90	0.87			1.25	6.79	6.0763		0.959	0.8582	
8	0.95	0.92			1.32	7.17	6.4116		1.012	0.9056	

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Seismic parameters :

For Ordinary reinforced concrete moment frames

Site Class :	B	Table 14.1.1
Occupancy Category :	II	Table 1.6-1
Spectral Acceleration, S_s :	0.1 g	
Spectral Acceleration, S_1 :	0.02 g	
Importance Factor ,I :	1	Table 9.5
Response Modification Coefficient ,R :	2.5	Table 10.2

Site Coefficient :

F_a :	1	Table 9.4.3a
F_v :	1	Table 9.4.3b

Maximum Spectral Response Accelerations for Short and 1-second Periods :

$S_{MS} = F_a S_s$:	0.1	Eq 9.4.3-1
$S_{M1} = F_v S_1$:	0.02	Eq 9.4.3-2

Design Spectral Response Accelerations for Short and 1-second Periods :

$S_{DS} = 2/3 S_{MS}$:	0.067	Eq 9.4.4-1
$S_{D1} = 2/3 S_{M1}$:	0.013	Eq 9.4.4-2

Period Coefficient, Ct : 0.044 Table 10.9.3.2

5.2.2 Vibration Check :

For vibrating grizzly screen
Motor power rating : 2 x 11 kW (as per dwg no MD - 216 -3110 - EG- ME -GAD - 1001)
Motor frequency : 60 Hz
Total weight of screen : 8.0 Mton
Operating frequency of screen : 15 Hz

Analysis method : static (as per doc no MD-101-SMEM-EG-ST-CRT-0001)
Additional requirement : weight ratio > 3
Weight of screen supporting structure : 127 Mton (self-weight of structure)
Weight ratio : 127 / 8 = 15.875 ok

Natural frequency of structure : 1.65 cps as per staad output
Remarks : The natural frequency values are found to be with in limit as per ACI 351.3R-18 clause no 6.5.1

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5.2.3. Load Combinations

Primary Load Cases are –

1 Seismic load (E +X)
2 Seismic load (E +Z)
3 Wind Load(W+X)
4 Wind Load(W+Z)
10 Structure Self Weight (D1)
11 Equipment Dead Weight (D2)
12 Live Load(L1)
13 Equipment Live Load(L2)
14 Equipment Reversable Load(L3)
15 Equipment Start & Stop_Condition(L4)

Load Combinations for Allowable Stress Design
SBC 301 clause 2.4.1

Load Case	Load Combination
101	D1 + D2
102	D1 + D2 + L1
103	D1 + D2 + L1 + L2
104	D1 + D2 + L1 + L2 + L3
105	D1 + D2 + L1 + L2 + L4
106	D1 + D2 + L1 + L2 - L3
107	D1 + D2 + L1 + L2 - L4
108	D1 + D2 + W(+X)
109	D1 + D2 + W(-X)
110	D1 + D2 + W(+Z)
111	D1 + D2 + W(-Z)
112	D1 + D2 + 0.7E(+X)
113	D1 + D2 + 0.7E(-X)
114	D1 + D2 + 0.7E(+Z)
115	D1 + D2 + 0.7E(-Z)
116	D1 + D2 + 0.75W(+X) + 0.75L1
117	D1 + D2 + 0.75W(-X) + 0.75L1
118	D1 + D2 + 0.75W(+Z) + 0.75L1
119	D1 + D2 + 0.75W(-Z) + 0.75L1
120	D1 + D2 + 0.75W(+X) + 0.75(L1+L2)
121	D1 + D2 + 0.75W(-X) + 0.75(L1+L2)
122	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2)
123	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2)

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124	D1 + D2 + 0.75W(+X) + 0.75(L1+L2+L3)
125	D1 + D2 + 0.75W(-X) + 0.75(L1+L2+L3)
126	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2+L3)
127	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2+L3)
128	D1 + D2 + 0.75W(+X) + 0.75(L1+L2-L3)
129	D1 + D2 + 0.75W(-X) + 0.75(L1+L2-L3)
130	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2-L3)
131	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2-L3)
132	D1 + D2 + 0.75W(+X) + 0.75(L1+L2+L4)
133	D1 + D2 + 0.75W(-X) + 0.75(L1+L2+L4)
134	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2+L4)
135	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2+L4)
136	D1 + D2 + 0.75W(+X) + 0.75(L1+L2-L4)
137	D1 + D2 + 0.75W(-X) + 0.75(L1+L2-L4)
138	D1 + D2 + 0.75W(+Z) + 0.75(L1+L2-L4)
139	D1 + D2 + 0.75W(-Z) + 0.75(L1+L2-L4)
140	D1 + D2 + 0.75*0.7E(+X) + 0.75(L1+L2)
141	D1 + D2 + 0.75*0.7E(-X) + 0.75(L1+L2)
142	D1 + D2 + 0.75*0.7E(+Z) + 0.75(L1+L2)
143	D1 + D2 + 0.75*0.7E(-Z) + 0.75(L1+L2)
144	0.6(D1 + D2) + W(+X)
145	0.6(D1 + D2) + W(-X)
146	0.6(D1 + D2) + W(+Z)
147	0.6(D1 + D2) + W(-Z)
148	0.6(D1 + D2) + 0.7E(+X)
149	0.6(D1 + D2) + 0.7E(-X)
150	0.6(D1 + D2) + 0.7E(+Z)
151	0.6(D1 + D2) + 0.7E(-Z)

Load Combinations for Strength Design
SBC 301 clause 2.3.2

Load Case	Load Combination
201	1.4(D1 + D2)
202	1.4(D1 + D2) +1.7 L1
203	1.4(D1 + D2) +1.7 (L1 + L2)
204	1.4(D1 + D2) +1.7 (L1 + L2 +L3)
205	1.4(D1 + D2) +1.7 (L1 + L2 +L4)
206	1.4(D1 + D2) +1.7 (L1 + L2 - L3)
207	1.4(D1 + D2) +1.7 (L1 + L2 - L4)

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208	1.2(D1 + D2) + 0.8W(+X)
209	1.2(D1 + D2) + 0.8W(-X)
210	1.2(D1 + D2) + 0.8W(+Z)
211	1.2(D1 + D2) + 0.8W(-Z)
212	1.2(D1 + D2) + 1.6W(+X) + L1
213	1.2(D1 + D2) + 1.6W(-X) + L1
214	1.2(D1 + D2) + 1.6W(+Z) + L1
215	1.2(D1 + D2) + 1.6W(-Z) + L1
216	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2)
217	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2)
218	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2)
219	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2)
220	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2 +L3)
221	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2 +L3)
222	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2 +L3)
223	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2 +L3)
224	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2 - L3)
225	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2 - L3)
226	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2 - L3)
227	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2 - L3)
228	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2 + L4)
229	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2 + L4)
230	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2 + L4)
231	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2 + L4)
232	1.2(D1 + D2) + 1.6W(+X) + (L1 + L2 - L4)
233	1.2(D1 + D2) + 1.6W(-X) + (L1 + L2 - L4)
234	1.2(D1 + D2) + 1.6W(+Z) + (L1 + L2 - L4)
235	1.2(D1 + D2) + 1.6W(-Z) + (L1 + L2 - L4)
236	1.2(D1 + D2) + 1.6E(+X) + (L1 + L2)
237	1.2(D1 + D2) + 1.6E(-X) + (L1 + L2)
238	1.2(D1 + D2) + 1.6E(+Z) + (L1 + L2)
239	1.2(D1 + D2) + 1.6E(-Z) + (L1 + L2)
240	0.9(D1 + D2) + 1.6W(+X)
241	0.9(D1 + D2) + 1.6W(-X)
242	0.9(D1 + D2) + 1.6W(+Z)
243	0.9(D1 + D2) + 1.6W(-Z)
244	0.9(D1 + D2) + 1.6E(+X)
245	0.9(D1 + D2) + 1.6E(-X)
246	0.9(D1 + D2) + 1.6E(+Z)
247	0.9(D1 + D2) + 1.6E(-Z)

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5.3. Design loads for Hopper Supporting Structure

As per STAAD_Pro support reaction from Hopper supporting structure.

5.3.1. Load Calculations

Dead Load (DL1) :

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	1	-13.17	-36.8	-12.974	0	0	0
2	1	-1.81	-508.542	-81.606	0	0	0
3	1	8.064	-15.674	0	0	0	0
4	1	0.115	-451.207	-2.891	0	0	0
5	1	-5.16	-24.176	12.184	0	0	0
6	1	3.443	-500.444	76.788	0	0	0
7	1	-0.062	-6.762	-2.618	0	0	0
8	1	8.579	-435.805	10.567	0	0	0
9	1	-13.088	-28.329	-0.029	0	0	0
10	1	13.087	-104.883	0.578	0	0	0

Equipment Dead Load (DL2) :

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	2	85.004	10.743	-93.184	0	0	0
2	2	53.615	-296.827	-91.87	0	0	0
3	2	87.811	-95.998	0	0	0	0
4	2	1.532	-229.333	-23.471	0	0	0
5	2	66.101	-59.537	107.416	0	0	0
6	2	45.132	-440.599	163.982	0	0	0
7	2	0.134	51.669	18.219	0	0	0
8	2	141.196	-283.579	55.05	0	0	0
9	2	-4.059	-14.625	0.361	0	0	0
10	2	4.06	-46.716	0.544	0	0	0

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Live Load (LL1) :

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	3	-2.642	-45.625	-34.613	0	0	0
2	3	-0.904	-141.465	-35.967	0	0	0
3	3	2.286	-15.925	0	0	0	0
4	3	0.05	-73.04	-5.356	0	0	0
5	3	-3.42	-51.437	26.746	0	0	0
6	3	0.216	-157.559	47.672	0	0	0
7	3	-0.006	-54.458	-3.153	0	0	0
8	3	4.419	-137.521	4.398	0	0	0
9	3	-6.961	-37.475	0.075	0	0	0
10	3	6.961	-42.256	0.198	0	0	0

Material Load (LL2) :

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	4	-5.236	-16.661	-9.19	0	0	0
2	4	0.223	-423.721	-91.227	0	0	0
3	4	10.046	-10.662	0	0	0	0
4	4	1.18	-1068.787	12.427	0	0	0
5	4	-0.772	-13.281	11.058	0	0	0
6	4	1.653	-405.012	101.471	0	0	0
7	4	-0.181	-21.365	-5.51	0	0	0
8	4	-11.34	-865.35	-19.727	0	0	0
9	4	-10.56	-1.623	-0.273	0	0	0
10	4	10.56	-112.969	0.972	0	0	0

Wind load WL+X

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	5	52.456	77.644	0.685	0	0	0
2	5	32.136	95.43	17.728	0	0	0
3	5	44.942	-44.535	0	0	0	0
4	5	2.817	-125.959	-0.727	0	0	0
5	5	35.489	49.672	-0.046	0	0	0
6	5	20.138	138.796	-8.248	0	0	0
7	5	2.77	-99.604	0.793	0	0	0
8	5	68.278	-100.045	-10.45	0	0	0
9	5	7.437	-15.629	0.227	0	0	0
10	5	6.866	24.23	0.039	0	0	0

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Wind load WL+Z

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	6	20.795	29.217	0.898	0	0	0
2	6	9.735	249.057	80.534	0	0	0
3	6	9.963	-12.364	1.246	0	0	0
4	6	0.058	91.901	8.224	0	0	0
5	6	-12.375	-16.006	1.241	0	0	0
6	6	-3.753	-276.732	88.62	0	0	0
7	6	0.027	159.605	55.638	0	0	0
8	6	-24.45	172.513	91.038	0	0	0
9	6	-33.944	-132.641	1.377	0	0	0
10	6	33.943	-264.55	0.604	0	0	0

Temperature Load (TL) :

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	7	-43.214	-45.001	-0.445	0	0	0
2	7	-3.912	41.156	-73.07	0	0	0
3	7	26.576	-26.594	-0.001	0	0	0
4	7	1.515	8.628	-16.162	0	0	0
5	7	-60.711	-73.021	0.759	0	0	0
6	7	-15.394	117.33	39.798	0	0	0
7	7	-0.02	70.686	18.336	0	0	0
8	7	95.16	-20.478	25.404	0	0	0
9	7	-91.77	-29.299	0.771	0	0	0
10	7	91.771	-43.406	4.609	0	0	0

EL+X

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	11	6.75	10.464	0.189	0	0	0
2	11	3.905	14.619	2.434	0	0	0
3	11	5.691	-5.944	0	0	0	0
4	11	0.151	-18.742	-0.119	0	0	0
5	11	4.051	6.055	-0.078	0	0	0
6	11	2.026	20.45	-1.178	0	0	0
7	11	-0.004	-14.894	-0.029	0	0	0
8	11	8.23	-13.33	-1.259	0	0	0
9	11	0.248	-3.735	0.033	0	0	0
10	11	0.14	5.057	0.006	0	0	0

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EL+Z

Node	L/C	Horizontal	Vertical	Horizontal	Moment		
		Fx	Fy	Fz	Mx	My	Mz
		kN			kN-m		
1	12	1.914	3.677	0.566	0	0	0
2	12	0.946	20.818	7.932	0	0	0
3	12	1.143	-1.283	0	0	0	0
4	12	0.015	6.039	0.729	0	0	0
5	12	-1.199	-2.555	0.565	0	0	0
6	12	-0.44	-22.775	8.307	0	0	0
7	12	0.002	12.743	4.468	0	0	0
8	12	-2.382	16.875	8.717	0	0	0
9	12	-2.818	-10.865	0.029	0	0	0
10	12	2.818	-22.675	0.012	0	0	0

5.3.2. Load Combinations

Primary Load Cases are –

1 Dead Load (DL1)
2 Equipment Dead Load (DL2)
3 Live Load(LL1)
4 Material Load (LL2)
5 Wind Load(WL+X)
6 Wind Load(WL+Z)
11 Seismic load (EL +X)
12 Seismic load (EL +Z)

Load Combinations for Strength Design

SBC 301 clause 2.3.2

Load Case	Load Combination
101	1.4(DL1 + DL2)
102	1.4(DL1 + DL2 + TL) +1.7 (LL1 + LL2)
103	1.2(DL1 + DL2) + 0.8WL(+X)
104	1.2(DL1 + DL2) + 0.8WL(-X)
105	1.2(DL1 + DL2) + 0.8WL(+Z)
106	1.2(DL1 + DL2) + 0.8WL(-Z)
107	1.2 (DL1+DL2)+1.6WL(+X)+0.5LL1+1.0LL2
108	1.2 (DL1+DL2)+1.6WL(-X)+0.5LL1+1.0LL2
109	1.2 (DL1+DL2)+1.6WL(+Z)+0.5LL1+1.0LL2
110	1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2

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111	1.2 (DL1+DL2)+1.6EL(+X)+0.5LL1+1.0LL2
112	1.2 (DL1+DL2)+1.6EL(-X)+0.5LL1+1.0LL2
113	1.2 (DL1+DL2)+1.6EL(+Z)+0.5LL1+1.0LL2
114	1.2 (DL1+DL2)+1.6EL(-Z)+0.5LL1+1.0LL2
115	0.9 (DL1+DL2)+1.6WL(+X)
116	0.9 (DL1+DL2)+1.6WL(-X)
117	0.9 (DL1+DL2)+1.6WL(+Z)
118	0.9 (DL1+DL2)+1.6WL(-Z)
119	0.9 (DL1+DL2)+1.6EL(+X)
120	0.9 (DL1+DL2)+1.6EL(-X)
121	0.9 (DL1+DL2)+1.6EL(+Z)
122	0.9 (DL1+DL2)+1.6EL(-Z)

Load Combinations for Allowable Stress Design
SBC 301 clause 2.4.1

Load Case	Load Combination
201	DL1+DL2
202	DL1+DL2+LL1+LL2+TL
203	DL1+DL2+0.75LL1+0.75LL2+0.75TL
204	DL1+DL2+WL(+X)
205	DL1+DL2+WL(-X)
206	DL1+DL2+WL(+Z)
207	DL1+DL2+WL(-Z)
208	DL1+DL2+0.7EL(+X)
209	DL1+DL2+0.7EL(-X)
210	DL1+DL2+0.7EL(+Z)
211	DL1+DL2+0.7EL(-Z)
212	DL1+DL2+0.75WL(+X)+0.75LL1+0.75LL2
213	DL1+DL2+0.75WL(-X)+0.75LL1+0.75LL2
214	DL1+DL2+0.75WL(+Z)+0.75LL1+0.75LL2
215	DL1+DL2+0.75WL(-Z)+0.75LL1+0.75LL2
216	DL1+DL2+0.75*0.7EL(+X)+0.75LL1+0.75LL2
217	DL1+DL2+0.75*0.7EL(-X)+0.75LL1+0.75LL2
218	DL1+DL2+0.75*0.7EL(+Z)+0.75LL1+0.75LL2
219	DL1+DL2+0.75*0.7EL(-Z)+0.75LL1+0.75LL2
220	0.6DL1+0.6DL2+WL(+X)
221	0.6DL1+0.6DL2+WL(-X)
222	0.6DL1+0.6DL2+WL(+Z)
223	0.6DL1+0.6DL2+WL(-Z)
224	0.6DL1+0.6DL2+0.7EL(+X)
225	0.6DL1+0.6DL2+0.7EL(-X)
226	0.6DL1+0.6DL2+0.7EL(+Z)
227	0.6DL1+0.6DL2+0.7EL(-Z)

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6. DESIGN CALCULATION OF REINFORCEMENT

6.1. Design calculation for crusher supporting structure

DESIGN OF TWO WAY SLAB AT CRUSHER FLOOR LEVEL:

A) Input Data :

Specified compressive strength of concrete, f_c'	=	35	Mpa
Specified yield strength of reinforcement, f_y	=	420	Mpa
Nornal weight concrete, λ	=	1	
Unit weight of concrete, γ_{conc}	=	24	KN/m^3
Effective span in longer direction (l_y)	=	4500	mm
Effective span in shorter direction (S)	=	2550	mm

B) Design Consideration :

Dia of main reinforcement in short direction	=	10	mm
Dia of main reinforcement in long direction	=	10	mm
Consider overall depth, h	=	200	mm
Clear cover to main reinforcement	=	50	mm
Effective depth in shorter direction, d_s	=	145	mm
Effective depth in longer direction, d_{ly}	=	135	mm

C) Load Data :

i)Live Load intensity w_{LL} = 5 KN/m^2

ii) Dead load calculation :

Total Dead load intensity = w_{DL} = 4.8 KN/m^2

Total load intensity, w = w_{LL} + w_{DL} = 9.80 KN/m^2

D) Structural Design :

TWO EDGES DISCONTINUOUS — Table B1,Appendix B,SBC-304:2018

Ratio of short span to long span ,m = S/L_y	=	0.57 two way slab
Bending Moment Co-efficient, C :		
Short Span continuous edge(-ve)	=	0.080
Short Span discontinuous edge(+ve)	=	0.041
Short Span midspan(+ve)	=	0.062
Long Span continuous edge(-ve)	=	0.049
Long Span discontinuous edge(+ve)	=	0.025
Long Span midspan(+ve)	=	0.037
Partial safety factor ,Y_m	=	1.4

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i)Bending moment due to Dead Load + Live load : [$M = CwS^2$]

Short Span continuous edge(-ve) =	1.4	*	0.080	*	9.800	*	2.55	^2	=	7.15	KN-m
Short Span discontinuous edge(+ve) =	1.4	*	0.041	*	9.800	*	2.55	^2	=	3.68	KN-m
Short Span midspan(+ve) =	1.4	*	0.062	*	9.800	*	2.55	^2	=	5.51	KN-m
Long Span continuous edge(-ve) =	1.4	*	0.049	*	9.800	*	2.55	^2	=	4.37	KN-m
Long Span discontinuous edge(+ve) =	1.4	*	0.025	*	9.800	*	2.55	^2	=	2.23	KN-m
Long Span midspan(+ve) =	1.4	*	0.037	*	9.800	*	2.55	^2	=	3.3	KN-m

E) Check for flexure :-

Factored moment in short direction , M_{uS} = 7.147 KN-m

β_1 = 0.8 steel strain ε_t = 0.005 and φ = 0.9

now, $C = 0.85 f_c' b a$; $T = A_s f_y$; $M_u = \varphi M_n = \varphi f_y A_s (d_s - a/2)$; $C = T$

$C = 0.85 * 35 * 1000 * a$; $T = 420 * A_s$; $a = 0.014 * A_s$

$M_{uS} = 0.9 * 420 * (145 - 0.014 A_s/2) * A_s = 7.15 * 10^6$

$A_{s\ required}$ = 131 mm^2

$A_{s,min}$ = 0.2 / 100 * 1000 * 145 = 261 mm^2 [Table 8.6.1.1 SBC 304:18]

Spacing required = 79 / 261 * 1000 = 302.7 mm c/c

Maximum spacing = = 400 mm c/c [cl 8.7.2.2 SBC 304:2018]

provide 10 φ main bar @ 200 mm c/c A_s provided = 395.0 mm^2

Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c (d_s - c)$ c = 0.014 * 395.0 / 0.800 = 6.97 mm

ε_t = 0.003 *(145 - 6.97)/ 6.97 = 0.059 > 0.005 tension controlled

Factored moment in long direction , $M_{u,Ly}$ = 4.372 KN-m

$M_{u,Ly} = 0.9 * 420 * (135 - 0.014 A_s/2) * A_s = 4.37 * 10^6$

$A_{s\ required}$ = 86 mm^2

$A_{s,min}$ = 0.2 / 100 * 1000 * 135 = 243 mm^2 [Table 8.6.1.1 SBC 304:18]

Spacing required = 79 / 243 * 1000 = 325.1 mm c/c

provide 10 φ main bar @ 200 mm c/c A_s provided = 395 mm^2

Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c (d_{ly} - c)$ c = 0.014 * 395.0 / 0.800 = 6.97 mm

ε_t = 0.003 *(135 - 6.97)/ 6.97 = 0.055 > 0.005 tension controlled

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F) Check for shear :

Shear force at a distance d from supporting beam

Effective depth in shorter direction ,d_s= 145 mm

UDL along shorter direction considering 1m width, w= 9.80 KN/m

Factored shear,V_u=1.4w*(0.5S-d_s)= 1.4 * 9.8 *(0.5 * 2.55 - 0.145)= 15.5 KN

$V_{c\,1}$=(0.16 λ $\sqrt{f_c'}$+17 ρ_w V_u d_s /M_{us}) $b_w d_s$

$V_{c\,2}$ = (0.16 λ $\sqrt{f_c'}$ +17 ρ_w) b_w d_s

$V_{c\,3}$ = 0.29 λ $\sqrt{f_c'}$ b_w d_s

V_c = Minimum of $V_{c\,1}$, $V_{c\,2}$ and $V_{c\,3}$

A_s=	395.0 mm²	M_{us}=	7.147 KN-m	=	139 KN
ρ_w=	0.0027	b_w =	1000 mm	=	144 KN
λ=	1	d_s =	145 mm	=	249 KN
[table 22.5.5.1 SBC 304 : 2018]				=	139 KN

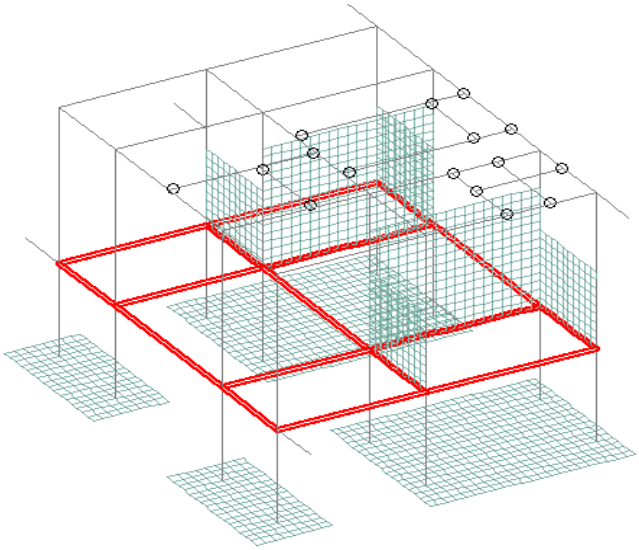
$V_u \leq \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$ = 529.2 KN ok [cl 22.5.5.1 SBC 304 : 2018]

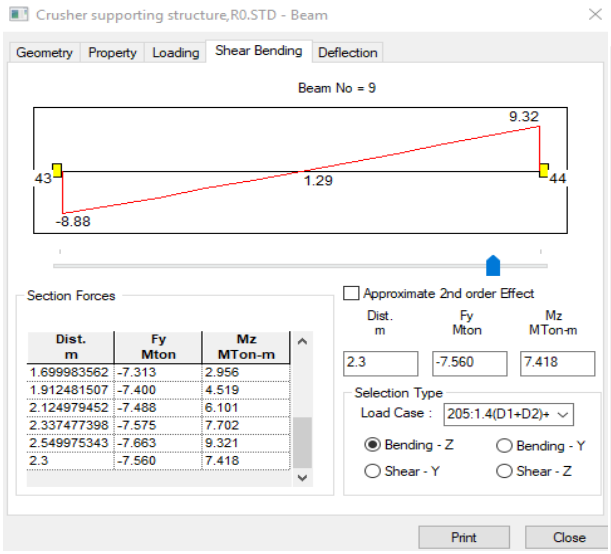
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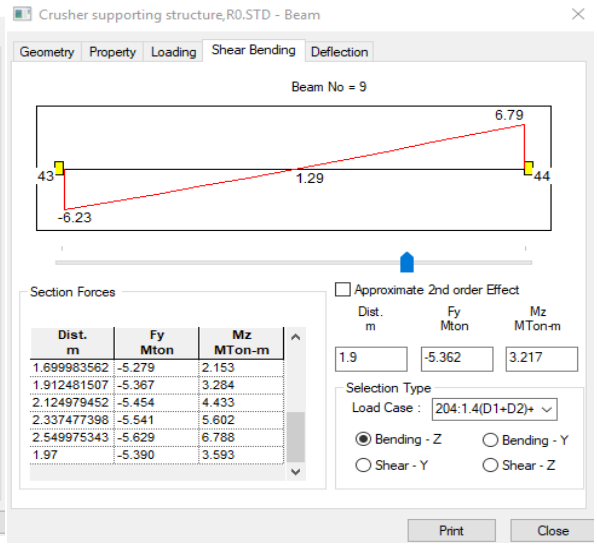
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Design of beam : TB1







Moment at the face of column

Shear at the distance d from face of column

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Design of beam = TB1

Refer staad file : Crusher Supporting Structure, R2

Beam End Forces : G8: _TB_1									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	2991	3161	10.7	-10.7	-0.21	0.249	0.06	1.086	224 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-ELL_RI
Min Fx	6	39	-18.1	2.164	0.585	-0.38	-0.15	-0.86	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Fy	4	37	8.52	9.168	0.27	-0.23	-0.04	3.193	225 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-ELL_RE
Min Fy	2991	40	8.93	-10.8	-0.26	0.293	-0.01	3.66	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Fz	3112	3262	-6.44	-2.18	0.895	0.013	0.226	-0.57	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Min Fz	2991	3161	4.17	-6.84	-0.73	0.266	-0.15	0.361	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Max Mx	15	39	-5.11	4.282	-0.15	0.403	-0.07	0.489	224 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-ELL_RI
Min Mx	6	39	-11.7	2.62	0.403	-0.46	0.045	-0.2	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max My	3112	42	-6.44	-2.27	0.895	0.013	0.455	-0	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Min My	4	37	2.24	4.061	0.828	-0.15	-0.43	1.255	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Max Mz	12	34	-0.72	2.756	0.039	-0.06	-0.07	4.727	223 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-ELL_RE
Min Mz	12	38	-0.72	1.382	0.039	-0.06	0.087	-3.34	223 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-ELL_RE

Max support Top Moment = 8 t-m at the face of the beam

Max support Bottom Moment = 8 t-m at the face of the beam

Max Shear = 6 t Corresponding Moment = 3.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	400	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 8.00 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 674 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 673.7 mm^2 ok

Max support Bottom Moment = 8 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 674 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 673.7 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 6 t Corresponding Moment, Mu = 3.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 10.28 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 10.97 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 16.99 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 7.708 t
Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.26* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.25* mm^2/m

Provide 2L-10dia stirrups @150 mm c/c

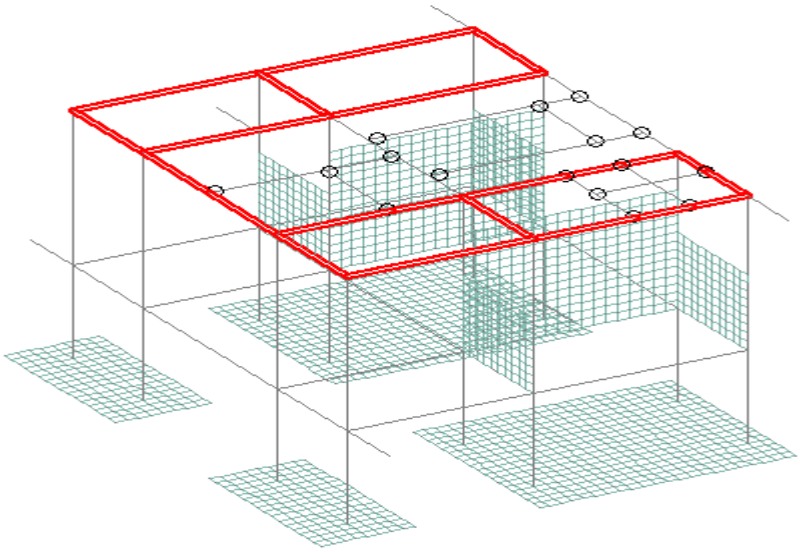
therefore, A_v/s provided = 1.05 mm^2/m ok

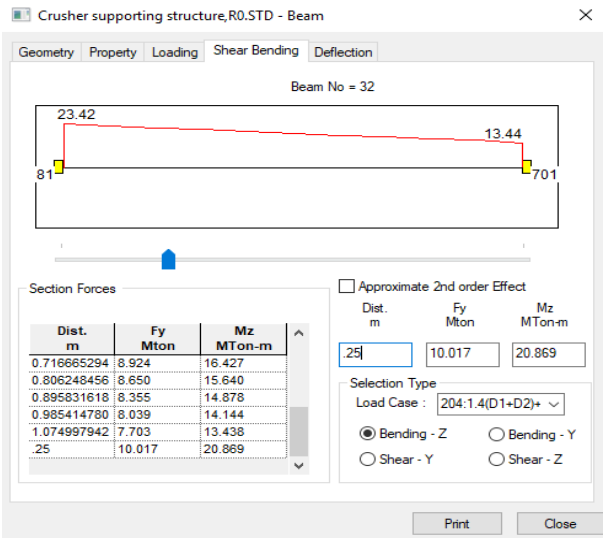
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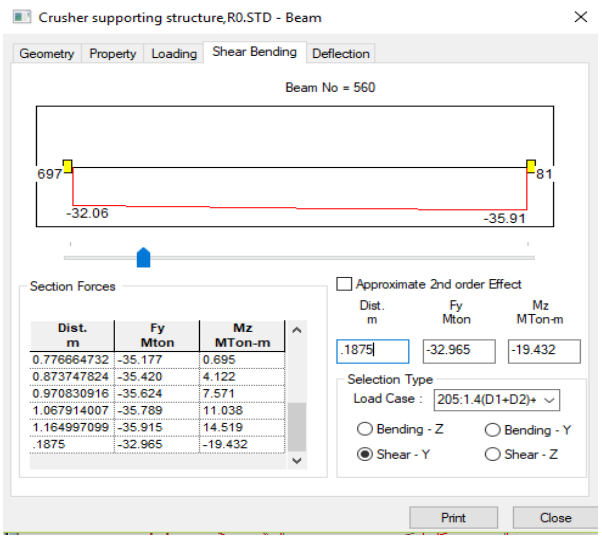
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Design of beam : FB1





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB1

Refer staad file : Crusher Supporting Structure, R1

Beam End Forces : G4: _FB_1									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	30	78	6.62	-0.29	-0.15	-1.43	0.225	1.789	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV
Min Fx	30	78	-0.4	1.384	0.097	-0.36	-0.13	5.361	233 0.9(D1+D2) + 1.6W(-Z)
Max Fy	40	77	2.41	16.85	-0.24	-1.23	0.416	8.582	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Min Fy	560	81	4.64	-35.9	-1.44	-1.56	-2	14.52	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV
Max Fz	27	73	3.17	6.323	2.362	1.471	-3.47	6.21	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Min Fz	35	70	1.88	11.88	-2.02	-3.39	4.071	7.508	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Max Mx	37	71	2.81	15.55	-0.82	4.012	1.749	15.37	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV
Min Mx	35	70	2.99	15.32	0.06	-4.29	-0.07	13.96	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV
Max My	35	70	1.88	11.88	-2.02	-3.39	4.071	7.508	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Min My	35	74	1.88	-2.67	-2.02	-3.39	-3.79	-10.5	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Max Mz	32	81	3.11	10.37	-0.39	-1.02	0.49	23.42	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV
Min Mz	563	699	3.07	-1	0.289	-1.56	-0.66	-27.9	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV

Max support Top Moment = 21 t-m at the face of the beam

Max support Bottom Moment = -27.9 t-m at the face of the beam

Max Shear = 33 t Corresponding Moment = 20 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete,f_c' =	35	Mpa	
Specified yield strength of main reinforcement,f_y =	420	Mpa	
Yield strength of transverse reinforcement,f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups,d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 21.00 t-m
Effective depth ,d = 727.5 mm
Area of steel required ,Ast,required = 778 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 1025 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 970 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer

Ast,provided = 1473 mm^2 > 1025 mm^2 ok

Max support Bottom Moment = 27.88 t-m
Effective depth ,d = 727.5 mm
Area of steel required ,Ast,required = 1040 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 1025 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 970 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer

Ast,provided = 1473 mm^2 > 1040 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 33 t Corresponding Moment, Mu = 20 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 30.55 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 30.05 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 49.93 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 22.54 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 13.95 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.457 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.35* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.33* mm^2/m

Provide 2L-10dia stirrups @200 mm c/c

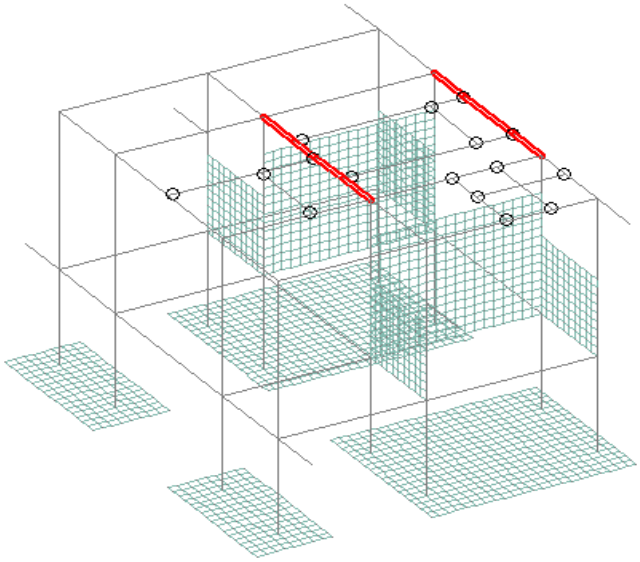
therefore, A_v/s provided = 0.79 mm^2/m ok

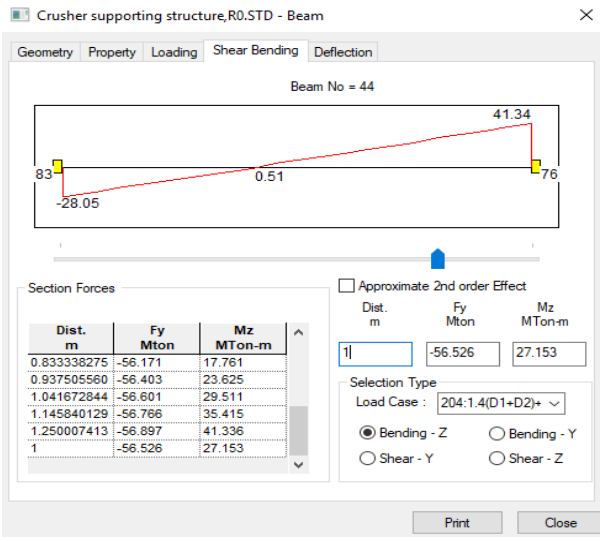
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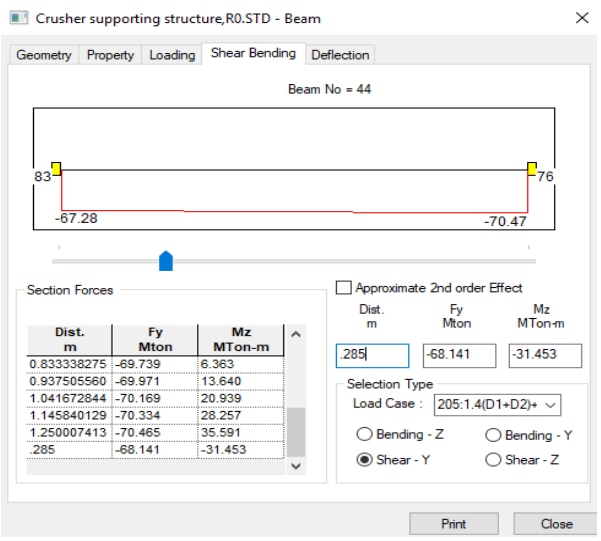
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Design of beam : FB1A





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB1A

Refer staad file : Crusher Supporting Structure, R2

Beam End Forces : G9: _FB_1A									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	31	79	13.7	33.15	-1.06	-1.12	0.241	9.857	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fx	42	80	0.23	3.664	-0.06	-0.23	0.092	-4.57	233 0.9(D1+D2) + 1.6W(-Z)
Max Fy	28	74	11.5	65.64	-1.51	0.579	0.553	34.28	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fy	44	76	10	-70.5	2.866	0.579	2.857	35.59	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Fz	28	74	6.83	40.17	3.307	-0.08	-4.24	8.007	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Fz	28	74	11.5	65.64	-1.51	0.579	0.553	34.28	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mx	28	74	11.5	65.64	-1.51	0.579	0.553	34.28	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	31	79	13.7	33.15	-1.06	-1.12	0.241	9.857	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	44	76	10	-70.5	2.866	0.579	2.857	35.59	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	28	74	6.83	40.17	3.307	-0.08	-4.24	8.007	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Mz	44	76	8.17	-56.9	-0.96	-0.08	1.014	41.34	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Mz	41	112	10.8	5.388	1.108	0.579	-0.53	-63.2	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

Max support Top Moment = 33 t-m at the face of the beam

Max span Bottom Moment = -63.2 t-m at the face of the beam

Max Shear = 68.5 t Corresponding Moment = 31.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 33.00 t-m
Effective depth ,d = 725.5 mm
Area of steel required ,Ast,required = 1241 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 1022 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 967.3 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer

Ast,provided = 1473 mm^2 > 1241 mm^2 ok

Max span Bottom Moment = 63.16 t-m
Effective depth ,d = 700.5 mm
Area of steel required ,Ast,required = 2549 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 986.7 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 934 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer
Provide 3 nos 25 dia at 2nd layer
Ast,provided = 2946 mm^2 > 2549 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 68.5 t Corresponding Moment, Mu = 31.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 30.88 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 29.5 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 48.93 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 22.13 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 61.83 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 2.065 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.35* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.33* mm^2/m

Provide 2L-12dia stirrups @100 mm c/c

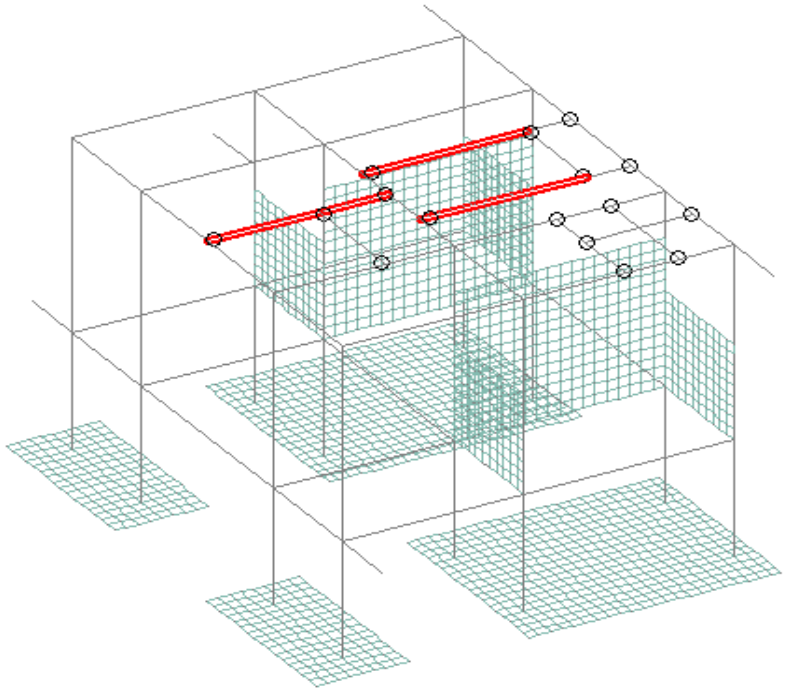
therefore, A_v/s provided = 2.26 mm^2/m ok

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Design of beam : FB2



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Design of beam = FB2

Refer staad file : Crusher Supporting Structure, R2

Beam End Forces : G5: _FB_2									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	46	83	3.03	54.28	0.639	-1.73	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fx	43	75	-3.11	40.37	-0.65	1.108	0	0	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Fy	43	75	2.62	54.42	0.728	1.401	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fy	43	1765	-1.54	-28.6	-0.31	1.401	0.496	-28.9	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Fz	43	75	2.62	54.42	0.728	1.401	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fz	46	83	-2.17	40.08	-0.71	-0.99	0	0	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Mx	43	75	2.62	54.42	0.728	1.401	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	46	83	3.03	54.28	0.639	-1.73	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	43	1765	-1.54	-28.6	-0.31	1.401	0.496	-28.9	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	46	1766	1.99	-21	0.327	-0.99	-0.44	-21.8	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Mz	43	75	0.01	17.69	0.01	0.423	0	0	201 1.4(D1+D2)
Min Mz	46	1766	-1.13	-28.3	-0.4	-1.73	0.181	-29.1	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

Max support Top Moment = 0 t-m at the face of the beam

Max span Bottom Moment = -29.1 t-m at the face of the beam

Max Shear = 54.42 t Corresponding Moment = 0.00 t-m

Design Input :

Specified compressive strength of concrete,f_c' =	35	Mpa	
Specified yield strength of main reinforcement,f_y =	420	Mpa	
Yield strength of transverse reinforcement,f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups,d_{tie} =	12	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

At top of beam
Provide 3 nos 16 dia bar althrough at 1st layer

Max span Bottom Moment = 29.09 t-m
Effective depth ,d = 705.5 mm
Area of steel required ,Ast,required = 1122 mm^2 As = $M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 993.8 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 940.7 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer
Provide 3 nos 20 dia at 2nd layer
Ast,provided = 1886 mm^2 > 1122 mm^2 ok

Design for shear reinforcement :
Max shear, Vu = 54.42 t Corresponding Moment, Mu = 0.001 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 40781 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 28.67 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 50.1 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 21.5 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 43.9 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 1.432 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.35* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.33* mm^2/m

Provide 2L-12dia stirrups @150 mm c/c

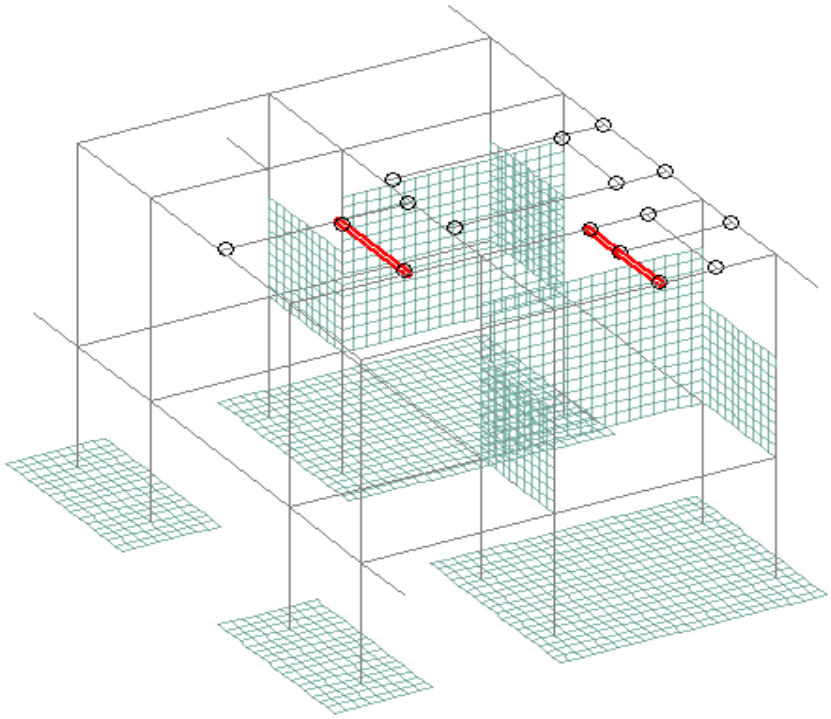
therefore, A_v/s provided = 1.51 mm^2/m ok

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Design of beam : FB3



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Design of beam = FB3

Refer staad file : Crusher Supporting Structure, R2

Beam End Forces : G6: _FB_3									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	568	703	0.52	2.745	0.067	0.092	-0.1	-1.22	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fx	565	699	-1.61	10.67	-0.09	0.092	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Fy	573	705	0.21	14.45	-0.6	-0.8	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Fy	568	700	0.08	-13	0.022	2.384	0	0	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Fz	573	705	0.04	-2.99	0.604	-0.79	0	0	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Fz	573	705	0.21	14.45	-0.6	-0.8	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mx	565	699	-0.54	6.158	-0.03	2.384	0	0	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Mx	573	705	0.21	14.45	-0.6	-0.8	0	0	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	565	703	-0.48	2.917	0.015	0.274	0.017	-4.18	231 0.9(D1+D2) + 1.6W(-X)
Min My	565	703	-0.56	5.363	-0.11	0.185	-0.12	-7.58	222 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-ELL_R
Max Mz	565	699	-0.1	7.06	-0.05	0.444	0	0	201 1.4(D1+D2)
Min Mz	568	703	0.08	-7.18	0.022	2.384	-0.03	-15.9	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)

Max support Top Moment = 0 t-m at the face of the beam

Max span Bottom Moment = -15.9 t-m at the face of the beam

Max Shear = 14.45 t Corresponding Moment = 0.00 t-m

Design Input :

Specified compressive strength of concrete,f_c' =	35	Mpa	
Specified yield strength of main reinforcement,f_y =	420	Mpa	
Yield strength of transverse reinforcement,f_{yt} =	420	Mpa	
Beam width,b =	400	mm	
Beam depth,h =	800	mm	
Cover =	50	mm	
Dia of stirrups,d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.035	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

At top of beam
Provide 3 nos 16 dia bar althrough at 1st layer

Max span Bottom Moment = 15.85 t-m
Effective depth ,d = 707.5 mm
Area of steel required ,Ast,required = 602 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 996.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 943.3 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer
Provide 3 nos 20 dia at 2nd layer
Ast,provided = 1886 mm^2 > 996.6 mm^2 ok

Design for shear reinforcement :
Max shear, Vu = 14.45 t Corresponding Moment, Mu = 0.00 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 10876 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 28.74 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 50.23 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 21.56 t
Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.35* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.33* mm^2/m

Provide 2L-10dia stirrups @200 mm c/c

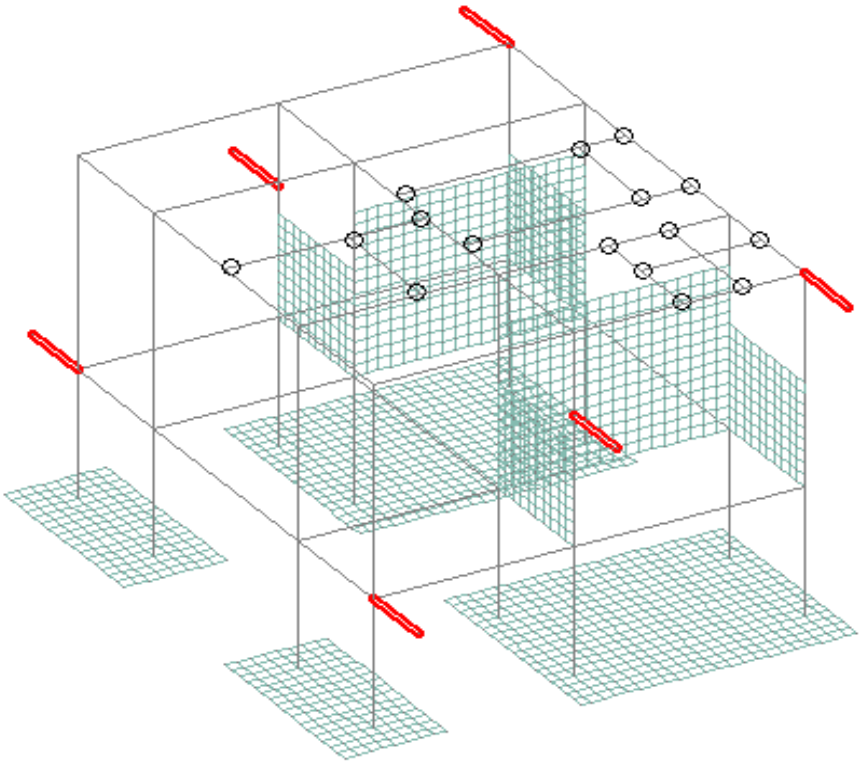
therefore, A_v/s provided = 0.79 mm^2/m ok

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Design of beam : FB4



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Design of beam = FB4

Refer staad file : Crusher Supporting Structure, R2

Beam End Forces : G7: _FB_4

	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	127	82	0	2.52	0	0	0	2.27	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Fx	125	119	0	-0.61	0	0	0	0	214 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)
Max Fy	126	116	0	4.169	0	0	0	3.837	202 1.4(D1+D2)+1.7L1
Min Fy	574	115	0	-4.17	0	0	0	3.837	202 1.4(D1+D2)+1.7L1
Max Fz	574	706	0	-1.13	0	0	0	0	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Min Fz	125	119	0	-0.15	0	0	0	0	206 1.2(D1+D2)+0.8W(+X)
Max Mx	574	706	0	-1.13	0	0	0	0	211 1.2(D1+D2)+1.6W(-X) + 1.0L1
Min Mx	127	82	0	1.74	0	0	0	1.547	214 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)
Max My	574	115	0	-2.79	0	0	0	2.545	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Min My	574	706	0	-1.13	0	0	0	0	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R
Max Mz	126	116	0	4.169	0	0	0	3.837	202 1.4(D1+D2)+1.7L1
Min Mz	121	113	0	0.171	0	0	0	0	201 1.4(D1+D2)

Max support Top Moment = 3.837 t-m at the face of the beam

Max support Bottom Moment = 0 t-m at the face of the beam

Max Shear = 4.169 t Corresponding Moment = 3.837 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	400	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 3.84 t-m

Effective depth ,d = 332.0 mm

Area of steel required ,Ast,required = 313 mm^2 $As = M_u / \varphi f_y (d - a/2)$

$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 350.7 mm^2 cl 9.6.1.2 SBC 304 : 2018

(b) $1.4/f_y$bd 332 mm^2

Provide 2 nos 16 dia bar althrough at 1st layer

Ast,provided = 402.3 mm^2 > 350.7 mm^2 ok 0.004

Design for shear reinforcement :

Max shear, Vu = 4.169 t Corresponding Moment, Mu = 3.837 t-m

V_c = lesser of table 22.5.5.1 SBC 304 : 2018

a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 9.675 t if $\varphi V_c < V_u$ shear reinforcement required

b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 10.11 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd

c) $0.29\lambda\sqrt{f_c'}bd$ 17.09 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 7.256 t

Remarks : provide minimum shear reinforcement

$V_s = V_u/\varphi - V_c$ - t

$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = - mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.26* mm^2/m table 9.6.3.3 SBC 304 : 2018

(b) $0.35b/f_{yt}$ *0.25* mm^2/m

Provide 2L-10dia stirrups @150 mm c/c

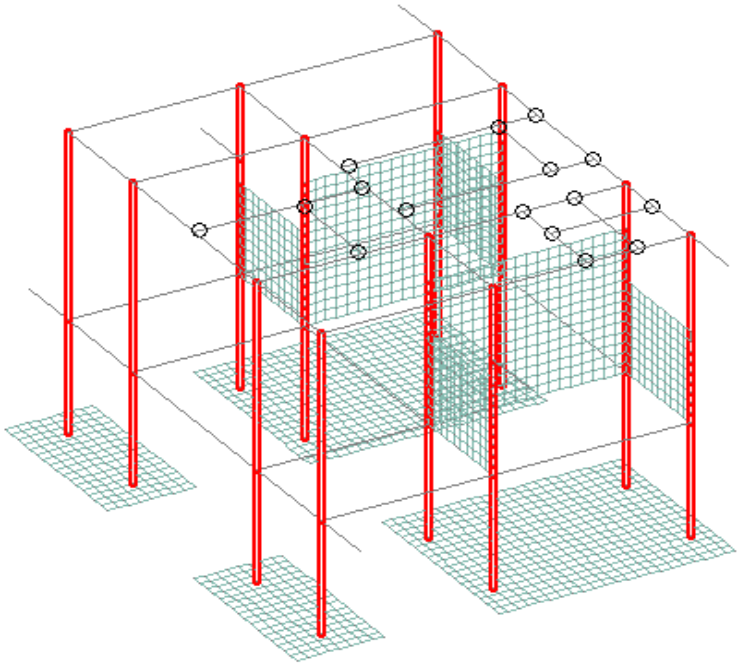
therefore, A_v/s provided = 1.05 mm^2/m ok

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Design of column C1 A03



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Design of column C1 A03

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.5 % i.e. ρ_g = 0.015

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.71

Y_b = (b-d')/b = 0.71

Unsupported length of column parallel to X-X axis, l_{ux} = 3.88 m kl_{ux}/b = 7.76

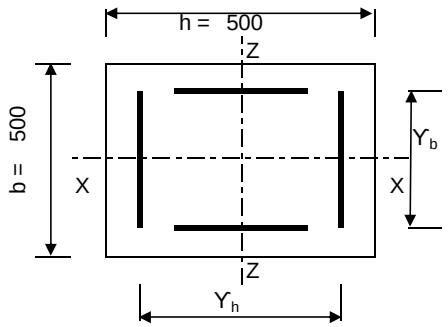
Unsupported length of column parallel to Z-Z axis, l_{uz} = 3.88 m kl_{uz}/h = 7.76

< 22 neglect slenderness effect

$P_{n,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 7177 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
205	1035	62.9	23.3	1593	97	35.8	0.182	0.090	393.8	0.25	0.090	393.8	0.09	1.61	0.13	safe
204	-13.2	27.2	24.2	-20	41.9	37.3	-0	0.090	393.8	0.11	0.090	393.8	0.09		0.05	safe
205	676.3	7.1	1.5	1040	11.0	2.3	0.119	0.090	393.8	0.03	0.090	393.8	0.01		0	safe
205	896.1	21.1	8.8	1379	32.5	13.5	0.158	0.090	393.8	0.08	0.090	393.8	0.03		0.02	safe
205	922.5	61	26.2	1419	93	40.3	0.162	0.090	393.8	0.24	0.090	393.8	0.10		0.12	safe
205	1035	63	23.3	1593	97	35.8	0.182	0.090	393.8	0.25	0.090	393.8	0.09		0.13	safe
205	670.0	34.5	10.8	1031	53.1	16.7	0.118	0.090	393.8	0.13	0.090	393.8	0.04		0.05	safe
205	770.5	43.0	16.5	1185	66.2	25.5	0.135	0.090	393.8	0.17	0.090	393.8	0.06		0.07	safe
205	551.8	112	9.5	849	173	14.6	0.097	0.090	393.8	0.44	0.090	393.8	0.04		0.27	safe
205	1033	116	47.7	1589	178	73	0.182	0.090	393.8	0.45	0.090	393.8	0.19		0.35	safe
218	500.0	9.6	102.4	769	14.8	158	0.088	0.090	393.8	0.04	0.090	393.8	0.40		0.24	safe
223	484.2	34	89.6	745	53	138	0.085	0.090	393.8	0.13	0.090	393.8	0.35		0.22	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3750	4	25	+	4	25	3929	1.57	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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Foundation Base Pressure for CF1 & CF2 : A03

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	1	101	0	92.05	0
Min Px	1	101	0	92.05	0
Max Py	2398	105	0	192.287	0
Min Py	2422	136	0	38.234	0
Max Pz	1	101	0	92.05	0
Min Pz	1	101	0	92.05	0

Gross Maximum Base pressure, p_{max} = 192.3 KN/m^2

Net Allowable Safe Bearing Capacity = 600 KN/m^2
Depth of soil above footing = 2.4 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 648 KN/m^2
> 192 KN/m^2 safe

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Design of Foundation CF1: A03

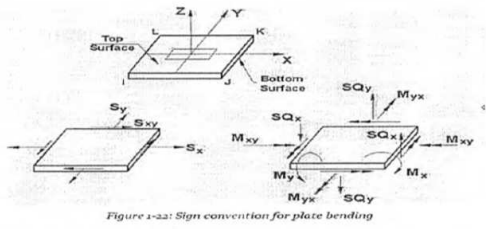
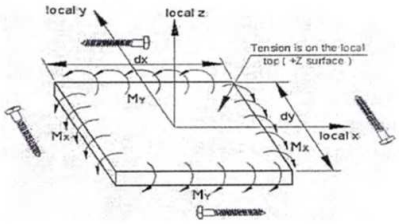


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G10 : CF_1_MOMENT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	983	0.395	-0.055	0	0	0	-14.07	-37.9	-3.712	-17.78	-41.61	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	944	-0.38	-0.05	0	0	0	-12.86	-36.85	0	-12.86	-36.85	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Qy	994	0.016	0.27	0	0	0	-25.2	-15.38	5.432	-30.63	-20.81	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	976	0.042	-0.332	0	0	0	-27.64	-22.3	-4.052	-31.69	-26.36	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sx	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Sy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Sxy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sxy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Mx	1582	0.006	-0.01	0	0	0	29.59	-10.61	-0.514	30.11	-11.13	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	975	-0.083	-0.297	0	0	0	-30.22	-22.24	5.179	-35.4	-27.41	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	2149	0.005	-0.006	0	0	0	-3.117	0.101	0.835	-3.952	0.936	230 0.9(D1+D2) + 1.6W(+X)
Min My	989	0.377	0.093	0	0	0	-13.66	-38.28	5.589	-19.25	-43.87	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mxy	995	0.194	0.193	0	0	0	-9.987	-17.1	10.02	-20	-27.12	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mxy	956	-0.186	0.184	0	0	0	-9.045	-16.49	-10.35	-19.39	-26.83	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

PLATE CENTER STRESS ; G11 : CF_1_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1558	0.129	-0.017	0	0	0	20.67	-16.7	-1.382	22.06	-18.09	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	1588	-0.116	-0.014	0	0	0	20.72	-15.71	0	20.72	-15.71	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Qy	1560	0.103	0.033	0	0	0	18.55	-11.66	4.96	23.51	-16.62	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	2155	-0.006	-0.041	0	0	0	-12.97	-0.9	1.166	-14.13	-2.066	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sx	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Sy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Sxy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Min Sxy	2172	-0.021	-0.007	0	0	0	15.32	-0.814	0.417	15.74	-1.231	201 1.4(D1+D2)
Max Mx	1582	0.006	-0.01	0	0	0	29.59	-10.61	-0.514	30.11	-11.13	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	2155	-0.006	-0.041	0	0	0	-12.97	-0.9	1.166	-14.13	-2.066	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	2149	0.005	-0.006	0	0	0	-3.117	0.101	0.835	-3.952	0.936	230 0.9(D1+D2) + 1.6W(+X)
Min My	1558	0.129	-0.017	0	0	0	20.67	-16.7	-1.382	22.06	-18.09	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mxy	2070	0.078	0.032	0	0	0	16.53	-5.673	5.229	21.76	-10.9	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mxy	2075	-0.07	0.029	0	0	0	16.86	-5.395	-5.392	22.25	-10.79	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

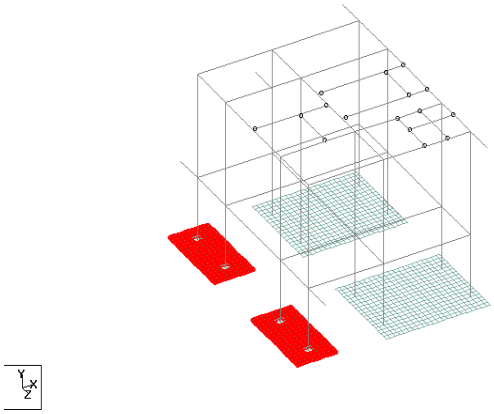
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	30.11	top face	50	342	30.11		234.0		16	279.37	200	1005.8	14.18	17.73	0.055	tension control
Min Mx	-35.40	bottom face	75	317	35.4	0.014	297.4	720	16	279.37	200	1005.8	14.18	17.73	0.051	tension control
Max Mxy	-20.00	bottom face	75	317	20		167.6		16	279.37	200	1005.8	14.18	17.73	0.051	tension control
Min Mxy	-19.39	bottom face	75	317	19.39	*As	162.4		16	279.37	200	1005.8	14.18	17.73	0.051	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.129	1	51.61	22.06	0.003	0.342	337.4	340.8	586.8	337.41	1255	> 51.611	ok	253.1	φVc>Vu	not required
	Min Qx	0.116	1	46.45	20.72			336.8			336.84	1254	> 46.445	ok	252.6	φVc>Vu	not required

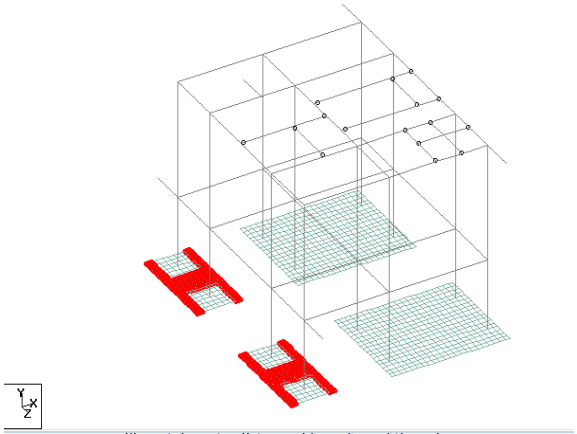
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	0.94	top face	50	342	0.936		7.2		16	279.37	200	1005.8	14.18	17.73	0.055	tension control
Min My	-43.87	bottom face	75	317	43.87	0.014	369.1	720	16	279.37	200	1005.8	14.18	17.73	0.051	tension control
Max Mxy	-27.12	bottom face	75	317	27.12		227.5		16	279.37	200	1005.8	14.18	17.73	0.051	tension control
Min Mxy	-26.83	bottom face	75	317	26.83	*As	225.0		16	279.37	200	1005.8	14.18	17.73	0.051	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.033	1	13.16	16.62	0.003	0.342	328.4	340.8	586.8	328.36	346	> 13.163	ok	246.3	φVc>Vu	not required
	Min Qy	0.041	1	16.55	2.066			370.6			340.83	356	> 16.551	ok	255.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation CF2: A03

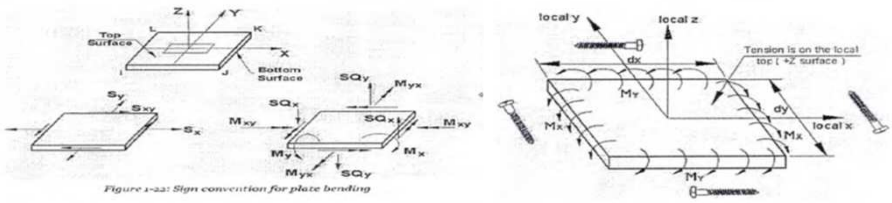


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G12 : CF_2_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1040	0.642	0.03	0	0	0	-81.87	-40.65	14.33	-96.2	-54.98	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	1057	-0.574	0.032	0	0	0	-67.44	-38.46	0	-67.44	-38.46	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Qy	1440	0.129	0.708	0	0	0	-99.92	16.23	24.12	-124	40.35	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	715	0.109	-0.621	0	0	0	-79.19	0.766	-9.429	-88.62	10.2	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sx	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Sy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Sxy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sxy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Mx	1065	0.042	0.018	0	0	0	63.75	1.346	5.407	69.16	6.753	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	1153	-0.055	-0.125	0	0	0	-148.1	1.106	15.46	-163.5	16.57	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	1813	0.002	0.008	0	0	0	0.485	193.1	3.514	3.999	196.6	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	1352	0.351	0.02	0	0	0	-49.83	-43.1	-17	-66.82	-60.09	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Mxy	883	-0.275	-0.343	0	0	0	-41.63	24.24	49.33	-90.96	73.57	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Mxy	1491	-0.251	0.207	0	0	0	-6.863	31.92	-40.56	-47.42	72.48	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

PLATE CENTER STRESS ; G13 : CF_2_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1062	0.392	0.011	0	0	0	26	0.214	12.97	38.97	13.18	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	1115	-0.369	-0.009	0	0	0	16.8	0.298	0	16.8	0.298	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Max Qy	1629	0.035	0.303	0	0	0	-37.89	119.2	19.68	-57.57	138.9	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	1981	0.026	-0.283	0	0	0	-31.03	99.93	-8.297	-39.32	108.2	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sx	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Sy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Sxy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Min Sxy	2260	0.017	-0.101	0	0	0	-1.27	39.53	-2.297	-3.567	41.82	201 1.4(D1+D2)
Max Mx	1065	0.042	0.018	0	0	0	63.75	1.346	5.407	69.16	6.753	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	1649	-0.022	0.3	0	0	0	-39.14	119.5	6.545	-45.68	126	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max My	1813	0.002	0.008	0	0	0	0.485	193.1	3.514	3.999	196.6	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	2222	0.038	-8E-04	0	0	0	-2.233	-6.438	-1.783	-4.016	-8.221	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_REV)
Max Mxy	881	-0.119	-0.12	0	0	0	14.15	34.52	43.39	57.55	77.91	204 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
Min Mxy	1488	-0.16	0.139	0	0	0	21.84	39.54	-35.03	56.87	74.57	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

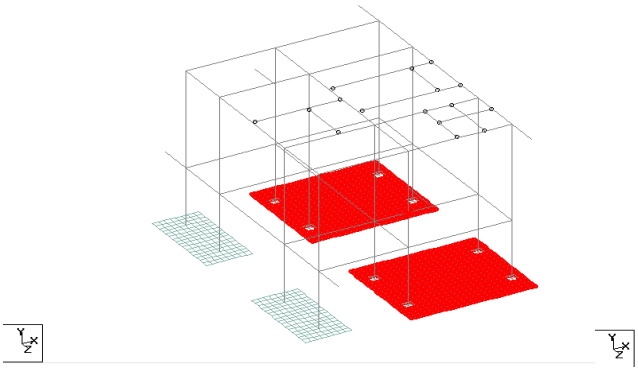
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	69.16	top face	50	540	69.16	0.014	340.3	1080	20	291.01	200	1571.5	22.16	27.7	0.055	tension control
Min Mx	-163.5	bottom face	75	515	163.5		849.9		20	291.01	200	1571.5	22.16	27.7	0.053	tension control
Max Mxy	-90.96	bottom face	75	515	90.96		470.3		20	291.01	200	1571.5	22.16	27.7	0.053	tension control
Min Mxy	-47.42	bottom face	75	515	47.42	*As	244.4		20	291.01	200	1571.5	22.16	27.7	0.053	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.392	1	235.2	38.97	0.003	0.54	598.2	537.9	926.5	537.86	1985	> 235.18	ok	403.4	φVc>Vu	not required
	Min Qx	0.369	1	221.7	16.8			701.5			537.86	1985	> 221.69	ok	403.4	φVc>Vu	not required

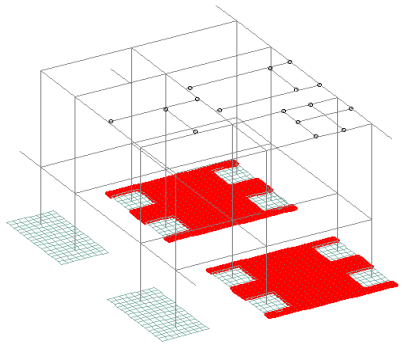
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	196.6	top face	50	540	196.6	0.014	975.5	1080	20	291.01	200	1571.5	22.16	27.7	0.055	tension control
Min My	-60.09	bottom face	75	515	60.09		310.0		20	291.01	200	1571.5	22.16	27.7	0.053	tension control
Max Mxy	73.57	top face	50	540	73.57		362.1		20	291.01	200	1571.5	22.16	27.7	0.055	tension control
Min Mxy	72.48	top face	50	540	72.48	*As	356.8		20	291.01	200	1571.5	22.16	27.7	0.055	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.303	1	182	138.9	0.003	0.54	530	537.9	926.5	530.05	556	> 181.96	ok	397.5	φVc>Vu	not required
	Min Qy	0.283	1	170	108.2			533.8			533.81	558	> 170.01	ok	400.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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6.2. Design calculation for screen supporting structure

DESIGN OF TWO WAY SLAB AT SCREEN FLOOR LEVEL:

A) Input Data :

Specified compressive strength of concrete, f_c'	=	35	Mpa
Specified yield strength of reinforcement, f_y	=	420	Mpa
Nornal weight concrete, λ	=	1	
Unit weight of concrete, γ_{conc}	=	24	KN/m^3
Effective span in longer direction (l_y)	=	2550	mm
Effective span in shorter direction (S)	=	2000	mm

B) Design Consideration :

Dia of main reinforcement in short direction		=	10	mm
Dia of main reinforcement in long direction		=	10	mm
Consider overall depth, h		=	150	mm
Clear cover to main reinforcement	[table 20.6.1.3.1 SBC 304:2018]	=	50	mm
Effective depth in shorter direction, d_s		=	95	mm
Effective depth in longer direction, d_{ly}		=	85	mm

C) Load Data :

i)Live Load intensity w_{LL} = 5 KN/m^2

ii) Dead load calculation :

Total Dead load intensity w_{DL} = 3.6 KN/m^2

Total load intensity, w = w_{LL} + w_{DL} = 8.60 KN/m^2

D) Structural Design :

ONE EDGE DISCONTINUOUS — Table B1,Appendix B,SBC-304:2018

Ratio of short span to long span ,m = S/L_y	=	0.79	two way slab
Bending Moment Co-efficient, C :			
Short Span continuous edge(-ve)	=	0.056	
Short Span discontinuous edge(+ve)	=	0.027	
Short Span midspan(+ve)	=	0.042	
Long Span continuous edge(-ve)	=	0.041	
Long Span discontinuous edge(+ve)	=	0.021	
Long Span midspan(+ve)	=	0.031	
Partial safety factor , γ_m	=	1.4	

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i)Bending moment due to Dead Load + Live load : [$M = CwS^2$]

Short Span continuous edge(-ve) =	1.4	*	0.056	*	8.600	*	2	^2	=	2.68	KN-m
Short Span discontinuous edge(+ve) =	1.4	*	0.027	*	8.600	*	2	^2	=	1.32	KN-m
Short Span midspan(+ve) =	1.4	*	0.042	*	8.600	*	2	^2	=	2	KN-m
Long Span continuous edge(-ve) =	1.4	*	0.041	*	8.600	*	2	^2	=	1.98	KN-m
Long Span discontinuous edge(+ve) =	1.4	*	0.021	*	8.600	*	2	^2	=	1.01	KN-m
Long Span midspan(+ve) =	1.4	*	0.031	*	8.600	*	2	^2	=	1.49	KN-m

E) Check for flexure :-

Factored moment in short direction , M_{uS} = 2.683 KN-m

β_1 = 0.8 steel strain ε_t = 0.005 and φ = 0.9

now, $C = 0.85 f_c' b a$; $T = A_s f_y$; $M_u = \varphi M_n = \varphi f_y A_s (d_s - a/2)$; $C = T$

$C = 0.85 * 35 * 1000 * a$; $T = 420 * A_s$; $a = 0.014 * A_s$

$M_{uS} = 0.9 * 420 * (95 - 0.014 A_s/2) * A_s = 2.68 * 10^6$

$A_{s\ required} = 75$ mm^2

$A_{s,min} = 0.2 / 100 * 1000 * 95 = 171$ mm^2 [Table 8.6.1.1 SBC 304:18]

Spacing required = 79 / 171 * 1000 = 462.0 mm c/c

Maximum spacing = 300 mm c/c [cl 8.7.2.2 SBC 304:2018]

provide 10 φ main bar @ 200 mm c/c A_s provided = 395.0 mm^2

Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c (d_s - c)$ c = 0.014 * 395.0 / 0.800 = 6.97 mm

ε_t = 0.003 *(95 - 6.97)/ 6.97 = 0.038 > 0.005 tension controlled

Factored moment in long direction , $M_{u,Ly}$ = 1.975 KN-m

$M_{u,Ly} = 0.9 * 420 * (85 - 0.014 A_s/2) * A_s = 1.98 * 10^6$

$A_{s\ required} = 62$ mm^2

$A_{s,min} = 0.2 / 100 * 1000 * 85 = 153$ mm^2 [Table 8.6.1.1 SBC 304:18]

Spacing required = 79 / 153 * 1000 = 516.34 mm c/c

provide 10 φ main bar @ 200 mm c/c A_s provided = 395 mm^2

Check for Tension controlled section :

$\varepsilon_t = \varepsilon_c / c (d_{ly} - c)$ c = 0.014 * 395.0 / 0.800 = 6.97 mm

ε_t = 0.003 *(85 - 6.97)/ 6.97 = 0.034 > 0.005 tension controlled

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F) Check for shear :

Shear force at a distance d from supporting beam

Effective depth in shorter direction ,d_s= 95 mm

UDL along shorter direction considering 1m width, w= 8.60 KN/m

Factored shear,V_u=1.6w*(0.5S-d_s)= 1.4 * 8.6 *(0.5 * 2 - 0.095)= 10.9 KN

$V_{c\,1}$=(0.16 λ $\sqrt{f_c'}$+17 ρ_w V_u d_s /M_{us}) $b_w d_s$

$V_{c\,2}$ = (0.16 λ $\sqrt{f_c'}$ +17 ρ_w) b_w d_s

$V_{c\,3}$ = 0.29 λ $\sqrt{f_c'}$ b_w d_s

V_c = Minimum of $V_{c\,1}$, $V_{c\,2}$ and $V_{c\,3}$

A_s=	395.0 mm²	M_{us}=	2.683 KN-m	=	92.5 KN
ρ_w=	0.0042	b_w =	1000 mm	=	96.6 KN
λ=	1	d_s =	95 mm	=	163 KN
[table 22.5.5.1 SBC 304 : 2018]				=	92.5 KN

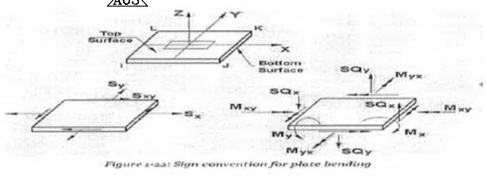
V_u ≤ φ (V_c + 0.66$\sqrt{f_c'}$$b_w d_s$) = 347.6 KN ok [cl 22.5.5.1 SBC 304 : 2018]

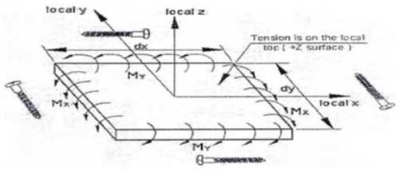
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Design of Wall : A03





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G14 : WALL_MT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	2838	0.197	-0.001	1.07	-0.08	0.056	-4.322	0.401	-6.605	-10.93	7.006	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	2853	-0.21	0.015	1.408	-0.1	-0.046	-0.724	0.283	0	-0.724	0.283	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Qy	2714	0.03	0.124	-0.18	-1.24	0.15	5.306	15.86	-8.063	13.37	23.92	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	2814	-0.004	-0.148	-0.01	-1.17	-0.349	2.246	12.26	11.92	14.16	24.17	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2852	-0.173	-0.025	1.429	-0.02	0.113	2.787	0.607	7.317	10.1	7.924	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sx	3075	-0.02	0.01	-2.11	-0.18	0.74	5.642	0.028	-0.899	6.541	0.927	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sy	3069	0.004	0.011	0.171	0.278	0.62	0.2	-0.69	-1.523	1.723	-2.213	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sy	2774	0.177	-0.067	-0.25	-1.41	-0.165	5.142	34.52	5.744	10.89	40.26	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sxy	3062	-0.023	0.017	-0.9	-1.27	1.403	4.88	-0.407	-1.58	6.46	-1.987	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sxy	2954	0.021	0.015	-0.79	-1.1	-1.208	4.165	-0.269	0.126	4.291	-0.395	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mx	2734	0.114	0.114	-0.2	-1.32	0.05	6.002	26.91	-7.307	13.31	34.22	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	2877	-0.056	-0.008	0.312	-0.19	0.005	-6.348	-7.058	-0.361	-6.709	-7.419	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_REV)
Max My	2754	0.19	0.03	-0.24	-1.39	-0.067	5.742	36.01	-1.934	7.676	37.95	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	2614	-0.074	0.013	-0.04	-0.74	0.417	-1.944	-7.791	-0.908	-2.852	-8.699	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mxy	2814	-0.004	-0.148	-0.01	-1.17	-0.349	2.246	12.26	11.92	14.16	24.17	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mxy	2504	0.008	-0.128	-0.06	-1.1	0.357	1.633	9.586	-11.05	12.68	20.64	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

PLATE CENTER STRESS ; G15 : WALL_SH At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	2838	0.197	-0.001	1.07	-0.08	0.056	-4.322	0.401	-6.605	-10.93	7.006	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qx	2853	-0.21	0.015	1.408	-0.1	-0.046	-0.724	0.283	0	-0.724	0.283	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Qy	2714	0.03	0.124	-0.18	-1.24	0.15	5.306	15.86	-8.063	13.37	23.92	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Qy	2814	-0.004	-0.148	-0.01	-1.17	-0.349	2.246	12.26	11.92	14.16	24.17	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sx	2852	-0.173	-0.025	1.429	-0.02	0.113	2.787	0.607	7.317	10.1	7.924	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sx	3075	-0.02	0.01	-2.11	-0.18	0.74	5.642	0.028	-0.899	6.541	0.927	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sy	3069	0.004	0.011	0.171	0.278	0.62	0.2	-0.69	-1.523	1.723	-2.213	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sy	2774	0.177	-0.067	-0.25	-1.41	-0.165	5.142	34.52	5.744	10.89	40.26	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Sxy	3062	-0.023	0.017	-0.9	-1.27	1.403	4.88	-0.407	-1.58	6.46	-1.987	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Sxy	2954	0.021	0.015	-0.79	-1.1	-1.208	4.165	-0.269	0.126	4.291	-0.395	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mx	2734	0.114	0.114	-0.2	-1.32	0.05	6.002	26.91	-7.307	13.31	34.22	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mx	2877	-0.056	-0.008	0.312	-0.19	0.005	-6.348	-7.058	-0.361	-6.709	-7.419	218 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_REV)
Max My	2754	0.19	0.03	-0.24	-1.39	-0.067	5.742	36.01	-1.934	7.676	37.95	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min My	2614	-0.074	0.013	-0.04	-0.74	0.417	-1.944	-7.791	-0.908	-2.852	-8.699	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Max Mxy	2814	-0.004	-0.148	-0.01	-1.17	-0.349	2.246	12.26	11.92	14.16	24.17	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
Min Mxy	2504	0.008	-0.128	-0.06	-1.1	0.357	1.633	9.586	-11.05	12.68	20.64	205 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	inner face	40	mm
	outer face	40	mm
Thicknes of slab =		300	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

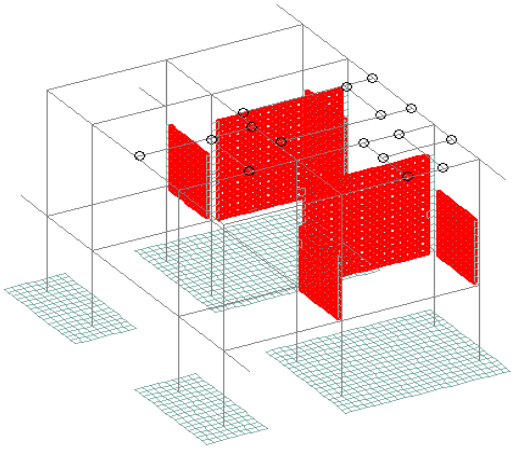
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	13.31	outer face	40	254	13.31		139.2		12	209.52	200	565.8	7.978	9.972	0.073	tension control
Min Mx	-6.71	inner face	40	254	6.709	0.014	70.0	540	12	209.52	200	565.8	7.978	9.972	0.073	tension control
Max Mxy	7.68	outer face	40	254	7.676		80.1		12	209.52	200	565.8	7.978	9.972	0.073	tension control
Min Mxy	-2.85	inner face	40	254	2.852	*As	29.7		12	209.52	200	565.8	7.978	9.972	0.073	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.197	1	59.03	10.93	0.002	0.254	253.6	250	fc'436	250.05	931	> 59.027	ok	187.5	φVc>Vu	not required
	Min Qx	0.21	1	63.15	0.724			453.5			250.05	931	> 63.147	ok	187.5	φVc>Vu	not required

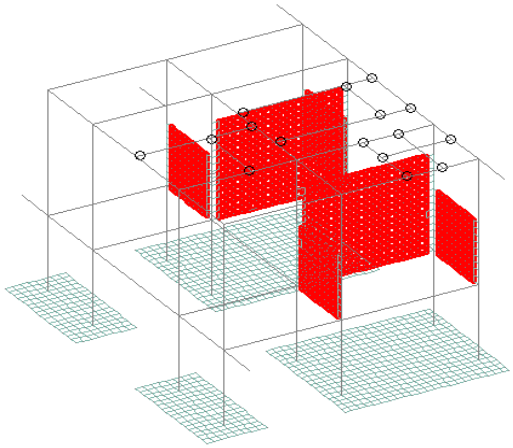
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	37.9	outer face	40	254	37.95		399.6		12	209.52	200	565.8	7.978	9.972	0.073	tension control
Min My	-8.7	inner face	40	254	8.699	0.014	90.8	540	12	209.52	200	565.8	7.978	9.972	0.073	tension control
Max Mxy	24.2	outer face	40	254	24.17		253.6		12	209.52	200	565.8	7.978	9.972	0.073	tension control
Min Mxy	20.6	outer face	40	254	20.64	*As	216.2		12	209.52	200	565.8	7.978	9.972	0.073	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.124	1	37.32	23.92	0.002	0.254	244.2	250	fc'436	244.24	258	> 37.323	ok	183.2	φVc>Vu	not required
	Min Qy	0.148	1	44.37	24.17			244.9			244.91	258	> 44.365	ok	183.7	φVc>Vu	not required



Moment taken at the face of the column

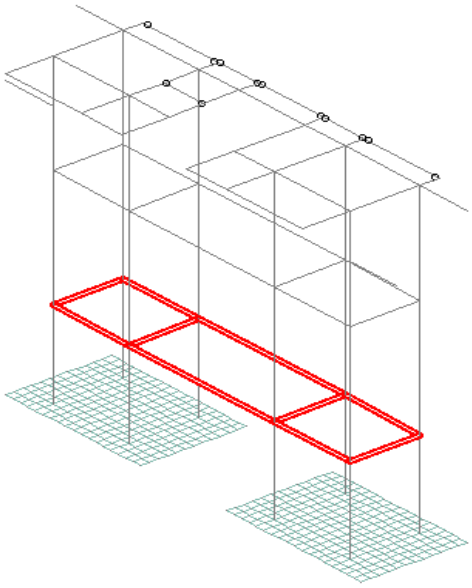


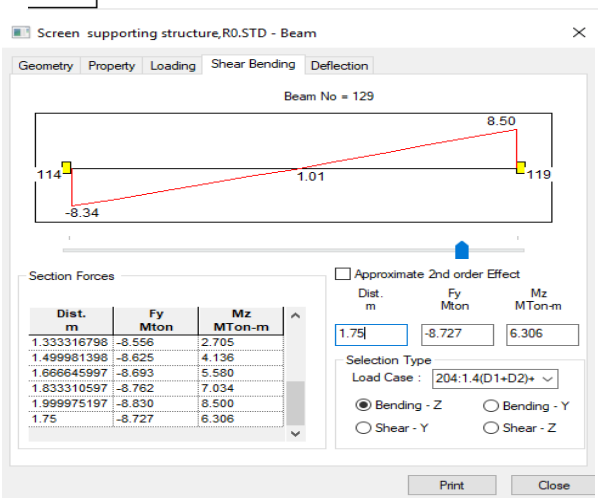
Shear taken at a face of the column

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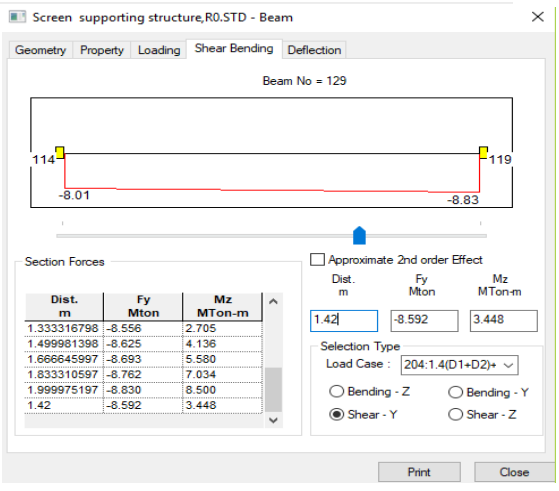
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Design of beam : TB2





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = TB2

Refer staad file : Screen Supporting Structure, R1

Beam End Forces : G3: _TB_2									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	136	119	3.47	0.981	-0	-0	-0.06	0.651	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Fx	129	114	-0.64	8.057	0.42	0.013	-0.42	7.726	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Fy	129	114	-0.61	8.195	0.385	0.027	-0.39	7.861	206 1.4(D1+D2)+1.7(L1+L2-L3)
Min Fy	129	119	-0.38	-8.83	-0.59	-0.04	-0.58	8.5	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Fz	125	116	-0.27	-5.83	0.691	-0.01	-0.7	-6.14	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Fz	131	113	-0.27	-6.44	-0.68	-0.04	0.685	-6.75	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Mx	132	113	0.21	1.611	0.192	0.157	-0.23	1.638	205 1.4(D1+D2)+1.7(L1+L2+L4)
Min Mx	134	115	0.23	-0.62	-0.18	-0.18	0.237	-1.21	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max My	131	113	-0.27	-6.44	-0.68	-0.04	0.685	-6.75	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min My	125	116	-0.27	-5.83	0.691	-0.01	-0.7	-6.14	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Mz	129	119	-0.38	-8.83	-0.59	-0.04	-0.58	8.5	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Mz	129	114	-0.38	-8.01	-0.59	-0.04	0.589	-8.34	204 1.4(D1+D2)+1.7(L1+L2+L3)

Max support Top Moment = 6.5 t-m at the face of the beam

Max support Bottom Moment = 6.5 t-m at the face of the beam

Max Shear = 8.83 t Corresponding Moment = 3.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	400	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 6.50 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 542 mm^2 As = M_u / φ f_y (d - a/2)
$A_{s,min}$ = greater of (a) 0.25√ f_c'/f_y bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) 1.4/$f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 542 mm^2 ok

Max support Bottom Moment = 6.5 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 542 mm^2 As = M_u / φ f_y (d - a/2)
$A_{s,min}$ = greater of (a) 0.25√ f_c'/f_y bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) 1.4/$f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 542 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 8.83 t Corresponding Moment, Mu = 3.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) (0.16λ√f_c' + 17$\rho_w V_u d/M_u$)bd 10.71 t if φV_c < V_u shear reinforcement required
b) (0.16λ√f_c' + 17ρ_w)bd 10.97 t if φV_c ≥ V_u ≥ 0.5φVc Minimum shear reinf reqd
c) 0.29λ√f_c'bd 16.99 t if 0.5φV_c > V_u Shear reinf not required

φV_c = 8.029 t
Remarks : Shear reinforcement required

V_s=V_u/φ-V_c 1.068 t
V_s = $A_v f_{yt} d/s$ therefore, A_v/s = $V_s/f_{yt}d$ = 0.077 mm^2/m

$A_{v,min}/s$ > max of (a) 0.062√$f_c'b/f_{yt}$ *0.26* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) 0.35b/f_{yt} *0.25* mm^2/m
Provide 2L-10dia stirrups @150 mm c/c

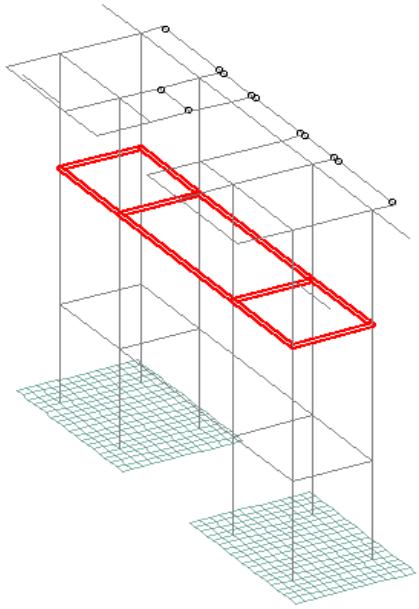
therefore, A_v/s provided = 1.05 mm^2/m ok

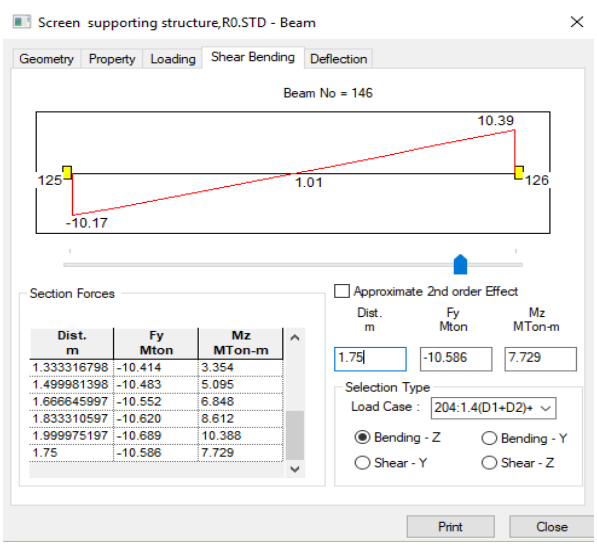
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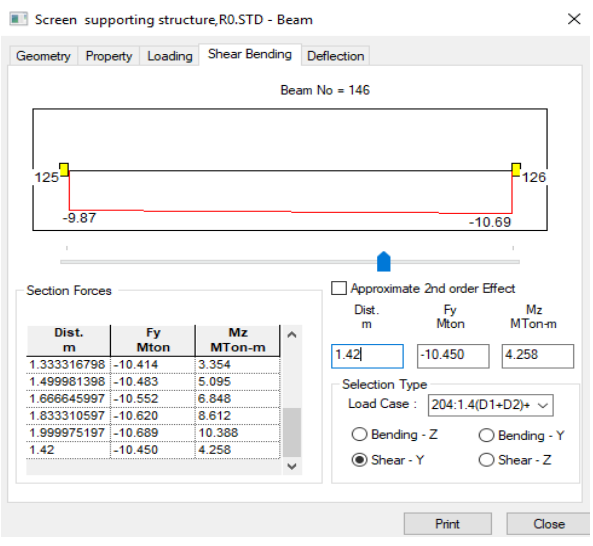
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Design of beam : FB5





Moment at the face of column



Shear at the distance d from face of column

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Design of beam = FB5

Refer staad file : Screen Supporting Structure, R1

Beam End Forces : G4: _FB_5									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	151	125	11.8	0.986	0.001	-0	0.389	0.377	205 1.4(D1+D2)+1.7(L1+L2+L4)
Min Fx	151	125	-11.9	0.987	0	-0	-0.26	1.116	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Fy	146	125	1.19	9.845	1.527	-0.09	-1.54	9.478	206 1.4(D1+D2)+1.7(L1+L2-L3)
Min Fy	146	126	0.78	-10.7	-2.22	0.253	-2.21	10.39	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Fz	143	123	0.81	-9.5	2.277	-0.29	-2.28	-9.8	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Fz	146	125	0.78	-9.87	-2.22	0.253	2.24	-10.2	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Mx	614	124	1.03	2.589	0.756	1.094	-0.94	-1.12	206 1.4(D1+D2)+1.7(L1+L2-L3)
Min Mx	612	126	1.12	-2.36	-0.77	-1.09	-0.97	-1.44	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max My	143	124	0.81	-10.3	2.277	-0.29	2.271	10.01	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min My	143	123	0.81	-9.5	2.277	-0.29	-2.28	-9.8	204 1.4(D1+D2)+1.7(L1+L2+L3)
Max Mz	146	126	0.78	-10.7	-2.22	0.253	-2.21	10.39	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Mz	146	125	0.78	-9.87	-2.22	0.253	2.24	-10.2	204 1.4(D1+D2)+1.7(L1+L2+L3)

Max support Top Moment = 8 t-m at the face of the beam

Max support Bottom Moment = 8 t-m at the face of the beam

Max Shear = 10.5 t Corresponding Moment = 4.5 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	400	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 8.00 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 674 mm^2 As = M_u / φ f_y (d - a/2)
$A_{s,min}$ = greater of (a) 0.25√ f_c'/f_y bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) 1.4/$f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 673.7 mm^2 ok

Max support Bottom Moment = 8 t-m
Effective depth ,d = 330.0 mm
Area of steel required ,Ast,required = 674 mm^2 As = M_u / φ f_y (d - a/2)
$A_{s,min}$ = greater of (a) 0.25√ f_c'/f_y bd 348.6 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) 1.4/$f_y$bd 330 mm^2

Provide 3 nos 20 dia bar althrough at 1st layer

Ast,provided = 942.9 mm^2 > 673.7 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 10.5 t Corresponding Moment, Mu = 4.5 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) (0.16λ√f_c' + 17$\rho_w V_u d/M_u$)bd 10.61 t if φV_c < V_u shear reinforcement required
b) (0.16λ√f_c' + 17ρ_w)bd 10.97 t if φV_c ≥ V_u ≥ 0.5φVc Minimum shear reinf reqd
c) 0.29λ√f_c'bd 16.99 t if 0.5φV_c > V_u Shear reinf not required

φV_c = 7.954 t
Remarks : Shear reinforcement required

V_s=V_u/φ-V_c 3.395 t
V_s = $A_v f_{yt} d/s$ therefore, A_v/s = $V_s/f_{yt}d$ = 0.245 mm^2/m

$A_{v,min}/s$ > max of (a) 0.062√$f_c'b/f_{yt}$ *0.26* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) 0.35b/f_{yt} *0.25* mm^2/m
Provide 2L-10dia stirrups @150 mm c/c

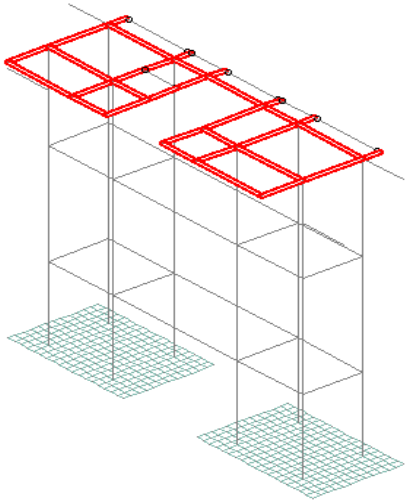
therefore, A_v/s provided = 1.05 mm^2/m ok

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Design of beam : FB6_FB7_FB8



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Design of beam = FB6_FB7_FB8

Refer staad file : Screen Supporting Structure, R1

Beam End Forces : G6: _FB6_FB7_FB8									
	Beam	Node	Fx	Fy	Fz	Mx	My	Mz	L/C
			Mton			MTon-m			
Max Fx	91	90	9.03	20.81	-7.09	1.343	4.246	15.84	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Fx	111	107	-6.07	-0.25	-4.82	0	2.648	-0.26	205 1.4(D1+D2)+1.7(L1+L2+L4)
Max Fy	91	90	8.74	21.36	-7.82	1.392	4.524	15.84	206 1.4(D1+D2)+1.7(L1+L2-L3)
Min Fy	103	91	8.48	-19.5	7.138	-1.33	4.208	15.54	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Fz	91	90	0.82	5.25	8.139	-1.09	-4.68	3.409	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Fz	91	90	8.74	21.36	-7.82	1.392	4.524	15.84	206 1.4(D1+D2)+1.7(L1+L2-L3)
Max Mx	79	93	-0.32	-0.28	-0.57	1.585	-0.4	1.178	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mx	85	96	-0.32	-0.27	0.556	-1.6	0.41	1.198	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max My	91	107	0.82	3.422	8.139	-1.09	6.309	-2.44	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min My	91	107	8.74	19.53	-7.82	1.392	-6.03	-11.8	206 1.4(D1+D2)+1.7(L1+L2-L3)
Max Mz	101	87	2.68	-16	4.306	0.435	2.892	18.6	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mz	109	107	7.97	-0.92	1.49	0.07	-3.24	-12.9	206 1.4(D1+D2)+1.7(L1+L2-L3)

Max support Top Moment = 18.6 t-m at the face of the beam

Max support Bottom Moment = -12.9 t-m at the face of the beam

Max Shear = 21.36 t Corresponding Moment = 15.84 t-m

Max shear at distance d from face of column

Design Input :

Specified compressive strength of concrete, f_c' =	35	Mpa	
Specified yield strength of main reinforcement, f_y =	420	Mpa	
Yield strength of transverse reinforcement, f_{yt} =	420	Mpa	
Beam width,b =	300	mm	
Beam depth,h =	600	mm	
Cover =	50	mm	
Dia of stirrups, d_{tie} =	10	mm	
Equivalent concrete compressive depth, a =	0.047	A_s	'a' = $f_y/0.85f_c'b * A_s$

Strength Reduction Factor,φ : *table 21.2.1*

Moment,φ [for εt 0.005]	0.9
Shear,φ	0.75

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Design for longitudinal reinforcement :

Max support Top Moment = 18.60 t-m
Effective depth ,d = 527.5 mm
Area of steel required ,Ast,required = 975 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 557.3 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 527.5 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer

Ast,provided = 1473 mm^2 > 975 mm^2 ok

Max support Bottom Moment = 12.94 t-m
Effective depth ,d = 527.5 mm
Area of steel required ,Ast,required = 669 mm^2 $As = M_u / \varphi f_y (d - a/2)$
$A_{s,min}$ = greater of (a) $0.25\sqrt{f_c'}/f_y$ bd 557.3 mm^2 cl 9.6.1.2 SBC 304 : 2018
(b) $1.4/f_y$bd 527.5 mm^2

Provide 3 nos 25 dia bar althrough at 1st layer

Ast,provided = 1473 mm^2 > 669 mm^2 ok

Design for shear reinforcement :

Max shear, Vu = 21.36 t Corresponding Moment, Mu = 15.84 t-m
V_c = lesser of table 22.5.5.1 SBC 304 : 2018
a) $(0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d/M_u)bd$ 16.76 t if $\varphi V_c < V_u$ shear reinforcement required
b) $(0.16\lambda\sqrt{f_c'} + 17\rho_w)bd$ 17.48 t if $\varphi V_c \geq V_u \geq 0.5\varphi Vc$ Minimum shear reinf reqd
c) $0.29\lambda\sqrt{f_c'}bd$ 27.15 t if $0.5\varphi V_c > V_u$ Shear reinf not required

φV_c = 12.57 t
Remarks : Shear reinforcement required

$V_s = V_u/\varphi - V_c$ 11.72 t
$V_s = A_v f_{yt} d/s$ therefore, $A_v/s = V_s/f_{yt}d$ = 0.529 mm^2/m

$A_{v,min}/s >$ max of (a) $0.062\sqrt{f_c'}b/f_{yt}$ *0.26* mm^2/m table 9.6.3.3 SBC 304 : 2018
(b) $0.35b/f_{yt}$ *0.25* mm^2/m

Provide 2L-10dia stirrups @200 mm c/c

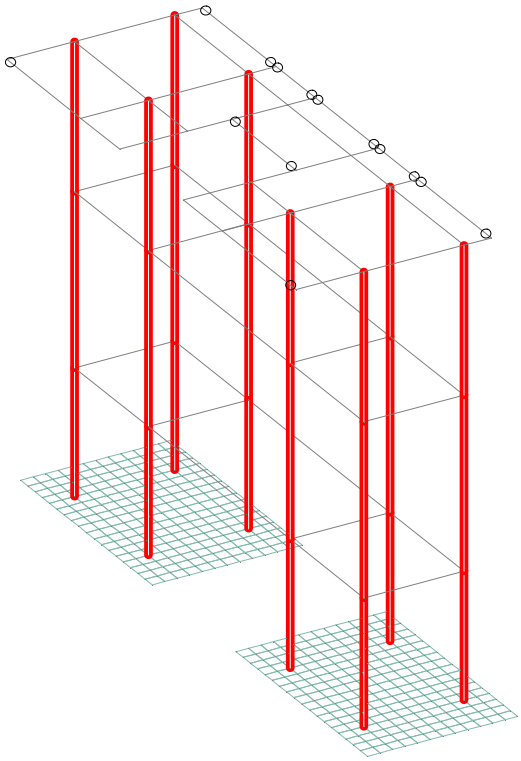
therefore, A_v/s provided = 0.79 mm^2/m ok

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Design of column C2 :



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Design of Column C2 :

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.2 % i.e. ρ_g = 0.012

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.71

Y_b = (b-d')/b = 0.71

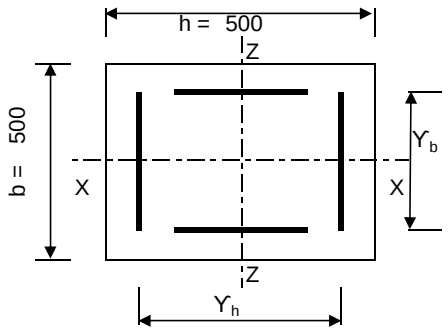
Unsupported length of column parallel to X-X axis, l_{ux} = 3.89 m kl_{ux}/b = 7.78 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 3.89 m kl_{uz}/h = 7.78

$P_{n,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 7177 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
207	690.7	36.9	64.3	1063	56.8	98.9	0.121	0.080	350	0.16	0.080	350	0.28		0.19	safe
204	160.9	43.9	50.9	247	67.6	78.3	0.028	0.080	350	0.19	0.080	350	0.22		0.16	safe
206	186.1	30.0	56.1	286	46.2	86.3	0.033	0.080	350	0.13	0.080	350	0.25		0.14	safe
207	455.2	63.0	36.1	700	97.0	55.5	0.08	0.080	350	0.28	0.080	350	0.16		0.18	safe
207	450.6	63.6	35.8	693	97.8	55.1	0.079	0.080	350	0.28	0.080	350	0.16		0.18	safe
204	163.7	46.2	47.7	252	71.1	73.4	0.029	0.080	350	0.20	0.080	350	0.21	1.61	0.16	safe
205	139.2	47.2	48.6	214	72.7	74.7	0.024	0.080	350	0.21	0.080	350	0.21		0.16	safe
207	427.8	143.1	50.2	658	220.1	77.3	0.075	0.080	350	0.63	0.080	350	0.22		0.56	safe
207	423.2	143.5	48.8	651	220.7	75.1	0.074	0.080	350	0.63	0.080	350	0.21		0.56	safe
206	158.7	95.8	106.3	244	147.3	163.6	0.028	0.080	350	0.42	0.080	350	0.47		0.54	safe
204	133.5	28.5	118.9	205	43.8	183.0	0.023	0.080	350	0.13	0.080	350	0.52		0.39	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
3000	4	25	+	4	25	3929	1.57	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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Foundation Base Pressure for CF3 :

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	13	101	0	105.403	0
Min Px	13	101	0	105.403	0
Max Py	582	107	0	261.603	0
Min Py	618	107	0	-10.721	0
Max Pz	13	101	0	105.403	0
Min Pz	13	101	0	105.403	0

Gross Maximum Base pressure, p_{max} = 261.6 KN/m^2

Net Allowable Safe Bearing Capacity = 600 KN/m^2

Depth of soil above footing = 2.4 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 648 KN/m^2

> 261.6 KN/m^2 safe

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Design of Foundation CF3:

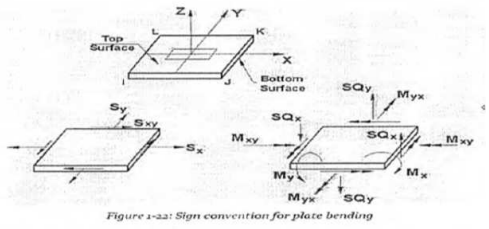
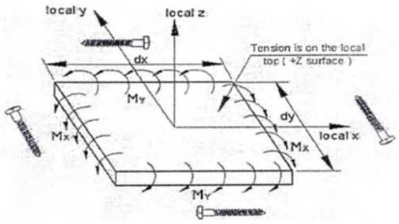


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G7 : fdn_MOMENT At the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	243	0.641	0.041	0	0	0	-56.95	-21.95	16.19	-73.14	-38.14	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Qx	300	-0.644	0.042	0	0	0	-57.33	-22.04	0	-57.33	-22.04	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Qy	313	-0.038	0.36	0	0	0	-71.41	33.57	-21.93	-93.33	55.5	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Qy	553	-0.26	-0.297	0	0	0	-111.8	-13.51	-4.956	-116.7	-18.47	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Sx	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sx	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Sy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Sxy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sxy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Mx	571	-0.055	0.013	0	0	0	63.31	0.993	-3.787	67.09	4.78	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mx	554	0.152	-0.277	0	0	0	-114.1	-11.85	-4.854	-119	-16.7	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max My	320	0.044	0.14	0	0	0	-44.53	44.25	0.961	-45.49	45.21	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min My	306	-0.541	0.188	0	0	0	-46.98	-22.38	-29.07	-76.04	-51.44	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Mxy	255	0.253	0.248	0	0	0	-35.25	21.85	35.74	-70.99	57.59	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mxy	312	-0.254	0.25	0	0	0	-35.56	22.17	-35.78	-71.34	57.95	207 1.4(D1+D2)+1.7(L1+L2-L4)

PLATE CENTER STRESS ; G8 : fdn_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	550	0.524	-0.001	0	0	0	-4.1	-3.595	11.66	-15.76	-15.25	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Qx	551	-0.526	0	0	0	0	-4.206	-3.609	0	-4.206	-3.609	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max Qy	319	-0.024	0.161	0	0	0	-21.17	6.165	3.31	-24.48	9.475	204 1.4(D1+D2)+1.7(L1+L2+L3)
Min Qy	337	-0.021	-0.17	0	0	0	-27.95	14.8	-6.978	-34.93	21.78	206 1.4(D1+D2)+1.7(L1+L2-L3)
Max Sx	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sx	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Sy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Sxy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Min Sxy	610	0.008	-0.007	0	0	0	17.15	0.24	-0.3	17.45	0.54	201 1.4(D1+D2)
Max Mx	571	-0.055	0.013	0	0	0	63.31	0.993	-3.787	67.09	4.78	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mx	319	-0.046	0.15	0	0	0	-45.94	43.97	-13.55	-59.49	57.52	207 1.4(D1+D2)+1.7(L1+L2-L4)
Max My	320	0.044	0.14	0	0	0	-44.53	44.25	0.961	-45.49	45.21	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min My	305	-0.07	0.028	0	0	0	-8.244	-9.124	5.872	-14.12	-15	220 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+L3)
Max Mxy	256	0.221	0.095	0	0	0	-3.306	17.55	30.28	-33.59	47.83	207 1.4(D1+D2)+1.7(L1+L2-L4)
Min Mxy	311	-0.222	0.096	0	0	0	-3.394	17.81	-30.27	-33.66	48.08	207 1.4(D1+D2)+1.7(L1+L2-L4)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

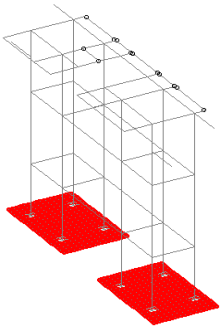
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	67.09	top face	50	442	67.09		404.2		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	45.00	top face	50	442	45	0.014	270.5	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Max Mxy	-70.99	bottom face	75	417	70.99		453.8		16	223.49	200	1005.8	14.18	17.73	0.068	tension control
Min Mxy	-71.34	bottom face	75	417	71.34	*As	456.1		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.524	1	262	15.76	0.002	0.442	544	435.5	fc'758	435.48	1621	> 262	ok	326.6	φVc>Vu	not required
	Min Qx	0.526	1	263	4.206			891			435.48	1621	> 263	ok	326.6	φVc>Vu	not required

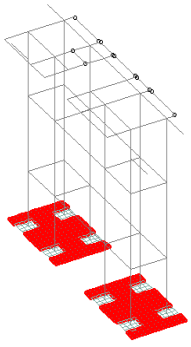
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	45.21	top face	50	442	45.21		271.8		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-51.44	bottom face	75	417	51.44	0.014	328.2	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control
Max Mxy	57.59	top face	50	442	57.59		346.6		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mxy	57.95	top face	50	442	57.95	*As	348.8		16	223.49	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.161	1	80.5	9.475	0.002	0.442	482.6	435.5	fc'758	435.48	456	> 80.5	ok	326.6	φVc>Vu	not required
	Min Qy	0.17	1	85	21.78			447.9			435.48	456	> 85	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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6.3. Design calculation for hopper supporting structure

Foundation Base Pressure :

refer staad file : Hopper supporting structure_fdn_R0

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	101	201	0	55.245	0
Min Px	101	201	0	55.245	0
Max Py	1734	214	0	356.165	0
Min Py	108	201	0	0	0
Max Pz	101	201	0	55.245	0
Min Pz	101	201	0	55.245	0

Gross Maximum Base pressure, p_{max} = 356.2 KN/m^2

Net Allowable Safe Bearing Capacity = 600 KN/m^2

Depth of soil above footing = 2.4 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 648 KN/m^2

> 356.2 KN/m^2 safe

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Design of Foundation CF4:

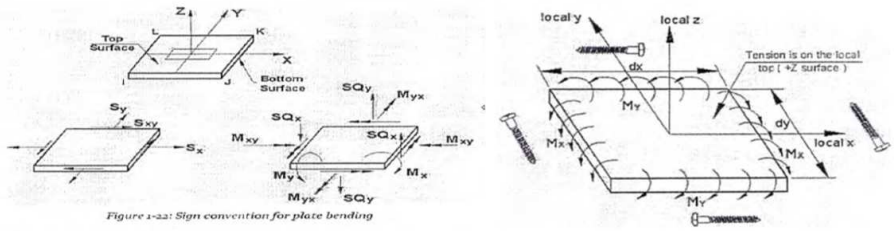


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : CF_4_MOMENT At the face of the column refer staad file : Hopper supporting strycture_fdn_R0

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1061	1.177	0.019	0	0	0	-300.8	-233.7	-81.64	-382.4	-315.3	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qx	1100	-0.976	0.214	0	0	0	-289.6	-226	0	-289.6	-226	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Qy	1432	0.308	1.206	0	0	0	-472.9	-261.4	-58.94	-531.9	-320.4	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	929	0.287	-0.549	0	0	0	-188.6	-106.9	-163.6	-352.2	-270.5	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Sx	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sx	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Sy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Sxy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sxy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Mx	1026	0.271	-0.058	0	0	0	239.9	-7.939	-56.7	296.6	-64.64	107 1.2 (DL1+DL2)+1.6WL(+X)+0.5LL1+1.0LL2
Min Mx	1432	0.308	1.206	0	0	0	-472.9	-261.4	-58.94	-531.9	-320.4	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max My	1170	0.109	0.453	0	0	0	1.568	177.5	-0.009	1.577	177.5	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min My	1017	1.068	-0.198	0	0	0	-273.2	-270.8	-126.8	-400	-397.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Mxy	968	0.075	-0.224	0	0	0	-149.6	-11.35	104	-253.6	-115.3	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mxy	973	0.083	-0.354	0	0	0	-169.7	-27.34	-186.3	-356.1	-213.7	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL

PLATE CENTER STRESS ; G2 : CF_4_SHEAR At a distance "d" from the face of the column

Load condition		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	L/C
Max Qx	1506	0.671	0.088	0	0	0	-126.7	-14.02	-28.37	-155	-42.39	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qx	1097	-0.348	0.029	0	0	0	-78.41	-90.62	0	-78.41	-90.62	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Qy	1483	-0.1	0.235	0	0	0	12.17	1.719	-29.02	41.19	30.74	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	1378	-0.064	-0.133	0	0	0	6.039	-12.82	6.319	12.36	-19.14	107 1.2 (DL1+DL2)+1.6WL(+X)+0.5LL1+1.0LL2
Max Sx	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sx	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Sy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Sxy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Min Sxy	1521	0.002	0.056	0	0	0	-36.38	-7.62	-16.32	-52.7	-23.94	101 1.4 (DL1+DL2)
Max Mx	1477	0.082	-0.003	0	0	0	168.9	-0.101	-70.68	239.6	-70.78	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mx	1501	0.09	0.225	0	0	0	-398.9	-18.78	-33.67	-432.6	-52.45	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max My	1379	-0.056	0.176	0	0	0	-4.065	25.25	2.663	-6.728	27.92	107 1.2 (DL1+DL2)+1.6WL(+X)+0.5LL1+1.0LL2
Min My	1064	0.483	0.012	0	0	0	-31.94	-97.59	-101.3	-133.3	-198.9	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Mxy	965	-0.143	-0.06	0	0	0	-121.9	-36.52	43.34	-165.2	-79.86	110 1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2
Min Mxy	976	0.269	-0.084	0	0	0	-48.6	-41.11	-140	-188.6	-181.1	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

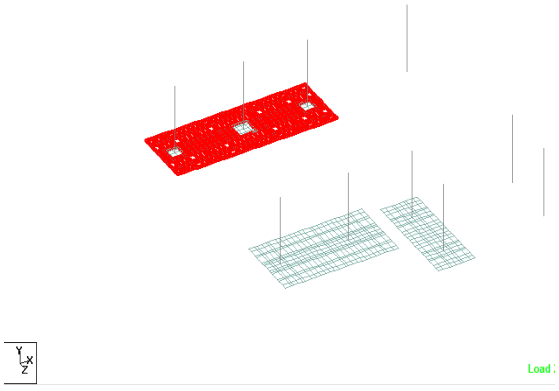
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	296.6	top face	50	640	296.6		1243.1		20	249.43	200	1571.5	22.16	27.7	0.066	tension control
Min Mx	-531.9	bottom face	75	613	531.9	0.014	2361.4	1260	25	207.96	200	2455.4	34.62	43.28	0.039	tension control
Max Mxy	-253.6	bottom face	75	613	253.6		1109.4		25	389.74	200	2455.4	34.62	43.28	0.039	tension control
Min Mxy	-356.1	bottom face	75	613	356.1	*As	1566.3		25	313.53	200	2455.4	34.62	43.28	0.039	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.671	1	469.7	155	0.004	0.64	686.7	647.5	1098	647.55	2360	> 469.7	ok	485.7	φVc>Vu	not required
	Min Qx	0.348	1	243.6	78.41			688.8			647.55	2360	> 243.6	ok	485.7	φVc>Vu	not required

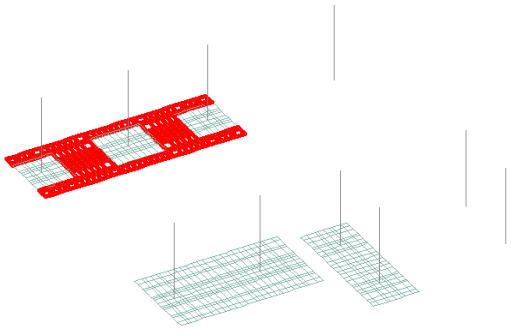
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	177.5	top face	50	640	177.5		739.8		20	249.43	200	1571.5	22.16	27.7	0.066	tension control
Min My	-397.6	bottom face	75	613	397.6	0.014	1752.8	1260	25	280.16	200	2455.4	34.62	43.28	0.039	tension control
Max Mxy	-115.3	bottom face	75	613	115.3		500.9		25	389.74	150	3273.9	46.16	57.7	0.029	tension control
Min Mxy	-213.7	bottom face	75	613	213.7	*As	933.0		25	389.74	150	3273.9	46.16	57.7	0.029	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.235	1	164.5	30.74	0.002	0.64	748.8	647.5	1098	647.55	673	> 164.5	ok	485.7	φVc>Vu	not required
	Min Qy	0.133	1	93.1	19.14			735.8			647.55	673	> 93.1	ok	485.7	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation CF5:

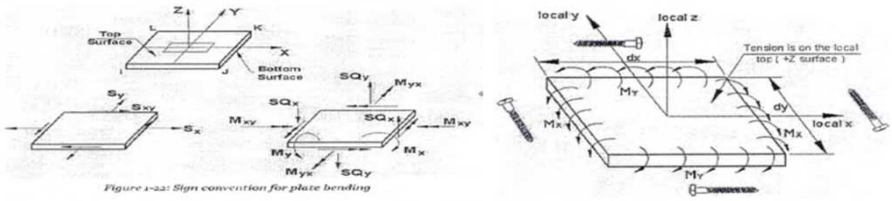


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G3 : CF_5_MOMENT At the face of the column refer staad file : Hopper supporting strycture_fdn_R0

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1191	1.127	0.017	0	0	0	-206.7	-297.8	5.948	-212.6	-303.7	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qx	997	-1.072	-0.264	0	0	0	-175.8	-286.4	0	-175.8	-286.4	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Qy	1200	0.128	0.617	0	0	0	-88.53	-132.9	91.78	-180.3	-224.7	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	1603	0.284	-1.388	0	0	0	-354.2	-340.6	-8.908	-363.1	-349.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Sx	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sx	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Sy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Sxy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sxy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Mx	2339	-0.045	-0.025	0	0	0	110.6	-3.183	-5.97	116.6	-9.153	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mx	1603	0.284	-1.388	0	0	0	-354.2	-340.6	-8.908	-355.1	-349.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max My	1536	0.101	-0.305	0	0	0	-106.2	174.6	-72.24	-178.4	246.9	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min My	1603	0.284	-1.388	0	0	0	-354.2	-340.6	-8.908	-363.1	-349.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Mxy	1203	-0.081	0.386	0	0	0	-70.64	-39.28	116.7	-187.3	-156	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mxy	1129	0.134	0.264	0	0	0	-36.34	-19.26	-173.2	-209.6	-192.5	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL

PLATE CENTER STRESS ; G4 : CF_5_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	2221	0.256	-0.23	0	0	0	-113.7	-50.12	-38.4	-152.1	-88.52	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qx	994	-0.34	-0.042	0	0	0	34.28	-127.6	0	34.28	-127.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Qy	2238	0.115	0.147	0	0	0	-18.74	-98.44	55.17	-73.9	-153.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	1645	0.034	-0.416	0	0	0	-248.5	-49.39	-18.82	-267.4	-68.21	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Sx	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sx	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Sy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Sxy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Min Sxy	2358	0.021	-0.028	0	0	0	-3.936	-4.372	-1.272	-5.208	-5.644	101 1.4 (DL1+DL2)
Max Mx	2339	-0.045	-0.025	0	0	0	110.6	-3.183	-5.97	116.6	-9.153	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mx	1645	0.034	-0.416	0	0	0	-248.5	-49.39	-18.82	-267.4	-68.21	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max My	1582	0.014	-0.091	0	0	0	-50.28	33.54	-21.04	-71.32	54.58	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min My	2235	0.249	-0.004	0	0	0	-13.77	-170.3	19.87	-33.65	-190.1	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Mxy	2238	0.115	0.147	0	0	0	-18.74	-98.44	55.17	-73.9	-153.6	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mxy	1126	-0.1	0.077	0	0	0	-1.886	-52.62	-94.04	-95.92	-146.7	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

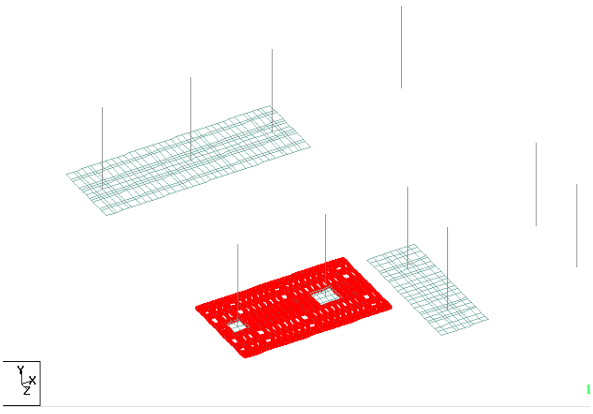
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	116.6	top face	50	640	116.6	0.014	484.6	1260	20	249.43	200	1571.5	22.16	27.7	0.066	tension control
Min Mx	-355.1	bottom face	75	615	355.1		1555.2		20	202.08	200	1571.5	22.16	27.7	0.064	tension control
Max Mxy	-187.3	bottom face	75	615	187.3		813.4		20	249.43	200	1571.5	22.16	27.7	0.064	tension control
Min Mxy	-209.6	bottom face	75	615	209.6	*As	911.0		20	249.43	200	1571.5	22.16	27.7	0.064	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.256	1	179.2	152.1	0.002	0.64	626	632.5	1098	625.96	2344	> 179.2	ok	469.5	φVc>Vu	not required
	Min Qx	0.34	1	238	34.28			724.5			632.52	2349	> 238	ok	474.4	φVc>Vu	not required

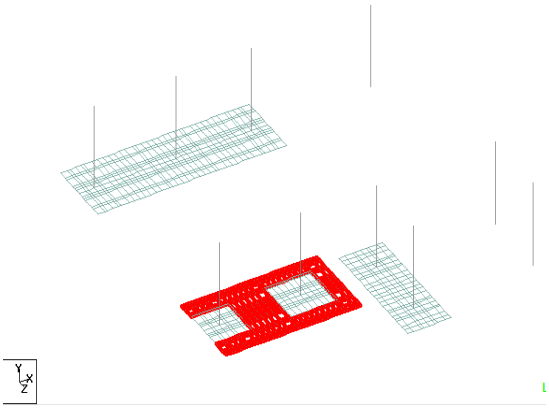
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	246.9	top face	50	640	246.9	0.014	1032.3	1260	20	249.43	200	1571.5	22.16	27.7	0.066	tension control
Min My	-349.6	bottom face	75	615	349.6		1530.5		20	205.35	200	1571.5	22.16	27.7	0.064	tension control
Max Mxy	-156.0	bottom face	75	615	156		676.1		20	249.43	200	1571.5	22.16	27.7	0.064	tension control
Min Mxy	-192.5	bottom face	75	615	192.5	*As	836.0		20	249.43	200	1571.5	22.16	27.7	0.064	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.147	1	102.9	153.6	0.002	0.64	617.3	632.5	1098	617.26	650	> 102.9	ok	462.9	φVc>Vu	not required
	Min Qy	0.416	1	291.2	68.21			678.8			632.52	662	> 291.2	ok	474.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation CF6:

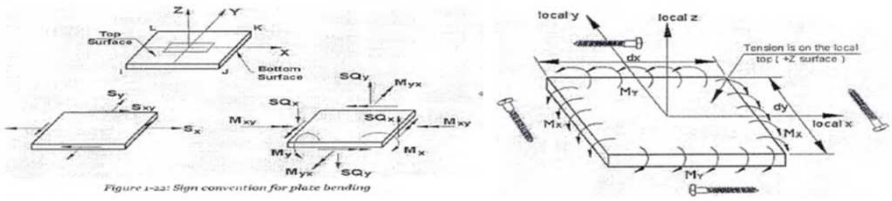


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G5 : CF_6_MOMENT At the face of the column refer staad file : Hopper supporting strycture_fdn_R0

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	1291	0.272	0.033	0	0	0	-3.924	-5.279	-11.39	-15.32	-16.67	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qx	1204	-0.617	0.238	0	0	0	-80.83	-92.63	0	-80.83	-92.63	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Qy	2176	-0.087	0.558	0	0	0	-6.839	-48.41	26.8	-33.64	-75.21	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	2141	-0.009	-0.559	0	0	0	2.566	-12.55	35.41	37.98	-47.96	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Sx	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sx	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Sy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Sxy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sxy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Mx	1281	-0.416	0.13	0	0	0	66.7	32.88	20.57	87.27	53.45	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mx	1276	-0.616	-0.197	0	0	0	-81.33	-91.58	23.52	-104.8	-115.1	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max My	1249	-0.043	0.063	0	0	0	-2.376	65.97	-10.05	-12.43	76.03	110 1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2
Min My	2178	-0.019	-0.07	0	0	0	-2.017	-182.5	-5.246	-7.263	-187.8	117 0.9 (DL1+DL2)+1.6WL(+Z)
Max Mxy	1242	0.258	-0.062	0	0	0	-3.688	-11.56	80.94	-84.63	-92.5	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mxy	1292	0.054	-0.2	0	0	0	-5.178	-55.42	-27.92	-33.1	-83.34	117 0.9 (DL1+DL2)+1.6WL(+Z)

PLATE CENTER STRESS ; G6 : CF_6_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	2290	0.086	-0.028	0	0	0	15.39	3.593	6.328	21.72	9.921	117 0.9 (DL1+DL2)+1.6WL(+Z)
Min Qx	2282	-0.163	-0.07	0	0	0	-4.508	-4.05	0	-4.508	-4.05	117 0.9 (DL1+DL2)+1.6WL(+Z)
Max Qy	2176	-0.087	0.558	0	0	0	-6.839	-48.41	26.8	-33.64	-75.21	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Qy	2141	-0.009	-0.559	0	0	0	2.566	-12.55	35.41	37.98	-47.96	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Max Sx	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sx	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Sy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Sxy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Min Sxy	2308	0	-0.002	0	0	0	0.432	0.484	0.408	0.84	0.892	101 1.4 (DL1+DL2)
Max Mx	2292	0.032	-0.096	0	0	0	34.73	4.06	2.497	37.22	6.557	117 0.9 (DL1+DL2)+1.6WL(+Z)
Min Mx	2292	-0.02	0.13	0	0	0	-27.87	-8.892	-3.845	-31.72	-12.74	110 1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2
Max My	1248	0	0.021	0	0	0	2.041	54.22	-6.342	8.383	60.56	110 1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2
Min My	2178	-0.019	-0.07	0	0	0	-2.017	-182.5	-5.246	-7.263	-187.8	117 0.9 (DL1+DL2)+1.6WL(+Z)
Max Mxy	1243	0.062	-0.006	0	0	0	-8.458	-13.86	78.8	-87.26	-92.66	102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
Min Mxy	1255	-0.012	0.031	0	0	0	0.349	24.95	-24.42	24.77	49.37	118 0.9 (DL1+DL2)+1.6WL(-Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

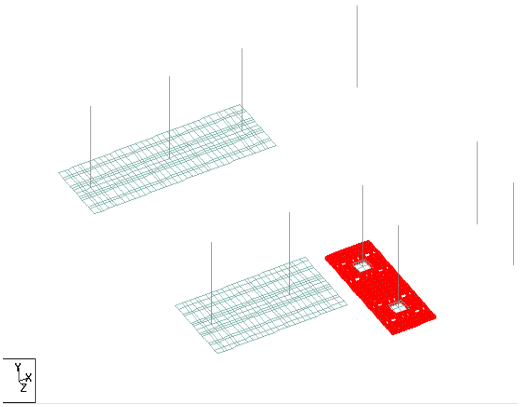
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	87.3	top face	50	442	87.27		526.8		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	-104.8	bottom face	75	417	104.8	0.014	672.8	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control
Max Mxy	-84.6	bottom face	75	417	84.63		541.8		16	223.49	200	1005.8	14.18	17.73	0.068	tension control
Min Mxy	-33.1	bottom face	75	417	33.1	*As	210.8		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.086	1	43	21.72	0.002	0.442	433.3	435.5	758.3	433.35	1619	> 43	ok	325	φVc>Vu	not required
	Min Qx	0.163	1	81.5	4.508			555			435.48	1621	> 81.5	ok	326.6	φVc>Vu	not required

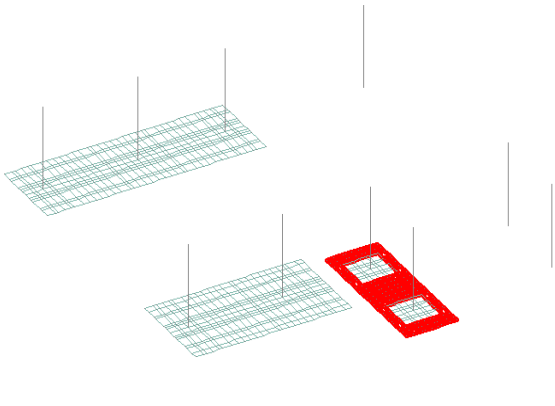
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	76.0	top face	50	442	76.03		458.4		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-187.8	bottom face	75	415	187.8	0.014	1222.5	900	20	257.08	200	1571.5	22.16	27.7	0.042	tension control
Max Mxy	-92.5	bottom face	75	417	92.5		592.8		16	223.49	200	1005.8	14.18	17.73	0.068	tension control
Min Mxy	-83.3	bottom face	75	417	83.34	*As	533.6		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.558	1	279	75.21	0.002	0.442	446.4	435.5	758.3	435.48	456	> 279	ok	326.6	φVc>Vu	not required
	Min Qy	0.559	1	279.5	47.96			462.4			435.48	456	> 279.5	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column

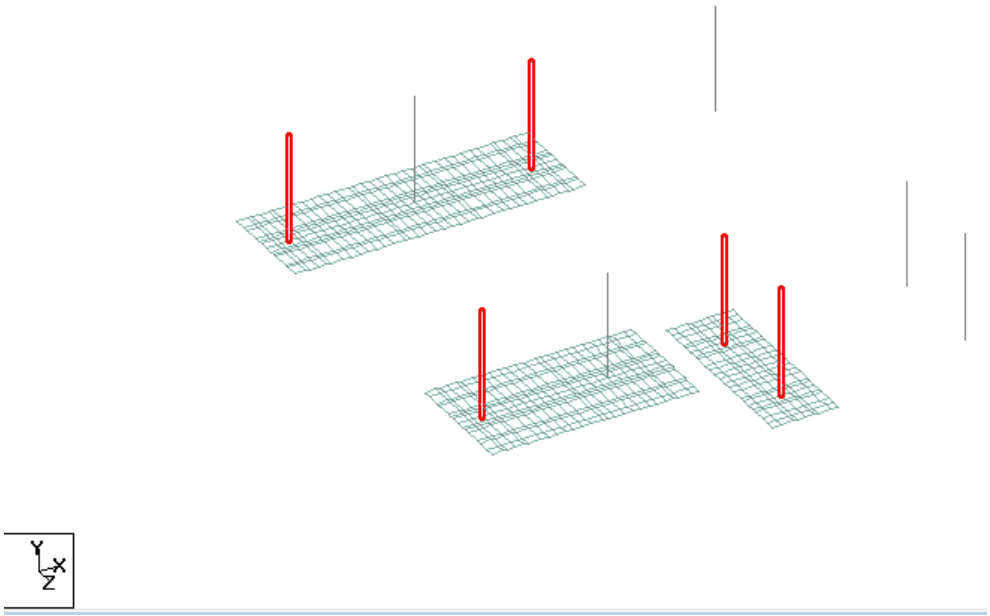


Shear taken at a distance d from face of the column

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Design of column C3



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Design of column C3

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 750 mm

Depth parallel to Z-Z axis, b = 750 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.81

Y_b = (b-d')/b = 0.81

Unsupported length of column parallel to X-X axis, l_{ux} = 2.4 m kl_{ux}/b = 6.4

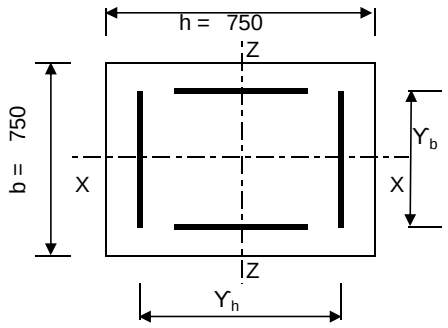
Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.4 m kl_{uz}/h = 6.4

< 22 neglect slenderness effect

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 15227.3 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx}= M_{ux}/\varphi$	$M_{nz}= M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx}= M_{nx}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^{\alpha} + (M_{nz}/M_{nz1})^{\alpha}$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	386.1	698.3	20	594	1074	31.4	0.03	0.080	1181	0.91	0.080	1181	0.03		0.86	safe
107	272.5	0.0	594	419	0	914.5	0.021	0.080	1181	0.00	0.080	1181	0.77		0.66	safe
102	224.3	3.6	547	345	6	841.2	0.018	0.080	1181	0.00	0.080	1181	0.71		0.58	safe
102	262.1	671.1	80	403	1032	123.1	0.02	0.080	1181	0.87	0.080	1181	0.10		0.83	safe
102	386.1	698.3	20	594	1074	31.4	0.03	0.080	1181	0.91	0.080	1181	0.03		0.86	safe
101	93.2	445.9	302	143	686	464.2	0.007	0.080	1181	0.58	0.080	1181	0.39	1.61	0.64	safe
101	93.2	445.9	302	143	686	464.2	0.007	0.080	1181	0.58	0.080	1181	0.39		0.64	safe
102	386.1	698.3	20	594	1074	31.4	0.03	0.080	1181	0.91	0.080	1181	0.03		0.86	safe
102	262.1	671.1	80	403	1032	123.1	0.02	0.080	1181	0.87	0.080	1181	0.10		0.83	safe
107	272.5	0.0	594	419	0	914.5	0.021	0.080	1181	0.00	0.080	1181	0.77		0.66	safe
102	224.3	3.6	547	345	6	841.2	0.018	0.080	1181	0.00	0.080	1181	0.71		0.58	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5625	12	25	+	0	0	5893	1.05	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

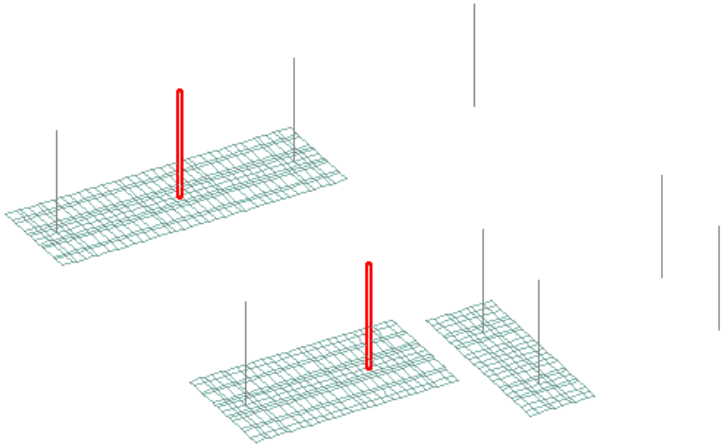
Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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Design of column C4





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Design of column C4

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.2 % i.e. ρ_g = 0.012

β = 0.65

k = 2

Column Dimensions :

Depth parallel to X-X axis, h = 1000 mm

Depth parallel to Z-Z axis, b = 1000 mm

Clear Cover = 50 mm

d' = 76 mm

Y_h = (h-d')/h = 0.85

Y_b = (b-d')/b = 0.85

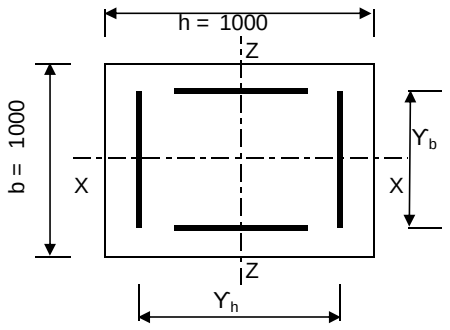
Unsupported length of column parallel to X-X axis, l_{ux} = 2.4 m kl_{ux}/b = 4.8 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.4 m kl_{uz}/h = 4.8

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 27819.3 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha=\log0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
102	2210	1939	148.9	3401	2983.1	229.1	0.097	0.100	3500	0.85	0.100	3500	0.07	1.61	0.79	safe
117	326.3	0.0	0.0	502	0.0	0.0	0.014	0.070	2450	0.00	0.070	2450	0.00		0	safe
107	1395	867.0	340	2146	1333.9	523	0.061	0.100	3500	0.38	0.100	3500	0.15		0.26	safe
116	942.3	553.5	14.4	1450	851.5	22.1	0.041	0.060	2100	0.41	0.060	2100	0.01		0.24	safe
102	2132	1684	197.7	3279	2591	304.1	0.094	0.090	3150	0.82	0.090	3150	0.10		0.75	safe
102	2210	1939	148.9	3401	2983	229.1	0.097	0.100	3500	0.85	0.100	3500	0.07		0.79	safe
101	1228	728.6	217.6	1890	1121	334.7	0.054	0.070	2450	0.46	0.070	2450	0.14		0.33	safe
101	1228	728.6	217.6	1890	1121	334.7	0.054	0.070	2450	0.46	0.070	2450	0.14		0.33	safe
102	2210	1939	148.9	3401	2983	229.1	0.097	0.100	3500	0.85	0.100	3500	0.07		0.79	safe
102	2132	1684	197.7	3279	2591	304.1	0.094	0.100	3500	0.74	0.100	3500	0.09		0.64	safe
107	1395	867.0	340	2146	1333.9	523	0.061	0.100	3500	0.38	0.100	3500	0.15		0.26	safe
116	942.3	553.5	14.4	1450	851.5	22.1	0.041	0.070	2450	0.35	0.070	2450	0.01		0.18	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db				A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
12000	16	32 +	0	0	12874	1.29	10	480	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.7

ACI DESIGN HANDBOOK-SP-17M(09)

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ATTACHMENT 1 STAAD OUTPUT FOR CRUSHER SUPPORTING STRUCTURE

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```
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*          STAAD.Pro CONNECT Edition             *
*          Version  22.06.00.138                 *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 12, 2022                 *
*          Time=    11:54:52                     *
*                                                *
*  Licensed to: Larsen & Toubro Ltd.             *
**************************************************
```

```
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INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\3.Primary Crusher\Civil\Super Structure_Screen,Crusher & .. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 14-FEB-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3.00001 0; 2 0 -3.00001 2.55001; 3 0 -3.00001 7.45002; 4 0 -3.00001 10
     9. 5 3.90001 -3.00001 0; 6 3.90001 -3.00001 2.55001; 7 3.90001 -3.00001 7.45002
    10. 8 3.90001 -3.00001 10; 9 8.40002 -3.00001 0; 10 8.40002 -3.00001 2.55001
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    12. 34 0 -0.2 2.55001; 35 0 -0.2 7.45002; 36 0 -0.2 10; 37 3.90001 -0.2 0
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    14. 41 8.40002 -0.2 0; 42 8.40002 -0.2 2.55001; 43 8.40002 -0.2 7.45002
    15. 44 8.40002 -0.2 10; 69 0 4.48001 0; 70 0 4.48001 2.55001; 71 0 4.48001 7.45002
    16. 72 0 4.48001 10; 73 3.90001 4.48001 0; 74 3.90001 4.48001 2.55001
    17. 75 3.90001 4.48001 4.00001; 76 3.90001 4.48001 7.45002; 77 3.90001 4.48001 10
    18. 78 8.40002 4.48001 0; 79 8.40002 4.48001 2.55001; 80 8.40002 4.48001 4.00001
    19. 81 8.40002 4.48001 7.45002; 82 8.40002 4.48001 10; 83 3.90001 4.48001 6.20001
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    23. 119 8.40002 4.48001 -1.5; 697 7.23502 4.48001 7.45002; 698 7.23502 4.48001 10
    24. 699 6.07002 4.48001 7.45002; 700 6.07002 4.48001 10
    25. 701 8.40002 4.48001 8.52502; 702 7.23502 4.48001 8.52502
    26. 703 6.07002 4.48001 8.52502; 704 2.41101 4.48001 7.45002
    27. 705 2.41101 4.48001 5.00001; 706 3.90001 2.69101 -1.5
    28. 707 3.90001 -3.00001 7.70502; 710 3.90001 -3.00001 7.96002
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    30. 716 3.90001 -3.00001 8.72501; 718 3.90001 -3.00001 8.98001
    31. 720 3.90001 -3.00001 9.23501; 722 3.90001 -3.00001 9.49
    32. 724 3.90001 -3.00001 9.745; 859 7.90002 -3.00001 7.70502
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109. 1204 0.250001 -3.00001 7.20002; 1205 0.250001 -3.00001 7.45002
110. 1206 0.250001 -3.00001 7.70002; 1207 0.250001 -3.00001 7.95002
111. 1208 0.250001 -3.00001 8.20002; 1209 0.500001 -3.00001 6.95002
112. 1210 0.500001 -3.00001 6.70002; 1211 0.500001 -3.00001 7.20002
113. 1212 0.500001 -3.00001 7.45002; 1213 0.500001 -3.00001 7.70002
114. 1214 0.500001 -3.00001 7.95002; 1215 0.500001 -3.00001 8.20002
115. 1222 -0.500001 -3.00001 9.5; 1223 -0.500001 -3.00001 9.25
116. 1225 -0.500001 -3.00001 9.75; 1227 -0.500001 -3.00001 10
117. 1229 -0.500001 -3.00001 10.25; 1231 -0.500001 -3.00001 10.5
118. 1232 -0.500001 -3.00001 10.75; 1233 -0.25 -3.00001 9.5
119. 1234 -0.25 -3.00001 9.25; 1235 -0.25 -3.00001 9.75; 1236 -0.25 -3.00001 10
120. 1237 -0.25 -3.00001 10.25; 1238 -0.25 -3.00001 10.5; 1239 -0.25 -3.00001 10.75
121. 1240 0 -3.00001 9.5; 1241 0 -3.00001 9.25; 1242 0 -3.00001 9.75
122. 1243 0 -3.00001 10.25; 1244 0 -3.00001 10.5; 1245 0 -3.00001 10.75
123. 1246 0.250001 -3.00001 9.5; 1247 0.250001 -3.00001 9.25
124. 1248 0.250001 -3.00001 9.75; 1249 0.250001 -3.00001 10
125. 1250 0.250001 -3.00001 10.25; 1251 0.250001 -3.00001 10.5
126. 1252 0.250001 -3.00001 10.75; 1253 0.500001 -3.00001 9.5
127. 1254 0.500001 -3.00001 9.25; 1255 0.500001 -3.00001 9.75
128. 1256 0.500001 -3.00001 10; 1257 0.500001 -3.00001 10.25
129. 1258 0.500001 -3.00001 10.5; 1259 0.500001 -3.00001 10.75
130. 1265 3.65001 -3.00001 0; 1266 3.65001 -3.00001 2.55001
131. 1267 3.65001 -3.00001 7.45002; 1268 3.65001 -3.00001 10
132. 1269 3.65001 -3.00001 7.70502; 1270 3.65001 -3.00001 7.96002
133. 1271 3.65001 -3.00001 8.21501; 1272 3.65001 -3.00001 8.47001
134. 1273 3.65001 -3.00001 8.72501; 1274 3.65001 -3.00001 8.98001
135. 1275 3.65001 -3.00001 9.23501; 1276 3.65001 -3.00001 9.49
136. 1277 3.65001 -3.00001 9.745; 1278 3.65001 -3.00001 0.255001
137. 1279 3.65001 -3.00001 0.510002; 1280 3.65001 -3.00001 0.765003
138. 1281 3.65001 -3.00001 1.02; 1282 3.65001 -3.00001 1.275
139. 1283 3.65001 -3.00001 1.53001; 1284 3.65001 -3.00001 1.78501
140. 1285 3.65001 -3.00001 2.04001; 1286 3.65001 -3.00001 2.29501
141. 1287 3.45001 -3.00001 0; 1288 3.45001 -3.00001 2.55001
142. 1289 3.45001 -3.00001 7.45002; 1290 3.45001 -3.00001 10
143. 1291 3.45001 -3.00001 7.70502; 1292 3.45001 -3.00001 7.96002
144. 1293 3.45001 -3.00001 8.21501; 1294 3.45001 -3.00001 8.47001
145. 1295 3.45001 -3.00001 8.72501; 1296 3.45001 -3.00001 8.98001
146. 1297 3.45001 -3.00001 9.23501; 1298 3.45001 -3.00001 9.49
147. 1299 3.45001 -3.00001 9.745; 1300 3.45001 -3.00001 0.255001
148. 1301 3.45001 -3.00001 0.510002; 1302 3.45001 -3.00001 0.765003
149. 1303 3.45001 -3.00001 1.02; 1304 3.45001 -3.00001 1.275
150. 1305 3.45001 -3.00001 1.53001; 1306 3.45001 -3.00001 1.78501

151. 1307 3.45001 -3.00001 2.04001; 1308 3.45001 -3.00001 2.29501
152. 1309 8.65002 -3.00001 0; 1310 8.65002 -3.00001 2.55001
153. 1311 8.65002 -3.00001 7.45002; 1312 8.65002 -3.00001 10
154. 1313 8.65002 -3.00001 7.70502; 1314 8.65002 -3.00001 7.96002
155. 1315 8.65002 -3.00001 8.21501; 1316 8.65002 -3.00001 8.47001
156. 1317 8.65002 -3.00001 8.72501; 1318 8.65002 -3.00001 8.98001
157. 1319 8.65002 -3.00001 9.23501; 1320 8.65002 -3.00001 9.49
158. 1321 8.65002 -3.00001 9.745; 1322 8.65002 -3.00001 0.255001
159. 1323 8.65002 -3.00001 0.510002; 1324 8.65002 -3.00001 0.765003
160. 1325 8.65002 -3.00001 1.02; 1326 8.65002 -3.00001 1.275
161. 1327 8.65002 -3.00001 1.53001; 1328 8.65002 -3.00001 1.78501
162. 1329 8.65002 -3.00001 2.04001; 1330 8.65002 -3.00001 2.29501
163. 1331 8.85002 -3.00001 0; 1332 8.85002 -3.00001 2.55001
164. 1333 8.85002 -3.00001 7.45002; 1334 8.85002 -3.00001 10
165. 1335 8.85002 -3.00001 7.70502; 1336 8.85002 -3.00001 7.96002
166. 1337 8.85002 -3.00001 8.21501; 1338 8.85002 -3.00001 8.47001
167. 1339 8.85002 -3.00001 8.72501; 1340 8.85002 -3.00001 8.98001
168. 1341 8.85002 -3.00001 9.23501; 1342 8.85002 -3.00001 9.49
169. 1343 8.85002 -3.00001 9.745; 1344 8.85002 -3.00001 0.255001
170. 1345 8.85002 -3.00001 0.510002; 1346 8.85002 -3.00001 0.765003
171. 1347 8.85002 -3.00001 1.02; 1348 8.85002 -3.00001 1.275
172. 1349 8.85002 -3.00001 1.53001; 1350 8.85002 -3.00001 1.78501
173. 1351 8.85002 -3.00001 2.04001; 1352 8.85002 -3.00001 2.29501
174. 1353 3.90001 -3.00001 -0.255001; 1354 3.90001 -3.00001 7.19502
175. 1355 8.40002 -3.00001 -0.255001; 1356 8.40002 -3.00001 7.19502
176. 1370 7.90002 -3.00001 7.19502; 1371 8.15002 -3.00001 7.19502
177. 1385 7.90002 -3.00001 -0.255001; 1386 8.15002 -3.00001 -0.255001
178. 1387 3.65001 -3.00001 -0.255001; 1388 3.65001 -3.00001 7.19502
179. 1389 3.45001 -3.00001 -0.255001; 1390 3.45001 -3.00001 7.19502
180. 1391 8.65002 -3.00001 -0.255001; 1392 8.65002 -3.00001 7.19502
181. 1393 8.85002 -3.00001 -0.255001; 1394 8.85002 -3.00001 7.19502
182. 1395 3.90001 -3.00001 -0.510001; 1396 3.90001 -3.00001 6.94002
183. 1397 8.40002 -3.00001 -0.510001; 1398 8.40002 -3.00001 6.94002
184. 1412 7.90002 -3.00001 6.94002; 1413 8.15002 -3.00001 6.94002
185. 1427 7.90002 -3.00001 -0.510001; 1428 8.15002 -3.00001 -0.510001
186. 1429 3.65001 -3.00001 -0.510001; 1430 3.65001 -3.00001 6.94002
187. 1431 3.45001 -3.00001 -0.510001; 1432 3.45001 -3.00001 6.94002
188. 1433 8.65002 -3.00001 -0.510001; 1434 8.65002 -3.00001 6.94002
189. 1435 8.85002 -3.00001 -0.510001; 1436 8.85002 -3.00001 6.94002
190. 1437 3.90001 -3.00001 -0.765002; 1438 3.90001 -3.00001 6.68502
191. 1439 8.40002 -3.00001 -0.765002; 1440 8.40002 -3.00001 6.68502
192. 1454 7.90002 -3.00001 6.68502; 1455 8.15002 -3.00001 6.68502
193. 1469 7.90002 -3.00001 -0.765002; 1470 8.15002 -3.00001 -0.765002
194. 1471 3.65001 -3.00001 -0.765002; 1472 3.65001 -3.00001 6.68502
195. 1473 3.45001 -3.00001 -0.765002; 1474 3.45001 -3.00001 6.68502
196. 1475 8.65002 -3.00001 -0.765002; 1476 8.65002 -3.00001 6.68502
197. 1477 8.85002 -3.00001 -0.765002; 1478 8.85002 -3.00001 6.68502
198. 1521 3.90001 -3.00001 2.80501; 1522 3.90001 -3.00001 10.255
199. 1523 8.40002 -3.00001 2.80501; 1524 8.40002 -3.00001 10.255
200. 1538 7.90002 -3.00001 10.255; 1539 8.15002 -3.00001 10.255
201. 1553 7.90002 -3.00001 2.80501; 1554 8.15002 -3.00001 2.80501
202. 1555 3.65001 -3.00001 2.80501; 1556 3.65001 -3.00001 10.255
203. 1557 3.45001 -3.00001 2.80501; 1558 3.45001 -3.00001 10.255
204. 1559 8.65002 -3.00001 2.80501; 1560 8.65002 -3.00001 10.255
205. 1561 8.85002 -3.00001 2.80501; 1562 8.85002 -3.00001 10.255
206. 1563 3.90001 -3.00001 3.06001; 1564 3.90001 -3.00001 10.51

207. 1565 8.40002 -3.00001 3.06001; 1566 8.40002 -3.00001 10.51
208. 1580 7.90002 -3.00001 10.51; 1581 8.15002 -3.00001 10.51
209. 1595 7.90002 -3.00001 3.06001; 1596 8.15002 -3.00001 3.06001
210. 1597 3.65001 -3.00001 3.06001; 1598 3.65001 -3.00001 10.51
211. 1599 3.45001 -3.00001 3.06001; 1600 3.45001 -3.00001 10.51
212. 1601 8.65002 -3.00001 3.06001; 1602 8.65002 -3.00001 10.51
213. 1603 8.85002 -3.00001 3.06001; 1604 8.85002 -3.00001 10.51
214. 1605 3.90001 -3.00001 3.31501; 1606 3.90001 -3.00001 10.765
215. 1607 8.40002 -3.00001 3.31501; 1608 8.40002 -3.00001 10.765
216. 1622 7.90002 -3.00001 10.765; 1623 8.15002 -3.00001 10.765
217. 1637 7.90002 -3.00001 3.31501; 1638 8.15002 -3.00001 3.31501
218. 1639 3.65001 -3.00001 3.31501; 1640 3.65001 -3.00001 10.765
219. 1641 3.45001 -3.00001 3.31501; 1642 3.45001 -3.00001 10.765
220. 1643 8.65002 -3.00001 3.31501; 1644 8.65002 -3.00001 10.765
221. 1645 8.85002 -3.00001 3.31501; 1646 8.85002 -3.00001 10.765
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223. 1691 4.40001 -3.00001 7.70502; 1692 4.40001 -3.00001 7.45002
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225. 1695 4.15001 -3.00001 8.21501; 1696 4.40001 -3.00001 8.21501
226. 1697 4.15001 -3.00001 8.47001; 1698 4.40001 -3.00001 8.47001
227. 1699 4.15001 -3.00001 8.72501; 1700 4.40001 -3.00001 8.72501
228. 1701 4.15001 -3.00001 8.98001; 1702 4.40001 -3.00001 8.98001
229. 1703 4.15001 -3.00001 9.23501; 1704 4.40001 -3.00001 9.23501
230. 1705 4.15001 -3.00001 9.49; 1706 4.40001 -3.00001 9.49
231. 1707 4.15001 -3.00001 9.745; 1708 4.40001 -3.00001 9.745
232. 1709 4.15001 -3.00001 10; 1710 4.40001 -3.00001 10
233. 1711 4.15001 -3.00001 0.255001; 1712 4.15001 -3.00001 0
234. 1713 4.40001 -3.00001 0.255001; 1714 4.40001 -3.00001 0
235. 1715 4.15001 -3.00001 0.510002; 1716 4.40001 -3.00001 0.510002
236. 1717 4.15001 -3.00001 0.765003; 1718 4.40001 -3.00001 0.765003
237. 1719 4.15001 -3.00001 1.02; 1720 4.40001 -3.00001 1.02
238. 1721 4.15001 -3.00001 1.275; 1722 4.40001 -3.00001 1.275
239. 1723 4.15001 -3.00001 1.53001; 1724 4.40001 -3.00001 1.53001
240. 1725 4.15001 -3.00001 1.78501; 1726 4.40001 -3.00001 1.78501
241. 1727 4.15001 -3.00001 2.04001; 1728 4.40001 -3.00001 2.04001
242. 1729 4.15001 -3.00001 2.29501; 1730 4.40001 -3.00001 2.29501
243. 1731 4.15001 -3.00001 2.55001; 1732 4.40001 -3.00001 2.55001
244. 1733 4.15001 -3.00001 7.19502; 1734 4.40001 -3.00001 7.19502
245. 1735 4.15001 -3.00001 -0.255001; 1736 4.40001 -3.00001 -0.255001
246. 1737 4.15001 -3.00001 6.94002; 1738 4.40001 -3.00001 6.94002
247. 1739 4.15001 -3.00001 -0.510001; 1740 4.40001 -3.00001 -0.510001
248. 1741 4.15001 -3.00001 6.68502; 1742 4.40001 -3.00001 6.68502
249. 1743 4.15001 -3.00001 -0.765002; 1744 4.40001 -3.00001 -0.765002
250. 1749 4.15001 -3.00001 10.255; 1750 4.40001 -3.00001 10.255
251. 1751 4.15001 -3.00001 2.80501; 1752 4.40001 -3.00001 2.80501
252. 1753 4.15001 -3.00001 10.51; 1754 4.40001 -3.00001 10.51
253. 1755 4.15001 -3.00001 3.06001; 1756 4.40001 -3.00001 3.06001
254. 1757 4.15001 -3.00001 10.765; 1758 4.40001 -3.00001 10.765
255. 1759 4.15001 -3.00001 3.31501; 1760 4.40001 -3.00001 3.31501
256. 1765 7.45002 4.48001 4.00001; 1766 7.45002 4.48001 6.20001
257. 1771 -0.500001 -3.00001 1; 1772 -0.25 -3.00001 1; 1773 0 -3.00001 1
258. 1774 0.250001 -3.00001 1; 1775 0.500001 -3.00001 1
259. 1776 -0.500001 -3.00001 8.45002; 1777 -0.25 -3.00001 8.45002
260. 1778 0 -3.00001 8.45002; 1779 0.250001 -3.00001 8.45002
261. 1780 0.500001 -3.00001 8.45002; 1785 -0.500001 -3.00001 1.25
262. 1786 -0.25 -3.00001 1.25; 1787 0 -3.00001 1.25; 1788 0.250001 -3.00001 1.25

263. 1789 0.500001 -3.00001 1.25; 1790 -0.500001 -3.00001 8.70002
264. 1791 -0.25 -3.00001 8.70002; 1792 0 -3.00001 8.70002
265. 1793 0.250001 -3.00001 8.70002; 1794 0.500001 -3.00001 8.70002
266. 1799 -0.500001 -3.00001 1.5; 1800 -0.25 -3.00001 1.5; 1801 0 -3.00001 1.5
267. 1802 0.250001 -3.00001 1.5; 1803 0.500001 -3.00001 1.5
268. 1804 -0.500001 -3.00001 8.95002; 1805 -0.25 -3.00001 8.95002
269. 1806 0 -3.00001 8.95002; 1807 0.250001 -3.00001 8.95002
270. 1808 0.500001 -3.00001 8.95002; 1809 4.65001 -3.00001 7.70502
271. 1810 4.65001 -3.00001 7.45002; 1811 4.65001 -3.00001 7.96002
272. 1812 4.65001 -3.00001 8.21501; 1813 4.65001 -3.00001 8.47001
273. 1814 4.65001 -3.00001 8.72501; 1815 4.65001 -3.00001 8.98001
274. 1816 4.65001 -3.00001 9.23501; 1817 4.65001 -3.00001 9.49
275. 1818 4.65001 -3.00001 9.745; 1819 4.65001 -3.00001 10
276. 1820 4.65001 -3.00001 0.255001; 1821 4.65001 -3.00001 0
277. 1822 4.65001 -3.00001 0.510002; 1823 4.65001 -3.00001 0.765003
278. 1824 4.65001 -3.00001 1.02; 1825 4.65001 -3.00001 1.275
279. 1826 4.65001 -3.00001 1.53001; 1827 4.65001 -3.00001 1.78501
280. 1828 4.65001 -3.00001 2.04001; 1829 4.65001 -3.00001 2.29501
281. 1830 4.65001 -3.00001 2.55001; 1831 4.65001 -3.00001 7.19502
282. 1832 4.65001 -3.00001 -0.255001; 1833 4.65001 -3.00001 6.94002
283. 1834 4.65001 -3.00001 -0.510001; 1835 4.65001 -3.00001 6.68502
284. 1836 4.65001 -3.00001 -0.765002; 1837 4.65001 -3.00001 10.255
285. 1838 4.65001 -3.00001 2.80501; 1839 4.65001 -3.00001 10.51
286. 1840 4.65001 -3.00001 3.06001; 1841 4.65001 -3.00001 10.765
287. 1842 4.65001 -3.00001 3.31501; 1843 4.90001 -3.00001 7.70502
288. 1844 4.90001 -3.00001 7.45002; 1845 4.90001 -3.00001 7.96002
289. 1846 4.90001 -3.00001 8.21501; 1847 4.90001 -3.00001 8.47001
290. 1848 4.90001 -3.00001 8.72501; 1849 4.90001 -3.00001 8.98001
291. 1850 4.90001 -3.00001 9.23501; 1851 4.90001 -3.00001 9.49
292. 1852 4.90001 -3.00001 9.745; 1853 4.90001 -3.00001 10
293. 1854 4.90001 -3.00001 0.255001; 1855 4.90001 -3.00001 0
294. 1856 4.90001 -3.00001 0.510002; 1857 4.90001 -3.00001 0.765003
295. 1858 4.90001 -3.00001 1.02; 1859 4.90001 -3.00001 1.275
296. 1860 4.90001 -3.00001 1.53001; 1861 4.90001 -3.00001 1.78501
297. 1862 4.90001 -3.00001 2.04001; 1863 4.90001 -3.00001 2.29501
298. 1864 4.90001 -3.00001 2.55001; 1865 4.90001 -3.00001 7.19502
299. 1866 4.90001 -3.00001 -0.255001; 1867 4.90001 -3.00001 6.94002
300. 1868 4.90001 -3.00001 -0.510001; 1869 4.90001 -3.00001 6.68502
301. 1870 4.90001 -3.00001 -0.765002; 1871 4.90001 -3.00001 10.255
302. 1872 4.90001 -3.00001 2.80501; 1873 4.90001 -3.00001 10.51
303. 1874 4.90001 -3.00001 3.06001; 1875 4.90001 -3.00001 10.765
304. 1876 4.90001 -3.00001 3.31501; 1877 5.15001 -3.00001 7.70502
305. 1878 5.15001 -3.00001 7.45002; 1879 5.15001 -3.00001 7.96002
306. 1880 5.15001 -3.00001 8.21501; 1881 5.15001 -3.00001 8.47001
307. 1882 5.15001 -3.00001 8.72501; 1883 5.15001 -3.00001 8.98001
308. 1884 5.15001 -3.00001 9.23501; 1885 5.15001 -3.00001 9.49
309. 1886 5.15001 -3.00001 9.745; 1887 5.15001 -3.00001 10
310. 1888 5.15001 -3.00001 0.255001; 1889 5.15001 -3.00001 0
311. 1890 5.15001 -3.00001 0.510002; 1891 5.15001 -3.00001 0.765003
312. 1892 5.15001 -3.00001 1.02; 1893 5.15001 -3.00001 1.275
313. 1894 5.15001 -3.00001 1.53001; 1895 5.15001 -3.00001 1.78501
314. 1896 5.15001 -3.00001 2.04001; 1897 5.15001 -3.00001 2.29501
315. 1898 5.15001 -3.00001 2.55001; 1899 5.15001 -3.00001 7.19502
316. 1900 5.15001 -3.00001 -0.255001; 1901 5.15001 -3.00001 6.94002
317. 1902 5.15001 -3.00001 -0.510001; 1903 5.15001 -3.00001 6.68502
318. 1904 5.15001 -3.00001 -0.765002; 1905 5.15001 -3.00001 10.255

319. 1906 5.15001 -3.00001 2.80501; 1907 5.15001 -3.00001 10.51
320. 1908 5.15001 -3.00001 3.06001; 1909 5.15001 -3.00001 10.765
321. 1910 5.15001 -3.00001 3.31501; 1911 5.40001 -3.00001 7.70502
322. 1912 5.40001 -3.00001 7.45002; 1913 5.40001 -3.00001 7.96002
323. 1914 5.40001 -3.00001 8.21501; 1915 5.40001 -3.00001 8.47001
324. 1916 5.40001 -3.00001 8.72501; 1917 5.40001 -3.00001 8.98001
325. 1918 5.40001 -3.00001 9.23501; 1919 5.40001 -3.00001 9.49
326. 1920 5.40001 -3.00001 9.745; 1921 5.40001 -3.00001 10
327. 1922 5.40001 -3.00001 0.255001; 1923 5.40001 -3.00001 0
328. 1924 5.40001 -3.00001 0.510002; 1925 5.40001 -3.00001 0.765003
329. 1926 5.40001 -3.00001 1.02; 1927 5.40001 -3.00001 1.275
330. 1928 5.40001 -3.00001 1.53001; 1929 5.40001 -3.00001 1.78501
331. 1930 5.40001 -3.00001 2.04001; 1931 5.40001 -3.00001 2.29501
332. 1932 5.40001 -3.00001 2.55001; 1933 5.40001 -3.00001 7.19502
333. 1934 5.40001 -3.00001 -0.255001; 1935 5.40001 -3.00001 6.94002
334. 1936 5.40001 -3.00001 -0.510001; 1937 5.40001 -3.00001 6.68502
335. 1938 5.40001 -3.00001 -0.765002; 1939 5.40001 -3.00001 10.255
336. 1940 5.40001 -3.00001 2.80501; 1941 5.40001 -3.00001 10.51
337. 1942 5.40001 -3.00001 3.06001; 1943 5.40001 -3.00001 10.765
338. 1944 5.40001 -3.00001 3.31501; 1945 5.65001 -3.00001 7.70502
339. 1946 5.65001 -3.00001 7.45002; 1947 5.65001 -3.00001 7.96002
340. 1948 5.65001 -3.00001 8.21501; 1949 5.65001 -3.00001 8.47001
341. 1950 5.65001 -3.00001 8.72501; 1951 5.65001 -3.00001 8.98001
342. 1952 5.65001 -3.00001 9.23501; 1953 5.65001 -3.00001 9.49
343. 1954 5.65001 -3.00001 9.745; 1955 5.65001 -3.00001 10
344. 1956 5.65001 -3.00001 0.255001; 1957 5.65001 -3.00001 0
345. 1958 5.65001 -3.00001 0.510002; 1959 5.65001 -3.00001 0.765003
346. 1960 5.65001 -3.00001 1.02; 1961 5.65001 -3.00001 1.275
347. 1962 5.65001 -3.00001 1.53001; 1963 5.65001 -3.00001 1.78501
348. 1964 5.65001 -3.00001 2.04001; 1965 5.65001 -3.00001 2.29501
349. 1966 5.65001 -3.00001 2.55001; 1967 5.65001 -3.00001 7.19502
350. 1968 5.65001 -3.00001 -0.255001; 1969 5.65001 -3.00001 6.94002
351. 1970 5.65001 -3.00001 -0.510001; 1971 5.65001 -3.00001 6.68502
352. 1972 5.65001 -3.00001 -0.765002; 1973 5.65001 -3.00001 10.255
353. 1974 5.65001 -3.00001 2.80501; 1975 5.65001 -3.00001 10.51
354. 1976 5.65001 -3.00001 3.06001; 1977 5.65001 -3.00001 10.765
355. 1978 5.65001 -3.00001 3.31501; 1979 5.90001 -3.00001 7.70502
356. 1980 5.90001 -3.00001 7.45002; 1981 5.90001 -3.00001 7.96002
357. 1982 5.90001 -3.00001 8.21501; 1983 5.90001 -3.00001 8.47001
358. 1984 5.90001 -3.00001 8.72501; 1985 5.90001 -3.00001 8.98001
359. 1986 5.90001 -3.00001 9.23501; 1987 5.90001 -3.00001 9.49
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1763. 1547 1548
1764. _CRUSHER_FLOOR2 32 560 561 563 TO 566 568
1765. MEMBER
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1773. _FB_1 24 TO 27 29 30 32 TO 40 115 560 561 563 564 566 571
1774. _FB_2 43 46 117 572
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1775. _FB_3 565 568 573
1776. _FB_4 120 121 125 TO 127 574
1777. _TB_1 1 TO 4 6 7 9 TO 17 118 119 2265 2268 2270 2272 2274 2276 2278 2280 2282 -
1778. 2284 2286 2288 2290 2292 2294 2296 2298 2560 2563 2565 2567 2569 2571 2573 -
1779. 2575 2577 2579 2581 2583 2585 2587 2589 2591 2593 2859 2862 2864 2866 2868 -
1780. 2870 2872 2874 2876 2975 2977 2979 2981 2983 2985 2987 2989 2991 3095 3098 -
1781. 3100 3102 3104 3106 3108 3110 3112 3211 3213 3215 3217 3219 3221 3223 3225 -
1782. 3227
1783. _FB_1A 28 31 41 42 44 45 116
1784. ELEMENT
1785. _CF_1_MOMENT 901 TO 908 911 TO 914 917 TO 924 937 TO 944 947 TO 950 -
1786. 953 TO 960 973 TO 980 983 TO 986 989 TO 996 1009 TO 1016 1019 TO 1022 1025 -
1787. 1026 TO 1032 1551 TO 1554 1557 TO 1560 1563 TO 1566 1569 TO 1572 1575 TO 1578 -
1788. 1581 TO 1584 1587 TO 1590 1593 TO 1596 2045 TO 2172
1789. _CF_1_SHEAR 1551 TO 1554 1557 TO 1560 1563 TO 1566 1569 TO 1572 1575 TO 1578 -
1790. 1581 TO 1584 1587 TO 1590 1593 TO 1596 2069 TO 2108 2133 TO 2172
1791. _CF_2_MOMENT 715 TO 724 726 TO 733 875 TO 884 886 TO 893 1040 TO 1047 1050 -
1792. 1051 TO 1057 1059 TO 1078 1080 TO 1087 1090 TO 1097 1099 TO 1118 1133 1149 -
1793. 1153 1154 1157 1158 1173 1174 1189 TO 1198 1213 1214 1229 TO 1238 1293 1309 -
1794. 1313 1314 1317 1318 1333 1334 1349 TO 1358 1373 1374 1389 TO 1398 1440 1442 -
1795. 1443 1445 1446 1448 1449 1451 1452 1454 1455 1457 1458 1460 1461 1463 1464 -
1796. 1467 1470 1472 1473 1475 1476 1478 1479 1481 1482 1484 1485 1487 1488 1490 -
1797. 1491 1493 1494 1497 1500 1503 1505 1506 1508 1509 1511 1512 1514 1515 1524 -
1798. 1527 1529 1530 1532 1533 1535 1536 1538 1539 1597 TO 2044 2173 TO 2260
1799. _CF_2_SHEAR 718 TO 721 728 TO 731 878 TO 881 888 TO 891 1042 TO 1045 1052 -
1800. 1053 TO 1055 1062 TO 1065 1072 TO 1075 1082 TO 1085 1092 TO 1095 1102 TO 1105 -
1801. 1112 TO 1115 1448 1449 1451 1452 1454 1455 1457 1458 1478 1479 1481 1482 -
1802. 1484 1485 1487 1488 1600 TO 1603 1610 TO 1613 1629 TO 2012 2016 TO 2019 2026 -
1803. 2027 TO 2029 2173 TO 2260
1804. _WALL_MT 2269 2271 2273 2275 2277 2279 2281 2283 2285 2287 2289 2291 2293 -
1805. 2295 2297 2299 2304 TO 2319 2324 TO 2339 2344 TO 2359 2364 TO 2379 -
1806. 2384 TO 2399 2404 TO 2419 2424 TO 2439 2444 TO 2459 2464 TO 2479 -
1807. 2484 TO 2499 2504 TO 2519 2524 TO 2539 2543 TO 2558 2564 2566 2568 2570 2572 -
1808. 2574 2576 2578 2580 2582 2584 2586 2588 2590 2592 2594 2599 TO 2614 2619 -
1809. 2620 TO 2634 2639 TO 2654 2659 TO 2674 2679 TO 2694 2699 TO 2714 2719 TO 2734 -
1810. 2739 TO 2754 2759 TO 2774 2779 TO 2794 2799 TO 2814 2819 TO 2834 -
1811. 2838 TO 2853 2863 2865 2867 2869 2871 2873 2875 2877 2881 TO 2888 -
1812. 2892 TO 2899 2903 TO 2910 2914 TO 2921 2925 TO 2932 2936 TO 2943 -
1813. 2947 TO 2954 2959 2961 2963 2965 2967 2969 2971 2973 2978 2980 2982 2984 -
1814. 2986 2988 2990 2992 2996 TO 3003 3007 TO 3014 3018 TO 3025 3029 TO 3036 3040 -
1815. 3041 TO 3047 3051 TO 3058 3062 TO 3069 3075 3077 3079 3081 3083 3085 3087 -
1816. 3089 3099 3101 3103 3105 3107 3109 3111 3113 3117 TO 3124 3128 TO 3135 3139 -
1817. 3140 TO 3146 3150 TO 3157 3161 TO 3168 3172 TO 3179 3183 TO 3190 3195 3197 -
1818. 3199 3201 3203 3205 3207 3209 3214 3216 3218 3220 3222 3224 3226 3228 3232 -
1819. 3233 TO 3239 3243 TO 3250 3254 TO 3261 3265 TO 3272 3276 TO 3283 3287 TO 3294 -
1820. 3298 TO 3305 3311 3313 3315 3317 3319 3321 3323 3325
1821. _WALL_SH 2269 2271 2273 2275 2277 2279 2281 2283 2285 2287 2289 2291 2293 -
1822. 2295 2297 2299 2304 TO 2319 2324 TO 2339 2344 TO 2359 2364 TO 2379 -
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1824. 2484 TO 2499 2504 TO 2519 2524 TO 2539 2543 TO 2558 2564 2566 2568 2570 2572 -
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1829. 2892 TO 2899 2903 TO 2910 2914 TO 2921 2925 TO 2932 2936 TO 2943 -
1830. 2947 TO 2954 2959 2961 2963 2965 2967 2969 2971 2973 2978 2980 2982 2984 -
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1831. 2986 2988 2990 2992 2996 TO 3003 3007 TO 3014 3018 TO 3025 3029 TO 3036 3040 -
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1837. 3298 TO 3305 3311 3313 3315 3317 3319 3321 3323 3325
1838. END GROUP DEFINITION
1839. ELEMENT PROPERTY
1840. 715 TO 734 875 TO 894 1039 TO 1118 1133 1134 1149 TO 1158 1173 1174 -
1841. 1189 TO 1198 1213 1214 1229 TO 1238 1293 1294 1309 TO 1318 1333 1334 1349 -
1842. 1350 TO 1358 1373 1374 1389 TO 1398 1439 1440 1442 1443 1445 1446 1448 1449 -
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1845. 1496 1497 1499 1500 1502 1503 1505 1506 1508 1509 1511 1512 1514 1515 1523 -
1846. 1524 1526 1527 1529 1530 1532 1533 1535 1536 1538 1539 1597 TO 2044 2173 -
1847. 2174 TO 2260 THICKNESS 0.6
1848. 901 TO 924 937 TO 960 973 TO 996 1009 TO 1032 1551 TO 1554 1557 TO 1560 1563 -
1849. 1564 TO 1566 1569 TO 1572 1575 TO 1578 1581 TO 1584 1587 TO 1590 1593 TO 1596 -
1850. 2045 TO 2172 THICKNESS 0.4
1851. 2267 2269 2271 2273 2275 2277 2279 2281 2283 2285 2287 2289 2291 2293 2295 -
1852. 2297 2299 2301 2303 TO 2319 2321 2323 TO 2339 2341 2343 TO 2359 2361 2363 -
1853. 2364 TO 2379 2381 2383 TO 2399 2401 2403 TO 2419 2421 2423 TO 2439 2441 2443 -
1854. 2444 TO 2459 2461 2463 TO 2479 2481 2483 TO 2499 2501 2503 TO 2519 2521 2523 -
1855. 2524 TO 2539 2541 TO 2559 2562 2564 2566 2568 2570 2572 2574 2576 2578 2580 -
1856. 2582 2584 2586 2588 2590 2592 2594 2596 2598 TO 2614 2616 2618 TO 2634 2636 -
1857. 2638 TO 2654 2656 2658 TO 2674 2676 2678 TO 2694 2696 2698 TO 2714 2716 2718 -
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1860. 2873 2875 2877 2878 2880 TO 2889 2891 TO 2900 2902 TO 2911 2913 TO 2922 2924 -
1861. 2925 TO 2933 2935 TO 2944 2946 TO 2955 2957 2959 2961 2963 2965 2967 2969 -
1862. 2971 2973 2974 2976 2978 2980 2982 2984 2986 2988 2990 2992 2994 TO 3003 -
1863. 3005 TO 3014 3016 TO 3025 3027 TO 3036 3038 TO 3047 3049 TO 3058 -
1864. 3060 TO 3069 3071 3073 3075 3077 3079 3081 3083 3085 3087 3089 3090 3097 -
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1866. 3139 TO 3147 3149 TO 3158 3160 TO 3169 3171 TO 3180 3182 TO 3191 3193 3195 -
1867. 3197 3199 3201 3203 3205 3207 3209 3210 3212 3214 3216 3218 3220 3222 3224 -
1868. 3226 3228 3230 TO 3239 3241 TO 3250 3252 TO 3261 3263 TO -
1869. 3269  THICKNESS 0.3
1870. 3270 THICKNESS 0.3
1871. 3271 THICKNESS 0.3
1872. 3272 3274 TO 3283 3285 TO 3294 3296 TO 3305 3307 3309 3311 3313 3315 3317 -
1873. 3319 3321 3323 3325 3326 THICKNESS 0.3
1874. MEMBER RELEASE
1875. 43 46 117 562 565 569 573 1549 START MY MZ
1876. 567 568 570 572 573 1547 TO 1549 END MY MZ
1877. DEFINE MATERIAL START
1878. ISOTROPIC CONCRETE
1879. E 3E+07
1880. POISSON 0.17
1881. DENSITY 24
1882. ALPHA 5.5E-06
1883. DAMP 0.05
1884. TYPE CONCRETE
1885. STRENGTH FCU 27578.9
1886. END DEFINE MATERIAL
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1887. MEMBER PROPERTY INDIAN
1888. 47 TO 70 122 123 2261 TO 2264 2266 2300 2302 2320 2322 2340 2342 2360 2362 -
1889. 2380 2382 2400 2402 2420 2422 2440 2442 2460 2462 2480 2482 2500 2502 2520 -
1890. 2522 2540 2561 2595 2597 2615 2617 2635 2637 2655 2657 2675 2677 2695 2697 -
1891. 2715 2717 2735 2737 2755 2757 2775 2777 2795 2797 2815 2817 2835 2855 2857 -
1892. 2860 2879 2890 2901 2912 2923 2934 2945 2993 3004 3015 3026 3037 3048 3059 -
1893. 3070 3091 3093 3096 3115 3126 3137 3148 3159 3170 3181 3229 3240 3251 3262 -
1894. 3273 3284 3295 3306 3327 TO 3338 PRIS YD 0.5 ZD 0.5
1895. 24 TO 46 115 TO 117 560 561 563 TO 566 568 571 572 1547 TO 1548 -
1896. 1549 PRIS YD 0.8 ZD 0.4
1897. 1 TO 4 6 7 9 TO 17 118 TO 121 125 TO 127 574 2265 2268 2270 2272 2274 2276 -
1898. 2278 2280 2282 2284 2286 2288 2290 2292 2294 2296 2298 2560 2563 2565 2567 -
1899. 2569 2571 2573 2575 2577 2579 2581 2583 2585 2587 2589 2591 2593 2859 2862 -
1900. 2864 2866 2868 2870 2872 2874 2876 2975 2977 2979 2981 2983 2985 2987 2989 -
1901. 2991 3095 3098 3100 3102 3104 3106 3108 3110 3112 3211 3213 3215 3217 3219 -
1902. 3221 3223 3225 3227 PRIS YD 0.4 ZD 0.3
1903. 562 567 569 570 573 PRIS YD 0.3 ZD 0.6
1904. CONSTANTS
1905. MATERIAL CONCRETE ALL
1906. SUPPORTS
1907. 715 TO 734 875 TO 894 901 TO 924 937 TO 960 973 TO 996 1009 TO 1032 -
1908. 1039 TO 1118 1133 1134 1149 TO 1158 1173 1174 1189 TO 1198 1213 1214 1229 -
1909. 1230 TO 1238 1293 1294 1309 TO 1318 1333 1334 1349 TO 1358 1373 1374 1389 -
1910. 1390 TO 1398 1439 1440 1442 1443 1445 1446 1448 1449 1451 1452 1454 1455 1457 -
1911. 1458 1460 1461 1463 1464 1466 1467 1469 1470 1472 1473 1475 1476 1478 1479 -
1912. 1481 1482 1484 1485 1487 1488 1490 1491 1493 1494 1496 1497 1499 1500 1502 -
1913. 1503 1505 1506 1508 1509 1511 1512 1514 1515 1523 1524 1526 1527 1529 1530 -
1914. 1532 1533 1535 1536 1538 1539 1551 TO 1554 1557 TO 1560 1563 TO 1566 1569 -
1915. 1570 TO 1572 1575 TO 1578 1581 TO 1584 1587 TO 1590 1593 TO 2259 -
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TOTAL	65.340012

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1918. LOAD R3 LOADTYPE WIND  TITLE WIND LOAD(W+X)
1919. *ROCK BREAKER
1920. JOINT LOAD
1921. *17
1922. 73 FX 6.721 FY 11.628 FZ 0.002 MX 0.004 MY -0 MZ -0
1923. *18
1924. 74 FX 6.72 FY 11.698 FZ -0.002 MX -0.005 MY -0 MZ -0
1925. *19
1926. 78 FX 6.721 FY -11.629 FZ -0.003 MX -0.003 MY -0 MZ -0
1927. *20
1928. 79 FX 6.72 FY -11.697 FZ 0.002 MX 0.006 MY -0 MZ -0
1929. MEMBER LOAD
1930. 48 54 3327 3330 UNI GX 1.39
1931. 50 52 58 60 66 68 2261 TO 2264 2266 2300 2302 2320 2322 2340 2342 2360 2362 -
1932. 2380 2382 2400 2402 2420 2422 2440 2442 2460 2462 2480 2482 2500 2502 2520 -
1933. 2522 2540 2561 2595 2597 2615 2617 2635 2637 2655 2657 2675 2677 2695 2697 -
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1934. 2715 2717 2735 2737 2755 2757 2775 2777 2795 2797 2815 2817 2835 3328 3329 -
1935. 3334 TO 3337 UNI GX 1.86
1936. 56 62 64 70 122 123 2855 2857 2860 2879 2890 2901 2912 2923 2934 2945 2993 -
1937. 3004 3015 3026 3037 3048 3059 3070 3091 3093 3096 3115 3126 3137 3148 3159 -
1938. 3170 3181 3229 3240 3251 3262 3273 3284 3295 3306 3331 TO 3333 -
1939. 3338 UNI GX 1.39
1940. LOAD R4 LOADTYPE WIND  TITLE WIND LOAD(W+Z)
1941. *ROCK BREAKER
1942. JOINT LOAD
1943. *17+B16:B23
1944. 73 FX 0.001 FY 9.003 FZ 7.176 MX 11.008 MY -0 MZ -0
1945. *18
1946. 74 FX -0.001 FY -9.003 FZ 7.175 MX 11.007 MY -0 MZ -0
1947. *19
1948. 78 FX -0.001 FY 9.27 FZ 7.177 MX 11.005 MY -0 MZ -0
1949. *20
1950. 79 FX 0.001 FY -9.27 FZ 7.174 MX 11 MY -0 MZ -0
1951. MEMBER LOAD
1952. 48 50 52 54 3327 TO 3330 UNI GZ 1
1953. 56 58 60 62 122 123 2261 2262 2266 2302 2322 2342 2362 2382 2402 2422 2442 -
1954. 2462 2482 2502 2522 2595 2615 2635 2655 2675 2695 2715 2735 2755 2775 2795 -
1955. 2815 2835 2855 2857 2860 2879 2890 2901 2912 2923 2934 2945 2993 3004 3015 -
1956. 3026 3037 3048 3059 3070 3331 3334 3337 3338 UNI GZ 3.09
1957. 64 66 68 70 2263 2264 2300 2320 2340 2360 2380 2400 2420 2440 2460 2480 2500 -
1958. 2520 2540 2561 2597 2617 2637 2657 2677 2697 2717 2737 2757 2777 2797 2817 -
1959. 3091 3093 3096 3115 3126 3137 3148 3159 3170 3181 3229 3240 3251 3262 3273 -
1960. 3284 3295 3306 3332 3333 3335 3336 UNI GZ 2.72
1962. LOAD R10 LOADTYPE DEAD  TITLE STRUCTURE SELF WEIGHT (D1)
1963. SELFWEIGHT Y -1
1964. JOINT LOAD
1965. *STAIR
1966. 113 114 117 119 FY -1.2
1967. 118 706 FY -2.24
1968. *ROCK BREAKER
1969. JOINT LOAD
1970. *17
1971. 73 FX -0.028 FY -7.934 FZ 0.005 MX 0.008 MY -0 MZ -0
1972. *18
1973. 74 FX -0.027 FY -7.973 FZ 0.01 MX 0.016 MY -0 MZ -0
1974. *19
1975. 78 FX 0.048 FY -10.994 FZ -0.011 MX -0.019 MY -0 MZ -0
1976. *20
1977. 79 FX 0.007 FY -10.243 FZ -0.004 MX -0.007 MY -0 MZ -0
1978. MEMBER LOAD
1979. *STAIR
1980. 120 125 CON GY -1.2 1.1
1981. 121 127 CON GY -1.2 0.4
1982. 126 CON GY -2.24 0.4
1983. 574 CON GY -2.24 1.1
1984. FLOOR LOAD
1985. **SELF WT OF FLOOR
1986. _CRUSHER_FLOOR FLOAD -4.8 GY
1987. _CRUSHER_FLOOR2 FLOAD -12 GY
1988. **SOIL PRESSURE
1989. ELEMENT LOAD
1990. 715 TO 734 875 TO 894 1039 TO 1118 1133 1134 1149 TO 1158 1173 1174 -
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1991. 1189 TO 1198 1213 1214 1229 TO 1238 1293 1294 1309 TO 1318 1333 1334 1349 -
1992. 1350 TO 1358 1373 1374 1389 TO 1398 1439 1440 1442 1443 1445 1446 1448 1449 -
1993. 1451 1452 1454 1455 1457 1458 1460 1461 1463 1464 1466 1467 1469 1470 1472 -
1994. 1473 1475 1476 1478 1479 1481 1482 1484 1485 1487 1488 1490 1491 1493 1494 -
1995. 1496 1497 1499 1500 1502 1503 1505 1506 1508 1509 1511 1512 1514 1515 1523 -
1996. 1524 1526 1527 1529 1530 1532 1533 1535 1536 1538 1539 1597 TO 2044 2173 -
1997. 2174 TO 2260 PR GY -48
1998. 901 TO 924 937 TO 960 973 TO 996 1009 TO 1032 1551 TO 1554 1557 TO 1560 1563 -
1999. 1564 TO 1566 1569 TO 1572 1575 TO 1578 1581 TO 1584 1587 TO 1590 1593 TO 1596 -
2000. 2045 TO 2172 PR GY -52
2001. LOAD R11 LOADTYPE DEAD  TITLE EQUIPMENT DEAD WEIGHT (D2)
2002. *ROCK BREAKER
2003. JOINT LOAD
2004. *17
2005. 73 FX -0 FY -2.466 FZ -0.003 MX 0.013 MY -0 MZ -0
2006. *18
2007. 74 FX -0.003 FY -8.243 FZ 0.005 MX 0.025 MY -0 MZ -0
2008. *19
2009. 78 FX 0.002 FY -7.931 FZ -0.007 MX 0.015 MY -0 MZ -0
2010. *20
2011. 79 FX 0.001 FY -16.36 FZ 0.005 MX 0.036 MY -0 MZ -0
2012. *JIB CRANE
2013. 702 FY -155
2014. MEMBER LOAD
2015. **CRUSHER
2016. 43 46 CON GY -75 0.552
2017. 43 46 CON GY -75 2.532
2018. *DRIVE
2019. 573 CON GY -45 0.554
2020. LOAD R12 LOADTYPE LIVE  TITLE LIVE LOAD(L1)
2021. *ROCK BREAKER
2022. JOINT LOAD
2023. *17
2024. 73 FX -0.096 FY -8.153 FZ 0.025 MX 0.036 MY -0 MZ -0
2025. *18
2026. 74 FX -0.09 FY -6.676 FZ 0.033 MX 0.05 MY -0 MZ -0
2027. *19
2028. 78 FX 0.172 FY -16.625 FZ -0.037 MX -0.071 MY -0 MZ -0
2029. *20
2030. 79 FX 0.014 FY -11.885 FZ -0.021 MX -0.046 MY -0 MZ -0
2031. *STAIR
2032. 113 114 117 119 FY -4.5
2033. 118 706 FY -8.4
2034. MEMBER LOAD
2035. *STAIR
2036. 120 125 CON GY -4.5 1.1
2037. 121 127 CON GY -4.5 0.4
2038. 126 CON GY -8.4 0.4
2039. 574 CON GY -8.4 1.1
2040. FLOOR LOAD
2041. **CRUSHER
2042. _CRUSHER_FLOOR FLOAD -5 GY
2043. _CRUSHER_FLOOR2 FLOAD -5 GY
2044. LOAD R13 LOADTYPE LIVE  TITLE EQUIPMENT LIVE LOAD(L2)
2045. JOINT LOAD
2046. *ROCK BREAKER
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2047. *17
2048. 73 FX 0.461 FY 44.513 FZ 11.814 MX 20.29 MY -0 MZ -0
2049. *18
2050. 74 FX -0.46 FY -44.513 FZ 11.836 MX 20.312 MY -0 MZ -0
2051. *19
2052. 78 FX 0.461 FY 62.727 FZ 25.696 MX 43.812 MY -0 MZ -0
2053. *20
2054. 79 FX -0.462 FY -62.727 FZ 25.654 MX 43.751 MY -0 MZ -0
2055. MEMBER LOAD
2056. **CRUSHER
2057. 43 46 CON GY -125 0.552
2058. 43 46 CON GY -125 2.532
2059. LOAD R14 LOADTYPE LIVE  TITLE EQUIPMENT LOAD REVERSIBLE (ELL_REV)
2060. JOINT LOAD
2061. *JIB CRANE
2062. 702 MX 370
2063. *ROCK BREAKER
2064. *17
2065. 73 FX 0.017 FY 31.465 FZ 0.003 MX 0.202 MY -0 MZ -0
2066. *18
2067. 74 FX -0.018 FY -31.465 FZ 0.013 MX 0.212 MY -0 MZ -0
2068. *19
2069. 78 FX 0.002 FY 46.606 FZ -0.004 MX 0.255 MY -0 MZ -0
2070. *20
2071. 79 FX -0.002 FY -46.606 FZ -0.011 MX 0.249 MY -0 MZ -0
2072. MEMBER LOAD
2073. **CRUSHER
2074. 43 46 CON GY 31 0.552
2075. 43 46 CON GX 12 0.552
2076. 43 46 CON GZ 3 0.552
2077. 43 46 CON GY 31 2.532
2078. 43 46 CON GX 12 2.532
2079. 43 46 CON GZ 3 2.532
2080. *DRIVE
2081. 573 CON GY 65 0.554
2082. 573 CON GX 4.5 0.554
2083. 573 CON GZ 2.5 0.554
2084. END DEFINE REFERENCE LOADS

 *** NOTE: NEITHER "MASS" NOR "GRAVITY" REFERENCE LOADTYPE IS PRESENT.
           MASS MODEL IS FORMED BY COMBINING ALL "DEAD" AND "LIVE" (IF ANY) REFERENCE LOADTYPES.

 *** NOTE: MASS MODEL FORMED WILL BE USED IN SEISMIC/RESPONSE/TIME HISTORY LOADING, IF ANY.
           IF MASS MODEL IS SEPARATELY PROVIDED IN INDIVIDUAL LOADING, THE GENERATED MASS
           WILL BE REPLACED BY THE MASS PROVIDED IN INDIVIDUAL LOADING.

2085. DEFINE IBC 2003
2086. SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 CT 0.044 PX 0.144 PZ -
2087. 0.144
2088. REFERENCE LOAD Y

 *** NOTE: SEISMIC WEIGHT TABLE IS PRESENT.THIS WEIGHT TABLE
           WILL BE CONSIDERED IN STATIC SEISMIC ANALYSIS.

2089. R10 1.0
2090. REFERENCE LOAD Y
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2091. R11 1.0
2092. REFERENCE LOAD Y
2093. R12 0.5
2094. REFERENCE LOAD Y
2095. R13 1.0
2096. LOAD 1 LOADTYPE SEISMIC  TITLE SEISMIC LOAD (E +X)
2097. IBC LOAD X 1
2098. LOAD 2 LOADTYPE SEISMIC  TITLE SEISMIC LOAD (E +Z)
2099. IBC LOAD Z 1
2100. LOAD 3 LOADTYPE WIND  TITLE WIND LOAD(W+X)
2101. REFERENCE LOAD
2102. R3 1.0
2103. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD(W+Z)
2104. REFERENCE LOAD
2105. R4 1.0
2106. LOAD 10 LOADTYPE DEAD  TITLE STRUCTURE SELF WEIGHT (D1)
2107. REFERENCE LOAD
2108. R10 1.0
2109. LOAD 11 LOADTYPE DEAD  TITLE EQUIPMENT DEAD WEIGHT (D2)
2110. REFERENCE LOAD
2111. R11 1.0
2112. LOAD 12 LOADTYPE LIVE  TITLE LIVE LOAD(L1)
2113. REFERENCE LOAD
2114. R12 1.0
2115. LOAD 13 LOADTYPE LIVE  TITLE EQUIPMENT LIVE LOAD(L2)
2116. REFERENCE LOAD
2117. R13 1.0
2118. LOAD 14 LOADTYPE LIVE  TITLE EQUIPMENT LOAD REVERSIBLE (ELL_REV)
2119. REFERENCE LOAD
2120. R14 1.0
2122. *FACTORED LOAD COMBINATIONS
2123. LOAD 201 1.4(D1+D2)
2124. REPEAT LOAD
2125. 10 1.4 11 1.4
2126. **1.4D + 1.7L
2127. LOAD 202 1.4(D1+D2)+1.7L1
2128. REPEAT LOAD
2129. 10 1.4 11 1.4 12 1.7
2130. LOAD 203 1.4(D1+D2)+1.7(L1+L2)
2131. REPEAT LOAD
2132. 10 1.4 11 1.4 12 1.7 13 1.7
2133. LOAD 204 LOADTYPE NONE  TITLE 1.4(D1+D2)+1.7(L1+L2+ELL_REV)
2134. REPEAT LOAD
2135. 10 1.4 11 1.4 12 1.7 13 1.7 14 1.7
2136. LOAD 205 LOADTYPE NONE  TITLE 1.4(D1+D2)+1.7(L1+L2-ELL_REV)
2137. REPEAT LOAD
2138. 10 1.4 11 1.4 12 1.7 13 1.7 14 -1.7
2139. **1.2D + 0.8W
2140. LOAD 206 1.2(D1+D2)+0.8W(+X)
2141. REPEAT LOAD
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2142. 10 1.2 11 1.2 3 0.8
2143. LOAD 207 1.2(D1+D2)+0.8W(-X)
2144. REPEAT LOAD
2145. 10 1.2 11 1.2 3 -0.8
2146. LOAD 208 1.2(D1+D2)+0.8W(+Z)
2147. REPEAT LOAD
2148. 10 1.2 11 1.2 4 0.8
2149. LOAD 209 1.2(D1+D2)+0.8W(-Z)
2150. REPEAT LOAD
2151. 10 1.2 11 1.2 4 -0.8
2152. **1.2D + 1.6W + 1.0L
2153. LOAD 210 1.2(D1+D2)+1.6W(+X) + 1.0L1
2154. REPEAT LOAD
2155. 10 1.2 11 1.2 3 1.6 12 1.0
2156. LOAD 211 1.2(D1+D2)+1.6W(-X) + 1.0L1
2157. REPEAT LOAD
2158. 10 1.2 11 1.2 3 -1.6 12 1.0
2159. LOAD 212 1.2(D1+D2)+1.6W(+Z) + 1.0L1
2160. REPEAT LOAD
2161. 10 1.2 11 1.2 4 1.6 12 1.0
2162. LOAD 213 1.2(D1+D2)+1.6W(-Z) + 1.0L1
2163. REPEAT LOAD
2164. 10 1.2 11 1.2 4 -1.6 12 1.0
2165. *
2166. LOAD 214 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)
2167. REPEAT LOAD
2168. 10 1.2 11 1.2 3 1.6 12 1.0 13 1.0
2169. LOAD 215 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2)
2170. REPEAT LOAD
2171. 10 1.2 11 1.2 3 -1.6 12 1.0 13 1.0
2172. LOAD 216 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2)
2173. REPEAT LOAD
2174. 10 1.2 11 1.2 4 1.6 12 1.0 13 1.0
2175. LOAD 217 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2)
2176. REPEAT LOAD
2177. 10 1.2 11 1.2 4 -1.6 12 1.0 13 1.0
2178. *
2179. LOAD 218 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_REV)
2180. REPEAT LOAD
2181. 10 1.2 11 1.2 3 1.6 12 1.0 13 1.0 14 1.0
2182. LOAD 219 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+ELL_REV)
2183. REPEAT LOAD
2184. 10 1.2 11 1.2 3 -1.6 12 1.0 13 1.0 14 1.0
2185. LOAD 220 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+ELL_REV)
2186. REPEAT LOAD
2187. 10 1.2 11 1.2 4 1.6 12 1.0 13 1.0 14 1.0
2188. LOAD 221 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+ELL_REV)
2189. REPEAT LOAD
2190. 10 1.2 11 1.2 4 -1.6 12 1.0 13 1.0 14 1.0
2191. *
2192. LOAD 222 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-ELL_REV)
2193. REPEAT LOAD
2194. 10 1.2 11 1.2 3 1.6 12 1.0 13 1.0 14 -1.0
2195. LOAD 223 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-ELL_REV)
2196. REPEAT LOAD
2197. 10 1.2 11 1.2 3 -1.6 12 1.0 13 1.0 14 -1.0
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2198. LOAD 224 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-ELL_REV)
2199. REPEAT LOAD
2200. 10 1.2 11 1.2 4 1.6 12 1.0 13 1.0 14 -1.0
2201. LOAD 225 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-ELL_REV)
2202. REPEAT LOAD
2203. 10 1.2 11 1.2 4 -1.6 12 1.0 13 1.0 14 -1.0
2204. **1.2D + 1.6E + 1.0L
2205. LOAD 226 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6 E(+X) + 1.0(L1+L2)
2206. REPEAT LOAD
2207. 10 1.2 11 1.2 1 1.6 12 1.0 13 1.0
2208. LOAD 227 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6 E(-X) + 1.0(L1+L2)
2209. REPEAT LOAD
2210. 10 1.2 11 1.2 1 -1.6 12 1.0 13 1.0
2211. LOAD 228 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6 E(+Z) + 1.0(L1+L2)
2212. REPEAT LOAD
2213. 10 1.2 11 1.2 2 1.6 12 1.0 13 1.0
2214. LOAD 229 LOADTYPE NONE  TITLE 1.2(D1+D2) + 1.6 E(-Z) + 1.0(L1+L2)
2215. REPEAT LOAD
2216. 10 1.2 11 1.2 2 -1.6 12 1.0 13 1.0
2217. *
2218. **0.9D + 1.6W
2219. LOAD 230 0.9(D1+D2) + 1.6W(+X)
2220. REPEAT LOAD
2221. 10 0.9 11 0.9 3 1.6
2222. LOAD 231 0.9(D1+D2) + 1.6W(-X)
2223. REPEAT LOAD
2224. 10 0.9 11 0.9 3 -1.6
2225. LOAD 232 0.9(D1+D2) + 1.6W(+Z)
2226. REPEAT LOAD
2227. 10 0.9 11 0.9 4 1.6
2228. LOAD 233 0.9(D1+D2) + 1.6W(-Z)
2229. REPEAT LOAD
2230. 10 0.9 11 0.9 4 -1.6
2231. **0.9D + 1.6E
2232. LOAD 234 0.9(D1+D2) + 1.6E(+X)
2233. REPEAT LOAD
2234. 10 0.9 11 0.9 1 1.6
2235. LOAD 235 0.9(D1+D2) + 1.6E(-X)
2236. REPEAT LOAD
2237. 10 0.9 11 0.9 1 -1.6
2238. LOAD 236 0.9(D1+D2) + 1.6E(+Z)
2239. REPEAT LOAD
2240. 10 0.9 11 0.9 2 1.6
2241. LOAD 237 0.9(D1+D2) + 1.6E(-Z)
2242. REPEAT LOAD
2243. 10 0.9 11 0.9 2 -1.6
2244. **UNFACTORED LOAD COMBINATIONS
2245. **D
2246. LOAD 101 1.0(D1+D2)
2247. REPEAT LOAD
2248. 10 1.0 11 1.0
2249. **D + L
2250. LOAD 102 1.0(D1+D2)+1.0L1
2251. REPEAT LOAD
2252. 10 1.0 11 1.0 12 1.0
2253. LOAD 103 1.0(D1+D2)+1.0(L1+L2)
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2254. REPEAT LOAD
2255. 10 1.0 11 1.0 12 1.0 13 1.0
2256. LOAD 104 LOADTYPE NONE  TITLE 1.0(D1+D2)+1.0(L1+L2+ELL_REV)
2257. REPEAT LOAD
2258. 10 1.0 11 1.0 12 1.0 13 1.0 14 1.0
2259. LOAD 105 LOADTYPE NONE  TITLE 1.0(D1+D2)+1.0(L1+L2-ELL_REV)
2260. REPEAT LOAD
2261. 10 1.0 11 1.0 12 1.0 13 1.0 14 -1.0
2262. **D + W
2263. LOAD 106 1.0(D1+D2)+1.0W(+X)
2264. REPEAT LOAD
2265. 10 1.0 11 1.0 3 1.0
2266. LOAD 107 1.0(D1+D2)+1.0W(-X)
2267. REPEAT LOAD
2268. 10 1.0 11 1.0 3 -1.0
2269. LOAD 108 1.0(D1+D2)+1.0W(+Z)
2270. REPEAT LOAD
2271. 10 1.0 11 1.0 4 1.0
2272. LOAD 109 1.0(D1+D2)+1.0W(-Z)
2273. REPEAT LOAD
2274. 10 1.0 11 1.0 4 -1.0
2275. **D + 0.7E
2276. LOAD 110 1.0(D1+D2)+0.7E(+X)
2277. REPEAT LOAD
2278. 10 1.0 11 1.0 1 0.7
2279. LOAD 111 1.0(D1+D2)+0.7E(-X)
2280. REPEAT LOAD
2281. 10 1.0 11 1.0 1 -0.7
2282. LOAD 112 1.0(D1+D2)+0.7E(+Z)
2283. REPEAT LOAD
2284. 10 1.0 11 1.0 2 0.7
2285. LOAD 113 1.0(D1+D2)+0.7E(-Z)
2286. REPEAT LOAD
2287. 10 1.0 11 1.0 2 -0.7
2288. **D + 0.75W + 0.75L
2289. LOAD 114 1.0(D1+D2)+0.75W(+X) + 0.75L1
2290. REPEAT LOAD
2291. 10 1.0 11 1.0 3 0.75 12 0.75
2292. LOAD 115 1.0(D1+D2)+0.75W(-X) + 0.75L1
2293. REPEAT LOAD
2294. 10 1.0 11 1.0 3 -0.75 12 0.75
2295. LOAD 116 1.0(D1+D2)+0.75W(+Z) + 0.75L1
2296. REPEAT LOAD
2297. 10 1.0 11 1.0 4 0.75 12 0.75
2298. LOAD 117 1.0(D1+D2)+0.75W(-Z) + 0.75L1
2299. REPEAT LOAD
2300. 10 1.0 11 1.0 4 -0.75 12 0.75
2301. *
2302. LOAD 118 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2)
2303. REPEAT LOAD
2304. 10 1.0 11 1.0 3 0.75 12 0.75 13 0.75
2305. LOAD 119 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2)
2306. REPEAT LOAD
2307. 10 1.0 11 1.0 3 -0.75 12 0.75 13 0.75
2308. LOAD 120 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2)
2309. REPEAT LOAD
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2310. 10 1.0 11 1.0 4 0.75 12 0.75 13 0.75
2311. LOAD 121 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2)
2312. REPEAT LOAD
2313. 10 1.0 11 1.0 4 -0.75 12 0.75 13 0.75
2314. *
2315. LOAD 122 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+ELL_REV)
2316. REPEAT LOAD
2317. 10 1.0 11 1.0 3 0.75 12 0.75 13 0.75 14 0.75
2318. LOAD 123 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+ELL_REV)
2319. REPEAT LOAD
2320. 10 1.0 11 1.0 3 -0.75 12 0.75 13 0.75 14 0.75
2321. LOAD 124 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+ELL_REV)
2322. REPEAT LOAD
2323. 10 1.0 11 1.0 4 0.75 12 0.75 13 0.75 14 0.75
2324. LOAD 125 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+ELL_REV)
2325. REPEAT LOAD
2326. 10 1.0 11 1.0 4 -0.75 12 0.75 13 0.75 14 0.75
2327. *
2328. LOAD 126 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-ELL_REV)
2329. REPEAT LOAD
2330. 10 1.0 11 1.0 3 0.75 12 0.75 13 0.75 14 -0.75
2331. LOAD 127 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-ELL_REV)
2332. REPEAT LOAD
2333. 10 1.0 11 1.0 3 -0.75 12 0.75 13 0.75 14 -0.75
2334. LOAD 128 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-ELL_REV)
2335. REPEAT LOAD
2336. 10 1.0 11 1.0 4 0.75 12 0.75 13 0.75 14 -0.75
2337. LOAD 129 LOADTYPE NONE  TITLE 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-ELL_REV)
2338. REPEAT LOAD
2339. 10 1.0 11 1.0 4 -0.75 12 0.75 13 0.75 14 -0.75
2340. **D + 0.75*0.7W + 0.75L
2341. LOAD 130 1.0(D1+D2)+0.75*0.7 E(+X) + 0.75(L1+L2)
2342. REPEAT LOAD
2343. 10 1.0 11 1.0 1 0.525 12 0.75 13 0.75
2344. LOAD 131 1.0(D1+D2)+0.75*0.7 E(-X) + 0.75(L1+L2)
2345. REPEAT LOAD
2346. 10 1.0 11 1.0 1 -0.525 12 0.75 13 0.75
2347. LOAD 132 1.0(D1+D2)+0.75*0.7 E(+Z) + 0.75(L1+L2)
2348. REPEAT LOAD
2349. 10 1.0 11 1.0 2 0.525 12 0.75 13 0.75
2350. LOAD 133 1.0(D1+D2)+0.75*0.7 E(-Z) + 0.75(L1+L2)
2351. REPEAT LOAD
2352. 10 1.0 11 1.0 2 -0.525 12 0.75 13 0.75
2353. *
2354. **0.6D + W
2355. LOAD 134 0.6(D1+D2) + 1.0W(+X)
2356. REPEAT LOAD
2357. 10 0.6 11 0.6 3 1.0
2358. LOAD 135 0.6(D1+D2) + 1.0W(-X)
2359. REPEAT LOAD
2360. 10 0.6 11 0.6 3 -1.0
2361. LOAD 136 0.6(D1+D2) + 1.0W(+Z)
2362. REPEAT LOAD
2363. 10 0.6 11 0.6 4 1.0
2364. LOAD 137 0.6(D1+D2) + 1.0W(-Z)
2365. REPEAT LOAD
```

```
2366. 10 0.6 11 0.6 4 -1.0
2367. **0.6D + 0.7E
2368. LOAD 138 0.6(D1+D2) + 0.7E(+X)
2369. REPEAT LOAD
2370. 10 0.6 11 0.6 1 0.7
2371. LOAD 139 0.6(D1+D2) + 0.7E(-X)
2372. REPEAT LOAD
2373. 10 0.6 11 0.6 1 -0.7
2374. LOAD 140 0.6(D1+D2) + 0.7E(+Z)
2375. REPEAT LOAD
2376. 10 0.6 11 0.6 2 0.7
2377. LOAD 141 0.6(D1+D2) + 0.7E(-Z)
2378. REPEAT LOAD
2379. 10 0.6 11 0.6 2 -0.7
2380. PERFORM ANALYSIS PRINT STATICS CHECK


          P R O B L E M   S T A T I S T I C S
          -----------------------------------

   NUMBER OF JOINTS       2203  NUMBER OF MEMBERS     266
   NUMBER OF PLATES       1912  NUMBER OF SOLIDS        0
   NUMBER OF SURFACES        0  NUMBER OF SUPPORTS   1180

         Using 64-bit analysis engine.

         SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

 TOTAL      PRIMARY LOAD CASES =    87, TOTAL DEGREES OF FREEDOM =    9678
 TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
```

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS & CHANGE.

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.044 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.199 T =  0.144 Tuser = 0.144                 *
*  TIME PERIOD USED (T) =  0.144                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.027 X      7813.92   *
*                       =       209.41  KN                 *
************************************************************
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.044 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.199 T =  0.144 Tuser = 0.144                 *
*  TIME PERIOD USED (T) =  0.144                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.027 X      7813.92   *
*                       =       209.41  KN                 *
************************************************************
```

*** LOADING 1 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 209.41 FORCE-Y = 0.00 FORCE-Z = 0.00
MOM-X = 0.00 MOM-Y = 1036.19 MOM-Z = -826.53

*** LOADING 2 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 0.00 FORCE-Y = 0.00 FORCE-Z = 209.41
MOM-X = 826.53 MOM-Y = -1088.00 MOM-Z = 0.00

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE SEISMIC TITLE SEISMIC LOAD (E +X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.519545377E+01
Y = 0.394684395E+01
Z = 0.494805305E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 209.41
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 1036.19 MZ= -826.53

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -209.41
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -1036.19 MZ= 826.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	1.60963E-01	112
Y =	-3.41337E-02	1561
Z =	4.24878E-03	3338
RX=	2.52555E-05	1328
RY=	-9.37732E-05	3353
RZ=	-2.40121E-04	34

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE SEISMIC TITLE SEISMIC LOAD (E +Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.519545377E+01
Y = 0.394684395E+01
Z = 0.494805305E+01

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 209.41

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 826.53 MY= -1088.00 MZ= 0.00

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TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -209.41

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -826.53 MY= 1088.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)
MAXIMUMS AT NODE
X = 3.68621E-03 114
Y = 5.75423E-02 706
Z = 1.76239E-01 1765
RX= 2.54762E-04 3008
RY= 8.84226E-05 1766
RZ= 3.95415E-05 1470

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE WIND TITLE WIND LOAD(W+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456646673E+01
Y = 0.267244819E+01
Z = 0.415240196E+01

TOTAL APPLIED LOAD 3

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 118.14
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.01 MY= 490.58 MZ= -420.70

TOTAL REACTION LOAD 3

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -118.14
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.01 MY= -490.58 MZ= 420.70

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	1.15134E-01	119
Y =	-2.28086E-02	1603
Z =	-1.16055E-02	69
RX=	-1.87889E-05	84
RY=	-7.26544E-05	112
RZ=	-1.63428E-04	33

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE WIND TITLE WIND LOAD(W+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531310607E+01
Y = 0.285186896E+01
Z = 0.431543598E+01

TOTAL APPLIED LOAD 4

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 492.02 MY= -829.83 MZ= 0.00

TOTAL REACTION LOAD 4

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -492.02 MY= 829.83 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	5.08977E-03	706
Y =	3.68675E-02	119
Z =	1.09009E-01	119
RX=	1.45758E-04	2991
RY=	-2.56116E-05	704
RZ=	2.76053E-05	1470

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
LOADTYPE DEAD TITLE STRUCTURE SELF WEIGHT (D1)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.464361677E+01
Y = -0.997603573E+00
Z = 0.502947242E+01

TOTAL APPLIED LOAD 10

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6322.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31800.04 MY= 0.02 MZ= -29360.37

TOTAL REACTION LOAD 10

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6322.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31800.04 MY= -0.02 MZ= 29360.37

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	-2.38552E-02	3442
Y =	-2.02189E-01	705
Z =	8.11955E-03	3141
RX=	1.35762E-04	118
RY=	-1.91199E-05	2950
RZ=	1.97136E-04	1733

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
LOADTYPE DEAD TITLE EQUIPMENT DEAD WEIGHT (D2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.580997513E+01
Y = 0.448000982E+01
Z = 0.591427741E+01

TOTAL APPLIED LOAD 11

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -535.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3164.14 MY= 0.00 MZ= -3108.34

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TOTAL REACTION LOAD 11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 535.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3164.14 MY= -0.00 MZ= 3108.34

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)
MAXIMUMS AT NODE
X = 8.15280E-03 3447
Y = -9.79399E-02 702
Z = 5.79563E-03 3139
RX= 1.59781E-04 75
RY= 1.72621E-05 2652
RZ= 2.43705E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
LOADTYPE LIVE TITLE LIVE LOAD(L1)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.421798134E+01
Y = 0.418141930E+01
Z = 0.463936427E+01

TOTAL APPLIED LOAD 12

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -483.44
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2242.85 MY= 0.07 MZ= -2039.14

TOTAL REACTION LOAD 12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 483.44
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2242.85 MY= -0.07 MZ= 2039.14

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	-1.79812E-02	3443
Y =	-4.96237E-02	706
Z =	-3.21849E-03	3442
RX=	-2.48583E-04	706
RY=	-6.99206E-06	75
RZ=	5.71150E-05	1751

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
LOADTYPE LIVE TITLE EQUIPMENT LIVE LOAD(L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.544200996E+01
Y = 0.448000982E+01
Z = 0.590326615E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.698101661E+01
Y = 0.618887651E+01
Z = 0.127466503E+01

TOTAL APPLIED LOAD 13

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -500.00
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3287.63 MY= -525.93 MZ= -2721.00

TOTAL REACTION LOAD 13

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 500.00
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3287.63 MY= 525.93 MZ= 2721.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	-3.83346E-02	117
Y =	-1.07348E-01	1765
Z =	1.17860E-01	119
RX=	-2.60465E-04	83
RY=	-9.01367E-05	699
RZ=	4.04640E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
LOADTYPE LIVE TITLE EQUIPMENT LOAD REVERSIBLE (ELL_REV)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.518223440E+01
Y = 0.549660867E+01
Z = 0.523902201E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.491469859E+01
Y = 0.300587975E+02
Z = 0.517828040E+01

TOTAL APPLIED LOAD 14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = 189.00
SUMMATION FORCE-Z = 14.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -358.45 MY= 198.47 MZ= 596.33

TOTAL REACTION LOAD 14

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = -189.00
SUMMATION FORCE-Z = -14.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 358.45 MY= -198.47 MZ= -596.33

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	5.80495E-02	112
Y =	4.36723E-02	705
Z =	6.48991E-02	1766
RX=	1.66901E-03	702
RY=	-4.97821E-05	701
RZ=	-1.62641E-04	81

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 201
1.4(D1+D2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460911E+01
Y = -0.570272707E+00
Z = 0.509849965E+01

TOTAL APPLIED LOAD 201

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 201)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9600.83
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 48949.85 MY= 0.03 MZ= -45456.19

TOTAL REACTION LOAD 201

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 201)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9600.83
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -48949.85 MY= -0.03 MZ= 45456.19

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 201)

	MAXIMUMS	AT NODE
X =	-3.93697E-02	3442
Y =	-3.89231E-01	702
Z =	1.94129E-02	3140
RX=	3.63383E-04	75
RY=	-4.99409E-05	2950
RZ=	5.10633E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 202
1.4(D1+D2)+1.7L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.469387205E+01
Y = -0.195592884E+00
Z = 0.506229598E+01

TOTAL APPLIED LOAD 202

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 202)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -10422.68
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 52762.70 MY= 0.14 MZ= -48922.73

TOTAL REACTION LOAD 202

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 202)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 10422.68
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -52762.70 MY= -0.14 MZ= 48922.73

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 202)

	MAXIMUMS	AT NODE
X =	-6.66121E-02	3442
Y =	-4.27723E-01	112
Z =	-2.24149E-02	3237
RX=	6.32708E-04	118
RY=	6.10428E-05	2652
RZ=	6.01966E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 203
1.4(D1+D2)+1.7(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475028429E+01
Y = 0.156964070E+00
Z = 0.512570809E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697669664E+01
Y = 0.624477045E+01
Z = 0.127552102E+01

TOTAL APPLIED LOAD 203

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 203)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -11272.68
SUMMATION FORCE-Z = 127.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 58351.68 MY= -893.93 MZ= -53548.44

TOTAL REACTION LOAD 203

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 203)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 11272.68
SUMMATION FORCE-Z = -127.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -58351.68 MY= 893.93 MZ= 53548.44

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 203)

	MAXIMUMS	AT NODE
X =	-1.05294E-01	3444
Y =	-5.82946E-01	112
Z =	2.00161E-01	78
RX=	8.75145E-04	118
RY=	-2.01708E-04	2950
RZ=	1.27335E-03	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 204
LOADTYPE NONE TITLE 1.4(D1+D2)+1.7(L1+L2+ELL_REV)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520218608E+01
Y = -0.155976295E+01
Z = 0.518966660E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476057287E+01
Y = 0.301312739E-01
Z = 0.521036313E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664261061E+01
Y = 0.101031316E+02
Z = 0.190784819E+01

TOTAL APPLIED LOAD 204

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 204)
SUMMATION FORCE-X = 89.25
SUMMATION FORCE-Y = -10951.38
SUMMATION FORCE-Z = 152.15

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 57742.32 MY= -556.53 MZ= -52534.68

TOTAL REACTION LOAD 204

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 204)
SUMMATION FORCE-X = -89.25
SUMMATION FORCE-Y = 10951.38
SUMMATION FORCE-Z = -152.15

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -57742.32 MY= 556.53 MZ= 52534.68

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 204)

	MAXIMUMS	AT NODE
X =	7.46288E-02	119
Y =	-5.43569E-01	1765
Z =	3.08510E-01	78
RX=	2.89210E-03	702
RY=	-2.32868E-04	2950
RZ=	1.02510E-03	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 205
LOADTYPE NONE TITLE 1.4(D1+D2)+1.7(L1+L2-ELL_REV)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.516228272E+01
Y = 0.125529800E+02
Z = 0.528837742E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474056597E+01
Y = 0.276767124E+00
Z = 0.504574509E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747093677E+01
Y = 0.536788742E+00
Z = 0.340068959E+00

TOTAL APPLIED LOAD 205

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 205)
SUMMATION FORCE-X = -89.25
SUMMATION FORCE-Y = -11593.98
SUMMATION FORCE-Z = 102.85

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 58961.04 MY= -1231.34 MZ= -54562.20

TOTAL REACTION LOAD 205

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 205)
SUMMATION FORCE-X = 89.25
SUMMATION FORCE-Y = 11593.98
SUMMATION FORCE-Z = -102.85

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -58961.04 MY= 1231.34 MZ= 54562.20

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 205)

	MAXIMUMS	AT NODE
X =	-1.77631E-01	3443
Y =	-6.53087E-01	112
Z =	9.18116E-02	119
RX=	-2.78252E-03	702
RY=	-1.70549E-04	2950
RZ=	1.53243E-03	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 206
1.2(D1+D2)+0.8W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456978049E+01
Y = 0.518370150E+00
Z = 0.415168973E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474481332E+01
Y = -0.570272616E+00
Z = 0.509849958E+01

TOTAL APPLIED LOAD 206

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 206)
SUMMATION FORCE-X = 94.51
SUMMATION FORCE-Y = -8229.29
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 41957.01 MY= 392.49 MZ= -39299.01

TOTAL REACTION LOAD 206

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 206)
SUMMATION FORCE-X = -94.51
SUMMATION FORCE-Y = 8229.29
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -41957.01 MY= -392.49 MZ= 39299.01

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 206)

	MAXIMUMS	AT NODE
X =	7.06760E-02	119
Y =	-3.37733E-01	702
Z =	1.74238E-02	3339
RX=	3.20403E-04	75
RY=	-4.93304E-05	2950
RZ=	3.81709E-04	1765

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STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 207
1.2(D1+D2)+0.8W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456315287E+01
Y = 0.482652617E+01
Z = 0.415311429E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.472440479E+01
Y = -0.570272616E+00
Z = 0.509849969E+01

TOTAL APPLIED LOAD 207

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 207)
SUMMATION FORCE-X = -94.51
SUMMATION FORCE-Y = -8229.29
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 41957.02 MY= -392.44 MZ= -38625.89

TOTAL REACTION LOAD 207

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 207)
SUMMATION FORCE-X = 94.51
SUMMATION FORCE-Y = 8229.29
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -41957.02 MY= 392.44 MZ= 38625.89

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 207)

	MAXIMUMS	AT NODE
X	-1.15091E-01	3441
Y	-3.30716E-01	112
Z	1.90197E-02	3140
RX	3.02540E-04	75
RY	8.41171E-05	112
RZ	4.93661E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 208
1.2(D1+D2)+0.8W(+Z)

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CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.510730879E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531237143E+01
Y = 0.288764638E+01
Z = 0.431582782E+01

TOTAL APPLIED LOAD 208

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 208)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -8229.29
SUMMATION FORCE-Z = 124.95

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 42350.62 MY= -663.84 MZ= -38962.45

TOTAL REACTION LOAD 208

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 208)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8229.29
SUMMATION FORCE-Z = -124.95

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -42350.62 MY= 663.84 MZ= 38962.45

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 208)

	MAXIMUMS	AT NODE
X	-3.34813E-02	3442
Y	-3.34927E-01	702
Z	9.00680E-02	119
RX	-3.11567E-04	83
RY	-5.56580E-05	2950
RZ	4.36827E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 209
1.2(D1+D2)+0.8W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.508969048E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531384084E+01
Y = 0.281609156E+01
Z = 0.431504413E+01

TOTAL APPLIED LOAD 209

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 209)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -8229.29
SUMMATION FORCE-Z = -124.95

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 41563.40 MY= 663.89 MZ= -38962.45

TOTAL REACTION LOAD 209

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 209)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8229.29
SUMMATION FORCE-Z = 124.95

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -41563.40 MY= -663.89 MZ= 38962.45

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 209)

	MAXIMUMS	AT NODE
X	-3.40098E-02	3442
Y	-3.32326E-01	702
Z	-8.75683E-02	117
RX	3.43834E-04	75
RY	5.39072E-05	2652
RZ	4.38544E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 210
1.2(D1+D2)+1.6W(+X) + 1.0L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.457255152E+01
Y = 0.151324906E+01
Z = 0.415102062E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.472521925E+01
Y = -0.306617195E+00
Z = 0.507302367E+01

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TOTAL APPLIED LOAD 210

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 210)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = -8712.73
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 44199.86 MY= 785.02 MZ= -41674.71

TOTAL REACTION LOAD 210

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 210)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = 8712.73
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -44199.86 MY= -785.02 MZ= 41674.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 210)

	MAXIMUMS	AT NODE
X =	1.50445E-01	119
Y =	-3.57037E-01	702
Z =	-2.59642E-02	3442
RX=	4.19408E-04	118
RY=	-1.01330E-04	75
RZ=	3.84136E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 211
1.2(D1+D2)+1.6W(-X) + 1.0L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456038184E+01
Y = 0.383164727E+01
Z = 0.415378340E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.468666700E+01
Y = -0.306617195E+00
Z = 0.507302387E+01

TOTAL APPLIED LOAD 211

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 211)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = -8712.73
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 44199.87 MY= -784.84 MZ= -40328.48

TOTAL REACTION LOAD 211

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 211)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = 8712.73
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -44199.87 MY= 784.84 MZ= 40328.48

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 211)

	MAXIMUMS	AT NODE
X =	-2.17984E-01	119
Y =	-3.71960E-01	112
Z =	2.72822E-02	3443
RX=	4.42337E-04	118
RY=	1.35583E-04	112
RZ=	6.03363E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 212
1.2(D1+D2)+1.6W(+Z) + 1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470594313E+01
Y = -0.306617195E+00
Z = 0.508966450E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531169435E+01
Y = 0.287425621E+01
Z = 0.431575436E+01

TOTAL APPLIED LOAD 212

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 212)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -8712.73
SUMMATION FORCE-Z = 249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 44987.09 MY= -1327.64 MZ= -41001.59

TOTAL REACTION LOAD 212

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 212)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8712.73
SUMMATION FORCE-Z = -249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -44987.09 MY= 1327.64 MZ= 41001.59

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 212)

	MAXIMUMS	AT NODE
X =	-4.92419E-02	3442
Y =	-3.54601E-01	112
Z =	1.75193E-01	78
RX=	6.25451E-04	118
RY=	-7.48143E-05	2950
RZ=	4.89694E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 213
1.2(D1+D2)+1.6W(-Z) + 1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470594313E+01
Y = -0.306617195E+00
Z = 0.505638303E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531451792E+01
Y = 0.282948172E+01
Z = 0.431511761E+01

TOTAL APPLIED LOAD 213

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 213)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -8712.73
SUMMATION FORCE-Z = -249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 43412.64 MY= 1327.82 MZ= -41001.59

TOTAL REACTION LOAD 213

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 213)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8712.73
SUMMATION FORCE-Z = 249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -43412.64 MY= -1327.82 MZ= 41001.59

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 213)

	MAXIMUMS	AT NODE
X =	-5.09463E-02	3441
Y =	-3.49933E-01	112
Z =	-1.76285E-01	82
RX=	-6.10523E-04	706
RY=	7.18286E-05	2652
RZ=	4.93128E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 214
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.457252771E+01
Y = 0.374772910E+00
Z = 0.413858270E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476412146E+01
Y = -0.468337614E-01
Z = 0.511808321E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697628222E+01
Y = 0.626355057E+01
Z = 0.127575305E+01

TOTAL APPLIED LOAD 214

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 214)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.49 MY= 259.10 MZ= -44395.71

TOTAL REACTION LOAD 214

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 214)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.49 MY= -259.10 MZ= 44395.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 214)

	MAXIMUMS	AT NODE
X =	1.71910E-01	119
Y =	-4.50554E-01	1765
Z =	1.28238E-01	78
RX=	5.62018E-04	118
RY=	-1.53472E-04	69
RZ=	7.88776E-04	1766

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STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 215
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456040564E+01
Y = 0.497012341E+01
Z = 0.416622132E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.472766155E+01
Y = -0.468337614E-01
Z = 0.511808340E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697634297E+01
Y = 0.626338914E+01
Z = 0.127569862E+01

TOTAL APPLIED LOAD 215

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 215)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.51 MY= -1310.77 MZ= -43049.48

TOTAL REACTION LOAD 215

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 215)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.51 MY= 1310.77 MZ= 43049.48

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 215)

	MAXIMUMS	AT NODE
X	-2.08051E-01	3442
Y	-4.63268E-01	112
Z	1.09040E-01	78
RX	-6.00616E-04	83
RY	1.26128E-04	112
RZ	9.98293E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 216
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.513382090E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.569704499E+01
Y = 0.363941167E+01
Z = 0.361374149E+01

TOTAL APPLIED LOAD 216

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 216)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 324.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 48274.73 MY= -1853.56 MZ= -43722.60

TOTAL REACTION LOAD 216

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 216)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -324.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -48274.73 MY= 1853.56 MZ= 43722.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 216)

	MAXIMUMS	AT NODE
X	-7.43847E-02	3444
Y	-4.45909E-01	112
Z	2.93053E-01	78
RX	7.68061E-04	118
RY	-1.57785E-04	2950
RZ	8.84624E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 217
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2)

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CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.510234571E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.459988257E+01
Y = 0.138889115E+01
Z = 0.561893796E+01

TOTAL APPLIED LOAD 217

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 217)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = -174.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 46700.27 MY= 801.89 MZ= -43722.60

TOTAL REACTION LOAD 217

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 217)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = 174.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -46700.27 MY= -801.89 MZ= 43722.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 217)

	MAXIMUMS	AT NODE
X	-6.22455E-02	3443
Y	-4.41241E-01	112
Z	-1.11180E-01	3442
RX	5.70982E-04	75
RY	-1.06378E-04	2950
RZ	8.88059E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 218
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470505558E+01
Y = 0.148807201E+01
Z = 0.437777783E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.477175622E+01
Y = -0.141647539E+00
Z = 0.517835828E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664225735E+01
Y = 0.101189383E+02
Z = 0.190805395E+01

TOTAL APPLIED LOAD 218

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 218)
SUMMATION FORCE-X = 241.53
SUMMATION FORCE-Y = -9023.73
SUMMATION FORCE-Z = 89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47129.04 MY= 457.57 MZ= -43799.38

TOTAL REACTION LOAD 218

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 218)
SUMMATION FORCE-X = -241.53
SUMMATION FORCE-Y = 9023.73
SUMMATION FORCE-Z = -89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47129.04 MY= -457.57 MZ= 43799.38

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 218)

	MAXIMUMS	AT NODE
X =	2.21390E-01	119
Y =	-4.33996E-01	1765
Z =	1.91973E-01	78
RX=	1.70747E-03	702
RY=	-1.66755E-04	69
RZ=	6.46134E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 219
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+ELL_R

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473453266E+01
Y = -0.141647539E+00
Z = 0.517835848E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664232020E+01
Y = 0.101186652E+02
Z = 0.190798573E+01

TOTAL APPLIED LOAD 219

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 219)
SUMMATION FORCE-X = -136.53
SUMMATION FORCE-Y = -9023.73
SUMMATION FORCE-Z = 89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47129.06 MY= -1112.29 MZ= -42453.15

TOTAL REACTION LOAD 219

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 219)
SUMMATION FORCE-X = 136.53
SUMMATION FORCE-Y = 9023.73
SUMMATION FORCE-Z = -89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47129.06 MY= 1112.29 MZ= 42453.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 219)

	MAXIMUMS	AT NODE
X =	-1.58587E-01	3442
Y =	-4.22008E-01	112
Z =	1.72775E-01	119
RX=	1.71136E-03	702
RY=	1.54784E-04	80
RZ=	8.45895E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 220
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520405772E+01
Y = -0.277638383E+01
Z = 0.518926430E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475314444E+01
Y = -0.141647539E+00
Z = 0.519442559E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.566361870E+01
Y = 0.476819796E+01
Z = 0.368058748E+01

TOTAL APPLIED LOAD 220

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 220)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = -9023.73
SUMMATION FORCE-Z = 339.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47916.28 MY= -1655.09 MZ= -43126.27

TOTAL REACTION LOAD 220

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 220)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = 9023.73
SUMMATION FORCE-Z = -339.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47916.28 MY= 1655.09 MZ= 43126.27

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 220)

	MAXIMUMS	AT NODE
X =	-5.19449E-02	118
Y =	-4.27503E-01	1765
Z =	3.56788E-01	78
RX=	1.85054E-03	702
RY=	-1.76113E-04	2950
RZ=	7.36346E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 221
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520405772E+01
Y = -0.277638383E+01
Z = 0.518926430E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475314444E+01
Y = -0.141647539E+00
Z = 0.516229116E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.457142080E+01
Y = -0.120308730E+01
Z = 0.565877677E+01

TOTAL APPLIED LOAD 221

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 221)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = -9023.73
SUMMATION FORCE-Z = -160.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 46341.82 MY= 1000.37 MZ= -43126.27

TOTAL REACTION LOAD 221

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 221)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = 9023.73
SUMMATION FORCE-Z = 160.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -46341.82 MY= -1000.37 MZ= 43126.27

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 221)

	MAXIMUMS	AT NODE
X	-3.65807E-02	118
Y	-4.02671E-01	1765
Z	-7.62364E-02	3442
RX	1.56829E-03	702
RY	-1.24707E-04	2950
RZ	7.35660E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 222
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-ELL_R

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475679365E+01
Y = 0.441679904E-01
Z = 0.506023152E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747043611E+01
Y = 0.559918803E+00
Z = 0.340331828E+00

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TOTAL APPLIED LOAD 222

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 222)
SUMMATION FORCE-X = 136.53
SUMMATION FORCE-Y = -9401.73
SUMMATION FORCE-Z = 60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47845.94 MY= 60.62 MZ= -44992.04

TOTAL REACTION LOAD 222

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 222)
SUMMATION FORCE-X = -136.53
SUMMATION FORCE-Y = 9401.73
SUMMATION FORCE-Z = -60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47845.94 MY= -60.62 MZ= 44992.04

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 222)

	MAXIMUMS	AT NODE
X =	1.22430E-01	119
Y =	-4.67111E-01	1765
Z =	6.45036E-02	119
RX=	-1.63054E-03	702
RY=	-1.60510E-04	75
RZ=	9.31418E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 223
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.469556840E+01
Y = 0.508456215E+01
Z = 0.439940883E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.472106669E+01
Y = 0.441679904E-01
Z = 0.506023171E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747048528E+01
Y = 0.560020361E+00
Z = 0.340313828E+00

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TOTAL APPLIED LOAD 223

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 223)
SUMMATION FORCE-X = -241.53
SUMMATION FORCE-Y = -9401.73
SUMMATION FORCE-Z = 60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47845.95 MY= -1509.24 MZ= -43645.81

TOTAL REACTION LOAD 223

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 223)
SUMMATION FORCE-X = 241.53
SUMMATION FORCE-Y = 9401.73
SUMMATION FORCE-Z = -60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47845.95 MY= 1509.24 MZ= 43645.81

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 223)

	MAXIMUMS	AT NODE
X =	-2.57515E-01	3442
Y =	-5.04527E-01	112
Z =	5.44788E-02	3140
RX=	-1.62665E-03	702
RY=	1.42230E-04	83
RZ=	1.15069E-03	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 224
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.516041109E+01
Y = 0.137696016E+02
Z = 0.528877972E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473893017E+01
Y = 0.441679904E-01
Z = 0.507565284E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.573359447E+01
Y = 0.240515660E+01
Z = 0.354064970E+01

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TOTAL APPLIED LOAD 224

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 224)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = -9401.73
SUMMATION FORCE-Z = 310.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 48633.17 MY= -2052.04 MZ= -44318.93

TOTAL REACTION LOAD 224

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 224)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = 9401.73
SUMMATION FORCE-Z = -310.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -48633.17 MY= 2052.04 MZ= 44318.93

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 224)

	MAXIMUMS	AT NODE
X =	-1.13237E-01	3443
Y =	-4.87168E-01	112
Z =	2.29318E-01	119
RX=	-1.48747E-03	702
RY=	-1.39456E-04	2950
RZ=	1.03702E-03	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 225
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-ELL_R

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.516041109E+01
Y = 0.137696016E+02
Z = 0.528877972E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473893017E+01
Y = 0.441679904E-01
Z = 0.504481039E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.462398607E+01
Y = 0.358396646E+01
Z = 0.558519957E+01

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TOTAL APPLIED LOAD 225

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 225)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = -9401.73
SUMMATION FORCE-Z = -189.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47058.72 MY= 603.42 MZ= -44318.93

TOTAL REACTION LOAD 225

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 225)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = 9401.73
SUMMATION FORCE-Z = 189.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47058.72 MY= -603.42 MZ= 44318.93

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 225)

	MAXIMUMS	AT NODE
X =	-1.11870E-01	3443
Y =	-4.82500E-01	112
Z =	-1.46123E-01	3442
RX=	-1.76973E-03	702
RY=	-8.80495E-05	2950
RZ=	1.04046E-03	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 226
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6 E(+X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.511808331E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697631259E+01
Y = 0.626346985E+01
Z = 0.127572584E+01

TOTAL APPLIED LOAD 226

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 226)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.50 MY= -525.84 MZ= -43722.60

TOTAL REACTION LOAD 226

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 226)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.50 MY= 525.84 MZ= 43722.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 226)
MAXIMUMS AT NODE
X = -6.82116E-02 3444
Y = -4.43575E-01 112
Z = 1.18639E-01 119
RX= -5.79418E-04 83
RY= -1.32081E-04 2950
RZ= 8.86341E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 227
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6 E(-X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.511808331E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697631259E+01
Y = 0.626346985E+01
Z = 0.127572584E+01

TOTAL APPLIED LOAD 227

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 227)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.50 MY= -525.84 MZ= -43722.60

TOTAL REACTION LOAD 227

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 227)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.50 MY= 525.84 MZ= 43722.60

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 227)

	MAXIMUMS	AT NODE
X =	-6.82116E-02	3444
Y =	-4.43575E-01	112
Z =	1.18639E-01	119
RX=	-5.79418E-04	83
RY=	-1.32081E-04	2950
RZ=	8.86341E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 228
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6 E(+Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.511808331E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697631259E+01
Y = 0.626346985E+01
Z = 0.127572584E+01

TOTAL APPLIED LOAD 228

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 228)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.50 MY= -525.84 MZ= -43722.60

TOTAL REACTION LOAD 228

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 228)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.50 MY= 525.84 MZ= 43722.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 228)

	MAXIMUMS	AT NODE
X =	-6.82116E-02	3444
Y =	-4.43575E-01	112
Z =	1.18639E-01	119
RX=	-5.79418E-04	83
RY=	-1.32081E-04	2950
RZ=	8.86341E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 229
LOADTYPE NONE TITLE 1.2(D1+D2) + 1.6 E(-Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474589150E+01
Y = -0.468337614E-01
Z = 0.511808331E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697631259E+01
Y = 0.626346985E+01
Z = 0.127572584E+01

TOTAL APPLIED LOAD 229

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 229)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -9212.73
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 47487.50 MY= -525.84 MZ= -43722.60

TOTAL REACTION LOAD 229

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 229)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 9212.73
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -47487.50 MY= 525.84 MZ= 43722.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 229)

	MAXIMUMS	AT NODE
X	-6.82116E-02	3444
Y	-4.43575E-01	112
Z	1.18639E-01	119
RX	-5.79418E-04	83
RY	-1.32081E-04	2950
RZ	8.86341E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 230
0.9(D1+D2) + 1.6W(+X)

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CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456770936E+01
Y = 0.186466892E+01
Z = 0.415213491E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476182054E+01
Y = -0.570272676E+00
Z = 0.509849951E+01

TOTAL APPLIED LOAD 230

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 230)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.75 MY= 784.95 MZ= -29894.95

TOTAL REACTION LOAD 230

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 230)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.75 MY= -784.95 MZ= 29894.95

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 230)

	MAXIMUMS	AT NODE
X =	1.68141E-01	119
Y =	-2.58433E-01	702
Z =	-2.14984E-02	3442
RX=	2.51466E-04	75
RY=	-9.92134E-05	75
RZ=	-3.06481E-04	10

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 231
0.9(D1+D2) + 1.6W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456522400E+01
Y = 0.348022741E+01
Z = 0.415266912E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470739781E+01
Y = -0.570272676E+00
Z = 0.509849979E+01

TOTAL APPLIED LOAD 231

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 231)
SUMMATION FORCE-X = -189.03
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.77 MY= -784.91 MZ= -28548.72

TOTAL REACTION LOAD 231

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 231)
SUMMATION FORCE-X = 189.03
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.77 MY= 784.91 MZ= 28548.72

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 231)

	MAXIMUMS	AT NODE
X =	-2.00288E-01	119
Y =	-2.60345E-01	112
Z =	2.47819E-02	3443
RX=	-2.31708E-04	83
RY=	1.35742E-04	112
RZ=	4.40216E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 232
0.9(D1+D2) + 1.6W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.512199074E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531283062E+01
Y = 0.286528546E+01
Z = 0.431558292E+01

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TOTAL APPLIED LOAD 232

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 232)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = 249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 32254.98 MY= -1327.71 MZ= -29221.84

TOTAL REACTION LOAD 232

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 232)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = -249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -32254.98 MY= 1327.71 MZ= 29221.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 232)

	MAXIMUMS	AT NODE
X =	-2.47806E-02	3442
Y =	-2.52821E-01	702
Z =	1.76560E-01	119
RX=	3.34887E-04	118
RY=	-5.78079E-05	2950
RZ=	3.26547E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 233
0.9(D1+D2) + 1.6W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.507500857E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531338165E+01
Y = 0.283845247E+01
Z = 0.431528904E+01

TOTAL APPLIED LOAD 233

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 233)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 30680.53 MY= 1327.75 MZ= -29221.84

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TOTAL REACTION LOAD 233

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 233)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 249.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -30680.53 MY= -1327.75 MZ= 29221.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 233)
MAXIMUMS AT NODE
X = -2.90005E-02 3441
Y = -2.47619E-01 702
Z = -1.73887E-01 82
RX= -3.17861E-04 706
RY= 5.45214E-05 2652
RZ= 3.29981E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 234
0.9(D1+D2) + 1.6E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.509849965E+01

TOTAL APPLIED LOAD 234

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 234)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.76 MY= 0.02 MZ= -29221.84

TOTAL REACTION LOAD 234

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 234)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.76 MY= -0.02 MZ= 29221.84

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 234)

	MAXIMUMS	AT NODE
X =	-2.53091E-02	3442
Y =	-2.50220E-01	702
Z =	1.24797E-02	3140
RX=	2.33603E-04	75
RY=	-3.21048E-05	2950
RZ=	3.28264E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 235
0.9(D1+D2) + 1.6E(-X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.509849965E+01

TOTAL APPLIED LOAD 235

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 235)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.76 MY= 0.02 MZ= -29221.84

TOTAL REACTION LOAD 235

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 235)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.76 MY= -0.02 MZ= 29221.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 235)

	MAXIMUMS	AT NODE
X =	-2.53091E-02	3442
Y =	-2.50220E-01	702
Z =	1.24797E-02	3140
RX=	2.33603E-04	75
RY=	-3.21048E-05	2950
RZ=	3.28264E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 236
0.9(D1+D2) + 1.6E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.509849965E+01

TOTAL APPLIED LOAD 236

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 236)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.76 MY= 0.02 MZ= -29221.84

TOTAL REACTION LOAD 236

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 236)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.76 MY= -0.02 MZ= 29221.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 236)

	MAXIMUMS	AT NODE
X =	-2.53091E-02	3442
Y =	-2.50220E-01	702
Z =	1.24797E-02	3140
RX=	2.33603E-04	75
RY=	-3.21048E-05	2950
RZ=	3.28264E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 237
0.9(D1+D2) + 1.6E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460918E+01
Y = -0.570272676E+00
Z = 0.509849965E+01

TOTAL APPLIED LOAD 237

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 237)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6171.96
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 31467.76 MY= 0.02 MZ= -29221.84

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TOTAL REACTION LOAD 237

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 237)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6171.96
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -31467.76 MY= -0.02 MZ= 29221.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 237)

	MAXIMUMS	AT NODE
X =	-2.53091E-02	3442
Y =	-2.50220E-01	702
Z =	1.24797E-02	3140
RX=	2.33603E-04	75
RY=	-3.21048E-05	2950
RZ=	3.28264E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 101
1.0(D1+D2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 101

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 101)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= 0.02 MZ= -32468.71

TOTAL REACTION LOAD 101

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 101)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= -0.02 MZ= 32468.71

STAAD SPACE -- PAGE NO. 107

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 101)

	MAXIMUMS	AT NODE
X =	-2.81212E-02	3442
Y =	-2.78022E-01	702
Z =	1.38664E-02	3140
RX=	2.59559E-04	75
RY=	-3.56720E-05	2950
RZ=	3.64738E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 102
1.0(D1+D2)+1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470058755E+01
Y = -0.257358710E+00
Z = 0.506826411E+01

TOTAL APPLIED LOAD 102

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 102)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -7341.18
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 37207.03 MY= 0.09 MZ= -34507.85

TOTAL REACTION LOAD 102

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 102)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 7341.18
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -37207.03 MY= -0.09 MZ= 34507.85

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 102)

	MAXIMUMS	AT NODE
X =	-4.41462E-02	3442
Y =	-2.98789E-01	112
Z =	-1.53965E-02	3237
RX=	3.99693E-04	118
RY=	4.21773E-05	2652
RZ=	4.18463E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 103
1.0(D1+D2)+1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474786505E+01
Y = 0.447239818E-01
Z = 0.512150879E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697651660E+01
Y = 0.625353575E+01
Z = 0.127561704E+01

TOTAL APPLIED LOAD 103

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 103)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -7841.18
SUMMATION FORCE-Z = 75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 40494.66 MY= -525.84 MZ= -37228.86

TOTAL REACTION LOAD 103

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 103)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 7841.18
SUMMATION FORCE-Z = -75.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -40494.66 MY= 525.84 MZ= 37228.86

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 103)

	MAXIMUMS	AT NODE
X	-6.48787E-02	3444
Y	-3.90096E-01	112
Z	1.18162E-01	78
RX	5.42303E-04	118
RY	-1.24947E-04	2950
RZ	8.13394E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 104
LOADTYPE NONE TITLE 1.0(D1+D2)+1.0(L1+L2+ELL_REV)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520306342E+01
Y = -0.213005384E+01
Z = 0.518947802E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475646672E+01
Y = -0.648224914E-01
Z = 0.519267195E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664245973E+01
Y = 0.101104771E+02
Z = 0.190792867E+01

TOTAL APPLIED LOAD 104

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 104)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = -7652.18
SUMMATION FORCE-Z = 89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 40136.22 MY= -327.37 MZ= -36632.53

TOTAL REACTION LOAD 104

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 104)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = 7652.18
SUMMATION FORCE-Z = -89.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -40136.22 MY= 327.37 MZ= 36632.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 104)

	MAXIMUMS	AT NODE
X	-4.12191E-02	118
Y	-3.64437E-01	1765
Z	1.81897E-01	78
RX	1.70507E-03	702
RY	-1.43276E-04	2950
RZ	6.64795E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 105
LOADTYPE NONE TITLE 1.0(D1+D2)+1.0(L1+L2-ELL_REV)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.516140539E+01
Y = 0.131232710E+02
Z = 0.528856600E+01

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CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473966828E+01
Y = 0.149113837E+00
Z = 0.505369546E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747071360E+01
Y = 0.547654380E+00
Z = 0.340187949E+00

TOTAL APPLIED LOAD 105

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 105)
SUMMATION FORCE-X = -52.50
SUMMATION FORCE-Y = -8030.18
SUMMATION FORCE-Z = 60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 40853.11 MY= -724.31 MZ= -37825.18

TOTAL REACTION LOAD 105

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 105)
SUMMATION FORCE-X = 52.50
SUMMATION FORCE-Y = 8030.18
SUMMATION FORCE-Z = -60.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -40853.11 MY= 724.31 MZ= 37825.18

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 105)

	MAXIMUMS	AT NODE
X	-1.08269E-01	3443
Y	-4.31356E-01	112
Z	5.44276E-02	78
RX	-1.63294E-03	702
RY	-1.06618E-04	2950
RZ	9.65792E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 106
1.0(D1+D2)+1.0W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456867594E+01
Y = 0.123639627E+01
Z = 0.415192710E+01

STAAD SPACE -- PAGE NO. 111

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474991551E+01
Y = -0.570272722E+00
Z = 0.509849956E+01

TOTAL APPLIED LOAD 106

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 106)
SUMMATION FORCE-X = 118.14
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.17 MY= 490.60 MZ= -32889.41

TOTAL REACTION LOAD 106

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 106)
SUMMATION FORCE-X = -118.14
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.17 MY= -490.60 MZ= 32889.41

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 106)

	MAXIMUMS	AT NODE
X =	9.72747E-02	119
Y =	-2.83155E-01	702
Z =	1.64448E-02	3339
RX=	2.70724E-04	75
RY=	-5.48985E-05	75
RZ=	2.94768E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 107
1.0(D1+D2)+1.0W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456425752E+01
Y = 0.410850011E+01
Z = 0.415287681E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.471930273E+01
Y = -0.570272719E+00
Z = 0.509849972E+01

STAAD SPACE -- PAGE NO. 112

TOTAL APPLIED LOAD 107

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 107)
SUMMATION FORCE-X = -118.14
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= -490.56 MZ= -32048.01

TOTAL REACTION LOAD 107

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 107)
SUMMATION FORCE-X = 118.14
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= 490.56 MZ= 32048.01

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 107)

	MAXIMUMS	AT NODE
X	-1.32993E-01	119
Y	-2.79700E-01	112
Z	1.89186E-02	3443
RX	2.48395E-04	75
RY	9.43157E-05	112
RZ	4.34708E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 108
1.0(D1+D2)+1.0W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.511171338E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531261627E+01
Y = 0.287572052E+01
Z = 0.431569721E+01

TOTAL APPLIED LOAD 108

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 108)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 35456.19 MY= -829.81 MZ= -32468.71

TOTAL REACTION LOAD 108

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 108)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -35456.19 MY= 829.81 MZ= 32468.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 108)
MAXIMUMS AT NODE
X = -2.77909E-02 3442
Y = -2.79648E-01 702
Z = 1.11393E-01 119
RX= 2.77510E-04 118
RY= -5.17364E-05 2950
RZ= 3.63664E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 109
1.0(D1+D2)+1.0W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.508528591E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531359588E+01
Y = 0.282801739E+01
Z = 0.431517475E+01

TOTAL APPLIED LOAD 109

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 109)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = -156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34472.16 MY= 829.85 MZ= -32468.71

TOTAL REACTION LOAD 109

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 109)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = 156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34472.16 MY= -829.85 MZ= 32468.71

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 109)

	MAXIMUMS	AT NODE
X =	-2.90359E-02	3441
Y =	-2.76397E-01	702
Z =	-1.09044E-01	117
RX=	3.00013E-04	75
RY=	4.96196E-05	2652
RZ=	3.65811E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 110
1.0(D1+D2)+0.7E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 110

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 110)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= 0.02 MZ= -32468.71

TOTAL REACTION LOAD 110

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 110)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= -0.02 MZ= 32468.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 110)

	MAXIMUMS	AT NODE
X =	-2.81212E-02	3442
Y =	-2.78022E-01	702
Z =	1.38664E-02	3140
RX=	2.59559E-04	75
RY=	-3.56720E-05	2950
RZ=	3.64738E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 111
1.0(D1+D2)+0.7E(-X)

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CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 111

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 111)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= 0.02 MZ= -32468.71

TOTAL REACTION LOAD 111

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 111)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= -0.02 MZ= 32468.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 111)

	MAXIMUMS	AT NODE
X =	-2.81212E-02	3442
Y =	-2.78022E-01	702
Z =	1.38664E-02	3140
RX=	2.59559E-04	75
RY=	-3.56720E-05	2950
RZ=	3.64738E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 112
1.0(D1+D2)+0.7E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 112

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 112)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= 0.02 MZ= -32468.71

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TOTAL REACTION LOAD 112

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 112)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= -0.02 MZ= 32468.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 112)
MAXIMUMS AT NODE
X = -2.81212E-02 3442
Y = -2.78022E-01 702
Z = 1.38664E-02 3140
RX= 2.59559E-04 75
RY= -3.56720E-05 2950
RZ= 3.64738E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 113
1.0(D1+D2)+0.7E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460912E+01
Y = -0.570272721E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 113

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 113)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -6857.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 34964.18 MY= 0.02 MZ= -32468.71

TOTAL REACTION LOAD 113

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 113)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6857.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -34964.18 MY= -0.02 MZ= 32468.71

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 113)

	MAXIMUMS	AT NODE
X =	-2.81212E-02	3442
Y =	-2.78022E-01	702
Z =	1.38664E-02	3140
RX=	2.59559E-04	75
RY=	-3.56720E-05	2950
RZ=	3.64738E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 114
1.0(D1+D2)+0.75W(+X) + 0.75L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.457649699E+01
Y = 0.626256105E+00
Z = 0.415012847E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.471956915E+01
Y = -0.331658798E+00
Z = 0.507544336E+01

TOTAL APPLIED LOAD 114

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 114)
SUMMATION FORCE-X = 88.61
SUMMATION FORCE-Y = -7220.32
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 36646.31 MY= 368.01 MZ= -34313.59

TOTAL REACTION LOAD 114

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 114)
SUMMATION FORCE-X = -88.61
SUMMATION FORCE-Y = 7220.32
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -36646.31 MY= -368.01 MZ= 34313.59

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 114)

	MAXIMUMS	AT NODE
X =	5.92376E-02	119
Y =	-2.93270E-01	702
Z =	-1.48949E-02	3442
RX=	3.33370E-04	118
RY=	-4.65168E-05	2950
RZ=	3.52554E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 115
1.0(D1+D2)+0.75W(-X) + 0.75L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.455643637E+01
Y = 0.471864019E+01
Z = 0.415467556E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.469776251E+01
Y = -0.331658796E+00
Z = 0.507544347E+01

TOTAL APPLIED LOAD 115

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 115)
SUMMATION FORCE-X = -88.61
SUMMATION FORCE-Y = -7220.32
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 36646.32 MY= -367.87 MZ= -33682.54

TOTAL REACTION LOAD 115

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 115)
SUMMATION FORCE-X = 88.61
SUMMATION FORCE-Y = 7220.32
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -36646.32 MY= 367.87 MZ= 33682.54

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 115)

	MAXIMUMS	AT NODE
X =	-1.15629E-01	3441
Y =	-3.00170E-01	112
Z =	1.68006E-02	3140
RX=	3.44118E-04	118
RY=	7.11592E-05	112
RZ=	4.57509E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 116
1.0(D1+D2)+0.75W(+Z) + 0.75L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470866583E+01
Y = -0.331658797E+00
Z = 0.508485606E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531078182E+01
Y = 0.289086869E+01
Z = 0.431598019E+01

TOTAL APPLIED LOAD 116

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 116)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7220.32
SUMMATION FORCE-Z = 117.14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 37015.33 MY= -622.30 MZ= -33998.07

TOTAL REACTION LOAD 116

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 116)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7220.32
SUMMATION FORCE-Z = -117.14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -37015.33 MY= 622.30 MZ= 33998.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 116)

	MAXIMUMS	AT NODE
X	-3.98922E-02	3442
Y	-2.92033E-01	112
Z	8.25794E-02	78
RX	4.29953E-04	118
RY	-5.24489E-05	2950
RZ	4.04227E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 117
1.0(D1+D2)+0.75W(-Z) + 0.75L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470866583E+01
Y = -0.331658797E+00
Z = 0.506603078E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531543015E+01
Y = 0.281286925E+01
Z = 0.431489173E+01

TOTAL APPLIED LOAD 117

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 117)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7220.32
SUMMATION FORCE-Z = -117.14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 36277.30 MY= 622.44 MZ= -33998.07

TOTAL REACTION LOAD 117

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 117)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7220.32
SUMMATION FORCE-Z = 117.14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -36277.30 MY= -622.44 MZ= 33998.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 117)

	MAXIMUMS	AT NODE
X	-4.03877E-02	3442
Y	-2.89845E-01	112
Z	-8.36024E-02	82
RX	-4.20378E-04	706
RY	5.10834E-05	2652
RZ	4.05837E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 118
1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.457645890E+01
Y = -0.119530580E+01
Z = 0.413022780E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475523787E+01
Y = -0.940945943E-01
Z = 0.511631505E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697615761E+01
Y = 0.627014282E+01
Z = 0.127581536E+01

TOTAL APPLIED LOAD 118

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 118)
SUMMATION FORCE-X = 88.61
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.04 MY= -26.44 MZ= -36354.34

TOTAL REACTION LOAD 118

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 118)
SUMMATION FORCE-X = -88.61
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.04 MY= 26.44 MZ= 36354.34

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 118)

	MAXIMUMS	AT NODE
X =	7.53364E-02	119
Y =	-3.57922E-01	1765
Z =	9.37175E-02	78
RX=	-4.48017E-04	83
RY=	-1.08744E-04	2950
RZ=	6.51543E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 119
1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.455647446E+01
Y = 0.654020210E+01
Z = 0.417457623E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473450788E+01
Y = -0.940945924E-01
Z = 0.511631516E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697619558E+01
Y = 0.627004175E+01
Z = 0.127578134E+01

TOTAL APPLIED LOAD 119

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 119)
SUMMATION FORCE-X = -88.61
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.05 MY= -762.31 MZ= -35723.30

TOTAL REACTION LOAD 119

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 119)
SUMMATION FORCE-X = 88.61
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.05 MY= 762.31 MZ= 35723.30

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 119)

	MAXIMUMS	AT NODE
X =	-1.16683E-01	3442
Y =	-3.68651E-01	112
Z =	8.47182E-02	119
RX=	-4.67890E-04	83
RY=	-9.65121E-05	2950
RZ=	7.53707E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 120
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.512526302E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.585263081E+01
Y = 0.396079142E+01
Z = 0.332933234E+01

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TOTAL APPLIED LOAD 120

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 120)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 173.39

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39481.05 MY= -1016.75 MZ= -36038.82

TOTAL REACTION LOAD 120

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 120)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -173.39

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39481.05 MY= 1016.75 MZ= 36038.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 120)

	MAXIMUMS	AT NODE
X =	-5.57188E-02	3444
Y =	-3.60514E-01	112
Z =	1.70974E-01	119
RX=	5.36910E-04	118
RY=	-1.14677E-04	2950
RZ=	7.00425E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 121
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.510736719E+01

TOTAL APPLIED LOAD 121

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 121)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = -60.89

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38743.03 MY= 228.00 MZ= -36038.82

TOTAL REACTION LOAD 121

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 121)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = 60.89

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38743.03 MY= -228.00 MZ= 36038.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 121)

	MAXIMUMS	AT NODE
X	-4.99314E-02	3444
Y	-3.58326E-01	112
Z	-4.27143E-02	3442
RX	4.35988E-04	75
RY	-9.05800E-05	2950
RZ	7.02035E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 122
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476283012E+01
Y = 0.863510097E+00
Z = 0.447135638E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476200124E+01
Y = -0.181083700E+00
Z = 0.517101082E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664215517E+01
Y = 0.101244363E+02
Z = 0.190810190E+01

TOTAL APPLIED LOAD 122

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 122)
SUMMATION FORCE-X = 127.98
SUMMATION FORCE-Y = -7453.57
SUMMATION FORCE-Z = 67.13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38843.20 MY= 122.42 MZ= -35907.09

TOTAL REACTION LOAD 122

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 122)
SUMMATION FORCE-X = -127.98
SUMMATION FORCE-Y = 7453.57
SUMMATION FORCE-Z = -67.13

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SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38843.20 MY= -122.42 MZ= 35907.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 122)

	MAXIMUMS	AT NODE
X =	1.12446E-01	119
Y =	-3.45504E-01	1765
Z =	1.41519E-01	119
RX=	1.28332E-03	702
RY=	-1.22491E-04	2950
RZ=	5.44562E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 123
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+ELL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474087702E+01
Y = -0.181083698E+00
Z = 0.517101093E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.664219445E+01
Y = 0.101242655E+02
Z = 0.190805927E+01

TOTAL APPLIED LOAD 123

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 123)
SUMMATION FORCE-X = -49.23
SUMMATION FORCE-Y = -7453.57
SUMMATION FORCE-Z = 67.13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38843.21 MY= -613.46 MZ= -35276.05

TOTAL REACTION LOAD 123

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 123)
SUMMATION FORCE-X = 49.23
SUMMATION FORCE-Y = 7453.57
SUMMATION FORCE-Z = -67.13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38843.21 MY= 613.46 MZ= 35276.05

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MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 123)

	MAXIMUMS	AT NODE
X =	-7.95849E-02	3442
Y =	-3.37707E-01	112
Z =	1.32519E-01	78
RX=	1.28515E-03	702
RY=	-1.10259E-04	2950
RZ=	6.39408E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 124
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520472058E+01
Y = -0.320727029E+01
Z = 0.518912180E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475143913E+01
Y = -0.181083699E+00
Z = 0.518012896E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.579727181E+01
Y = 0.550115784E+01
Z = 0.343846166E+01

TOTAL APPLIED LOAD 124

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 124)
SUMMATION FORCE-X = 39.37
SUMMATION FORCE-Y = -7453.57
SUMMATION FORCE-Z = 184.26

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39212.22 MY= -867.89 MZ= -35591.57

TOTAL REACTION LOAD 124

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 124)
SUMMATION FORCE-X = -39.37
SUMMATION FORCE-Y = 7453.57
SUMMATION FORCE-Z = -184.26

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39212.22 MY= 867.89 MZ= 35591.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 124)

	MAXIMUMS	AT NODE
X =	-3.83199E-02	118
Y =	-3.42461E-01	1765
Z =	2.18776E-01	119
RX=	1.35039E-03	702
RY=	-1.28423E-04	2950
RZ=	5.86849E-04	1766

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 125
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.520472058E+01
Y = -0.320727029E+01
Z = 0.518912180E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.475143913E+01
Y = -0.181083699E+00
Z = 0.516189279E+01

TOTAL APPLIED LOAD 125

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 125)
SUMMATION FORCE-X = 39.37
SUMMATION FORCE-Y = -7453.57
SUMMATION FORCE-Z = -50.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 38474.19 MY= 376.85 MZ= -35591.57

TOTAL REACTION LOAD 125

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 125)
SUMMATION FORCE-X = -39.37
SUMMATION FORCE-Y = 7453.57
SUMMATION FORCE-Z = 50.01

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -38474.19 MY= -376.85 MZ= 35591.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 125)

	MAXIMUMS	AT NODE
X =	-3.11180E-02	118
Y =	-3.30820E-01	1765
Z =	5.52623E-02	78
RX=	1.21808E-03	702
RY=	-1.04327E-04	2950
RZ=	5.87736E-04	1765

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STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 126
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-ELL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474872230E+01
Y = -0.102929248E-01
Z = 0.506362344E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747027671E+01
Y = 0.568148192E+00
Z = 0.340418400E+00

TOTAL APPLIED LOAD 126

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 126)
SUMMATION FORCE-X = 49.23
SUMMATION FORCE-Y = -7737.07
SUMMATION FORCE-Z = 45.37

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39380.87 MY= -175.29 MZ= -36801.59

TOTAL REACTION LOAD 126

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 126)
SUMMATION FORCE-X = -49.23
SUMMATION FORCE-Y = 7737.07
SUMMATION FORCE-Z = -45.37

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39380.87 MY= 175.29 MZ= 36801.59

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 126)

	MAXIMUMS	AT NODE
X =	3.82265E-02	119
Y =	-3.81134E-01	112
Z =	4.66968E-02	3338
RX=	-1.22018E-03	702
RY=	-9.49978E-05	2950
RZ=	7.63051E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 127
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474899404E+01
Y = 0.621913318E+01
Z = 0.450206070E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.472837211E+01
Y = -0.102929229E-01
Z = 0.506362354E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.747030745E+01
Y = 0.568211395E+00
Z = 0.340407146E+00

TOTAL APPLIED LOAD 127

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 127)
SUMMATION FORCE-X = -127.98
SUMMATION FORCE-Y = -7737.07
SUMMATION FORCE-Z = 45.38

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39380.88 MY= -911.17 MZ= -36170.54

TOTAL REACTION LOAD 127

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 127)
SUMMATION FORCE-X = 127.98
SUMMATION FORCE-Y = 7737.07
SUMMATION FORCE-Z = -45.38

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39380.88 MY= 911.17 MZ= 36170.54

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 127)

	MAXIMUMS	AT NODE
X	-1.53781E-01	3442
Y	-3.99596E-01	112
Z	4.09069E-02	3140
RX	-1.21836E-03	702
RY	8.30989E-05	83
RZ	8.68006E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 128
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.515974820E+01
Y = 0.142004877E+02
Z = 0.528892220E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473854721E+01
Y = -0.102929239E-01
Z = 0.507240747E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.591539930E+01
Y = 0.221425559E+01
Z = 0.320559667E+01

TOTAL APPLIED LOAD 128

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 128)
SUMMATION FORCE-X = -39.37
SUMMATION FORCE-Y = -7737.07
SUMMATION FORCE-Z = 162.51

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39749.89 MY= -1165.60 MZ= -36486.07

TOTAL REACTION LOAD 128

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 128)
SUMMATION FORCE-X = 39.37
SUMMATION FORCE-Y = 7737.07
SUMMATION FORCE-Z = -162.51

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39749.89 MY= 1165.60 MZ= 36486.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 128)

	MAXIMUMS	AT NODE
X	-8.68776E-02	3443
Y	-3.91459E-01	112
Z	1.23173E-01	78
RX	-1.15312E-03	702
RY	-1.00930E-04	2950
RZ	8.14724E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 129
LOADTYPE NONE TITLE 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-ELL

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.515974820E+01
Y = 0.142004877E+02
Z = 0.528892220E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473854721E+01
Y = -0.102929239E-01
Z = 0.505483951E+01

TOTAL APPLIED LOAD 129

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 129)
SUMMATION FORCE-X = -39.37
SUMMATION FORCE-Y = -7737.07
SUMMATION FORCE-Z = -71.76

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39011.86 MY= 79.14 MZ= -36486.07

TOTAL REACTION LOAD 129

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 129)
SUMMATION FORCE-X = 39.37
SUMMATION FORCE-Y = 7737.07
SUMMATION FORCE-Z = 71.76

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39011.86 MY= -79.14 MZ= 36486.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 129)

	MAXIMUMS	AT NODE
X	-8.62371E-02	3443
Y	-3.89271E-01	112
Z	-6.89220E-02	3442
RX	-1.28543E-03	702
RY	-7.68333E-05	2950
RZ	8.16333E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 130
1.0(D1+D2)+0.75*0.7 E(+X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.511631511E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697617659E+01
Y = 0.627009228E+01
Z = 0.127579835E+01

TOTAL APPLIED LOAD 130

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 130)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.04 MY= -394.37 MZ= -36038.82

TOTAL REACTION LOAD 130

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 130)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.04 MY= 394.37 MZ= 36038.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 130)

	MAXIMUMS	AT NODE
X =	-5.28251E-02	3444
Y =	-3.59420E-01	112
Z =	8.92178E-02	119
RX=	-4.57954E-04	83
RY=	-1.02628E-04	2950
RZ=	7.01230E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 131
1.0(D1+D2)+0.75*0.7 E(-X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.511631511E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697617659E+01
Y = 0.627009228E+01
Z = 0.127579835E+01

STAAD SPACE -- PAGE NO. 133

TOTAL APPLIED LOAD 131

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 131)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.04 MY= -394.37 MZ= -36038.82

TOTAL REACTION LOAD 131

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 131)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.04 MY= 394.37 MZ= 36038.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 131)

	MAXIMUMS	AT NODE
X =	-5.28251E-02	3444
Y =	-3.59420E-01	112
Z =	8.92178E-02	119
RX=	-4.57954E-04	83
RY=	-1.02628E-04	2950
RZ=	7.01230E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 132
1.0(D1+D2)+0.75*0.7 E(+Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.511631511E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697617659E+01
Y = 0.627009228E+01
Z = 0.127579835E+01

TOTAL APPLIED LOAD 132

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 132)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.04 MY= -394.37 MZ= -36038.82

STAAD SPACE -- PAGE NO. 134

TOTAL REACTION LOAD 132

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 132)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.04 MY= 394.37 MZ= 36038.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 132)
MAXIMUMS AT NODE
X = -5.28251E-02 3444
Y = -3.59420E-01 112
Z = 8.92178E-02 119
RX= -4.57954E-04 83
RY= -1.02628E-04 2950
RZ= 7.01230E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 133
1.0(D1+D2)+0.75*0.7 E(-Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.474487287E+01
Y = -0.940945933E-01
Z = 0.511631511E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.697617659E+01
Y = 0.627009228E+01
Z = 0.127579835E+01

TOTAL APPLIED LOAD 133

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 133)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -7595.32
SUMMATION FORCE-Z = 56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 39112.04 MY= -394.37 MZ= -36038.82

TOTAL REACTION LOAD 133

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 133)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 7595.32
SUMMATION FORCE-Z = -56.25

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -39112.04 MY= 394.37 MZ= 36038.82

STAAD SPACE -- PAGE NO. 135

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 133)

	MAXIMUMS	AT NODE
X =	-5.28251E-02	3444
Y =	-3.59420E-01	112
Z =	8.92178E-02	119
RX=	-4.57954E-04	83
RY=	-1.02628E-04	2950
RZ=	7.01230E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 134
0.6(D1+D2) + 1.0W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456779226E+01
Y = 0.181081696E+01
Z = 0.415211704E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.476011971E+01
Y = -0.570272619E+00
Z = 0.509849951E+01

TOTAL APPLIED LOAD 134

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 134)
SUMMATION FORCE-X = 118.14
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.50 MY= 490.59 MZ= -19901.92

TOTAL REACTION LOAD 134

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 134)
SUMMATION FORCE-X = -118.14
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.50 MY= -490.59 MZ= 19901.92

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 134)

	MAXIMUMS	AT NODE
X =	1.04418E-01	119
Y =	-1.71946E-01	702
Z =	-1.35924E-02	3442
RX=	1.66900E-04	75
RY=	-6.13990E-05	75
RZ=	-1.96734E-04	10

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 135
0.6(D1+D2) + 1.0W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.456514121E+01
Y = 0.353407941E+01
Z = 0.415268687E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.470909840E+01
Y = -0.570272613E+00
Z = 0.509849977E+01

TOTAL APPLIED LOAD 135

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 135)
SUMMATION FORCE-X = -118.14
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.51 MY= -490.57 MZ= -19060.53

TOTAL REACTION LOAD 135

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 135)
SUMMATION FORCE-X = 118.14
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.51 MY= 490.57 MZ= 19060.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 135)

	MAXIMUMS	AT NODE
X =	-1.25850E-01	119
Y =	-1.72743E-01	112
Z =	1.57827E-02	3443
RX=	-1.53589E-04	83
RY=	8.56512E-05	112
RZ=	2.88813E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 136
0.6(D1+D2) + 1.0W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.512052253E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531281219E+01
Y = 0.286617992E+01
Z = 0.431559271E+01

TOTAL APPLIED LOAD 136

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 136)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = 156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 21470.52 MY= -829.82 MZ= -19481.23

TOTAL REACTION LOAD 136

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 136)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = -156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -21470.52 MY= 829.82 MZ= 19481.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 136)

	MAXIMUMS	AT NODE
X	-1.65424E-02	3442
Y	-1.68439E-01	702
Z	1.10439E-01	119
RX	2.15150E-04	118
RY	-3.74676E-05	2950
RZ	2.17769E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 137
0.6(D1+D2) + 1.0W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.507647674E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.531339996E+01
Y = 0.283755799E+01
Z = 0.431527924E+01

TOTAL APPLIED LOAD 137

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 137)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20486.49 MY= 829.84 MZ= -19481.23

TOTAL REACTION LOAD 137

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 137)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 156.19

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20486.49 MY= -829.84 MZ= 19481.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 137)

	MAXIMUMS	AT NODE
X	-1.90605E-02	3441
Y	-1.65188E-01	702
Z	-1.08711E-01	117
RX	-2.04173E-04	706
RY	3.54082E-05	2652
RZ	2.19916E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 138
0.6(D1+D2) + 0.7E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 138

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 138)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.51 MY= 0.01 MZ= -19481.23

STAAD SPACE -- PAGE NO. 139

TOTAL REACTION LOAD 138

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 138)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.51 MY= -0.01 MZ= 19481.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 138)
MAXIMUMS AT NODE
X = -1.68728E-02 3442
Y = -1.66813E-01 702
Z = 8.31982E-03 3140
RX= 1.55736E-04 75
RY= -2.14032E-05 2950
RZ= 2.18843E-04 1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 139
0.6(D1+D2) + 0.7E(-X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 139

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 139)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.51 MY= 0.01 MZ= -19481.23

TOTAL REACTION LOAD 139

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 139)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.51 MY= -0.01 MZ= 19481.23

STAAD SPACE -- PAGE NO. 140

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 139)

	MAXIMUMS	AT NODE
X =	-1.68728E-02	3442
Y =	-1.66813E-01	702
Z =	8.31982E-03	3140
RX=	1.55736E-04	75
RY=	-2.14032E-05	2950
RZ=	2.18843E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 140
0.6(D1+D2) + 0.7E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 140

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 140)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.51 MY= 0.01 MZ= -19481.23

TOTAL REACTION LOAD 140

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 140)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.51 MY= -0.01 MZ= 19481.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 140)

	MAXIMUMS	AT NODE
X =	-1.68728E-02	3442
Y =	-1.66813E-01	702
Z =	8.31982E-03	3140
RX=	1.55736E-04	75
RY=	-2.14032E-05	2950
RZ=	2.18843E-04	1765

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 141
0.6(D1+D2) + 0.7E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.473460905E+01
Y = -0.570272616E+00
Z = 0.509849964E+01

TOTAL APPLIED LOAD 141

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 141)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -4114.64
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20978.51 MY= 0.01 MZ= -19481.23

TOTAL REACTION LOAD 141

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 141)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 4114.64
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20978.51 MY= -0.01 MZ= 19481.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 141)

	MAXIMUMS	AT NODE
X =	-1.68728E-02	3442
Y =	-1.66813E-01	702
Z =	8.31982E-03	3140
RX=	1.55736E-04	75
RY=	-2.14032E-05	2950
RZ=	2.18843E-04	1765

********** END OF DATA FROM INTERNAL STORAGE **********

2381. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 12,2022 TIME= 11:58: 0 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://www.bentley.com/en/support/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 2 STAAD OUTPUT FOR SCREEN SUPPORTING STRUCTURE

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 31, 2020                *
*           Time=    16:39:41                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

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     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\3.Primary Crusher\STAAD Final\C01\Screen  supporting stru.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 01-NOV-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 13 7.75148E-07 -3.00001 0; 14 7.75148E-07 -3.00001 2.55001
     9. 15 7.75148E-07 -3.00001 7.45002; 16 7.75148E-07 -3.00001 10
    10. 17 1.99998 -3.00001 0; 18 1.99998 -3.00001 2.55001
    11. 19 1.99998 -3.00001 7.45002; 20 1.99998 -3.00001 10; 85 7.75148E-07 6.89001 0
    12. 86 7.75148E-07 6.89001 2.55001; 87 7.75148E-07 6.89001 7.45002
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247. MEMBER INCIDENCES
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408. START GROUP DEFINITION
409. FLOOR
410. _SCREEN_FLOOR_1 79 TO 103 105 108 111 156 157 159 160 621 TO 625
411. MEMBER
412. _COLUMN2 71 TO 78 120 TO 124 126 128 130 138 139 141 142 144 145 147 148 615 -
413. 616
414. _TB2 125 127 129 131 TO 137
415. _FB5 140 143 146 149 151 154 611 TO 614
416. _FB9 623
417. _FB4 617 TO 620
418. _FB6_TB7_TB8 79 TO 103 105 108 109 111 621 622 624 625
419. ELEMENT
420. _FDN_MOMENT 179 180 183 TO 186 189 TO 228 231 TO 234 237 TO 240 243 TO 246 -
421. 249 TO 288 291 TO 294 297 TO 300 303 TO 306 309 TO 348 351 TO 354 -
422. 357 TO 360 363 TO 366 369 TO 408 411 TO 414 417 TO 610
423. _FDN_SHEAR 179 184 185 190 191 196 TO 221 226 227 232 233 238 239 244 245 -
424. 250 251 256 TO 281 286 287 292 293 298 299 304 305 310 311 316 TO 341 346 -
425. 347 352 353 358 359 364 365 370 371 376 TO 401 406 407 412 413 418 TO 539 -
426. 544 545 550 551 556 557 562 TO 575 580 581 586 587 592 593 598 TO 610
427. END GROUP DEFINITION
428. ELEMENT PROPERTY
429. 179 TO 610 THICKNESS 0.5
430. MEMBER RELEASE
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431. 156 TO 160 623 START MY MZ
432. 156 TO 160 623 END MY MZ
433. DEFINE MATERIAL START
434. ISOTROPIC CONCRETE
435. E 3E+07
436. POISSON 0.17
437. DENSITY 24
438. ALPHA 5.5E-06
439. DAMP 0.05
440. TYPE CONCRETE
441. STRENGTH FCU 27578.9
442. END DEFINE MATERIAL
443. MEMBER PROPERTY INDIAN
444. 71 TO 78 120 TO 124 126 128 130 138 139 141 142 144 145 147 148 615 -
445. 616 PRIS YD 0.5 ZD 0.5
446. 79 TO 103 105 108 109 111 621 TO 625 PRIS YD 0.6 ZD 0.3
447. MEMBER PROPERTY INDIAN
448. 156 TO 160 PRIS YD 0.1 ZD 0.1
449. MEMBER PROPERTY INDIAN
450. 125 127 129 131 TO 137 140 143 146 149 151 154 611 TO 614 PRIS YD 0.4 ZD 0.3
451. MEMBER PROPERTY INDIAN
452. 617 TO 620 PRIS YD 0.4 ZD 0.3
453. CONSTANTS
454. MATERIAL CONCRETE ALL
455. SUPPORTS
456. 179 TO 610 PLATE MAT DIRECT Y SUBGRADE 68400 PRINT
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600	0.031251
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       608          0.031250
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       615          0.031250
       616          0.031250
       617          0.031250
       618          0.015625
       619          0.032812
       620          0.015625
       621          0.032812
       622          0.032812
       623          0.015625
       624          0.032812
       625          0.015625
       626          0.034375
       627          0.034375
    TOTAL          27.299973
457. DEFINE REFERENCE LOADS
458. LOAD R3 LOADTYPE WIND  TITLE WIND LOAD (W +X)
459. MEMBER LOAD
460. 120 123 124 130 UNI GX 0.81
461. 121 122 126 128 UNI GX 1.25
462. 138 139 147 148 615 616 UNI GX 0.9
463. 141 142 144 145 UNI GX 1.35
464. LOAD R4 LOADTYPE WIND  TITLE WIND LOAD (W +Z)
465. MEMBER LOAD
466. 120 TO 124 126 128 130 UNI GZ 0.92
467. 138 139 141 142 144 145 147 148 615 616 UNI GZ 1.05
468. *
469. LOAD R7 SEISMIC_WEIGHT _X
470. SELFWEIGHT X 1 LIST ALL
471. *STAIR
472. JOINT LOAD
473. 630 TO 633 FX 0.55
474. MEMBER LOAD
475. 617 TO 620 CON GX 0.55 0.4
476. *FLOOR LOAD
477. FLOOR LOAD
478. _SCREEN_FLOOR_1 FLOAD 3.6 GX
479. MEMBER LOAD
480. *OPERATION PLATFORM
481. 611 UNI GX 9.85 1.25 2.55
482. 612 UNI GX 11.45 1.25 2.55
483. *
484. 613 UNI GX 9.85 0 1.3
485. 614 UNI GX 11.45 0 1.3
486. MEMBER LOAD
487. **VIBRATING GRIZZLYSCREEN
488. 102 108 CON GX 20 0.558
489. 158 CON GX 20 0.657
490. 158 CON GX 20 1.542
491. JOINT LOAD
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107 FX 15
MEMBER LOAD
**VIBRATING GRIZZLYSCREEN
102 108 CON GX 23 0.558
158 CON GX 23 0.657
158 CON GX 23 1.542
**STAIR
JOINT LOAD
630 TO 633 FX 1
MEMBER LOAD
617 TO 620 CON GX 1 0.4
FLOOR LOAD
**SCREEN FLOOR
_SCREEN_FLOOR_1 FLOAD 2.5 GX
MEMBER LOAD
*OPERATION PLATFORM
611 UNI GX 5.45 1.25 2.55
612 UNI GX 6.36 1.25 2.55
*
613 UNI GX 5.45 0 1.3
614 UNI GX 6.36 0 1.3
*
LOAD R8 SEISMIC_WEIGHT _Y
SELFWEIGHT Y 1 LIST ALL
*STAIR
JOINT LOAD
630 TO 633 FY 0.55
MEMBER LOAD
617 TO 620 CON GY 0.55 0.4
*FLOOR LOAD
FLOOR LOAD
_SCREEN_FLOOR_1 FLOAD 3.6 GY
MEMBER LOAD
*OPERATION PLATFORM
611 UNI GY 9.85 1.25 2.55
612 UNI GY 11.45 1.25 2.55
*
613 UNI GY 9.85 0 1.3
614 UNI GY 11.45 0 1.3
MEMBER LOAD
**VIBRATING GRIZZLYSCREEN
102 108 CON GY 20 0.558
158 CON GY 20 0.657
158 CON GY 20 1.542
JOINT LOAD
107 FY 15
MEMBER LOAD
**VIBRATING GRIZZLYSCREEN
102 108 CON GY 23 0.558
158 CON GY 23 0.657
158 CON GY 23 1.542
**STAIR
JOINT LOAD
630 TO 633 FY 1
MEMBER LOAD
617 TO 620 CON GY 1 0.4
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548. FLOOR LOAD
549. **SCREEN FLOOR
550. _SCREEN_FLOOR_1 FLOAD 2.5 GY
551. MEMBER LOAD
552. *OPERATION PLATFORM
553. 611 UNI GY 5.45 1.25 2.55
554. 612 UNI GY 6.36 1.25 2.55
555. *
556. 613 UNI GY 5.45 0 1.3
557. 614 UNI GY 6.36 0 1.3
558. *
559. LOAD R9 SEISMIC_WEIGHT _Z
560. SELFWEIGHT Z 1 LIST ALL
561. *STAIR
562. JOINT LOAD
563. 630 TO 633 FZ 0.55
564. MEMBER LOAD
565. 617 TO 620 CON GZ 0.55 0.4
566. *FLOOR LOAD
567. FLOOR LOAD
568. _SCREEN_FLOOR_1 FLOAD 3.6 GZ
569. MEMBER LOAD
570. *OPERATION PLATFORM
571. 611 UNI GZ 9.85 1.25 2.55
572. 612 UNI GZ 11.45 1.25 2.55
573. *
574. 613 UNI GZ 9.85 0 1.3
575. 614 UNI GZ 11.45 0 1.3
576. MEMBER LOAD
577. **VIBRATING GRIZZLYSCREEN
578. 102 108 CON GZ 20 0.558
579. 158 CON GZ 20 0.657
580. 158 CON GZ 20 1.542
581. JOINT LOAD
582. 107 FZ 15
583. MEMBER LOAD
584. **VIBRATING GRIZZLYSCREEN
585. 102 108 CON GZ 23 0.558
586. 158 CON GZ 23 0.657
587. 158 CON GZ 23 1.542
588. **STAIR
589. JOINT LOAD
590. 630 TO 633 FZ 1
591. MEMBER LOAD
592. 617 TO 620 CON GZ 1 0.4
593. FLOOR LOAD
594. **SCREEN FLOOR
595. _SCREEN_FLOOR_1 FLOAD 2.5 GZ
596. MEMBER LOAD
597. *OPERATION PLATFORM
598. 611 UNI GZ 5.45 1.25 2.55
599. 612 UNI GZ 6.36 1.25 2.55
600. *
601. 613 UNI GZ 5.45 0 1.3
602. 614 UNI GZ 6.36 0 1.3
603. LOAD R10 LOADTYPE DEAD  TITLE STRUCTURE SELF WEIGHT (D1)
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604. SELFWEIGHT Y -1 LIST ALL
605. *STAIR DEAD LOAD
606. JOINT LOAD
607. 630 TO 633 FY -0.55
608. MEMBER LOAD
609. 617 TO 620 CON GY -0.55 0.4
610. *FLOOR LOAD
611. FLOOR LOAD
612. _SCREEN_FLOOR_1 FLOAD -3.6 GY
613. MEMBER LOAD
614. *OPERATION PLATFORM
615. 611 UNI GY -9.85 1.25 2.55
616. 611 UMOM GZ 0.97 1.25 2.55
617. 612 UNI GY -11.45 1.25 2.55
618. 612 UMOM GZ -2.51 1.25 2.55
619. *
620. 613 UNI GY -9.85 0 1.3
621. 613 UMOM GZ 0.97 0 1.3
622. 614 UNI GY -11.45 0 1.3
623. 614 UMOM GZ -2.51 0 1.3
624. *SOIL PRESSURE
625. ELEMENT LOAD
626. 179 TO 610 PR GY -50
627. LOAD R11 LOADTYPE DEAD  TITLE EQUIPMENT DEAD WEIGHT (D2)
628. MEMBER LOAD
629. **VIBRATING GRIZZLYSCREEN
630. 102 108 CON GY -20 0.558
631. 158 CON GY -20 0.657
632. 158 CON GY -20 1.542
633. JOINT LOAD
634. 107 FY -15
635. LOAD R12 LOADTYPE LIVE  TITLE LIVE LOAD(L1)
636. **STAIR
637. JOINT LOAD
638. 630 TO 633 FY -2
639. MEMBER LOAD
640. 617 TO 620 CON GY -2 0.4
641. FLOOR LOAD
642. **SCREEN FLOOR
643. _SCREEN_FLOOR_1 FLOAD -5 GY
644. MEMBER LOAD
645. *OPERATION PLATFORM
646. 611 UNI GY -10.9 1.25 2.55
647. 611 UMOM GZ 1.08 1.25 2.55
648. 612 UNI GY -12.72 1.25 2.55
649. 612 UMOM GZ -2.78 1.25 2.55
650. *
651. 613 UNI GY -10.9 0 1.3
652. 613 UMOM GZ 1.08 0 1.3
653. 614 UNI GY -12.72 0 1.3
654. 614 UMOM GZ -2.78 0 1.3
655. LOAD R13 LOADTYPE LIVE  TITLE EQUIPMENT LIVE LOAD(L2)
656. MEMBER LOAD
657. **VIBRATING GRIZZLYSCREEN
658. 102 108 CON GY -23 0.558
659. 158 CON GY -23 0.657
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660. 158 CON GY -23 1.542
661. LOAD R14 LOADTYPE DEAD  TITLE EQUIPMENT REVERSABLE LOAD(L3)
662. MEMBER LOAD
663. **VIBRATING GRIZZLYSCREEN
664. 102 108 CON GY 33 0.558
665. 158 CON GY 33 0.657
666. 158 CON GY 33 1.542
667. 102 108 CON GX 29 0.558
668. 158 CON GX 29 0.657
669. 158 CON GX 29 1.542
670. JOINT LOAD
671. 107 FX 23 FY 23
672. LOAD R15 LOADTYPE DEAD  TITLE EQUIPMENT START _STOP_CONDITION(L4)
673. MEMBER LOAD
674. **VIBRATING GRIZZLYSCREEN
675. 102 108 CON GY 45 0.558
676. 158 CON GY 45 0.657
677. 158 CON GY 45 1.542
678. 102 108 CON GX 35 0.558
679. 158 CON GX 35 0.657
680. 158 CON GX 35 1.542
681. END DEFINE REFERENCE LOADS
682. DEFINE IBC 2003
683. SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 CT 0.044
684. REFERENCE LOAD Y
685. R10 1.0
686. REFERENCE LOAD Y
687. R11 1.0
688. REFERENCE LOAD Y
689. R12 0.5
690. REFERENCE LOAD Y
691. R13 1.0
692. LOAD 1 LOADTYPE SEISMIC  TITLE SEISMIC LOAD(E +X)
693. IBC LOAD X 1
694. LOAD 2 LOADTYPE SEISMIC  TITLE SEISMIC LOAD(E +Z)
695. IBC LOAD Z 1
696. LOAD 3 LOADTYPE WIND  TITLE WIND LOAD (W +X)
697. REFERENCE LOAD
698. R3 1.0
699. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD (W +Z)
700. REFERENCE LOAD
701. R4 1.0
702. LOAD 10 LOADTYPE DEAD  TITLE STRUCTURE SELF WEIGHT (D1)
703. REFERENCE LOAD
704. R10 1.0
705. LOAD 11 LOADTYPE DEAD  TITLE EQUIPMENT DEAD WEIGHT (D2)
706. REFERENCE LOAD
707. R11 1.0
708. LOAD 12 LOADTYPE LIVE  TITLE LIVE LOAD(L1)
709. REFERENCE LOAD
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710. R12 1.0
711. LOAD 13 LOADTYPE LIVE  TITLE EQUIPMENT LIVE LOAD(L2)
712. REFERENCE LOAD
713. R13 1.0
714. LOAD 14 LOADTYPE DEAD  TITLE EQUIPMENT REVERSABLE LOAD(L3)
715. REFERENCE LOAD
716. R14 1.0
717. LOAD 15 LOADTYPE DEAD  TITLE EQUIPMENT START _STOP_CONDITION(L4)
718. REFERENCE LOAD
719. R15 1.0
720. LOAD 30 MODAL_ANALYSIS
721. REFERENCE LOAD
722. R7 1.0
723. REFERENCE LOAD
724. R8 1.0
725. REFERENCE LOAD
726. R9 1.0
727. MODAL CALCULATION REQUESTED
728. **D
729. LOAD 101 1.0(D1+D2)
730. REFERENCE LOAD
731. R10 1.0 R11 1.0
732. *D + L
733. LOAD 102 1.0(D1+D2)+1.0L1
734. REFERENCE LOAD
735. R10 1.0 R11 1.0 R12 1.0
736. LOAD 103 1.0(D1+D2)+1.0(L1+L2)
737. REFERENCE LOAD
738. R10 1.0 R11 1.0 R12 1.0 R13 1.0
739. LOAD 104 1.0(D1+D2)+1.0(L1+L2+L3)
740. REFERENCE LOAD
741. R10 1.0 R11 1.0 R12 1.0 R13 1.0 R14 1.0
742. LOAD 105 1.0(D1+D2)+1.0(L1+L2+L4)
743. REFERENCE LOAD
744. R10 1.0 R11 1.0 R12 1.0 R13 1.0 R15 1.0
745. LOAD 106 1.0(D1+D2)+1.0(L1+L2-L3)
746. REFERENCE LOAD
747. R10 1.0 R11 1.0 R12 1.0 R13 1.0 R14 -1.0
748. LOAD 107 1.0(D1+D2)+1.0(L1+L2-L4)
749. REFERENCE LOAD
750. R10 1.0 R11 1.0 R12 1.0 R13 1.0 R15 -1.0
751. **1.0(D + W)
752. LOAD 108 1.0(D1+D2)+1.0W(+X)
753. REFERENCE LOAD
754. R10 1.0 R11 1.0 R3 1.0
755. LOAD 109 1.0(D1+D2)+1.0W(-X)
756. REFERENCE LOAD
757. R10 1.0 R11 1.0 R3 -1.0
758. LOAD 110 1.0(D1+D2)+1.0W(+Z)
759. REFERENCE LOAD
760. R10 1.0 R11 1.0 R4 1.0
761. LOAD 111 1.0(D1+D2)+1.0W(-Z)
762. REFERENCE LOAD
763. R10 1.0 R11 1.0 R4 -1.0
764. ***1.0D + 0.7E
765. LOAD 112 1.0(D1+D2)+0.7E(+X)
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766. REFERENCE LOAD
767. R10 1.0 R11 1.0
768. REPEAT LOAD
769. 1 0.7
770. LOAD 113 1.0(D1+D2)+0.7E(-X)
771. REFERENCE LOAD
772. R10 1.0 R11 1.0
773. REPEAT LOAD
774. 1 -0.7
775. LOAD 114 1.0(D1+D2)+0.7E(+Z)
776. REFERENCE LOAD
777. R10 1.0 R11 1.0
778. REPEAT LOAD
779. 2 0.7
780. LOAD 115 1.0(D1+D2)+0.7E(-Z)
781. REFERENCE LOAD
782. R10 1.0 R11 1.0
783. REPEAT LOAD
784. 2 -0.7
785. *1.0D + 0.75W + 0.75L
786. LOAD 116 1.0(D1+D2)+0.75W(+X) + 0.75L1
787. REFERENCE LOAD
788. R10 1.0 R11 1.0 R3 0.75 R12 0.75
789. LOAD 117 1.0(D1+D2)+0.75W(-X) + 0.75L1
790. REFERENCE LOAD
791. R10 1.0 R11 1.0 R3 -0.75 R12 0.75
792. LOAD 118 1.0(D1+D2)+0.75W(+Z) + 0.75L1
793. REFERENCE LOAD
794. R10 1.0 R11 1.0 R4 0.75 R12 0.75
795. LOAD 119 1.0(D1+D2)+0.75W(-Z) + 0.75L1
796. REFERENCE LOAD
797. R10 1.0 R11 1.0 R4 -0.75 R12 0.75
798. *
799. LOAD 120 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2)
800. REFERENCE LOAD
801. R10 1.0 R11 1.0 R3 0.75 R12 0.75 R13 0.75
802. LOAD 121 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2)
803. REFERENCE LOAD
804. R10 1.0 R11 1.0 R3 -0.75 R12 0.75 R13 0.75
805. LOAD 122 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2)
806. REFERENCE LOAD
807. R10 1.0 R11 1.0 R4 0.75 R12 0.75 R13 0.75
808. LOAD 123 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2)
809. REFERENCE LOAD
810. R10 1.0 R11 1.0 R4 -0.75 R12 0.75 R13 0.75
811. *
812. LOAD 124 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+L3)
813. REFERENCE LOAD
814. R10 1.0 R11 1.0 R3 0.75 R12 0.75 R13 0.75 R14 0.75
815. LOAD 125 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+L3)
816. REFERENCE LOAD
817. R10 1.0 R11 1.0 R3 -0.75 R12 0.75 R13 0.75 R14 0.75
818. LOAD 126 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+L3)
819. REFERENCE LOAD
820. R10 1.0 R11 1.0 R4 0.75 R12 0.75 R13 0.75 R14 0.75
821. LOAD 127 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+L3)
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822. REFERENCE LOAD
823. R10 1.0 R11 1.0 R4 -0.75 R12 0.75 R13 0.75 R14 0.75
824. *
825. LOAD 128 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-L3)
826. REFERENCE LOAD
827. R10 1.0 R11 1.0 R3 0.75 R12 0.75 R13 0.75 R14 -0.75
828. LOAD 129 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-L3)
829. REFERENCE LOAD
830. R10 1.0 R11 1.0 R3 -0.75 R12 0.75 R13 0.75 R14 -0.75
831. LOAD 130 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-L3)
832. REFERENCE LOAD
833. R10 1.0 R11 1.0 R4 0.75 R12 0.75 R13 0.75 R14 -0.75
834. LOAD 131 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-L3)
835. REFERENCE LOAD
836. R10 1.0 R11 1.0 R4 -0.75 R12 0.75 R13 0.75 R14 -0.75
837. *
838. LOAD 132 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+L4)
839. REFERENCE LOAD
840. R10 1.0 R11 1.0 R3 0.75 R12 0.75 R13 0.75 R15 0.75
841. LOAD 133 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+L4)
842. REFERENCE LOAD
843. R10 1.0 R11 1.0 R3 -0.75 R12 0.75 R13 0.75 R15 0.75
844. LOAD 134 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+L4)
845. REFERENCE LOAD
846. R10 1.0 R11 1.0 R4 0.75 R12 0.75 R13 0.75 R15 0.75
847. LOAD 135 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+L4)
848. REFERENCE LOAD
849. R10 1.0 R11 1.0 R4 -0.75 R12 0.75 R13 0.75 R15 0.75
850. *
851. LOAD 136 1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-L4)
852. REFERENCE LOAD
853. R10 1.0 R11 1.0 R3 0.75 R12 0.75 R13 0.75 R15 -0.75
854. LOAD 137 1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-L4)
855. REFERENCE LOAD
856. R10 1.0 R11 1.0 R3 -0.75 R12 0.75 R13 0.75 R15 -0.75
857. LOAD 138 1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-L4)
858. REFERENCE LOAD
859. R10 1.0 R11 1.0 R4 0.75 R12 0.75 R13 0.75 R15 -0.75
860. LOAD 139 1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-L4)
861. REFERENCE LOAD
862. R10 1.0 R11 1.0 R4 -0.75 R12 0.75 R13 0.75 R15 -0.75
863. *
864. **D + 0.75*0.7E + 0.75L
865. LOAD 140 1.0(D1+D2)+0.75*0.7 E(+X) + 0.75(L1+L2)
866. REFERENCE LOAD
867. R10 1.0 R11 1.0 R12 0.75 R13 0.75
868. REPEAT LOAD
869. 1 0.525
870. LOAD 141 1.0(D1+D2)+0.75*0.7 E(-X) + 0.75(L1+L2)
871. REFERENCE LOAD
872. R10 1.0 R11 1.0 R12 0.75 R13 0.75
873. REPEAT LOAD
874. 1 -0.525
875. LOAD 142 1.0(D1+D2)+0.75*0.7 E(+Z) + 0.75(L1+L2)
876. REFERENCE LOAD
877. R10 1.0 R11 1.0 R12 0.75 R13 0.75
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878. REPEAT LOAD
879. 2 0.525
880. LOAD 143 1.0(D1+D2)+0.75*0.7 E(-Z) + 0.75(L1+L2)
881. REFERENCE LOAD
882. R10 1.0 R11 1.0 R12 0.75 R13 0.75
883. REPEAT LOAD
884. 2 -0.525
885. *
886. **0.6D + 1.0W
887. LOAD 144 0.6(D1+D2) + 1.0W(+X)
888. REFERENCE LOAD
889. R10 0.6 R11 0.6 R3 1.0
890. LOAD 145 0.6(D1+D2) + 1.0W(-X)
891. REFERENCE LOAD
892. R10 0.6 R11 0.6 R3 -1.0
893. LOAD 146 0.6(D1+D2) + 1.0W(+Z)
894. REFERENCE LOAD
895. R10 0.6 R11 0.6 R4 1.0
896. LOAD 147 0.6(D1+D2) + 1.0W(-Z)
897. REFERENCE LOAD
898. R10 0.6 R11 0.6 R4 -1.0
899. **0.6D + 0.7E
900. LOAD 148 0.6(D1+D2) + 0.7E(+X)
901. REFERENCE LOAD
902. R10 0.6 R11 0.6
903. REPEAT LOAD
904. 1 0.7
905. LOAD 149 0.6(D1+D2) + 0.7E(-X)
906. REFERENCE LOAD
907. R10 0.6 R11 0.6
908. REPEAT LOAD
909. 1 -0.7
910. LOAD 150 0.6(D1+D2) + 0.7E(+Z)
911. REFERENCE LOAD
912. R10 0.6 R11 0.6
913. REPEAT LOAD
914. 2 0.7
915. LOAD 151 0.6(D1+D2) + 0.7E(-Z)
916. REFERENCE LOAD
917. R10 0.6 R11 0.6
918. REPEAT LOAD
919. 2 -0.7
920. **FACTORED LOAD COMBINATIONS
921. **1.4D
922. LOAD 201 1.4(D1+D2)
923. REFERENCE LOAD
924. R10 1.4 R11 1.4
925. **1.4D + 1.7L
926. LOAD 202 1.4(D1+D2)+1.7L1
927. REFERENCE LOAD
928. R10 1.4 R11 1.4 R12 1.7
929. LOAD 203 1.4(D1+D2)+1.7(L1+L2)
930. REFERENCE LOAD
931. R10 1.4 R11 1.4 R12 1.7 R13 1.7
932. LOAD 204 1.4(D1+D2)+1.7(L1+L2+L3)
933. REFERENCE LOAD
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934. R10 1.4 R11 1.4 R12 1.7 R13 1.7 R14 1.7
935. LOAD 205 1.4(D1+D2)+1.7(L1+L2+L4)
936. REFERENCE LOAD
937. R10 1.4 R11 1.4 R12 1.7 R13 1.7 R15 1.7
938. LOAD 206 1.4(D1+D2)+1.7(L1+L2-L3)
939. REFERENCE LOAD
940. R10 1.4 R11 1.4 R12 1.7 R13 1.7 R14 -1.7
941. LOAD 207 1.4(D1+D2)+1.7(L1+L2-L4)
942. REFERENCE LOAD
943. R10 1.4 R11 1.4 R12 1.7 R13 1.7 R15 -1.7
944. **1.2D + 0.8W
945. LOAD 208 1.2(D1+D2)+0.8W(+X)
946. REFERENCE LOAD
947. R10 1.2 R11 1.2 R3 0.8
948. LOAD 209 1.2(D1+D2)+0.8W(-X)
949. REFERENCE LOAD
950. R10 1.2 R11 1.2 R3 -0.8
951. LOAD 210 1.2(D1+D2)+0.8W(+Z)
952. REFERENCE LOAD
953. R10 1.2 R11 1.2 R4 0.8
954. LOAD 211 1.2(D1+D2)+0.8W(-Z)
955. REFERENCE LOAD
956. R10 1.2 R11 1.2 R4 -0.8
957. **1.2D + 1.6W + 1.0L
958. LOAD 212 1.2(D1+D2)+1.6W(+X) + 1.01L1
959. REFERENCE LOAD
960. R10 1.2 R11 1.2 R3 1.6 R12 1.0
961. LOAD 213 1.2(D1+D2)+1.6W(-X) + 1.0L1
962. REFERENCE LOAD
963. R10 1.2 R11 1.2 R3 -1.6 R12 1.0
964. LOAD 214 1.2(D1+D2)+1.6W(+Z) + 1.0L1
965. REFERENCE LOAD
966. R10 1.2 R11 1.2 R4 1.6 R12 1.0
967. LOAD 215 1.2(D1+D2)+1.6W(-Z) + 1.0L1
968. REFERENCE LOAD
969. R10 1.2 R11 1.2 R4 -1.6 R12 1.0
970. *
971. LOAD 216 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)
972. REFERENCE LOAD
973. R10 1.2 R11 1.2 R3 1.6 R12 1.0 R13 1.0
974. LOAD 217 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2)
975. REFERENCE LOAD
976. R10 1.2 R11 1.2 R3 -1.6 R12 1.0 R13 1.0
977. LOAD 218 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2)
978. REFERENCE LOAD
979. R10 1.2 R11 1.2 R4 1.6 R12 1.0 R13 1.0
980. LOAD 219 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2)
981. REFERENCE LOAD
982. R10 1.2 R11 1.2 R4 -1.6 R12 1.0 R13 1.0
983. *
984. LOAD 220 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+L3)
985. REFERENCE LOAD
986. R10 1.2 R11 1.2 R3 1.6 R12 1.0 R13 1.0 R14 1.0
987. LOAD 221 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+L3)
988. REFERENCE LOAD
989. R10 1.2 R11 1.2 R3 -1.6 R12 1.0 R13 1.0 R14 1.0
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 990. LOAD 222 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+L3)
 991. REFERENCE LOAD
 992. R10 1.2 R11 1.2 R4 1.6 R12 1.0 R13 1.0 R14 1.0
 993. LOAD 223 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+L3)
 994. REFERENCE LOAD
 995. R10 1.2 R11 1.2 R4 -1.6 R12 1.0 R13 1.0 R14 1.0
 996. *
 997. LOAD 224 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-L3)
 998. REFERENCE LOAD
 999. R10 1.2 R11 1.2 R3 1.6 R12 1.0 R13 1.0 R14 -1.0
1000. LOAD 225 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-L3)
1001. REFERENCE LOAD
1002. R10 1.2 R11 1.2 R3 -1.6 R12 1.0 R13 1.0 R14 -1.0
1003. LOAD 226 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-L3)
1004. REFERENCE LOAD
1005. R10 1.2 R11 1.2 R4 1.6 R12 1.0 R13 1.0 R14 -1.0
1006. LOAD 227 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-L3)
1007. REFERENCE LOAD
1008. R10 1.2 R11 1.2 R4 -1.6 R12 1.0 R13 1.0 R14 -1.0
1009. *
1010. LOAD 228 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+L4)
1011. REFERENCE LOAD
1012. R10 1.2 R11 1.2 R3 1.6 R12 1.0 R13 1.0 R15 1.0
1013. LOAD 229 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+L4)
1014. REFERENCE LOAD
1015. R10 1.2 R11 1.2 R3 -1.6 R12 1.0 R13 1.0 R15 1.0
1016. LOAD 230 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+L4)
1017. REFERENCE LOAD
1018. R10 1.2 R11 1.2 R4 1.6 R12 1.0 R13 1.0 R15 1.0
1019. LOAD 231 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+L4)
1020. REFERENCE LOAD
1021. R10 1.2 R11 1.2 R4 -1.6 R12 1.0 R13 1.0 R15 1.0
1022. *
1023. LOAD 232 1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-L4)
1024. REFERENCE LOAD
1025. R10 1.2 R11 1.2 R3 1.6 R12 1.0 R13 1.0 R15 -1.0
1026. LOAD 233 1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-L4)
1027. REFERENCE LOAD
1028. R10 1.2 R11 1.2 R3 -1.6 R12 1.0 R13 1.0 R15 -1.0
1029. LOAD 234 1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-L4)
1030. REFERENCE LOAD
1031. R10 1.2 R11 1.2 R4 1.6 R12 1.0 R13 1.0 R15 -1.0
1032. LOAD 235 1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-L4)
1033. REFERENCE LOAD
1034. R10 1.2 R11 1.2 R4 -1.6 R12 1.0 R13 1.0 R15 -1.0
1035. *
1036. **1.2D + 1.0E + 1.0L
1037. LOAD 236 1.2(D1+D2)+1.0 E(+X) + 1.0(L1+L2)
1038. REFERENCE LOAD
1039. R10 1.2 R11 1.2 R12 1.0 R13 1.0
1040. REPEAT LOAD
1041. 1 1.0
1042. LOAD 237 1.2(D1+D2)+1.0 E(-X) + 1.0(L1+L2)
1043. REFERENCE LOAD
1044. R10 1.2 R11 1.2 R12 1.0 R13 1.0
1045. REPEAT LOAD
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1046. 1 -1.0
1047. LOAD 238 1.2(D1+D2)+1.0 E(+Z) + 1.0(L1+L2)
1048. REFERENCE LOAD
1049. R10 1.2 R11 1.2 R12 1.0 R13 1.0
1050. REPEAT LOAD
1051. 2 1.0
1052. LOAD 239 1.2(D1+D2)+1.0 E(-Z) + 1.0(L1+L2)
1053. REFERENCE LOAD
1054. R10 1.2 R11 1.2 R12 1.0 R13 1.0
1055. REPEAT LOAD
1056. 2 -1.0
1057. **0.9D + 1.6W
1058. LOAD 240 0.9(D1+D2) + 1.6W(+X)
1059. REFERENCE LOAD
1060. R10 0.9 R11 0.9 R3 1.6
1061. LOAD 241 0.9(D1+D2) + 1.6W(-X)
1062. REFERENCE LOAD
1063. R10 0.9 R11 0.9 R3 -1.6
1064. LOAD 242 0.9(D1+D2) + 1.6W(+Z)
1065. REFERENCE LOAD
1066. R10 0.9 R11 0.9 R4 1.6
1067. LOAD 243 0.9(D1+D2) + 1.6W(-Z)
1068. REFERENCE LOAD
1069. R10 0.9 R11 0.9 R4 -1.6
1070. **0.9D + 1.0E
1071. LOAD 244 0.9(D1+D2) + 1.0E(+X)
1072. REFERENCE LOAD
1073. R10 0.9 R11 0.9
1074. REPEAT LOAD
1075. 1 1.0
1076. LOAD 245 0.9(D1+D2) + 1.0E(-X)
1077. REFERENCE LOAD
1078. R10 0.9 R11 0.9
1079. REPEAT LOAD
1080. 1 -1.0
1081. LOAD 246 0.9(D1+D2) + 1.0E(+Z)
1082. REFERENCE LOAD
1083. R10 0.9 R11 0.9
1084. REPEAT LOAD
1085. 2 1.0
1086. LOAD 247 0.9(D1+D2) + 1.0E(-Z)
1087. REFERENCE LOAD
1088. R10 0.9 R11 0.9
1089. REPEAT LOAD
1090. 2 -1.0
1091. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	542	NUMBER OF MEMBERS	89
NUMBER OF PLATES	432	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	494

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 531/ 75/ 264 DOF
TOTAL PRIMARY LOAD CASES = 109, TOTAL DEGREES OF FREEDOM = 1770
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 468 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 26.5/ 195872.7 MB

***NOTE: MASSES DEFINED UNDER LOAD# 30 WILL FORM
THE FINAL MASS MATRIX FOR DYNAMIC ANALYSIS.

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.245 T =  0.600 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.417                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.012 X      3037.99   *
*                       =        37.87  KN                 *
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************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.018 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.245 T =  0.383 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.383                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.014 X      3037.99   *
*                       =        41.27  KN                 *
************************************************************
```

*** LOADING 1 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 37.87 FORCE-Y = 0.00 FORCE-Z = 0.00
MOM-X = 0.00 MOM-Y = 188.32 MOM-Z = -213.58

*** LOADING 2 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X = 0.00 FORCE-Y = 0.00 FORCE-Z = 41.27
MOM-X = 232.76 MOM-Y = -33.59 MOM-Z = 0.00

EIGEN METHOD : SUBSPACE

NUMBER OF MODES REQUESTED = 6
NUMBER OF EXISTING MASSES IN THE MODEL = 638
NUMBER OF MODES THAT WILL BE USED = 6

CALCULATED FREQUENCIES FOR LOAD CASE 30

MODE	FREQUENCY(CYCLES/SEC)	PERIOD(SEC)	ACCURACY
1	1.652	0.60522	7.911E-16
2	2.559	0.39083	2.199E-16
3	2.837	0.35249	1.789E-15
4	6.386	0.15660	1.140E-12
5	10.009	0.09991	8.208E-07
6	12.238	0.08171	1.505E-07

The following Frequencies are estimates that were calculated. These are for information only and will not be used. Remaining values are either above the cut off mode/freq values or are of low accuracy. To use these frequencies, rerun with a higher cutoff mode (or mode + freq) value.

CALCULATED FREQUENCIES FOR LOAD CASE 30

MODE	FREQUENCY(CYCLES/SEC)	PERIOD(SEC)	ACCURACY
7	12.800	0.07812	2.186E-07
8	14.155	0.07065	2.398E-06
9	15.442	0.06476	3.382E-06
10	15.638	0.06395	1.955E-05

MODE	MODAL WEIGHT (MODAL MASS TIMES g) IN KN X	Y	Z	GENERALIZED WEIGHT
1	1.108182E+03	2.360417E-01	1.597551E-04	8.354081E+02
2	1.028123E-02	2.048129E-04	9.274884E+02	6.701026E+02
3	4.397003E-02	1.458786E-03	1.951254E+02	3.172062E+02
4	5.567305E+00	3.778802E+00	6.697729E-04	3.047110E+02
5	5.205435E-02	4.306808E-02	7.169666E-01	6.564525E+02
6	2.788078E+01	5.638427E+02	6.124221E-02	1.399472E+02

MASS PARTICIPATION FACTORS IN PERCENT

MODE	X	Y	Z	SUMM-X	SUMM-Y	SUMM-Z
1	86.70	0.01	0.00	86.699	0.014	0.000
2	0.00	0.00	72.56	86.700	0.014	72.563
3	0.00	0.00	15.27	86.703	0.014	87.828
4	0.44	0.23	0.00	87.139	0.240	87.828
5	0.00	0.00	0.06	87.143	0.243	87.884
6	2.18	33.70	0.00	89.324	33.945	87.889

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE SEISMIC TITLE SEISMIC LOAD(E +X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.813972791E+00
Y = 0.563990157E+01
Z = 0.497292200E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 37.87
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 188.32 MZ= -213.58

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -37.87
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= -188.32 MZ= 213.58

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	3.50039E-01	105
Y =	6.36873E-02	93
Z =	-1.24529E-02	101
RX=	-1.91910E-05	628
RY=	-5.13743E-05	96
RZ=	-3.46568E-04	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE SEISMIC TITLE SEISMIC LOAD(E +Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.813972796E+00
Y = 0.563990158E+01
Z = 0.497292197E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 41.27

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 232.76 MY= -33.59 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -41.27

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -232.76 MY= 33.59 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-1.97952E-02	630
Y =	3.21606E-02	631
Z =	1.57815E-01	105
RX=	1.54536E-04	116
RY=	3.70308E-05	105
RZ=	4.58681E-05	635

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE WIND TITLE WIND LOAD (W +X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.342291385E+01
Z = 0.500000900E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 60.90
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 304.50 MZ= -208.45

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -60.90
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= -304.50 MZ= 208.45

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	4.02941E-01	101
Y =	7.31095E-02	93
Z =	3.23664E-03	93
RX=	9.18282E-06	360
RY=	-1.45883E-05	96
RZ=	-4.17385E-04	118

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE WIND TITLE WIND LOAD (W +Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990398E+00
Y = 0.346183573E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 192.38 MY= -55.57 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -192.38 MY= 55.57 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	5.75113E-03	632
Y =	-2.78014E-02	633
Z =	1.52532E-01	101
RX=	1.73232E-04	113
RY=	1.05339E-05	103
RZ=	3.73342E-05	635

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
LOADTYPE DEAD TITLE STRUCTURE SELF WEIGHT (D1)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.948084589E+00
Y = -0.462692069E+00
Z = 0.499621287E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2736.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 13670.82 MY= 0.00 MZ= -2594.18

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2736.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -13670.82 MY= -0.00 MZ= 2594.18

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	-1.26191E-01	630
Y =	-2.36145E-01	105
Z =	1.63432E-02	105
RX=	1.82860E-04	106
RY=	5.71578E-05	635
RZ=	2.74939E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
LOADTYPE DEAD TITLE EQUIPMENT DEAD WEIGHT (D2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.105598843E+01
Y = 0.689000959E+01
Z = 0.482611531E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -95.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 458.48 MY= 0.00 MZ= -100.32

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 95.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -458.48 MY= -0.00 MZ= 100.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)

	MAXIMUMS	AT NODE
X =	1.05043E-02	106
Y =	-3.88042E-02	108
Z =	1.26988E-02	105
RX=	1.54382E-04	106
RY=	-3.75231E-05	102
RZ=	-8.46453E-05	108

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
LOADTYPE LIVE TITLE LIVE LOAD(L1)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.774782105E+00
Y = 0.596821099E+01
Z = 0.500000157E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -229.51
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1147.56 MY= 0.00 MZ= -177.82

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 229.51
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1147.56 MY= -0.00 MZ= 177.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	-4.34819E-02	630
Y =	-4.26536E-02	101
Z =	-3.30923E-03	101
RX=	-6.00540E-05	631
RY=	1.24314E-05	635
RZ=	9.94300E-05	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
LOADTYPE LIVE TITLE EQUIPMENT LIVE LOAD(L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990075E+00
Y = 0.689000959E+01
Z = 0.499976002E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -92.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 459.98 MY= 0.00 MZ= -80.87

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 92.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -459.98 MY= -0.00 MZ= 80.87

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	-1.72149E-02	630
Y =	-4.55559E-02	101
Z =	-1.16818E-02	101
RX=	-1.68822E-04	101
RY=	4.30315E-05	635
RZ=	8.51183E-05	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
LOADTYPE DEAD TITLE EQUIPMENT REVERSABLE LOAD(L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.686306707E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104533049E+01
Y = 0.689000959E+01
Z = 0.483657133E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = 155.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -749.67 MY= 669.67 MZ= -795.69

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = -155.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 749.67 MY= -669.67 MZ= 795.69

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	1.57730E+00	105
Y =	3.29643E-01	105
Z =	-1.41524E-01	101
RX=	-2.32328E-04	106
RY=	-5.95964E-04	95
RZ=	-1.55096E-03	126

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 15
LOADTYPE DEAD TITLE EQUIPMENT START _STOP_CONDITION(L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.685353222E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990032E+00
Y = 0.689000959E+01
Z = 0.499976001E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = 180.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -899.96 MY= 699.97 MZ= -806.38

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = -180.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 899.96 MY= -699.97 MZ= 806.38

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 15)

	MAXIMUMS	AT NODE
X =	1.59182E+00	101
Y =	3.51715E-01	101
Z =	-1.23141E-01	101
RX=	2.93037E-04	102
RY=	-5.49926E-04	95
RZ=	-1.56121E-03	124

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 30
MODAL_ANALYSIS

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.892455314E+00
Y = 0.286756994E+01
Z = 0.498100969E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.892455316E+00
Y = 0.287048677E+01
Z = 0.498392656E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.892455416E+00
Y = 0.287048693E+01
Z = 0.498392657E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 30)
SUMMATION FORCE-X = 1672.99
SUMMATION FORCE-Y = 1672.99
SUMMATION FORCE-Z = 1672.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3535.77 MY= 6845.00 MZ= -3309.23

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 30)
SUMMATION FORCE-X = -1672.99
SUMMATION FORCE-Y = -1672.99
SUMMATION FORCE-Z = -1672.99

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3535.77 MY= -6845.00 MZ= 3309.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 30)

	MAXIMUMS	AT NODE
X =	1.05841E+01	104
Y =	2.64914E+00	101
Z =	4.55808E+00	105
RX=	4.37627E-03	116
RY=	2.10042E-03	93
RZ=	-1.05115E-02	124

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 101
1.0(D1+D2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 101)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 0.00 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 101)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -0.00 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 101)

	MAXIMUMS	AT NODE
X =	-1.32550E-01	632
Y =	-2.74212E-01	105
Z =	2.90420E-02	105
RX=	3.37241E-04	106
RY=	9.43011E-05	635
RZ=	3.35097E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 102
1.0(D1+D2)+1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938438591E+00
Y = 0.247745619E+00
Z = 0.499121743E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 102)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3060.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15276.86 MY= 0.00 MZ= -2872.32

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 102)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3060.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15276.86 MY= -0.00 MZ= 2872.32

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 102)

	MAXIMUMS	AT NODE
X =	-1.75940E-01	632
Y =	-3.16575E-01	105
Z =	3.22116E-02	105
RX=	3.93951E-04	106
RY=	1.06732E-04	635
RZ=	4.34527E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 103
1.0(D1+D2)+1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936703827E+00
Y = 0.441572805E+00
Z = 0.499146672E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 103)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3152.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15736.84 MY= 0.00 MZ= -2953.19

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 103)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3152.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15736.84 MY= -0.00 MZ= 2953.19

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 103)

	MAXIMUMS	AT NODE
X =	-1.93064E-01	632
Y =	-3.61947E-01	105
Z =	4.37713E-02	105
RX=	5.53814E-04	106
RY=	1.49764E-04	635
RZ=	5.19645E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 104
1.0(D1+D2)+1.0(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.695877832E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.931087233E+00
Y = 0.108153263E+00
Z = 0.499947566E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 104)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = -2997.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14987.17 MY= 669.67 MZ= -3748.88

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 104)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = 2997.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14987.17 MY= -669.67 MZ= 3748.88

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 104)

	MAXIMUMS	AT NODE
X =	1.43470E+00	105
Y =	-3.66992E-01	108
Z =	-1.80784E-01	101
RX=	-3.26297E-04	102
RY=	-6.90467E-04	95
RZ=	-1.39525E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 105
1.0(D1+D2)+1.0(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.694855980E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940198400E+00
Y = 0.511197212E-01
Z = 0.499096456E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 105)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = -2972.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14836.88 MY= 699.97 MZ= -3759.57

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 105)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = 2972.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14836.88 MY= -699.97 MZ= 3759.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 105)

	MAXIMUMS	AT NODE
X =	1.44832E+00	105
Y =	-3.67432E-01	108
Z =	1.66882E-01	105
RX=	2.65124E-04	106
RY=	-6.44429E-04	95
RZ=	-1.40016E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 106
1.0(D1+D2)+1.0(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.676735583E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.941794041E+00
Y = 0.743744488E+00
Z = 0.498420837E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 106)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = -3307.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16486.51 MY= -669.67 MZ= -2157.51

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 106)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = 3307.75
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16486.51 MY= 669.67 MZ= 2157.51

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 106)

	MAXIMUMS	AT NODE
X =	-1.71990E+00	105
Y =	-6.91591E-01	105
Z =	1.02264E-01	101
RX=	7.86142E-04	106
RY=	5.01461E-04	95
RZ=	1.80019E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 107
1.0(D1+D2)+1.0(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.675850462E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.933586737E+00
Y = 0.789849577E+00
Z = 0.499191463E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 107)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = -3332.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16636.79 MY= -699.97 MZ= -2146.81

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 107)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = 3332.75
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16636.79 MY= 699.97 MZ= 2146.81

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 107)

	MAXIMUMS	AT NODE
X =	-1.73538E+00	101
Y =	-7.12756E-01	105
Z =	8.38812E-02	101
RX=	8.42504E-04	106
RY=	-4.63614E-04	93
RZ=	1.87045E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 108
1.0(D1+D2)+1.0W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.352593156E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 108)
SUMMATION FORCE-X = 60.90
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 304.50 MZ= -2902.96

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 108)
SUMMATION FORCE-X = -60.90
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -304.50 MZ= 2902.96

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 108)

	MAXIMUMS	AT NODE
X =	3.04452E-01	106
Y =	-2.45979E-01	108
Z =	3.22622E-02	105
RX=	3.36178E-04	106
RY=	1.02190E-04	635
RZ=	-3.25107E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 109
1.0(D1+D2)+1.0W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.331989614E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 109)
SUMMATION FORCE-X = -60.90
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= -304.50 MZ= -2486.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 109)
SUMMATION FORCE-X = 60.90
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= 304.50 MZ= 2486.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 109)

	MAXIMUMS	AT NODE
X =	-5.27969E-01	632
Y =	-3.46183E-01	105
Z =	2.58218E-02	105
RX=	3.38304E-04	106
RY=	8.64121E-05	635
RZ=	6.47678E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 110
1.0(D1+D2)+1.0W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990398E+00
Y = 0.346183392E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 110)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14321.69 MY= -55.57 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 110)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = -55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14321.69 MY= 55.57 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 110)

	MAXIMUMS	AT NODE
X =	-1.32519E-01	630
Y =	-2.90940E-01	105
Z =	1.81517E-01	105
RX=	3.78666E-04	106
RY=	1.03475E-04	635
RZ=	3.41027E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 111
1.0(D1+D2)+1.0W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990398E+00
Y = 0.346183754E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 111)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = -55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 13936.92 MY= 55.57 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 111)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -13936.92 MY= -55.57 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 111)

	MAXIMUMS	AT NODE
X	-1.38302E-01	632
Y	-2.89333E-01	101
Z	-1.76801E-01	101
RX	-3.71895E-04	102
RY	-8.74601E-05	102
RZ	3.50719E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 112
1.0(D1+D2)+0.7E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 112)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 0.00 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 112)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -0.00 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 112)

	MAXIMUMS	AT NODE
X =	-1.32550E-01	632
Y =	-2.74212E-01	105
Z =	2.90420E-02	105
RX=	3.37241E-04	106
RY=	9.43011E-05	635
RZ=	3.35097E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 113
1.0(D1+D2)+0.7E(-X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 113)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 0.00 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 113)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -0.00 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 113)

	MAXIMUMS	AT NODE
X =	-1.32550E-01	632
Y =	-2.74212E-01	105
Z =	2.90420E-02	105
RX=	3.37241E-04	106
RY=	9.43011E-05	635
RZ=	3.35097E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 114
1.0(D1+D2)+0.7E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 114)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 0.00 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 114)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -0.00 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 114)

	MAXIMUMS	AT NODE
X =	-1.32550E-01	632
Y =	-2.74212E-01	105
Z =	2.90420E-02	105
RX=	3.37241E-04	106
RY=	9.43011E-05	635
RZ=	3.35097E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 115
1.0(D1+D2)+0.7E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705220E+00
Y = -0.215977729E+00
Z = 0.499050537E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 115)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14129.30 MY= 0.00 MZ= -2694.50

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 115)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2831.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14129.30 MY= -0.00 MZ= 2694.50

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 115)

	MAXIMUMS	AT NODE
X =	-1.32550E-01	632
Y =	-2.74212E-01	105
Z =	2.90420E-02	105
RX=	3.37241E-04	106
RY=	9.43011E-05	635
RZ=	3.35097E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 116
1.0(D1+D2)+0.75W(+X) + 0.75L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.363284982E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.941565156E+00
Y = 0.138459245E+00
Z = 0.499104962E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 116)
SUMMATION FORCE-X = 45.67
SUMMATION FORCE-Y = -3003.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14989.97 MY= 228.37 MZ= -2984.21

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 116)
SUMMATION FORCE-X = -45.67
SUMMATION FORCE-Y = 3003.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14989.97 MY= -228.37 MZ= 2984.21

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 116)

	MAXIMUMS	AT NODE
X =	1.73587E-01	106
Y =	-2.52006E-01	105
Z =	3.38344E-02	105
RX=	3.78976E-04	106
RY=	1.09541E-04	635
RZ=	-1.92859E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 117
1.0(D1+D2)+0.75W(-X) + 0.75L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.321297810E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.941565156E+00
Y = 0.138459245E+00
Z = 0.499104962E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 117)
SUMMATION FORCE-X = -45.67
SUMMATION FORCE-Y = -3003.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14989.97 MY= -228.37 MZ= -2671.53

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 117)
SUMMATION FORCE-X = 45.67
SUMMATION FORCE-Y = 3003.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14989.97 MY= 228.37 MZ= 2671.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 117)

	MAXIMUMS	AT NODE
X =	-4.61657E-01	632
Y =	-3.59962E-01	105
Z =	2.90041E-02	105
RX=	3.80571E-04	106
RY=	9.77079E-05	635
RZ=	6.44105E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 118
1.0(D1+D2)+0.75W(+Z) + 0.75L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.941565156E+00
Y = 0.138459245E+00
Z = 0.499104962E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 118)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3003.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15134.26 MY= -41.68 MZ= -2827.87

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 118)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3003.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15134.26 MY= 41.68 MZ= 2827.87

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 118)

	MAXIMUMS	AT NODE
X =	-1.63713E-01	630
Y =	-3.18530E-01	105
Z =	1.45775E-01	105
RX=	4.10842E-04	106
RY=	1.10505E-04	635
RZ=	4.11455E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 119
1.0(D1+D2)+0.75W(-Z) + 0.75L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.941565156E+00
Y = 0.138459245E+00
Z = 0.499104962E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 119)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3003.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14845.68 MY= 41.68 MZ= -2827.87

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 119)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3003.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14845.68 MY= -41.68 MZ= 2827.87

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 119)

	MAXIMUMS	AT NODE
X =	-1.69406E-01	632
Y =	-3.17030E-01	101
Z =	-1.41150E-01	101
RX=	-4.04539E-04	102
RY=	9.67442E-05	635
RZ=	4.21386E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 120
1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.367571006E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 120)
SUMMATION FORCE-X = 45.67
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= 228.37 MZ= -3044.86

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 120)
SUMMATION FORCE-X = -45.67
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= -228.37 MZ= 3044.86

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 120)

	MAXIMUMS	AT NODE
X =	1.70634E-01	106
Y =	-2.86036E-01	105
Z =	4.25041E-02	105
RX=	4.98874E-04	106
RY=	1.41815E-04	635
RZ=	2.39073E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 121
1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.317011785E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 121)
SUMMATION FORCE-X = -45.67
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= -228.37 MZ= -2732.18

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 121)
SUMMATION FORCE-X = 45.67
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= 228.37 MZ= 2732.18

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 121)

	MAXIMUMS	AT NODE
X =	-4.74499E-01	632
Y =	-3.93992E-01	105
Z =	3.76739E-02	105
RX=	5.00468E-04	106
RY=	1.29981E-04	635
RZ=	7.07944E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 122
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 122)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15479.24 MY= -41.68 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 122)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15479.24 MY= 41.68 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 122)

	MAXIMUMS	AT NODE
X =	-1.76624E-01	630
Y =	-3.52560E-01	105
Z =	1.54445E-01	105
RX=	5.30740E-04	106
RY=	1.42779E-04	635
RZ=	4.75008E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 123
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 123)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15190.67 MY= 41.68 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 123)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15190.67 MY= -41.68 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 123)

	MAXIMUMS	AT NODE
X	-1.82249E-01	632
Y	-3.51197E-01	101
Z	-1.49911E-01	101
RX	-5.25244E-04	102
RY	1.29018E-04	635
RZ	4.85225E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 124
1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104483160E+01
Y = 0.589203929E+01
Z = 0.487330088E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936023972E+00
Y = 0.305438771E-01
Z = 0.499732782E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 124)
SUMMATION FORCE-X = 149.92
SUMMATION FORCE-Y = -2956.12
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14772.70 MY= 730.63 MZ= -3641.62

```
***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   124 )
    SUMMATION FORCE-X =        -149.92
    SUMMATION FORCE-Y =        2956.12
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=      -14772.70  MY=        -730.63  MZ=        3641.62


MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING    124)
          MAXIMUMS    AT NODE
   X =  1.35359E+00     105
   Y = -3.60454E-01     108
   Z = -1.44058E-01     101
   RX= -3.25789E-04     102
   RY= -5.43109E-04      95
   RZ= -1.31162E-03     125


       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.   125
       1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+L3)


        CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.936023972E+00
                    Y =  0.305438771E-01
                    Z =  0.499732782E+01

***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING   125 )
    SUMMATION FORCE-X =          58.58
    SUMMATION FORCE-Y =       -2956.12
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=       14772.70  MY=         273.88  MZ=       -3328.94


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   125 )
    SUMMATION FORCE-X =         -58.58
    SUMMATION FORCE-Y =        2956.12
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=      -14772.70  MY=        -273.88  MZ=        3328.94
```

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 125)

	MAXIMUMS	AT NODE
X =	7.49216E-01	105
Y =	-2.89309E-01	108
Z =	-1.39252E-01	101
RX=	-3.27862E-04	102
RY=	-5.23981E-04	95
RZ=	-7.30866E-04	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 126
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447763E+01
Y = 0.697382321E+01
Z = 0.481778697E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936023972E+00
Y = 0.305438771E-01
Z = 0.499732782E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 126)
SUMMATION FORCE-X = 104.25
SUMMATION FORCE-Y = -2956.12
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14916.99 MY= 460.58 MZ= -3485.28

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 126)
SUMMATION FORCE-X = -104.25
SUMMATION FORCE-Y = 2956.12
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14916.99 MY= -460.58 MZ= 3485.28

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 126)

	MAXIMUMS	AT NODE
X =	1.05036E+00	105
Y =	-3.27464E-01	108
Z =	2.36802E-01	105
RX=	3.56493E-04	106
RY=	-5.27208E-04	95
RZ=	-1.01919E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 127
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447763E+01
Y = 0.697382321E+01
Z = 0.481778697E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936023972E+00
Y = 0.305438771E-01
Z = 0.499732782E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 127)
SUMMATION FORCE-X = 104.25
SUMMATION FORCE-Y = -2956.12
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14628.41 MY= 543.93 MZ= -3485.28

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 127)
SUMMATION FORCE-X = -104.25
SUMMATION FORCE-Y = 2956.12
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14628.41 MY= -543.93 MZ= 3485.28

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 127)

	MAXIMUMS	AT NODE
X =	1.05244E+00	105
Y =	-3.23938E-01	104
Z =	-2.56054E-01	101
RX=	-3.59440E-04	102
RY=	-5.39882E-04	95
RZ=	-1.02329E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 128
1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-L3)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943994113E+00
Y = 0.530705634E+00
Z = 0.498560617E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 128)
SUMMATION FORCE-X = -58.58
SUMMATION FORCE-Y = -3188.62
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15897.20 MY= -273.88 MZ= -2448.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 128)
SUMMATION FORCE-X = 58.58
SUMMATION FORCE-Y = 3188.62
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15897.20 MY= 273.88 MZ= 2448.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 128)

	MAXIMUMS	AT NODE
X =	-1.01235E+00	105
Y =	-5.33268E-01	105
Z =	6.82271E-02	101
RX=	6.73120E-04	106
RY=	3.50836E-04	95
RZ=	1.19747E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 129
1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104483160E+01
Y = 0.573801071E+01
Z = 0.487330088E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943994113E+00
Y = 0.530705634E+00
Z = 0.498560617E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 129)
SUMMATION FORCE-X = -149.92
SUMMATION FORCE-Y = -3188.62
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15897.20 MY= -730.63 MZ= -2135.42

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 129)
SUMMATION FORCE-X = 149.92
SUMMATION FORCE-Y = 3188.62
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15897.20 MY= 730.63 MZ= 2135.42

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 129)

	MAXIMUMS	AT NODE
X =	-1.61673E+00	105
Y =	-6.41224E-01	105
Z =	7.30337E-02	101
RX=	6.74715E-04	106
RY=	3.71106E-04	96
RZ=	1.66613E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 130
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447763E+01
Y = 0.675231095E+01
Z = 0.481778697E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943994113E+00
Y = 0.530705634E+00
Z = 0.498560617E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 130)
SUMMATION FORCE-X = -104.25
SUMMATION FORCE-Y = -3188.62
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16041.49 MY= -543.93 MZ= -2291.76

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 130)
SUMMATION FORCE-X = 104.25
SUMMATION FORCE-Y = 3188.62
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16041.49 MY= 543.93 MZ= 2291.76

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 130)

	MAXIMUMS	AT NODE
X	-1.31558E+00	105
Y	-5.99792E-01	105
Z	1.85029E-01	101
RX	7.04986E-04	106
RY	3.66737E-04	95
RZ	1.44259E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 131
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447763E+01
Y = 0.675231095E+01
Z = 0.481778697E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943994113E+00
Y = 0.530705634E+00
Z = 0.498560617E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 131)
SUMMATION FORCE-X = -104.25
SUMMATION FORCE-Y = -3188.62
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15752.92 MY= -460.58 MZ= -2291.76

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 131)
SUMMATION FORCE-X = 104.25
SUMMATION FORCE-Y = 3188.62
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15752.92 MY= 460.58 MZ= 2291.76

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 131)

	MAXIMUMS	AT NODE
X	-1.31350E+00	105
Y	-5.88774E-01	101
Z	-1.56627E-01	94
RX	-6.91048E-04	102
RY	3.54556E-04	96
RZ	1.42194E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 132
1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.915669314E+00
Y = 0.589022818E+01
Z = 0.499983548E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942971163E+00
Y = -0.132418755E-01
Z = 0.499085390E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 132)
SUMMATION FORCE-X = 150.67
SUMMATION FORCE-Y = -2937.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14659.99 MY= 753.35 MZ= -3649.65

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 132)
SUMMATION FORCE-X = -150.67
SUMMATION FORCE-Y = 2937.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14659.99 MY= -753.35 MZ= 3649.65

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 132)

	MAXIMUMS	AT NODE
X =	1.36381E+00	105
Y =	-3.60784E-01	108
Z =	1.34837E-01	105
RX=	2.82356E-04	106
RY=	-5.08581E-04	95
RZ=	-1.31531E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 133
1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2+L4)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942971163E+00
Y = -0.132418755E-01
Z = 0.499085390E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 133)
SUMMATION FORCE-X = 59.33
SUMMATION FORCE-Y = -2937.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14659.99 MY= 296.60 MZ= -3336.97

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 133)
SUMMATION FORCE-X = -59.33
SUMMATION FORCE-Y = 2937.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14659.99 MY= -296.60 MZ= 3336.97

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 133)

	MAXIMUMS	AT NODE
X =	7.59434E-01	105
Y =	-2.89639E-01	108
Z =	1.30007E-01	105
RX=	2.83951E-04	106
RY=	-4.89453E-04	95
RZ=	-7.34554E-04	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 134
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990070E+00
Y = 0.696349724E+01
Z = 0.499976000E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942971163E+00
Y = -0.132418755E-01
Z = 0.499085390E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 134)
SUMMATION FORCE-X = 105.00
SUMMATION FORCE-Y = -2937.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14804.27 MY= 483.30 MZ= -3493.31

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 134)
SUMMATION FORCE-X = -105.00
SUMMATION FORCE-Y = 2937.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -14804.27 MY= -483.30 MZ= 3493.31

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 134)

	MAXIMUMS	AT NODE
X =	1.06237E+00	101
Y =	-3.27794E-01	108
Z =	2.46778E-01	105
RX=	3.14222E-04	106
RY=	4.92985E-04	94
RZ=	-1.02530E-03	123

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 135
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990070E+00
Y = 0.696349724E+01
Z = 0.499976000E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942971163E+00
Y = -0.132418755E-01
Z = 0.499085390E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 135)
SUMMATION FORCE-X = 105.00
SUMMATION FORCE-Y = -2937.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 14515.70 MY= 566.65 MZ= -3493.31

```
***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   135 )
    SUMMATION FORCE-X =        -105.00
    SUMMATION FORCE-Y =        2937.37
    SUMMATION FORCE-Z =          41.68

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=      -14515.70  MY=        -566.65  MZ=        3493.31


MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING    135)
          MAXIMUMS    AT NODE
   X =  1.06266E+00     105
   Y = -3.22629E-01     108
   Z = -2.42267E-01     101
   RX= -3.05466E-04     102
   RY= -5.05354E-04      95
   RZ= -1.02698E-03     125


       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.   136
       1.0(D1+D2) + 0.75W(+X) + 0.75(L1+L2-L4)


        CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.937585155E+00
                    Y =  0.567881546E+00
                    Z =  0.499160363E+01

***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING   136 )
    SUMMATION FORCE-X =         -59.33
    SUMMATION FORCE-Y =       -3207.37
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=       16009.92  MY=        -296.60  MZ=       -2440.07


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   136 )
    SUMMATION FORCE-X =          59.33
    SUMMATION FORCE-Y =        3207.37
    SUMMATION FORCE-Z =          -0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=      -16009.92  MY=         296.60  MZ=        2440.07
```

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 136)

	MAXIMUMS	AT NODE
X =	-1.02423E+00	101
Y =	-5.49142E-01	105
Z =	5.44404E-02	101
RX=	7.15391E-04	106
RY=	-3.24127E-04	93
RZ=	1.25218E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 137
1.0(D1+D2) + 0.75W(-X) + 0.75(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.915669314E+00
Y = 0.573696629E+01
Z = 0.499983548E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.937585155E+00
Y = 0.567881546E+00
Z = 0.499160363E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 137)
SUMMATION FORCE-X = -150.67
SUMMATION FORCE-Y = -3207.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16009.92 MY= -753.35 MZ= -2127.39

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 137)
SUMMATION FORCE-X = 150.67
SUMMATION FORCE-Y = 3207.37
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16009.92 MY= 753.35 MZ= 2127.39

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 137)

	MAXIMUMS	AT NODE
X =	-1.62864E+00	101
Y =	-6.57098E-01	105
Z =	5.92469E-02	101
RX=	7.16986E-04	106
RY=	-3.45919E-04	93
RZ=	1.72105E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 138
1.0(D1+D2) + 0.75W(+Z) + 0.75(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990070E+00
Y = 0.674356721E+01
Z = 0.499976000E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.937585155E+00
Y = 0.567881546E+00
Z = 0.499160363E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346183114E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 138)
SUMMATION FORCE-X = -105.00
SUMMATION FORCE-Y = -3207.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16154.21 MY= -566.65 MZ= -2283.73

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 138)
SUMMATION FORCE-X = 105.00
SUMMATION FORCE-Y = 3207.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16154.21 MY= 566.65 MZ= 2283.73

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 138)

	MAXIMUMS	AT NODE
X =	-1.32580E+00	105
Y =	-6.15666E-01	105
Z =	1.71243E-01	101
RX=	7.47257E-04	106
RY=	3.33527E-04	96
RZ=	1.48646E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 139
1.0(D1+D2) + 0.75W(-Z) + 0.75(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990070E+00
Y = 0.674356721E+01
Z = 0.499976000E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.937585155E+00
Y = 0.567881546E+00
Z = 0.499160363E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.346184000E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 139)
SUMMATION FORCE-X = -105.00
SUMMATION FORCE-Y = -3207.37
SUMMATION FORCE-Z = -41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15865.63 MY= -483.30 MZ= -2283.73

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 139)
SUMMATION FORCE-X = 105.00
SUMMATION FORCE-Y = 3207.37
SUMMATION FORCE-Z = 41.68

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15865.63 MY= 483.30 MZ= 2283.73

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 139)

	MAXIMUMS	AT NODE
X	-1.32751E+00	101
Y	-6.14983E-01	101
Z	-1.66600E-01	105
RX	-7.45022E-04	102
RY	-3.40597E-04	93
RZ	1.49833E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 140
1.0(D1+D2)+0.75*0.7 E(+X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 140)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= 0.00 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 140)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= -0.00 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 140)

	MAXIMUMS	AT NODE
X	-1.77935E-01	632
Y	-3.40014E-01	105
Z	4.00890E-02	105
RX	4.99671E-04	106
RY	1.35898E-04	635
RZ	4.73508E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 141
1.0(D1+D2)+0.75*0.7 E(-X) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 141)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= 0.00 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 141)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= -0.00 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 141)

	MAXIMUMS	AT NODE
X =	-1.77935E-01	632
Y =	-3.40014E-01	105
Z =	4.00890E-02	105
RX=	4.99671E-04	106
RY=	1.35898E-04	635
RZ=	4.73508E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 142
1.0(D1+D2)+0.75*0.7 E(+Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 142)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= 0.00 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 142)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= -0.00 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 142)

	MAXIMUMS	AT NODE
X =	-1.77935E-01	632
Y =	-3.40014E-01	105
Z =	4.00890E-02	105
RX=	4.99671E-04	106
RY=	1.35898E-04	635
RZ=	4.73508E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 143
1.0(D1+D2)+0.75*0.7 E(-Z) + 0.75(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940159828E+00
Y = 0.290087120E+00
Z = 0.499124523E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 143)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 15334.95 MY= 0.00 MZ= -2888.52

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 143)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3072.37
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -15334.95 MY= -0.00 MZ= 2888.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 143)

	MAXIMUMS	AT NODE
X =	-1.77935E-01	632
Y =	-3.40014E-01	105
Z =	4.00890E-02	105
RX=	4.99671E-04	106
RY=	1.35898E-04	635
RZ=	4.73508E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 144
0.6(D1+D2) + 1.0W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.348472449E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 144)
SUMMATION FORCE-X = 60.90
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= 304.50 MZ= -1825.15

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 144)
SUMMATION FORCE-X = -60.90
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8477.58 MY= -304.50 MZ= 1825.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 144)

	MAXIMUMS	AT NODE
X =	3.43835E-01	106
Y =	-1.66559E-01	108
Z =	2.06454E-02	105
RX=	2.01282E-04	106
RY=	6.44696E-05	635
RZ=	-3.62011E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 145
0.6(D1+D2) + 1.0W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990403E+00
Y = 0.336110321E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 145)
SUMMATION FORCE-X = -60.90
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= -304.50 MZ= -1408.25

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 145)
SUMMATION FORCE-X = 60.90
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8477.58 MY= 304.50 MZ= 1408.25

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 145)

	MAXIMUMS	AT NODE
X =	-4.74949E-01	632
Y =	-2.36498E-01	105
Z =	1.42050E-02	105
RX=	2.03408E-04	106
RY=	4.86917E-05	635
RZ=	5.13639E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 146
0.6(D1+D2) + 1.0W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990398E+00
Y = 0.346183463E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 146)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8669.97 MY= -55.57 MZ= -1616.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 146)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = -55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8669.97 MY= 55.57 MZ= 1616.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 146)

	MAXIMUMS	AT NODE
X =	-8.17790E-02	630
Y =	-1.81255E-01	105
Z =	1.69900E-01	105
RX=	2.43770E-04	106
RY=	6.57546E-05	635
RZ=	2.10368E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 147
0.6(D1+D2) + 1.0W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990398E+00
Y = 0.346183683E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 147)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = -55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8285.20 MY= 55.57 MZ= -1616.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 147)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = 55.57

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8285.20 MY= -55.57 MZ= 1616.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 147)

	MAXIMUMS	AT NODE
X	-8.52814E-02	632
Y	-1.80468E-01	101
Z	-1.67093E-01	101
RX	-2.40532E-04	102
RY	-5.60784E-05	102
RZ	2.16680E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 148
0.6(D1+D2) + 0.7E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 148)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= 0.00 MZ= -1616.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 148)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8477.58 MY= -0.00 MZ= 1616.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 148)

	MAXIMUMS	AT NODE
X =	-7.95303E-02	632
Y =	-1.64527E-01	105
Z =	1.74252E-02	105
RX=	2.02345E-04	106
RY=	5.65806E-05	635
RZ=	2.01058E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 149
0.6(D1+D2) + 0.7E(-X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 149)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= 0.00 MZ= -1616.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 149)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8477.58 MY= -0.00 MZ= 1616.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 149)

	MAXIMUMS	AT NODE
X =	-7.95303E-02	632
Y =	-1.64527E-01	105
Z =	1.74252E-02	105
RX=	2.02345E-04	106
RY=	5.65806E-05	635
RZ=	2.01058E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 150
0.6(D1+D2) + 0.7E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 150)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= 0.00 MZ= -1616.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 150)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8477.58 MY= -0.00 MZ= 1616.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 150)

	MAXIMUMS	AT NODE
X =	-7.95303E-02	632
Y =	-1.64527E-01	105
Z =	1.74252E-02	105
RX=	2.02345E-04	106
RY=	5.65806E-05	635
RZ=	2.01058E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 151
0.6(D1+D2) + 0.7E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 151)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1698.74
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8477.58 MY= 0.00 MZ= -1616.70

```
***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   151 )
    SUMMATION FORCE-X =          -0.00
    SUMMATION FORCE-Y =        1698.74
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=       -8477.58  MY=          -0.00  MZ=        1616.70


MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING    151)
         MAXIMUMS    AT NODE
   X = -7.95303E-02     632
   Y = -1.64527E-01     105
   Z =  1.74252E-02     105
   RX=  2.02345E-04     106
   RY=  5.65806E-05     635
   RZ=  2.01058E-04      95


       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.   201
       1.4(D1+D2)


        CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.951705228E+00
                    Y = -0.215977807E+00
                    Z =  0.499050539E+01

***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING   201 )
    SUMMATION FORCE-X =           0.00
    SUMMATION FORCE-Y =       -3963.73
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=       19781.02  MY=           0.00  MZ=       -3772.30


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING   201 )
    SUMMATION FORCE-X =          -0.00
    SUMMATION FORCE-Y =        3963.73
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=      -19781.02  MY=          -0.00  MZ=        3772.30
```

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 201)

	MAXIMUMS	AT NODE
X =	-1.85571E-01	632
Y =	-3.83897E-01	105
Z =	4.06589E-02	105
RX=	4.72138E-04	106
RY=	1.32021E-04	635
RZ=	4.69136E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 202
1.4(D1+D2)+1.7L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.935850497E+00
Y = 0.338210008E+00
Z = 0.499135637E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 202)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -4353.90
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 21731.87 MY= 0.00 MZ= -4074.60

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 202)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 4353.90
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -21731.87 MY= -0.00 MZ= 4074.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 202)

	MAXIMUMS	AT NODE
X =	-2.59333E-01	632
Y =	-4.55913E-01	105
Z =	4.60472E-02	105
RX=	5.68544E-04	106
RY=	1.53155E-04	635
RZ=	6.38167E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 203
1.4(D1+D2)+1.7(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.933878796E+00
Y = 0.565401415E+00
Z = 0.499164777E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 203)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -4510.30
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 22513.83 MY= 0.00 MZ= -4212.07

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 203)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 4510.30
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -22513.83 MY= -0.00 MZ= 4212.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 203)

	MAXIMUMS	AT NODE
X =	-2.88443E-01	632
Y =	-5.33047E-01	105
Z =	6.56986E-02	105
RX=	8.40312E-04	106
RY=	2.26308E-04	635
RZ=	7.82868E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 204
1.4(D1+D2)+1.7(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447762E+01
Y = 0.695081343E+01
Z = 0.481778700E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.926963583E+00
Y = 0.172980266E+00
Z = 0.500126976E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 204)
SUMMATION FORCE-X = 236.30
SUMMATION FORCE-Y = -4246.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 21239.40 MY= 1138.44 MZ= -5564.74

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 204)
SUMMATION FORCE-X = -236.30
SUMMATION FORCE-Y = 4246.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -21239.40 MY= -1138.44 MZ= 5564.74

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 204)

	MAXIMUMS	AT NODE
X =	2.46852E+00	105
Y =	-5.64322E-01	108
Z =	-3.00051E-01	101
RX=	-4.56183E-04	102
RY=	-1.15496E-03	95
RZ=	-2.40215E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 205
1.4(D1+D2)+1.7(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990032E+00
Y = 0.694065184E+01
Z = 0.499976003E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.937873747E+00
Y = 0.105079881E+00
Z = 0.499105734E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 205)
SUMMATION FORCE-X = 238.00
SUMMATION FORCE-Y = -4204.30
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 20983.91 MY= 1189.94 MZ= -5582.93

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 205)
SUMMATION FORCE-X = -238.00
SUMMATION FORCE-Y = 4204.30
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -20983.91 MY= -1189.94 MZ= 5582.93

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 205)

	MAXIMUMS	AT NODE
X =	2.49192E+00	101
Y =	-5.65070E-01	108
Z =	2.74986E-01	105
RX=	4.20723E-04	629
RY=	-1.07670E-03	95
RZ=	-2.41051E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 206
1.4(D1+D2)+1.7(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447762E+01
Y = 0.677532073E+01
Z = 0.481778700E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940030609E+00
Y = 0.914501524E+00
Z = 0.498308800E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 206)
SUMMATION FORCE-X = -236.30
SUMMATION FORCE-Y = -4773.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 23788.27 MY= -1138.44 MZ= -2859.41

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 206)
SUMMATION FORCE-X = 236.30
SUMMATION FORCE-Y = 4773.80
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -23788.27 MY= 1138.44 MZ= 2859.41

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 206)

	MAXIMUMS	AT NODE
X =	-2.89428E+00	105
Y =	-1.09344E+00	105
Z =	1.81129E-01	101
RX=	1.23527E-03	106
RY=	8.71317E-04	95
RZ=	2.96233E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 207
1.4(D1+D2)+1.7(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990032E+00
Y = 0.676641265E+01
Z = 0.499976003E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.930391474E+00
Y = 0.967230595E+00
Z = 0.499216318E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 207)
SUMMATION FORCE-X = -238.00
SUMMATION FORCE-Y = -4816.30
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 24043.76 MY= -1189.94 MZ= -2841.22

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 207)
SUMMATION FORCE-X = 238.00
SUMMATION FORCE-Y = 4816.30
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -24043.76 MY= 1189.94 MZ= 2841.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 207)

	MAXIMUMS	AT NODE
X =	-2.92026E+00	101
Y =	-1.12942E+00	105
Z =	1.49879E-01	101
RX=	1.33108E-03	106
RY=	-8.03369E-04	93
RZ=	3.07924E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 208
1.2(D1+D2)+0.8W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.357744059E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 208)
SUMMATION FORCE-X = 48.72
SUMMATION FORCE-Y = -3397.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16955.16 MY= 243.60 MZ= -3400.17

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 208)
SUMMATION FORCE-X = -48.72
SUMMATION FORCE-Y = 3397.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16955.16 MY= -243.60 MZ= 3400.17

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 208)

	MAXIMUMS	AT NODE
X =	2.04178E-01	106
Y =	-2.76203E-01	108
Z =	3.74266E-02	105
RX=	4.03839E-04	106
RY=	1.19472E-04	635
RZ=	-2.23182E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 209
1.2(D1+D2)+0.8W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.326838738E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 209)
SUMMATION FORCE-X = -48.72
SUMMATION FORCE-Y = -3397.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16955.16 MY= -243.60 MZ= -3066.64

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 209)
SUMMATION FORCE-X = 48.72
SUMMATION FORCE-Y = 3397.48
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16955.16 MY= 243.60 MZ= 3066.64

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 209)

	MAXIMUMS	AT NODE
X =	-4.75396E-01	632
Y =	-3.86631E-01	105
Z =	3.22743E-02	105
RX=	4.05540E-04	106
RY=	1.06850E-04	635
RZ=	6.52181E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 210
1.2(D1+D2)+0.8W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183312E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 210)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3397.48
SUMMATION FORCE-Z = 44.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17109.07 MY= -44.46 MZ= -3233.40

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 210)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3397.48
SUMMATION FORCE-Z = -44.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17109.07 MY= 44.46 MZ= 3233.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 210)

	MAXIMUMS	AT NODE
X =	-1.56755E-01	630
Y =	-3.42437E-01	105
Z =	1.56830E-01	105
RX=	4.37830E-04	106
RY=	1.20500E-04	635
RZ=	4.03480E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 211
1.2(D1+D2)+0.8W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705214E+00
Y = -0.215977681E+00
Z = 0.499050536E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183861E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 211)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3397.48
SUMMATION FORCE-Z = -44.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 16801.26 MY= 44.46 MZ= -3233.40

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 211)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3397.48
SUMMATION FORCE-Z = 44.46

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -16801.26 MY= -44.46 MZ= 3233.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 211)

	MAXIMUMS	AT NODE
X =	-1.63661E-01	632
Y =	-3.40331E-01	101
Z =	-1.51148E-01	101
RX=	-4.28880E-04	102
RY=	1.05822E-04	635
RZ=	4.14614E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 212
1.2(D1+D2)+1.6W(+X) + 1.01L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.354553910E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940509774E+00
Y = 0.175349066E+00
Z = 0.499110626E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 212)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = -3627.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18102.72 MY= 487.20 MZ= -3744.75

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 212)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = 3627.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18102.72 MY= -487.20 MZ= 3744.75

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 212)

	MAXIMUMS	AT NODE
X =	4.86323E-01	106
Y =	-3.28477E-01	108
Z =	4.31724E-02	105
RX=	4.59699E-04	106
RY=	1.38215E-04	635
RZ=	-5.19867E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 213
1.2(D1+D2)+1.6W(-X) + 1.0L1

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.330028886E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940509774E+00
Y = 0.175349066E+00
Z = 0.499110626E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 213)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = -3627.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18102.72 MY= -487.20 MZ= -3077.70

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 213)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = 3627.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18102.72 MY= 487.20 MZ= 3077.70

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 213)

	MAXIMUMS	AT NODE
X =	-8.35120E-01	632
Y =	-4.86570E-01	105
Z =	3.28677E-02	105
RX=	4.63100E-04	106
RY=	1.12970E-04	635
RZ=	1.00168E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 214
1.2(D1+D2)+1.6W(+Z) + 1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940509774E+00
Y = 0.175349066E+00
Z = 0.499110626E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 214)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3627.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18410.53 MY= -88.92 MZ= -3411.22

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 214)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3627.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18410.53 MY= 88.92 MZ= 3411.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 214)

	MAXIMUMS	AT NODE
X =	-2.04772E-01	630
Y =	-3.98182E-01	105
Z =	2.81980E-01	105
RX=	5.27679E-04	106
RY=	1.40271E-04	635
RZ=	5.13681E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 215
1.2(D1+D2)+1.6W(-Z) + 1.0L1

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.940509774E+00
Y = 0.175349066E+00
Z = 0.499110626E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 215)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3627.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17794.91 MY= 88.92 MZ= -3411.22

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 215)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3627.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17794.91 MY= -88.92 MZ= 3411.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 215)

	MAXIMUMS	AT NODE
X	-2.11652E-01	632
Y	-3.96721E-01	101
Z	-2.76483E-01	101
RX	-5.21690E-04	102
RY	-1.17378E-04	102
RZ	5.26542E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 216
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.357232673E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 216)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= 487.20 MZ= -3825.62

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 216)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= -487.20 MZ= 3825.62

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 216)

	MAXIMUMS	AT NODE
X =	4.82386E-01	106
Y =	-3.59773E-01	108
Z =	5.47320E-02	105
RX=	6.19562E-04	106
RY=	1.81247E-04	635
RZ=	-5.13349E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 217
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.327350124E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 217)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= -487.20 MZ= -3158.57

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 217)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= 487.20 MZ= 3158.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 217)

	MAXIMUMS	AT NODE
X =	-8.52244E-01	632
Y =	-5.31943E-01	105
Z =	4.44274E-02	105
RX=	6.22963E-04	106
RY=	1.56002E-04	635
RZ=	1.08679E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 218
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 218)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18870.51 MY= -88.92 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 218)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18870.51 MY= 88.92 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 218)

	MAXIMUMS	AT NODE
X =	-2.21987E-01	630
Y =	-4.43555E-01	105
Z =	2.93539E-01	105
RX=	6.87542E-04	106
RY=	1.83302E-04	635
RZ=	5.98419E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 219
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 219)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18254.88 MY= 88.92 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 219)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18254.88 MY= -88.92 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 219)

	MAXIMUMS	AT NODE
X =	-2.28775E-01	632
Y =	-4.42277E-01	101
Z =	-2.88165E-01	101
RX=	-6.82629E-04	102
RY=	1.53946E-04	635
RZ=	6.11660E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 220
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.103790175E+01
Y = 0.550691955E+01
Z = 0.489288255E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.934363009E+00
Y = 0.566553965E-01
Z = 0.499805045E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 220)
SUMMATION FORCE-X = 236.44
SUMMATION FORCE-Y = -3564.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17813.03 MY= 1156.87 MZ= -4621.30

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 220)
SUMMATION FORCE-X = -236.44
SUMMATION FORCE-Y = 3564.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17813.03 MY= -1156.87 MZ= 4621.30

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 220)

	MAXIMUMS	AT NODE
X =	2.05967E+00	105
Y =	-4.82590E-01	108
Z =	-1.90764E-01	101
RX=	-3.89768E-04	102
RY=	-7.23426E-04	95
RZ=	-1.99456E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 221
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+L3)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.934363009E+00
Y = 0.566553965E-01
Z = 0.499805045E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 221)
SUMMATION FORCE-X = 41.56
SUMMATION FORCE-Y = -3564.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17813.03 MY= 182.48 MZ= -3954.25

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 221)
SUMMATION FORCE-X = -41.56
SUMMATION FORCE-Y = 3564.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17813.03 MY= -182.48 MZ= 3954.25

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 221)

	MAXIMUMS	AT NODE
X =	7.70334E-01	105
Y =	-3.30815E-01	108
Z =	-1.80510E-01	101
RX=	-3.94190E-04	102
RY=	-6.82620E-04	95
RZ=	-7.55624E-04	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 222
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.696780525E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.934363009E+00
Y = 0.566553965E-01
Z = 0.499805045E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 222)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = -3564.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18120.84 MY= 580.76 MZ= -4287.78

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 222)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = 3564.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18120.84 MY= -580.76 MZ= 4287.78

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 222)

	MAXIMUMS	AT NODE
X =	1.41278E+00	105
Y =	-4.12212E-01	108
Z =	4.03349E-01	105
RX=	4.65806E-04	105
RY=	-6.89504E-04	95
RZ=	-1.37072E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 223
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.696780525E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.934363009E+00
Y = 0.566553965E-01
Z = 0.499805045E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 223)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = -3564.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17505.21 MY= 758.59 MZ= -4287.78

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 223)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = 3564.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17505.21 MY= -758.59 MZ= 4287.78

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 223)

	MAXIMUMS	AT NODE
X =	1.41722E+00	105
Y =	-4.07999E-01	104
Z =	-4.29688E-01	101
RX=	-4.65757E-04	101
RY=	-7.16541E-04	95
RZ=	-1.37947E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 224
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-L3)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943242708E+00
Y = 0.603465507E+00
Z = 0.498512877E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 224)
SUMMATION FORCE-X = -41.56
SUMMATION FORCE-Y = -3874.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19312.37 MY= -182.48 MZ= -3029.93

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 224)
SUMMATION FORCE-X = 41.56
SUMMATION FORCE-Y = 3874.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19312.37 MY= 182.48 MZ= 3029.93

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 224)

	MAXIMUMS	AT NODE
X =	-1.09492E+00	105
Y =	-6.31280E-01	105
Z =	9.22827E-02	101
RX=	8.51890E-04	106
RY=	4.68501E-04	95
RZ=	1.36562E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 225
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.103790175E+01
Y = 0.538377051E+01
Z = 0.489288255E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943242708E+00
Y = 0.603465507E+00
Z = 0.498512877E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 225)
SUMMATION FORCE-X = -236.44
SUMMATION FORCE-Y = -3874.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19312.37 MY= -1156.87 MZ= -2362.88

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 225)
SUMMATION FORCE-X = 236.44
SUMMATION FORCE-Y = 3874.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19312.37 MY= 1156.87 MZ= 2362.88

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 225)

	MAXIMUMS	AT NODE
X =	-2.38426E+00	105
Y =	-8.61586E-01	105
Z =	1.02537E-01	101
RX=	8.55292E-04	106
RY=	5.11438E-04	96
RZ=	2.36542E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 226
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.675832888E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943242708E+00
Y = 0.603465507E+00
Z = 0.498512877E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 226)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = -3874.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19620.18 MY= -758.59 MZ= -2696.41

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 226)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = 3874.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19620.18 MY= 758.59 MZ= 2696.41

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 226)

	MAXIMUMS	AT NODE
X =	-1.74181E+00	105
Y =	-7.73198E-01	105
Z =	3.41461E-01	101
RX=	9.19871E-04	106
RY=	5.02423E-04	95
RZ=	1.88853E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 227
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-L3)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106447757E+01
Y = 0.675832888E+01
Z = 0.481778702E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.943242708E+00
Y = 0.603465507E+00
Z = 0.498512877E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 227)
SUMMATION FORCE-X = -139.00
SUMMATION FORCE-Y = -3874.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19004.55 MY= -580.76 MZ= -2696.41

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 227)
SUMMATION FORCE-X = 139.00
SUMMATION FORCE-Y = 3874.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19004.55 MY= 580.76 MZ= 2696.41

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 227)

	MAXIMUMS	AT NODE
X =	-1.73737E+00	105
Y =	-7.59046E-01	101
Z =	-3.04189E-01	105
RX=	-9.03701E-04	102
RY=	4.76131E-04	96
RZ=	1.86062E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 228
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.928645484E+00
Y = 0.550700918E+01
Z = 0.499986221E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942039508E+00
Y = 0.838356112E-02
Z = 0.499089108E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 228)
SUMMATION FORCE-X = 237.44
SUMMATION FORCE-Y = -3539.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17662.74 MY= 1187.16 MZ= -4632.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 228)
SUMMATION FORCE-X = -237.44
SUMMATION FORCE-Y = 3539.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17662.74 MY= -1187.16 MZ= 4632.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 228)

	MAXIMUMS	AT NODE
X =	2.07329E+00	105
Y =	-4.83030E-01	108
Z =	1.77842E-01	105
RX=	3.30872E-04	106
RY=	-6.77388E-04	95
RZ=	-1.99948E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 229
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2+L4)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942039508E+00
Y = 0.838356112E-02
Z = 0.499089108E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 229)
SUMMATION FORCE-X = 42.56
SUMMATION FORCE-Y = -3539.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17662.74 MY= 212.77 MZ= -3964.95

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 229)
SUMMATION FORCE-X = -42.56
SUMMATION FORCE-Y = 3539.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17662.74 MY= -212.77 MZ= 3964.95

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 229)

	MAXIMUMS	AT NODE
X =	7.83957E-01	105
Y =	-3.31255E-01	108
Z =	1.67538E-01	105
RX=	3.34273E-04	106
RY=	-6.36582E-04	95
RZ=	-7.60541E-04	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 230
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.695752226E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942039508E+00
Y = 0.838356112E-02
Z = 0.499089108E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 230)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = -3539.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17970.56 MY= 611.05 MZ= -4298.47

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 230)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = 3539.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17970.56 MY= -611.05 MZ= 4298.48

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 230)

	MAXIMUMS	AT NODE
X =	1.43064E+00	101
Y =	-4.12652E-01	108
Z =	4.16650E-01	105
RX=	4.35843E-04	629
RY=	6.55233E-04	94
RZ=	-1.38247E-03	123

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 231
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2+L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.695752226E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.942039508E+00
Y = 0.838356112E-02
Z = 0.499089108E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 231)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = -3539.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17354.93 MY= 788.88 MZ= -4298.47

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 231)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = 3539.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17354.93 MY= -788.88 MZ= 4298.48

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 231)

	MAXIMUMS	AT NODE
X =	1.43084E+00	105
Y =	-4.06168E-01	104
Z =	-4.11306E-01	101
RX=	-4.26789E-04	628
RY=	-6.70504E-04	95
RZ=	-1.38438E-03	125

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 232
1.2(D1+D2) + 1.6W(+X) + 1.0(L1+L2-L4)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936218060E+00
Y = 0.643774260E+00
Z = 0.499170996E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 232)
SUMMATION FORCE-X = -42.56
SUMMATION FORCE-Y = -3899.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19462.65 MY= -212.77 MZ= -3019.23

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 232)
SUMMATION FORCE-X = 42.56
SUMMATION FORCE-Y = 3899.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19462.65 MY= 212.77 MZ= 3019.23

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 232)

	MAXIMUMS	AT NODE
X =	-1.11059E+00	101
Y =	-6.52446E-01	105
Z =	7.39004E-02	101
RX=	9.08252E-04	106
RY=	-4.30443E-04	94
RZ=	1.43734E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 233
1.2(D1+D2) + 1.6W(-X) + 1.0(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.928645484E+00
Y = 0.538437882E+01
Z = 0.499986221E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936218060E+00
Y = 0.643774260E+00
Z = 0.499170996E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 233)
SUMMATION FORCE-X = -237.44
SUMMATION FORCE-Y = -3899.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19462.65 MY= -1187.16 MZ= -2352.18

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 233)
SUMMATION FORCE-X = 237.44
SUMMATION FORCE-Y = 3899.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19462.65 MY= 1187.16 MZ= 2352.18

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 233)

	MAXIMUMS	AT NODE
X =	-2.40000E+00	101
Y =	-8.82752E-01	105
Z =	8.41544E-02	101
RX=	9.11653E-04	106
RY=	-4.76709E-04	93
RZ=	2.43760E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 234
1.2(D1+D2) + 1.6W(+Z) + 1.0(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.674954217E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936218060E+00
Y = 0.643774260E+00
Z = 0.499170996E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183326E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 234)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = -3899.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19770.47 MY= -788.88 MZ= -2685.71

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 234)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = 3899.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19770.47 MY= 788.88 MZ= 2685.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 234)

	MAXIMUMS	AT NODE
X =	-1.75543E+00	105
Y =	-7.94363E-01	105
Z =	3.23078E-01	101
RX=	9.76232E-04	106
RY=	4.57067E-04	96
RZ=	1.94702E-03	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 235
1.2(D1+D2) + 1.6W(-Z) + 1.0(L1+L2-L4)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.878990006E+00
Y = 0.674954217E+01
Z = 0.499976004E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.936218060E+00
Y = 0.643774260E+00
Z = 0.499170996E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183847E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 235)
SUMMATION FORCE-X = -140.00
SUMMATION FORCE-Y = -3899.00
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 19154.84 MY= -611.05 MZ= -2685.71

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 235)
SUMMATION FORCE-X = 140.00
SUMMATION FORCE-Y = 3899.00
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -19154.84 MY= 611.05 MZ= 2685.71

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 235)

	MAXIMUMS	AT NODE
X	-1.75760E+00	101
Y	-7.93992E-01	101
Z	-3.17490E-01	105
RX	-9.75666E-04	102
RY	-4.65356E-04	93
RZ	1.96247E-03	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 236
1.2(D1+D2)+1.0 E(+X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 236)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= 0.00 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 236)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= -0.00 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 236)

	MAXIMUMS	AT NODE
X =	-2.19574E-01	632
Y =	-4.16790E-01	105
Z =	4.95797E-02	105
RX=	6.21262E-04	106
RY=	1.68624E-04	635
RZ=	5.86665E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 237
1.2(D1+D2)+1.0 E(-X) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 237)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= 0.00 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 237)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= -0.00 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 237)

	MAXIMUMS	AT NODE
X =	-2.19574E-01	632
Y =	-4.16790E-01	105
Z =	4.95797E-02	105
RX=	6.21262E-04	106
RY=	1.68624E-04	635
RZ=	5.86665E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 238
1.2(D1+D2)+1.0 E(+Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 238)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= 0.00 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 238)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= -0.00 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 238)

	MAXIMUMS	AT NODE
X =	-2.19574E-01	632
Y =	-4.16790E-01	105
Z =	4.95797E-02	105
RX=	6.21262E-04	106
RY=	1.68624E-04	635
RZ=	5.86665E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 239
1.2(D1+D2)+1.0 E(-Z) + 1.0(L1+L2)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.938987903E+00
Y = 0.341455409E+00
Z = 0.499132034E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 239)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 18562.70 MY= 0.00 MZ= -3492.09

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 239)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3719.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -18562.70 MY= -0.00 MZ= 3492.09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 239)

	MAXIMUMS	AT NODE
X =	-2.19574E-01	632
Y =	-4.16790E-01	105
Z =	4.95797E-02	105
RX=	6.21262E-04	106
RY=	1.68624E-04	635
RZ=	5.86665E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 240
0.9(D1+D2) + 1.6W(+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.348086146E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 240)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= 487.20 MZ= -2758.58

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 240)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= -487.20 MZ= 2758.58

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 240)

	MAXIMUMS	AT NODE
X =	5.56044E-01	105
Y =	-2.54582E-01	108
Z =	3.12901E-02	105
RX=	3.01816E-04	106
RY=	9.74933E-05	635
RZ=	-5.84754E-04	114

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 241
0.9(D1+D2) + 1.6W(-X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990406E+00
Y = 0.336496651E+01
Z = 0.500000900E+01

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 241)
SUMMATION FORCE-X = -97.44
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= -487.20 MZ= -2091.53

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 241)
SUMMATION FORCE-X = 97.44
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= 487.20 MZ= 2091.53

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 241)

	MAXIMUMS	AT NODE
X =	-7.51965E-01	632
Y =	-3.61944E-01	105
Z =	2.09855E-02	105
RX=	3.05218E-04	106
RY=	7.22486E-05	635
RZ=	8.01716E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 242
0.9(D1+D2) + 1.6W(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183485E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 242)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 13024.19 MY= -88.92 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 242)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -13024.19 MY= 88.92 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 242)

	MAXIMUMS	AT NODE
X =	-1.23235E-01	630
Y =	-2.73556E-01	105
Z =	2.70098E-01	105
RX=	3.69797E-04	106
RY=	9.95492E-05	635
RZ=	3.16989E-04	94

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 243
0.9(D1+D2) + 1.6W(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.999990397E+00
Y = 0.346183688E+01
Z = 0.500000749E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 243)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = -88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12408.56 MY= 88.92 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 243)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 88.92

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12408.56 MY= -88.92 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 243)

	MAXIMUMS	AT NODE
X =	-1.28497E-01	632
Y =	-2.72419E-01	101
Z =	-2.65893E-01	101
RX=	-3.65146E-04	102
RY=	-8.50181E-05	102
RZ=	3.26582E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 244
0.9(D1+D2) + 1.0E(+X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 244)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= 0.00 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 244)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= -0.00 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 244)

	MAXIMUMS	AT NODE
X	-1.19295E-01	632
Y	-2.46791E-01	105
Z	2.61378E-02	105
RX	3.03517E-04	106
RY	8.48710E-05	635
RZ	3.01587E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 245
0.9(D1+D2) + 1.0E(-X)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 245)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= 0.00 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 245)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= -0.00 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 245)

	MAXIMUMS	AT NODE
X =	-1.19295E-01	632
Y =	-2.46791E-01	105
Z =	2.61378E-02	105
RX=	3.03517E-04	106
RY=	8.48710E-05	635
RZ=	3.01587E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 246
0.9(D1+D2) + 1.0E(+Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 246)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= 0.00 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 246)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= -0.00 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 246)

	MAXIMUMS	AT NODE
X =	-1.19295E-01	632
Y =	-2.46791E-01	105
Z =	2.61378E-02	105
RX=	3.03517E-04	106
RY=	8.48710E-05	635
RZ=	3.01587E-04	95

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 247
0.9(D1+D2) + 1.0E(-Z)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.951705230E+00
Y = -0.215977727E+00
Z = 0.499050535E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 247)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12716.37 MY= 0.00 MZ= -2425.05

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 247)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 2548.11
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12716.37 MY= -0.00 MZ= 2425.05

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 247)

	MAXIMUMS	AT NODE
X =	-1.19295E-01	632
Y =	-2.46791E-01	105
Z =	2.61378E-02	105
RX=	3.03517E-04	106
RY=	8.48710E-05	635
RZ=	3.01587E-04	95

********** END OF DATA FROM INTERNAL STORAGE **********

1092. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 31,2020 TIME= 16:40:19 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

ATTACHMENT 3 STAAD OUTPUT FOR HOPPER SUPPORTING STRUCTURE FDN

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```
*************************************************
*                                               *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                *
*           Proprietary Program of              *
*           Bentley Systems, Inc.               *
*           Date=    MAR 31, 2020               *
*           Time=    15:55:18                   *
*                                               *
*      USER ID: LNT ECC DIV                     *
*************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\3.Primary Crusher\STAAD Final\C01\Hopper supporting struc.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 14-FEB-20
     4. JOB NAME MAADEN
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 913 0; 2 3.775 913 0; 3 7.275 913 0; 4 12.775 913 0; 5 0 913 10
    10. 6 3.775 913 10; 7 7.275 913 9.85; 8 12.775 913 10; 9 7.275 913 13.15
    11. 10 12.775 913 13; 101 0 910 0; 102 0 910 10; 103 3.775 910 0; 104 3.775 910 10
    12. 105 7.275 910 0; 106 7.275 910 9.85; 107 7.275 910 13.15; 108 12.775 910 0
    13. 109 12.775 910 10; 110 12.775 910 13; 111 0 910 0.250001; 112 0 910 10.25
    14. 113 3.775 910 0.250001; 114 3.775 910 10.25; 115 7.275 910 0.250001
    15. 116 0.251667 910 0.250001; 117 0.251667 910 0; 118 0.378333 910 0.250001
    16. 119 0.378333 910 0; 120 0.755 910 0.250001; 121 0.755 910 0
    17. 122 1.00667 910 0.250001; 123 1.00667 910 0; 124 1.25833 910 0.250001
    18. 125 1.25833 910 0; 126 1.51 910 0.250001; 127 1.51 910 0
    19. 128 1.76167 910 0.250001; 129 1.76167 910 0; 130 2.01333 910 0.250001
    20. 131 2.01333 910 0; 132 2.265 910 0.250001; 133 2.265 910 0
    21. 134 2.64167 910 0.250001; 135 2.64167 910 0; 136 2.76833 910 0.250001
    22. 137 2.76833 910 0; 138 3.02 910 0.250001; 139 3.02 910 0
    23. 140 3.27167 910 0.250001; 141 3.27167 910 0; 142 3.52333 910 0.250001
    24. 143 3.52333 910 0; 144 4.025 910 0.250001; 145 4.025 910 0
    25. 146 4.272 910 0.250001; 147 4.272 910 0; 148 4.525 910 0.250001
    26. 149 4.525 910 0; 150 4.775 910 0.250001; 151 4.775 910 0; 152 4.9 910 0.250001
    27. 153 4.9 910 0; 154 5.275 910 0.250001; 155 5.275 910 0; 156 5.525 910 0.250001
    28. 157 5.525 910 0; 158 5.775 910 0.250001; 159 5.775 910 0
    29. 160 6.025 910 0.250001; 161 6.025 910 0; 162 6.275 910 0.250001
    30. 163 6.275 910 0; 164 6.525 910 0.250001; 165 6.525 910 0; 166 6.9 910 0.250001
    31. 167 6.9 910 0; 168 7.025 910 0.250001; 169 7.025 910 0; 170 0.251667 910 10.25
    32. 171 0.251667 910 10; 172 0.375333 910 10.25; 173 0.375333 910 10
    33. 174 0.755 910 10.25; 175 0.755 910 10; 176 1.00667 910 10.25
    34. 177 1.00667 910 10; 178 1.25833 910 10.25; 179 1.25833 910 10
    35. 180 1.51 910 10.25; 181 1.51 910 10; 182 1.76167 910 10.25; 183 1.76167 910 10
    36. 184 2.01333 910 10.25; 185 2.01333 910 10; 186 2.265 910 10.25
    37. 187 2.265 910 10; 188 2.64167 910 10.25; 189 2.64167 910 10
    38. 190 2.76833 910 10.25; 191 2.76833 910 10; 192 3.02 910 10.25; 193 3.02 910 10
```

39. 194 3.27167 910 10.25; 195 3.27167 910 10; 196 3.52333 910 10.25
40. 197 3.52333 910 10; 198 0 910 0.375001; 199 0 910 10.375
41. 200 3.775 910 0.375001; 201 3.775 910 10.375; 202 7.275 910 0.375001
42. 203 0.251667 910 0.375001; 204 0.378333 910 0.375001; 205 0.755 910 0.375001
43. 206 1.00667 910 0.375001; 207 1.25833 910 0.375001; 208 1.51 910 0.375001
44. 209 1.76167 910 0.375001; 210 2.01333 910 0.375001; 211 2.265 910 0.375001
45. 212 2.64167 910 0.375001; 213 2.76833 910 0.375001; 214 3.02 910 0.375001
46. 215 3.27167 910 0.375001; 216 3.52333 910 0.375001; 217 4.025 910 0.375001
47. 218 4.272 910 0.375001; 219 4.525 910 0.375001; 220 4.775 910 0.375001
48. 221 4.9 910 0.375001; 222 5.275 910 0.375001; 223 5.525 910 0.375001
49. 224 5.775 910 0.375001; 225 6.025 910 0.375001; 226 6.275 910 0.375001
50. 227 6.525 910 0.375001; 228 6.9 910 0.375001; 229 7.025 910 0.375001
51. 230 0.251667 910 10.375; 231 0.375333 910 10.375; 232 0.755 910 10.375
52. 233 1.00667 910 10.375; 234 1.25833 910 10.375; 235 1.51 910 10.375
53. 236 1.76167 910 10.375; 237 2.01333 910 10.375; 238 2.265 910 10.375
54. 239 2.64167 910 10.375; 240 2.76833 910 10.375; 241 3.02 910 10.375
55. 242 3.27167 910 10.375; 243 3.52333 910 10.375; 244 0 910 0.500002
56. 245 0 910 10.5; 246 3.775 910 0.500002; 247 3.775 910 10.5
57. 248 7.275 910 0.500002; 249 0.251667 910 0.500002; 250 0.378333 910 0.500002
58. 251 0.755 910 0.500002; 252 1.00667 910 0.500002; 253 1.25833 910 0.500002
59. 254 1.51 910 0.500002; 255 1.76167 910 0.500002; 256 2.01333 910 0.500002
60. 257 2.265 910 0.500002; 258 2.64167 910 0.500002; 259 2.76833 910 0.500002
61. 260 3.02 910 0.500002; 261 3.27167 910 0.500002; 262 3.52333 910 0.500002
62. 263 4.025 910 0.500002; 264 4.272 910 0.500002; 265 4.525 910 0.500002
63. 266 4.775 910 0.500002; 267 4.9 910 0.500002; 268 5.275 910 0.500002
64. 269 5.525 910 0.500002; 270 5.775 910 0.500002; 271 6.025 910 0.500002
65. 272 6.275 910 0.500002; 273 6.525 910 0.500002; 274 6.9 910 0.500002
66. 275 7.025 910 0.500002; 276 0.251667 910 10.5; 277 0.375333 910 10.5
67. 278 0.755 910 10.5; 279 1.00667 910 10.5; 280 1.25833 910 10.5
68. 281 1.51 910 10.5; 282 1.76167 910 10.5; 283 2.01333 910 10.5
69. 284 2.265 910 10.5; 285 2.64167 910 10.5; 286 2.76833 910 10.5
70. 287 3.02 910 10.5; 288 3.27167 910 10.5; 289 3.52333 910 10.5
71. 290 0 910 -0.250001; 291 0 910 9.75; 292 3.775 910 -0.250001
72. 293 3.775 910 9.75; 294 7.275 910 -0.250001; 295 0.251667 910 -0.250001
73. 296 0.378333 910 -0.250001; 297 0.755 910 -0.250001; 298 1.00667 910 -0.250001
74. 299 1.25833 910 -0.250001; 300 1.51 910 -0.250001; 301 1.76167 910 -0.250001
75. 302 2.01333 910 -0.250001; 303 2.265 910 -0.250001; 304 2.64167 910 -0.250001
76. 305 2.76833 910 -0.250001; 306 3.02 910 -0.250001; 307 3.27167 910 -0.250001
77. 308 3.52333 910 -0.250001; 309 4.025 910 -0.250001; 310 4.272 910 -0.250001
78. 311 4.525 910 -0.250001; 312 4.775 910 -0.250001; 313 4.9 910 -0.250001
79. 314 5.275 910 -0.250001; 315 5.525 910 -0.250001; 316 5.775 910 -0.250001
80. 317 6.025 910 -0.250001; 318 6.275 910 -0.250001; 319 6.525 910 -0.250001
81. 320 6.9 910 -0.250001; 321 7.025 910 -0.250001; 322 0.251667 910 9.75
82. 323 0.375333 910 9.75; 324 0.755 910 9.75; 325 1.00667 910 9.75
83. 326 1.25833 910 9.75; 327 1.51 910 9.75; 328 1.76167 910 9.75
84. 329 2.01333 910 9.75; 330 2.265 910 9.75; 331 2.64167 910 9.75
85. 332 2.76833 910 9.75; 333 3.02 910 9.75; 334 3.27167 910 9.75
86. 335 3.52333 910 9.75; 336 0 910 -0.375001; 337 0 910 9.625
87. 338 3.775 910 -0.375001; 339 3.775 910 9.625; 340 7.275 910 -0.375001
88. 341 0.251667 910 -0.375001; 342 0.378333 910 -0.375001
89. 343 0.755 910 -0.375001; 344 1.00667 910 -0.375001; 345 1.25833 910 -0.375001
90. 346 1.51 910 -0.375001; 347 1.76167 910 -0.375001; 348 2.01333 910 -0.375001
91. 349 2.265 910 -0.375001; 350 2.64167 910 -0.375001; 351 2.76833 910 -0.375001
92. 352 3.02 910 -0.375001; 353 3.27167 910 -0.375001; 354 3.52333 910 -0.375001
93. 355 4.025 910 -0.375001; 356 4.272 910 -0.375001; 357 4.525 910 -0.375001
94. 358 4.775 910 -0.375001; 359 4.9 910 -0.375001; 360 5.275 910 -0.375001

95. 361 5.525 910 -0.375001; 362 5.775 910 -0.375001; 363 6.025 910 -0.375001
96. 364 6.275 910 -0.375001; 365 6.525 910 -0.375001; 366 6.9 910 -0.375001
97. 367 7.025 910 -0.375001; 368 0.251667 910 9.625; 369 0.375333 910 9.625
98. 370 0.755 910 9.625; 371 1.00667 910 9.625; 372 1.25833 910 9.625
99. 373 1.51 910 9.625; 374 1.76167 910 9.625; 375 2.01333 910 9.625
100. 376 2.265 910 9.625; 377 2.64167 910 9.625; 378 2.76833 910 9.625
101. 379 3.02 910 9.625; 380 3.27167 910 9.625; 381 3.52333 910 9.625
102. 382 0 910 -0.500002; 383 0 910 9.5; 384 3.775 910 -0.500002; 385 3.775 910 9.5
103. 386 7.275 910 -0.500002; 387 0.251667 910 -0.500002
104. 388 0.378333 910 -0.500002; 389 0.755 910 -0.500002; 390 1.00667 910 -0.500002
105. 391 1.25833 910 -0.500002; 392 1.51 910 -0.500002; 393 1.76167 910 -0.500002
106. 394 2.01333 910 -0.500002; 395 2.265 910 -0.500002; 396 2.64167 910 -0.500002
107. 397 2.76833 910 -0.500002; 398 3.02 910 -0.500002; 399 3.27167 910 -0.500002
108. 400 3.52333 910 -0.500002; 401 4.025 910 -0.500002; 402 4.272 910 -0.500002
109. 403 4.525 910 -0.500002; 404 4.775 910 -0.500002; 405 4.9 910 -0.500002
110. 406 5.275 910 -0.500002; 407 5.525 910 -0.500002; 408 5.775 910 -0.500002
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113. 415 0.375333 910 9.5; 416 0.755 910 9.5; 417 1.00667 910 9.5
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115. 421 2.01333 910 9.5; 422 2.265 910 9.5; 423 2.64167 910 9.5
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119. 433 7.525 910 -0.375001; 434 7.525 910 -0.500002; 435 7.65 910 0
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121. 439 7.65 910 -0.250001; 440 7.65 910 -0.375001; 441 7.65 910 -0.500002
122. 442 8.025 910 0; 443 8.025 910 0.250001; 444 8.025 910 0.375001
123. 445 8.025 910 0.500002; 446 8.025 910 -0.250001; 447 8.025 910 -0.375001
124. 448 8.025 910 -0.500002; 449 -0.700002 910 0.500002; 450 -0.700002 910 10.5
125. 451 -0.700002 910 -0.500002; 452 -0.700002 910 9.5; 453 -0.375001 910 10.5
126. 454 -0.375001 910 10.375; 455 -0.700001 910 10.375; 456 -0.25 910 10.5
127. 457 -0.25 910 10.375; 458 -0.375001 910 10.25; 459 -0.700001 910 10.25
128. 460 -0.25 910 10.25; 461 -0.375001 910 10; 462 -0.700002 910 10
129. 463 -0.25 910 10; 464 -0.375001 910 9.75; 465 -0.700001 910 9.75
130. 466 -0.25 910 9.75; 467 -0.375001 910 9.625; 468 -0.700002 910 9.625
131. 469 -0.25 910 9.625; 470 -0.375001 910 9.5; 471 -0.25 910 9.5
132. 472 -0.375001 910 0.500002; 473 -0.375001 910 0.375001
133. 474 -0.700001 910 0.375001; 475 -0.25 910 0.500002; 476 -0.25 910 0.375001
134. 477 -0.375001 910 0.25; 478 -0.700001 910 0.25; 479 -0.25 910 0.25
135. 480 -0.375001 910 0; 481 -0.700002 910 0; 482 -0.25 910 0
136. 483 -0.375001 910 -0.250001; 484 -0.700001 910 -0.250001
137. 485 -0.25 910 -0.250001; 486 -0.375001 910 -0.375001
138. 487 -0.700002 910 -0.375001; 488 -0.25 910 -0.375001
139. 489 -0.375001 910 -0.500002; 490 -0.25 910 -0.500002; 491 4.525 910 10.5
140. 492 4.525 910 9.5; 493 4.025 910 10.5; 494 4.025 910 10.375
141. 495 4.272 910 10.5; 496 4.272 910 10.375; 497 4.525 910 10.375
142. 498 4.025 910 10.25; 499 4.272 910 10.25; 500 4.525 910 10.25
143. 501 4.025 910 10; 502 4.272 910 10; 503 4.525 910 10; 504 4.025 910 9.75
144. 505 4.272 910 9.75; 506 4.525 910 9.75; 507 4.025 910 9.625
145. 508 4.272 910 9.625; 509 4.525 910 9.625; 510 4.025 910 9.5; 511 4.272 910 9.5
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147. 516 6.525 910 12.9; 517 7.025 910 13.15; 518 7.025 910 12.9
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344. MEMBER INCIDENCES
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347. ELEMENT INCIDENCES SHELL
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641. 2288 1660 1652 1641 1649; 2290 637 638 1664 1663; 2291 638 639 1665 1664
642. 2292 639 636 1662 1665; 2293 636 640 1666 1662; 2294 640 641 1667 1666
643. 2295 641 642 1668 1667; 2296 642 1517 1669 1668; 2298 1555 637 1663 1671
644. 2300 1663 1664 1675 1674; 2301 1664 1665 1676 1675; 2302 1665 1662 1673 1676
645. 2303 1662 1666 1677 1673; 2304 1666 1667 1678 1677; 2305 1667 1668 1679 1678
646. 2306 1668 1669 1680 1679; 2308 1671 1663 1674 1682; 2309 903 905 1685 1683
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648. 2313 908 909 1689 1688; 2314 909 910 1690 1689; 2315 910 911 1691 1690
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653. 2328 923 924 1704 1703; 2329 924 922 1702 1704; 2330 922 1584 1705 1702
654. 2331 1584 1597 1706 1705; 2332 1597 1610 1707 1706; 2333 1610 1623 1708 1707

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655. 2334 1709 1711 971 969; 2335 1711 1712 972 971; 2336 1712 1713 973 972
656. 2337 1713 1714 974 973; 2338 1714 1715 975 974; 2339 1715 1716 976 975
657. 2340 1716 1717 977 976; 2341 1717 1718 978 977; 2342 1718 1719 979 978
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659. 2346 1722 1723 983 982; 2347 1723 1724 984 983; 2348 1724 1710 970 984
660. 2349 1725 1726 986 985; 2350 1726 1727 987 986; 2351 1727 1709 969 987
661. 2352 1710 1729 989 970; 2353 1729 1730 990 989; 2354 1730 1728 988 990
662. 2355 1728 1731 1587 988; 2356 1731 1732 1600 1587; 2357 1732 1733 1613 1600
663. 2358 1733 1734 1626 1613
664. START GROUP DEFINITION
665. ELEMENT
666. _CF4_MOMENT 870 TO 880 885 TO 894 914 TO 924 929 TO 938 956 TO 968 -
667. 973 TO 984 1002 TO 1012 1017 TO 1026 1046 TO 1056 1061 TO 1070 1088 TO 1100 -
668. 1105 TO 1116 1134 1137 1140 1143 TO 1149 1168 TO 1171 1174 1177 1180 1183 -
669. 1184 TO 1185 1312 TO 1521
670. _CF4_SHEAR 872 TO 877 888 TO 892 916 TO 921 932 TO 936 960 TO 965 976 TO 980 -
671. 1004 TO 1009 1020 TO 1024 1048 TO 1053 1064 TO 1068 1092 TO 1097 -
672. 1108 TO 1112 1316 TO 1321 1332 TO 1336 1347 TO 1356 1367 TO 1416 -
673. 1421 TO 1426 1437 TO 1441 1452 TO 1461 1472 TO 1521
674. _CF5_MOMENT 899 TO 909 943 TO 953 985 TO 997 1031 TO 1041 1075 TO 1085 1117 -
675. 1118 TO 1129 1150 TO 1153 1156 1159 1162 1165 TO 1167 1188 1191 1194 1197 -
676. 1200 1203 1522 TO 1647 2210 TO 2257 2309 TO 2358
677. _CF5_SHEAR 901 TO 906 945 TO 950 989 TO 994 1033 TO 1038 1077 TO 1082 1121 -
678. 1122 TO 1126 1526 TO 1531 1543 TO 1552 1558 TO 1560 1564 TO 1584 1589 TO 1594 -
679. 1606 TO 1615 1621 TO 1623 1627 TO 1647 2218 2221 2230 2233 TO 2257 -
680. 2309 TO 2358
681. _CF6_MOMENT 1204 1207 1210 TO 1234 1237 1242 TO 1249 1254 TO 1261 -
682. 1264 TO 1276 1281 1282 1287 TO 1294 1299 1300 1305 TO 1311 2138 TO 2155 2174 -
683. 2175 TO 2191 2270 TO 2276 2278 2280 TO 2286 2288 2290 TO 2296 2298 -
684. 2300 TO 2306 2308
685. _CF6_SHEAR 1213 TO 1230 1243 TO 1248 1255 TO 1260 1267 TO 1272 2138 TO 2155 -
686. 2174 TO 2191 2270 TO 2276 2278 2280 TO 2286 2288 2290 TO 2296 2298 -
687. 2300 TO 2306 2308
688. MEMBER
689. _COLUMN_C3 858 860 862 864 865
690. _COLUMN_C4 859 863
691. END GROUP DEFINITION
692. ELEMENT PROPERTY
693. 897 TO 911 941 TO 955 985 TO 999 1029 TO 1043 1073 TO 1087 1117 TO 1131 1150 -
694. 1151 TO 1167 1186 TO 1203 1522 TO 1647 2210 TO 2257 2309 TO 2357 -
695. 2358 THICKNESS 0.7
696. 868 TO 896 912 TO 940 956 TO 984 1000 TO 1028 1044 TO 1072 1088 TO 1116 1132 -
697. 1133 TO 1149 1168 TO 1185 1312 TO 1521 THICKNESS 0.7
698. 1204 TO 1311 2138 TO 2155 2174 TO 2191 2270 TO 2276 2278 2280 TO 2286 2288 -
699. 2290 TO 2296 2298 2300 TO 2306 2308 THICKNESS 0.5
700. DEFINE MATERIAL START
701. ISOTROPIC CONCRETE
702. E 3E+007
703. POISSON 0.17
704. DENSITY 24
705. ALPHA 5.5E-006
706. DAMP 0.05
707. TYPE CONCRETE
708. STRENGTH FCU 27578.9
709. END DEFINE MATERIAL
710. MEMBER PROPERTY AMERICAN
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711. 859 861 863 866 867 PRIS YD 1 ZD 1
712. MEMBER PROPERTY AMERICAN
713. 858 860 862 864 865 PRIS YD 0.75 ZD 0.75
714. CONSTANTS
715. MATERIAL CONCRETE ALL
716. SUPPORTS
717. 108 TO 110 FIXED
718. 868 TO 1647 2138 TO 2155 2174 TO 2191 2210 TO 2257 2270 TO 2276 2278 2280 -
719. 2281 TO 2286 2288 2290 TO 2296 2298 2300 TO 2306 2308 TO 2357 -
720. 2358 PLATE MAT DIRECT Y SUBGRADE 68400 PRINT COMPRESSION
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106	0.062500
107	0.062500
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112	0.047031
113	0.047032
114	0.047032
115	0.046875
116	0.035469
117	0.047292
118	0.047188
119	0.062917
120	0.058907
121	0.078542
122	0.047187
123	0.062916
124	0.047187
125	0.062916
126	0.047188
127	0.062918
128	0.047187
129	0.062917
130	0.047187
131	0.062916
132	0.058907
133	0.078543
134	0.047187
135	0.062917
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137	0.047291
138	0.047188
139	0.062918
140	0.047187
141	0.062916
142	0.047187
143	0.062917
144	0.046594
145	0.062125

146	0.046875
147	0.062500
148	0.047156
149	0.062875
150	0.035156
151	0.046875
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153	0.062500
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823	0.047031
824	0.047032
825	0.046875
826	0.035469
827	0.047187
828	0.058907
829	0.047187
830	0.047187
831	0.047188
832	0.047187
833	0.047187
834	0.058907
835	0.047187
836	0.035468
837	0.047188
838	0.047187
839	0.047187
840	0.046594
841	0.046875
842	0.047156
843	0.035156
844	0.046875
845	0.058594
846	0.046875
847	0.046875
848	0.046875
849	0.046875
850	0.058594
851	0.046875
852	0.035156
853	0.035156
854	0.046875
855	0.035156
856	0.030469
857	0.042188
858	0.035156
859	0.078385
860	0.078386
861	0.058646
862	0.078646
863	0.098646
864	0.078645
865	0.078645
866	0.078647
867	0.078645
868	0.078645
869	0.098178
870	0.078645
871	0.059114
872	0.078647
873	0.078645

874	0.078645
875	0.050781
876	0.070313
877	0.058594
878	0.078594
879	0.077656
880	0.078125
881	0.062708
882	0.062709
883	0.046917
884	0.062917
885	0.078917
886	0.062916
887	0.062916
888	0.062918
889	0.062916
890	0.062916
891	0.078543
892	0.062916
893	0.047291
894	0.062918
895	0.062916
896	0.062916
897	0.040625
898	0.056250
899	0.046875
900	0.062875
901	0.062125
902	0.062500
903	0.078385
904	0.078386
905	0.058646
906	0.078646
907	0.098646
908	0.078645
909	0.078645
910	0.078647
911	0.078645
912	0.078645
913	0.098178
914	0.078645
915	0.059114
916	0.078647
917	0.078645
918	0.078645
919	0.050781
920	0.070313
921	0.058594
922	0.078594
923	0.077656
924	0.078125
925	0.078385
926	0.078386
927	0.058646
928	0.078646
929	0.098646

930	0.078645
931	0.078645
932	0.078647
933	0.078645
934	0.078645
935	0.098178
936	0.078645
937	0.059114
938	0.078647
939	0.078645
940	0.078645
941	0.050781
942	0.070313
943	0.058594
944	0.078594
945	0.077656
946	0.078125
947	0.062708
948	0.062709
949	0.046917
950	0.062917
951	0.078917
952	0.062916
953	0.062916
954	0.062918
955	0.062916
956	0.062916
957	0.078543
958	0.062916
959	0.047291
960	0.062918
961	0.062916
962	0.062916
963	0.040625
964	0.056250
965	0.046875
966	0.062875
967	0.062125
968	0.062500
969	0.078385
970	0.078386
971	0.058646
972	0.078646
973	0.098646
974	0.078645
975	0.078645
976	0.078647
977	0.078645
978	0.078645
979	0.098178
980	0.078645
981	0.059114
982	0.078647
983	0.078645
984	0.078645
985	0.050781

986	0.070313
987	0.058594
988	0.078594
989	0.077656
990	0.078125
1499	0.031250
1500	0.031250
1501	0.023438
1502	0.031250
1503	0.039063
1504	0.031250
1505	0.040625
1506	0.050000
1507	0.040625
1508	0.031250
1509	0.039063
1510	0.031250
1511	0.023437
1512	0.023438
1513	0.031250
1514	0.039062
1515	0.023438
1516	0.031250
1517	0.039063
1537	0.031250
1538	0.031250
1539	0.023438
1540	0.031250
1541	0.039063
1542	0.031250
1543	0.040625
1544	0.050000
1545	0.040625
1546	0.031250
1547	0.039063
1548	0.031250
1549	0.023437
1550	0.023438
1551	0.031250
1552	0.039062
1553	0.023438
1554	0.031250
1555	0.039063
1575	0.058594
1576	0.058594
1577	0.023438
1578	0.035156
1579	0.046875
1580	0.035156
1581	0.023438
1582	0.058594
1583	0.046875
1584	0.058594
1585	0.058594
1586	0.046875
1587	0.058594

1588	0.078125
1589	0.078125
1590	0.031250
1591	0.046875
1592	0.062500
1593	0.046875
1594	0.031250
1595	0.078125
1596	0.062500
1597	0.078125
1598	0.078125
1599	0.062500
1600	0.078125
1601	0.097656
1602	0.097656
1603	0.039063
1604	0.058594
1605	0.078125
1606	0.058594
1607	0.039063
1608	0.097656
1609	0.078125
1610	0.097656
1611	0.097656
1612	0.078125
1613	0.097656
1614	0.039062
1615	0.039062
1616	0.015625
1617	0.023437
1618	0.031250
1619	0.023438
1620	0.015625
1621	0.039062
1622	0.031250
1623	0.039062
1624	0.039062
1625	0.031250
1626	0.039062
1640	0.062500
1641	0.078125
1642	0.062500
1643	0.046875
1644	0.046875
1645	0.062500
1646	0.078125
1647	0.031250
1649	0.031250
1651	0.031250
1652	0.039063
1653	0.031250
1654	0.023438
1655	0.023437
1656	0.031250
1657	0.039063
1658	0.015625

1660	0.015625
1662	0.062500
1663	0.078125
1664	0.062500
1665	0.046875
1666	0.046875
1667	0.062500
1668	0.078125
1669	0.031250
1671	0.031250
1673	0.031250
1674	0.039063
1675	0.031250
1676	0.023438
1677	0.023437
1678	0.031250
1679	0.039063
1680	0.015625
1682	0.015625
1683	0.031354
1684	0.031354
1685	0.023458
1686	0.031458
1687	0.039458
1688	0.031458
1689	0.031458
1690	0.031459
1691	0.031458
1692	0.031458
1693	0.039271
1694	0.031458
1695	0.023646
1696	0.031459
1697	0.031458
1698	0.031458
1699	0.020313
1700	0.028125
1701	0.023438
1702	0.031437
1703	0.031062
1704	0.031250
1705	0.023437
1706	0.031250
1707	0.039062
1708	0.015625
1709	0.031354
1710	0.031354
1711	0.023458
1712	0.031458
1713	0.039458
1714	0.031458
1715	0.031458
1716	0.031459
1717	0.031458
1718	0.031458
1719	0.039271

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      1720          0.031458
      1721          0.023646
      1722          0.031459
      1723          0.031458
      1724          0.031458
      1725          0.020313
      1726          0.028125
      1727          0.023438
      1728          0.031437
      1729          0.031062
      1730          0.031250
      1731          0.023437
      1732          0.031250
      1733          0.039062
      1734          0.015625
    TOTAL          59.562489
721. ***
722. DEFINE REFERENCE LOADS
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
723. LOAD R1 LOADTYPE DEAD  TITLE DEAD LOAD(DL)
724. JOINT LOAD
725. 1 FX -13.17 FY -36.8 FZ -12.974
726. 2 FX -1.81 FY -508.542 FZ -81.606
727. 3 FX 8.064 FY -15.674 FZ 0
728. 4 FX 0.115 FY -451.207 FZ -2.891
729. 5 FX -5.16 FY -24.176 FZ 12.184
730. 6 FX 3.443 FY -500.444 FZ 76.788
731. 7 FX -0.062 FY -6.762 FZ -2.618
732. 8 FX 8.579 FY -435.805 FZ 10.567
733. 9 FX -13.088 FY -28.329 FZ -0.029
734. 10 FX 13.087 FY -104.883 FZ 0.578
735. SELFWEIGHT Y -1
736. ELEMENT LOAD
737. 868 TO 1647 2138 TO 2155 2174 TO 2191 2210 TO 2257 2270 TO 2276 2278 2280 -
738. 2281 TO 2286 2288 2290 TO 2296 2298 2300 TO 2306 2308 TO 2358 PR GY -38
739. LOAD R2 LOADTYPE DEAD  TITLE EQUIPMENT DEAD LOAD(EQL)
740. JOINT LOAD
741. 1 FX 85.004 FY 10.743 FZ -93.184
742. 2 FX 53.615 FY -296.827 FZ -91.87
743. 3 FX 87.811 FY -95.998 FZ 0
744. 4 FX 1.532 FY -229.333 FZ -23.471
745. 5 FX 66.101 FY -59.537 FZ 107.416
746. 6 FX 45.132 FY -440.599 FZ 163.982
747. 7 FX 0.134 FY 51.669 FZ 18.219
748. 8 FX 141.196 FY -283.579 FZ 55.05
749. 9 FX -4.059 FY -14.625 FZ 0.361
750. 10 FX 4.06 FY -46.716 FZ 0.544
751. LOAD R3 LOADTYPE LIVE  TITLE LIVE LOAD(LL)
752. JOINT LOAD
753. 1 FX -2.642 FY -45.625 FZ -34.613
754. 2 FX -0.904 FY -141.465 FZ -35.967
755. 3 FX 2.286 FY -15.925 FZ 0
756. 4 FX 0.05 FY -73.04 FZ -5.356
757. 5 FX -3.42 FY -51.437 FZ 26.746
758. 6 FX 0.216 FY -157.559 FZ 47.672
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759. 7 FX -0.006 FY -54.458 FZ -3.153
760. 8 FX 4.419 FY -137.521 FZ 4.398
761. 9 FX -6.961 FY -37.475 FZ 0.075
762. 10 FX 6.961 FY -42.256 FZ 0.198
763. LOAD R4 LOADTYPE LIVE  TITLE MATERIAL LOAD(MATL)
764. JOINT LOAD
765. 1 FX -5.236 FY -16.661 FZ -9.19
766. 2 FX 0.223 FY -423.721 FZ -91.227
767. 3 FX 10.046 FY -10.662 FZ 0
768. 4 FX 1.18 FY -1068.79 FZ 12.427
769. 5 FX -0.772 FY -13.281 FZ 11.058
770. 6 FX 1.653 FY -405.012 FZ 101.471
771. 7 FX -0.181 FY -21.365 FZ -5.51
772. 8 FX -11.34 FY -865.35 FZ -19.727
773. 9 FX -10.56 FY -1.623 FZ -0.273
774. 10 FX 10.56 FY -112.969 FZ 0.972
775. LOAD R5 LOADTYPE WIND  TITLE WIND LOAD WL+X
776. JOINT LOAD
777. 1 FX 52.456 FY 77.644 FZ 0.685
778. 2 FX 32.136 FY 95.43 FZ 17.728
779. 3 FX 44.942 FY -44.535 FZ 0
780. 4 FX 2.817 FY -125.959 FZ -0.727
781. 5 FX 35.489 FY 49.672 FZ -0.046
782. 6 FX 20.138 FY 138.796 FZ -8.248
783. 7 FX 2.77 FY -99.604 FZ 0.793
784. 8 FX 68.278 FY -100.045 FZ -10.45
785. 9 FX 7.437 FY -15.629 FZ 0.227
786. 10 FX 6.866 FY 24.23 FZ 0.039
787. LOAD R6 LOADTYPE WIND  TITLE WIND LOAD WL+Z
788. JOINT LOAD
789. 1 FX 20.795 FY 29.217 FZ 0.898
790. 2 FX 9.735 FY 249.057 FZ 80.534
791. 3 FX 9.963 FY -12.364 FZ 1.246
792. 4 FX 0.058 FY 91.901 FZ 8.224
793. 5 FX -12.375 FY -16.006 FZ 1.241
794. 6 FX -3.753 FY -276.732 FZ 88.62
795. 7 FX 0.027 FY 159.605 FZ 55.638
796. 8 FX -24.45 FY 172.513 FZ 91.038
797. 9 FX -33.944 FY -132.641 FZ 1.377
798. 10 FX 33.943 FY -264.55 FZ 0.604
799. LOAD R7 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD(TL)
800. JOINT LOAD
801. 1 FX -43.214 FY -45.001 FZ -0.445
802. 2 FX -3.912 FY 41.156 FZ -73.07
803. 3 FX 26.576 FY -26.594 FZ -0.001
804. 4 FX 1.515 FY 8.628 FZ -16.162
805. 5 FX -60.711 FY -73.021 FZ 0.759
806. 6 FX -15.394 FY 117.33 FZ 39.798
807. 7 FX -0.02 FY 70.686 FZ 18.336
808. 8 FX 95.16 FY -20.478 FZ 25.404
809. 9 FX -91.77 FY -29.299 FZ 0.771
810. 10 FX 91.771 FY -43.406 FZ 4.609
811. LOAD R11 LOADTYPE SEISMIC  TITLE E +X
812. JOINT LOAD
813. 1 FX 6.75 FY 10.464 FZ 0.189
814. 2 FX 3.905 FY 14.619 FZ 2.434
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815. 3 FX 5.691 FY -5.944 FZ 0
816. 4 FX 0.151 FY -18.742 FZ -0.119
817. 5 FX 4.051 FY 6.055 FZ -0.078
818. 6 FX 2.026 FY 20.45 FZ -1.178
819. 7 FX -0.004 FY -14.894 FZ -0.029
820. 8 FX 8.23 FY -13.33 FZ -1.259
821. 9 FX 0.248 FY -3.735 FZ 0.033
822. 10 FX 0.14 FY 5.057 FZ 0.006
823. LOAD R12 LOADTYPE SEISMIC  TITLE E +Z
824. JOINT LOAD
825. 1 FX 1.914 FY 3.677 FZ 0.566
826. 2 FX 0.946 FY 20.818 FZ 7.932
827. 3 FX 1.143 FY -1.283 FZ 0
828. 4 FX 0.015 FY 6.039 FZ 0.729
829. 5 FX -1.199 FY -2.555 FZ 0.565
830. 6 FX -0.44 FY -22.775 FZ 8.307
831. 7 FX 0.002 FY 12.743 FZ 4.468
832. 8 FX -2.382 FY 16.875 FZ 8.717
833. 9 FX -2.818 FY -10.865 FZ 0.029
834. 10 FX 2.818 FY -22.675 FZ 0.012
835. END DEFINE REFERENCE LOADS
836. *FOR STRENGTH DESIGN
837. LOAD 101 1.4 (DL1+DL2)
838. REFERENCE LOAD
839. R1 1.4 R2 1.4
840. LOAD 102 1.4 (DL1+DL2)+1.7(LL1+LL2)+1.4TL
841. REFERENCE LOAD
842. R1 1.4 R2 1.4 R3 1.7 R4 1.7 R7 1.4
843. LOAD 103 1.2 (DL1+DL2)+0.8WL(+X)
844. REFERENCE LOAD
845. R1 1.2 R2 1.2 R5 0.8
846. LOAD 104 1.2 (DL1+DL2)+0.8WL(-X)
847. REFERENCE LOAD
848. R1 1.2 R2 1.2 R5 -0.8
849. LOAD 105 1.2 (DL1+DL2)+0.8WL(+Z)
850. REFERENCE LOAD
851. R1 1.2 R2 1.2 R6 0.8
852. LOAD 106 1.2 (DL1+DL2)+0.8WL(-Z)
853. REFERENCE LOAD
854. R1 1.2 R2 1.2 R6 -0.8
855. LOAD 107 1.2 (DL1+DL2)+1.6WL(+X)+0.5LL1+1.0LL2
856. REFERENCE LOAD
857. R1 1.2 R2 1.2 R5 1.6 R3 0.5 R4 1.0
858. LOAD 108 1.2 (DL1+DL2)+1.6WL(-X)+0.5LL1+1.0LL2
859. REFERENCE LOAD
860. R1 1.2 R2 1.2 R5 -1.6 R3 0.5 R4 1.0
861. LOAD 109 1.2 (DL1+DL2)+1.6WL(+Z)+0.5LL1+1.0LL2
862. REFERENCE LOAD
863. R1 1.2 R2 1.2 R6 1.6 R3 0.5 R4 1.0
864. LOAD 110 1.2 (DL1+DL2)+1.6WL(-Z)+0.5LL1+1.0LL2
865. REFERENCE LOAD
866. R1 1.2 R2 1.2 R6 -1.6 R3 0.5 R4 1.0
867. LOAD 111 1.2 (DL1+DL2)+1.6EL(+X)+0.5LL1+1.0LL2
868. REFERENCE LOAD
869. R1 1.2 R2 1.2 R11 1.6 R3 0.5 R4 1.0
870. LOAD 112 1.2 (DL1+DL2)+1.6EL(-X)+0.5LL1+1.0LL2
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871. REFERENCE LOAD
872. R1 1.2 R2 1.2 R11 -1.6 R3 0.5 R4 1.0
873. LOAD 113 1.2 (DL1+DL2)+1.6EL(+Z)+0.5LL1+1.0LL2
874. REFERENCE LOAD
875. R1 1.2 R2 1.2 R12 1.6 R3 0.5 R4 1.0
876. LOAD 114 1.2 (DL1+DL2)+1.6EL(-Z)+0.5LL1+1.0LL2
877. REFERENCE LOAD
878. R1 1.2 R2 1.2 R12 -1.6 R3 0.5 R4 1.0
879. LOAD 115 0.9 (DL1+DL2)+1.6WL(+X)
880. REFERENCE LOAD
881. R1 0.9 R2 0.9 R5 1.6
882. LOAD 116 0.9 (DL1+DL2)+1.6WL(-X)
883. REFERENCE LOAD
884. R1 0.9 R2 0.9 R5 -1.6
885. LOAD 117 0.9 (DL1+DL2)+1.6WL(+Z)
886. REFERENCE LOAD
887. R1 0.9 R2 0.9 R6 1.6
888. LOAD 118 0.9 (DL1+DL2)+1.6WL(-Z)
889. REFERENCE LOAD
890. R1 0.9 R2 0.9 R6 -1.6
891. LOAD 119 0.9 (DL1+DL2)+1.6EL(+X)
892. REFERENCE LOAD
893. R1 0.9 R2 0.9 R11 1.6
894. LOAD 120 0.9 (DL1+DL2)+1.6EL(-X)
895. REFERENCE LOAD
896. R1 0.9 R2 0.9 R11 -1.6
897. LOAD 121 0.9 (DL1+DL2)+1.6EL(+Z)
898. REFERENCE LOAD
899. R1 0.9 R2 0.9 R12 1.6
900. LOAD 122 0.9 (DL1+DL2)+1.6EL(-Z)
901. REFERENCE LOAD
902. R1 0.9 R2 0.9 R12 -1.6
903. **************
904. *FOR ALLOWABLE STRESS DESIGN
905. LOAD 201 DL1+DL2
906. REFERENCE LOAD
907. R1 1.0 R2 1.0
908. LOAD 202 DL1+DL2+LL1+LL2+TL
909. REFERENCE LOAD
910. R1 1.0 R2 1.0 R3 1.0 R4 1.0 R7 1.0
911. LOAD 203 DL1+DL2+0.75LL1+0.75LL2+0.75TL
912. REFERENCE LOAD
913. R1 1.0 R2 1.0 R3 0.75 R4 0.75 R7 0.75
914. LOAD 204 DL1+DL2+WL(+X)
915. REFERENCE LOAD
916. R1 1.0 R2 1.0 R5 1.0
917. LOAD 205 DL1+DL2+WL(-X)
918. REFERENCE LOAD
919. R1 1.0 R2 1.0 R5 -1.0
920. LOAD 206 DL1+DL2+WL(+Z)
921. REFERENCE LOAD
922. R1 1.0 R2 1.0 R6 1.0
923. LOAD 207 DL1+DL2+WL(-Z)
924. REFERENCE LOAD
925. R1 1.0 R2 1.0 R6 -1.0
926. LOAD 208 DL1+DL2+0.7EL(+X)
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927. REFERENCE LOAD
928. R1 1.0 R2 1.0 R11 0.7
929. LOAD 209 DL1+DL2+0.7EL(-X)
930. REFERENCE LOAD
931. R1 1.0 R2 1.0 R11 -0.7
932. LOAD 210 DL1+DL2+0.7EL(+Z)
933. REFERENCE LOAD
934. R1 1.0 R2 1.0 R12 0.7
935. LOAD 211 DL1+DL2+0.7EL(-Z)
936. REFERENCE LOAD
937. R1 1.0 R2 1.0 R12 -0.7
938. LOAD 212 DL1+DL2+0.75WL(+X)+0.75LL1+0.75LL2
939. REFERENCE LOAD
940. R1 1.0 R2 1.0 R5 0.75 R3 0.75 R4 0.75
941. LOAD 213 DL1+DL2+0.75WL(-X)+0.75LL1+0.75LL2
942. REFERENCE LOAD
943. R1 1.0 R2 1.0 R5 -0.75 R3 0.75 R4 0.75
944. LOAD 214 DL1+DL2+0.75WL(+Z)+0.75LL1+0.75LL2
945. REFERENCE LOAD
946. R1 1.0 R2 1.0 R6 0.75 R3 0.75 R4 0.75
947. LOAD 215 DL1+DL2+0.75WL(-Z)+0.75LL1+0.75LL2
948. REFERENCE LOAD
949. R1 1.0 R2 1.0 R6 -0.75 R3 0.75 R4 0.75
950. LOAD 216 DL1+DL2+0.75*0.7EL(+X)+0.75LL1+0.75LL2
951. REFERENCE LOAD
952. R1 1.0 R2 1.0 R11 0.525 R3 0.75 R4 0.75
953. LOAD 217 DL1+DL2+0.75*0.7EL(-X)+0.75LL1+0.75LL2
954. REFERENCE LOAD
955. R1 1.0 R2 1.0 R11 -0.525 R3 0.75 R4 0.75
956. LOAD 218 DL1+DL2+0.75*0.7EL(+Z)+0.75LL1+0.75LL2
957. REFERENCE LOAD
958. R1 1.0 R2 1.0 R12 0.525 R3 0.75 R4 0.75
959. LOAD 219 DL1+DL2+0.75*0.7EL(-Z)+0.75LL1+0.75LL2
960. REFERENCE LOAD
961. R1 1.0 R2 1.0 R12 -0.525 R3 0.75 R4 0.75
962. LOAD 220 0.6DL1+0.6DL2+WL(+X)
963. REFERENCE LOAD
964. R1 0.6 R2 0.6 R5 1.0
965. LOAD 221 0.6DL1+0.6DL2+WL(-X)
966. REFERENCE LOAD
967. R1 0.6 R2 0.6 R5 -1.0
968. LOAD 222 0.6DL1+0.6DL2+WL(+Z)
969. REFERENCE LOAD
970. R1 0.6 R2 0.6 R6 1.0
971. LOAD 223 0.6DL1+0.6DL2+WL(-Z)
972. REFERENCE LOAD
973. R1 0.6 R2 0.6 R6 -1.0
974. LOAD 224 0.6DL1+0.6DL2+0.7EL(+X)
975. REFERENCE LOAD
976. R1 0.6 R2 0.6 R11 0.7
977. LOAD 225 0.6DL1+0.6DL2+0.7EL(-X)
978. REFERENCE LOAD
979. R1 0.6 R2 0.6 R11 -0.7
980. LOAD 226 0.6DL1+0.6DL2+0.7EL(+Z)
981. REFERENCE LOAD
982. R1 0.6 R2 0.6 R12 0.7
```

983. LOAD 227 0.6DL1+0.6DL2+0.7EL(-Z)
984. REFERENCE LOAD
985. R1 0.6 R2 0.6 R12 -0.7
986. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 1078 NUMBER OF MEMBERS 10
NUMBER OF PLATES 946 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1068

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 698/ 35/ 108 DOF
TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 3255
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 352 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 22.9/ 195894.0 MB

*** LOAD CASE 101 -- START ITERATION NO. 2
*** LOAD CASE 101 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 101

*** LOAD CASE 102 -- START ITERATION NO. 2
*** LOAD CASE 102 -- START ITERATION NO. 3
*** LOAD CASE 102 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 102

*** LOAD CASE 103 -- START ITERATION NO. 2
*** LOAD CASE 103 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 103

*** LOAD CASE 104 -- START ITERATION NO. 2
*** LOAD CASE 104 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 104

*** LOAD CASE 105 -- START ITERATION NO. 2
*** LOAD CASE 105 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 105

*** LOAD CASE 106 -- START ITERATION NO. 2
*** LOAD CASE 106 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 106

*** LOAD CASE 107 -- START ITERATION NO. 2
*** LOAD CASE 107 -- START ITERATION NO. 3
*** LOAD CASE 107 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 107

*** LOAD CASE 108 -- START ITERATION NO. 2
*** LOAD CASE 108 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 108

*** LOAD CASE 109 -- START ITERATION NO. 2
*** LOAD CASE 109 -- START ITERATION NO. 3
*** LOAD CASE 109 -- START ITERATION NO. 4
*** LOAD CASE 109 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 109

*** LOAD CASE 110 -- START ITERATION NO. 2
*** LOAD CASE 110 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 110

*** LOAD CASE 111 -- START ITERATION NO. 2
*** LOAD CASE 111 -- START ITERATION NO. 3
*** LOAD CASE 111 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 111

*** LOAD CASE 112 -- START ITERATION NO. 2
*** LOAD CASE 112 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 112

*** LOAD CASE 113 -- START ITERATION NO. 2
*** LOAD CASE 113 -- START ITERATION NO. 3
*** LOAD CASE 113 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 113

```
*** LOAD CASE   114 -- START ITERATION NO.   2
*** LOAD CASE   114 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  114


*** LOAD CASE   115 -- START ITERATION NO.   2
*** LOAD CASE   115 -- START ITERATION NO.   3
*** LOAD CASE   115 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  115


*** LOAD CASE   116 -- START ITERATION NO.   2
*** LOAD CASE   116 -- START ITERATION NO.   3
*** LOAD CASE   116 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  116


*** LOAD CASE   117 -- START ITERATION NO.   2
*** LOAD CASE   117 -- START ITERATION NO.   3
*** LOAD CASE   117 -- START ITERATION NO.   4
*** LOAD CASE   117 -- START ITERATION NO.   5
*** LOAD CASE   117 -- START ITERATION NO.   6

  **NOTE-Tension/Compression converged after  6 iterations, Case=  117


*** LOAD CASE   118 -- START ITERATION NO.   2
*** LOAD CASE   118 -- START ITERATION NO.   3
*** LOAD CASE   118 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  118


*** LOAD CASE   119 -- START ITERATION NO.   2
*** LOAD CASE   119 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  119


*** LOAD CASE   120 -- START ITERATION NO.   2
*** LOAD CASE   120 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  120


*** LOAD CASE   121 -- START ITERATION NO.   2
*** LOAD CASE   121 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  121


*** LOAD CASE   122 -- START ITERATION NO.   2
```

```
*** LOAD CASE   122 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  122



*** LOAD CASE   201 -- START ITERATION NO.   2
*** LOAD CASE   201 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  201



*** LOAD CASE   202 -- START ITERATION NO.   2
*** LOAD CASE   202 -- START ITERATION NO.   3
*** LOAD CASE   202 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  202



*** LOAD CASE   203 -- START ITERATION NO.   2
*** LOAD CASE   203 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  203



*** LOAD CASE   204 -- START ITERATION NO.   2
*** LOAD CASE   204 -- START ITERATION NO.   3
*** LOAD CASE   204 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  204



*** LOAD CASE   205 -- START ITERATION NO.   2
*** LOAD CASE   205 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  205



*** LOAD CASE   206 -- START ITERATION NO.   2
*** LOAD CASE   206 -- START ITERATION NO.   3
*** LOAD CASE   206 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  206



*** LOAD CASE   207 -- START ITERATION NO.   2
*** LOAD CASE   207 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  207



*** LOAD CASE   208 -- START ITERATION NO.   2
*** LOAD CASE   208 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  208
```

```
*** LOAD CASE   209 -- START ITERATION NO.   2
*** LOAD CASE   209 -- START ITERATION NO.   3
*** LOAD CASE   209 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  209



*** LOAD CASE   210 -- START ITERATION NO.   2
*** LOAD CASE   210 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  210



*** LOAD CASE   211 -- START ITERATION NO.   2
*** LOAD CASE   211 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  211



*** LOAD CASE   212 -- START ITERATION NO.   2
*** LOAD CASE   212 -- START ITERATION NO.   3
*** LOAD CASE   212 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  212



*** LOAD CASE   213 -- START ITERATION NO.   2
*** LOAD CASE   213 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  213



*** LOAD CASE   214 -- START ITERATION NO.   2
*** LOAD CASE   214 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  214



*** LOAD CASE   215 -- START ITERATION NO.   2
*** LOAD CASE   215 -- START ITERATION NO.   3

  **NOTE-Tension/Compression converged after  3 iterations, Case=  215



*** LOAD CASE   216 -- START ITERATION NO.   2
*** LOAD CASE   216 -- START ITERATION NO.   3
*** LOAD CASE   216 -- START ITERATION NO.   4

  **NOTE-Tension/Compression converged after  4 iterations, Case=  216



*** LOAD CASE   217 -- START ITERATION NO.   2
*** LOAD CASE   217 -- START ITERATION NO.   3
*** LOAD CASE   217 -- START ITERATION NO.   4
```

**NOTE-Tension/Compression converged after 4 iterations, Case= 217

*** LOAD CASE 218 -- START ITERATION NO. 2
*** LOAD CASE 218 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 218

*** LOAD CASE 219 -- START ITERATION NO. 2
*** LOAD CASE 219 -- START ITERATION NO. 3
*** LOAD CASE 219 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 219

*** LOAD CASE 220 -- START ITERATION NO. 2
*** LOAD CASE 220 -- START ITERATION NO. 3
*** LOAD CASE 220 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 220

*** LOAD CASE 221 -- START ITERATION NO. 2
*** LOAD CASE 221 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 221

*** LOAD CASE 222 -- START ITERATION NO. 2
*** LOAD CASE 222 -- START ITERATION NO. 3
*** LOAD CASE 222 -- START ITERATION NO. 4
*** LOAD CASE 222 -- START ITERATION NO. 5
*** LOAD CASE 222 -- START ITERATION NO. 6
*** LOAD CASE 222 -- START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 222

*** LOAD CASE 223 -- START ITERATION NO. 2
*** LOAD CASE 223 -- START ITERATION NO. 3
*** LOAD CASE 223 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 223

*** LOAD CASE 224 -- START ITERATION NO. 2
*** LOAD CASE 224 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 224

*** LOAD CASE 225 -- START ITERATION NO. 2
*** LOAD CASE 225 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 225

*** LOAD CASE 226 -- START ITERATION NO. 2
*** LOAD CASE 226 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 226

*** LOAD CASE 227 -- START ITERATION NO. 2
*** LOAD CASE 227 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 227

987. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 31,2020 TIME= 15:56:36 ****

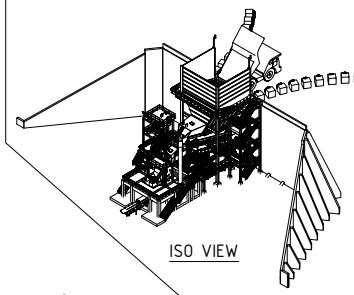
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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

ATTACHMENT 4 L&T INPUT DRAWINGS(PRIMARY CRUSHER HOUSE)

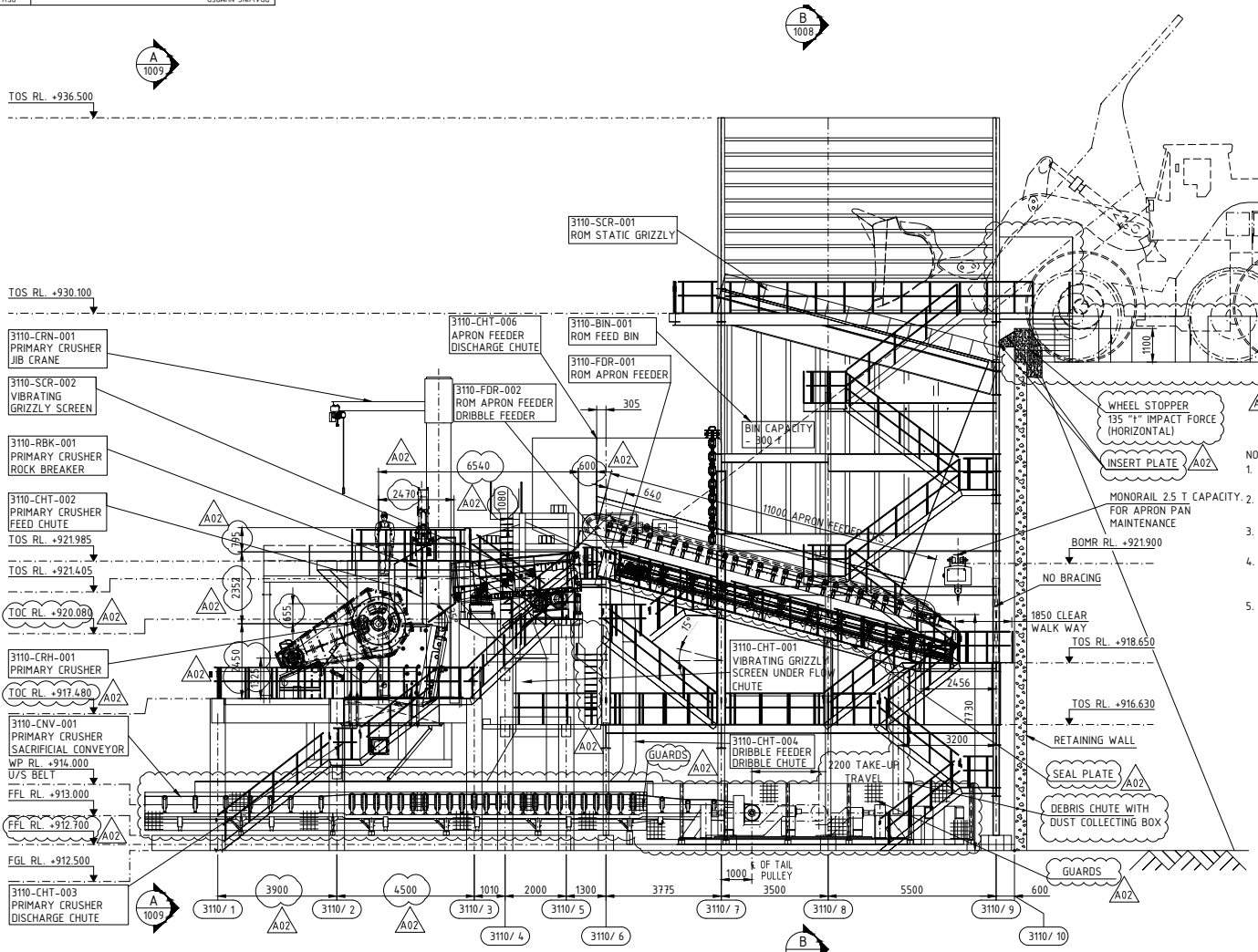
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DRAWING NUMBER MD-216-3110-EG-ME-ELV-1007 REV A02



ISO VIEW

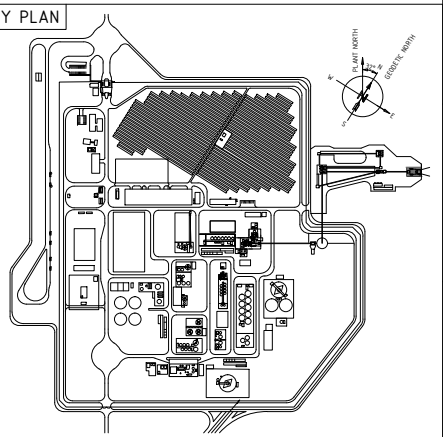


ELEVATION
SCALE 1:75

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3100-EG-PI-GAD-1007	PIPING - DUST SUPPRESSION SYSTEM FOR PRIMARY CRUSHING AREA								
MD-216-3110-EG-ME-ELV-1009	SECTION								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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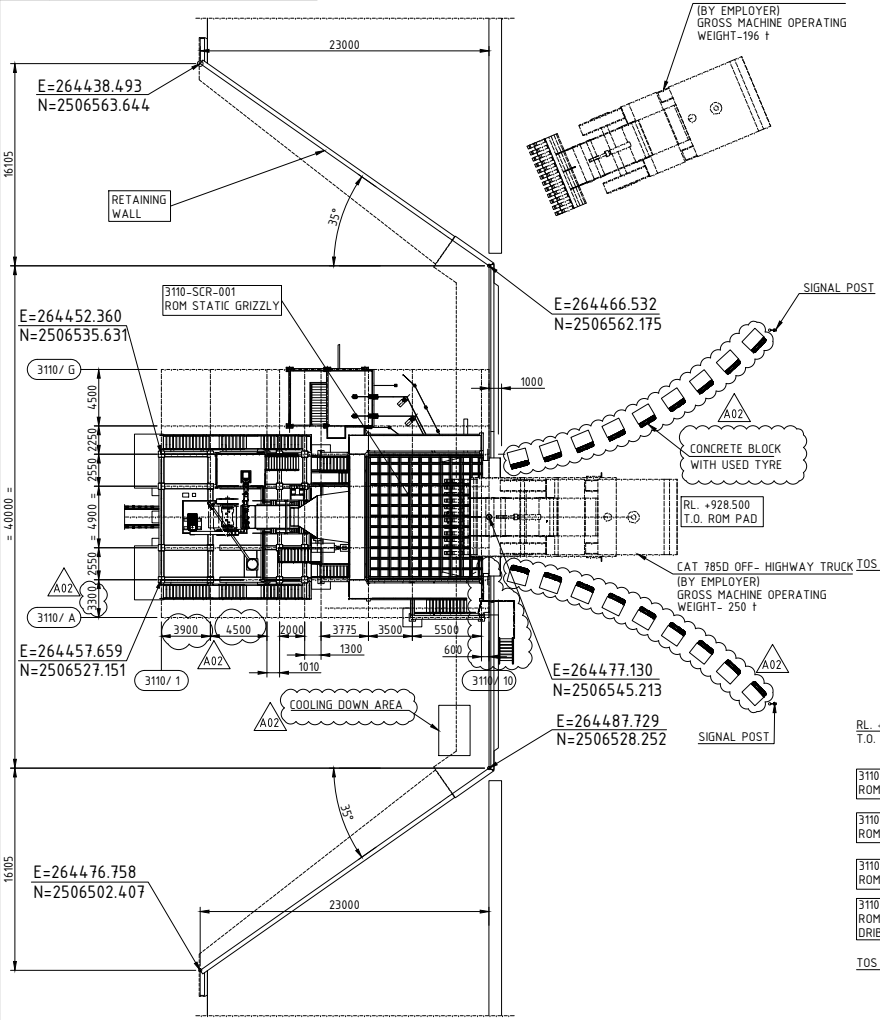
LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_100_H_G_16123_001	1
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707__GAU01__ZC01__MTL01__31106	OU100666770 / 1

DRAWN	SKH	29/11/2019
CHECKED	BRD	30/11/2019
ENGINEER	BD	02/12/2019
APPROVED	BGR	02/12/2019
EMPLOYER		
SCALE	1:75	A1

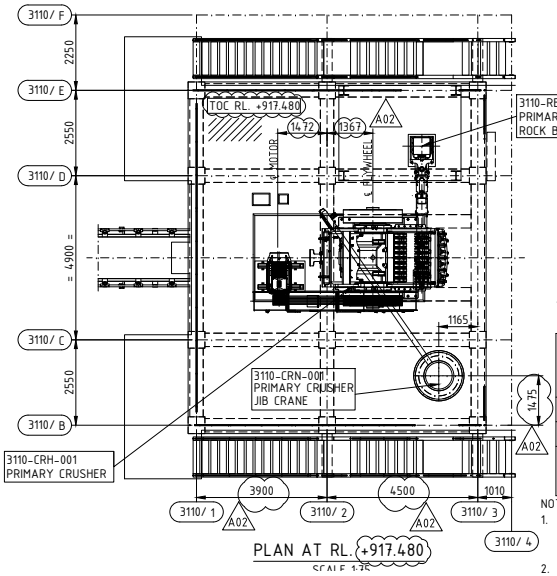
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CRUSHING
PRIMARY CRUSHER HOUSE - ELEVATION

DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1007 REV A02

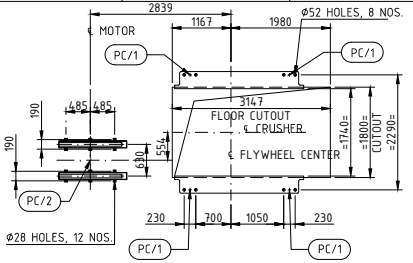
DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1008 | REV: A02



PLAN AT RL. (+) 928.500
SCALE 1:200
(T.O. ROM PAD)



PLAN AT RL. +917.480
SCALE 1:25



PRIMARY CRUSHER FOUNDATION PLAN
SCALE 1:50

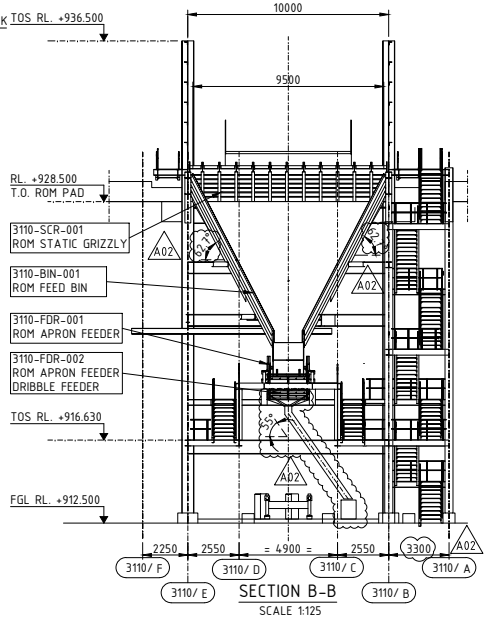
TABLE-5 : LOAD TABLE FOR PRIMARY CRUSHER
3110-CRH-001

POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")	DYNAMIC LOAD f=3.3 Hz (IN "t")	HORIZONTAL LOAD f=3.3 Hz (IN "t")	
				HORIZONTAL LONGITUDINAL	HORIZONTAL TRANSVERSAL
PC/1	(+)916.985	(+)20.0	(±)3.10	(±)1.2	(±)0.3
PC/2		(+)4.5	(±)6.5	(±)0.45	(±)0.25

NOTE: The dynamic load frequency is 3.3 Hz for the vertical and longitudinal load using the 90 degree method settings. The forces are derived based on a rigid foundation assumption. A weak foundation or resonance increase the dynamic loads.

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

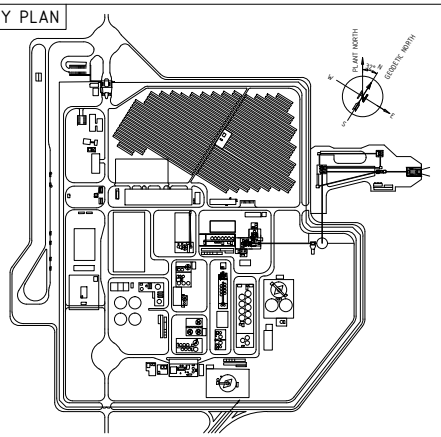
CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL



SECTION B-B
SCALE 1:125

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS HEAD PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1009	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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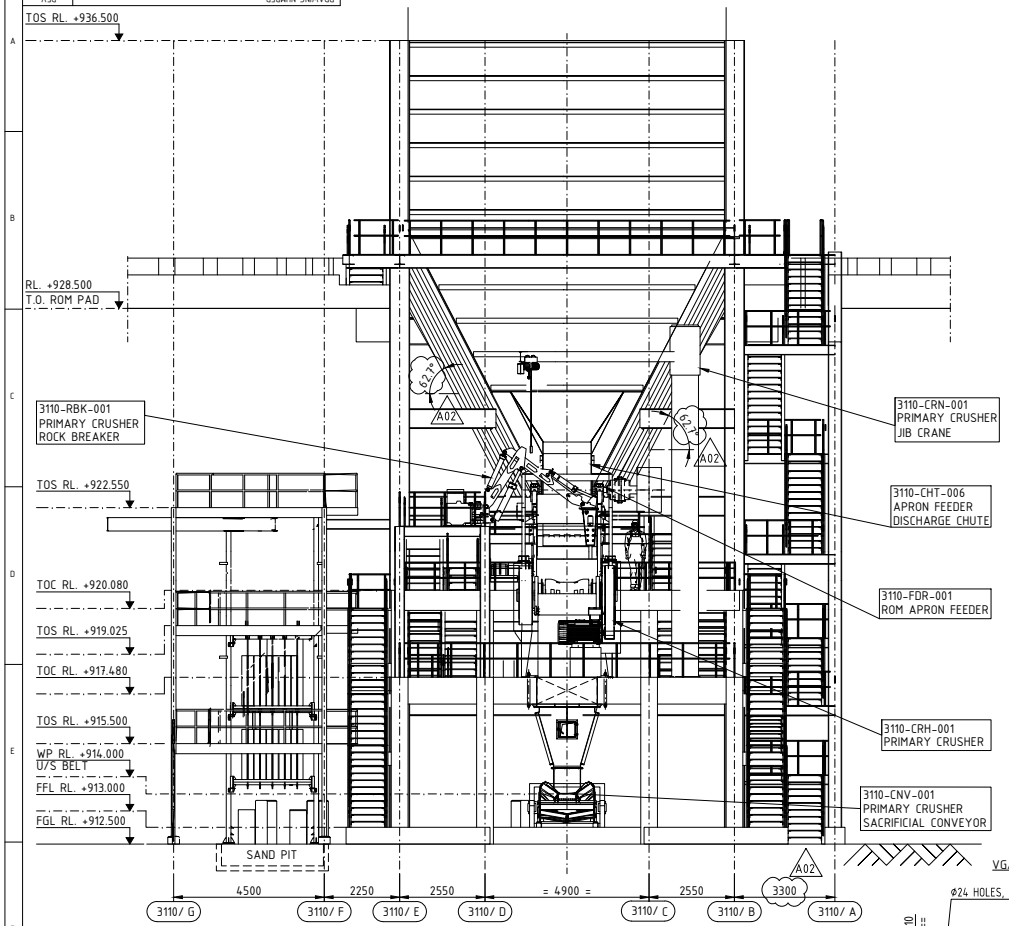
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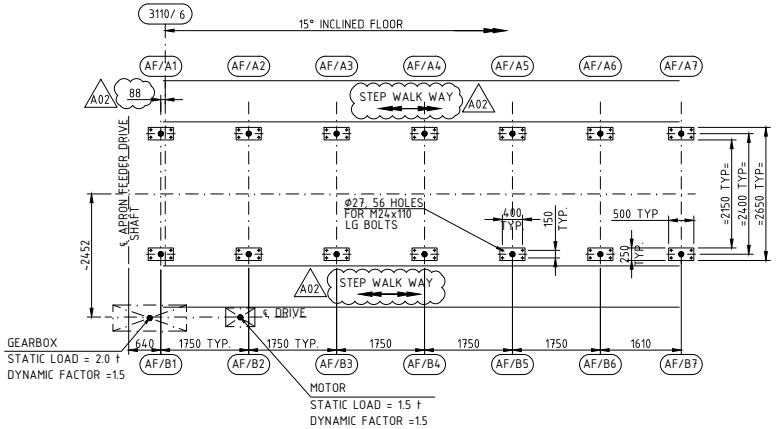
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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16123_004		REVISION 1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31109	DOCUMENT ID 0U100666773	REVISION 1
DRAWN	SKH	29/11/2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	30/11/2019	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	02/12/2019	CRUSHING
APPROVED	BGR	02/12/2019	PRIMARY CRUSHER HOUSE - SECTION
EMPLOYER			
SCALE	1:200, 1:75, 1:50	A1	DRAWING NUMBER MD-216-3110-EG-ME-ELV-1008 REV A02

DRAWING NUMBER MD-216-3110-EG-ME-ELV-1009 REV A02



VIEW A-A
SCALE 1:75



ROM APRON FEEDER FOUNDATION PLAN
SCALE 1:50

TABLE-7 :
LOAD TABLE FOR ROM APRON FEEDER
3110-FDR-001

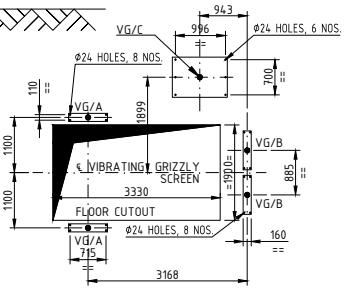
POINT	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t") ALONG THE CONV.
AF/A1	(+)10.5	(-)3.5
AF/B1	(+)12.5	(-)3.5
AF/A2	(+)3.25	(-)3.5
AF/B2	(+)5.00	(-)3.5
AF/A3	(+)10.00	(-)3.5
AF/B3	(+)10.00	(-)3.5
AF/A4	(+)13.75	(-)3.5
AF/B4	(+)13.75	(-)3.5
AF/A5	(+)13.75	(-)3.5
AF/B5	(+)13.75	(-)3.5
AF/A6	(+)11.50	(-)3.5
AF/B6	(+)11.50	(-)3.5
AF/A7	(+)5.25	(-)3.5
AF/B7	(+)5.25	(-)3.5

TABLE-6 : LOAD TABLE FOR VIBRATING GRIZZLYSCREEN
3110-SCR-002

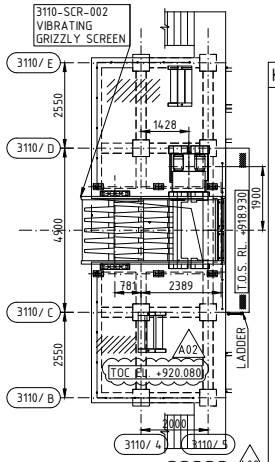
POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")	DYNAMIC LOAD (IN "t")		DYNAMIC LOAD (IN "t") (START & STOP)	
			VERTICAL	HORIZONTAL	VERTICAL	HORIZONTAL
VG/A	(+)919.540	(+)4.3	(±)3.3	(±)2.9	(±)4.5	(±)3.5
VG/B		(+)4.3	(±)3.3	(±)2.9	(±)4.5	(±)3.5
VG/C		(+)1.5	(±)2.3	(±)2.3	-	-

NOTE:
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2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS HEAD PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

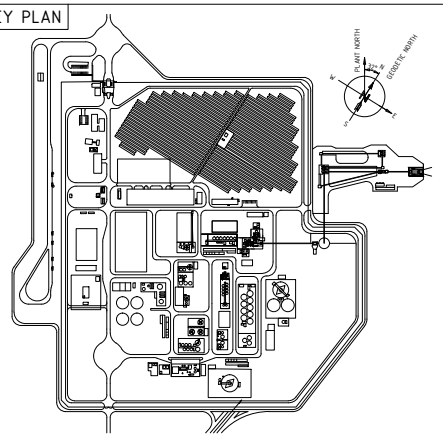


VIBRATING GRIZZLY SCREEN FOUNDATION PLAN
SCALE 1:50



PLAN AT RL. +920.080
SCALE 1:75

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

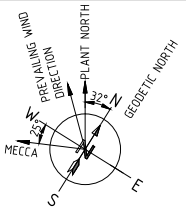
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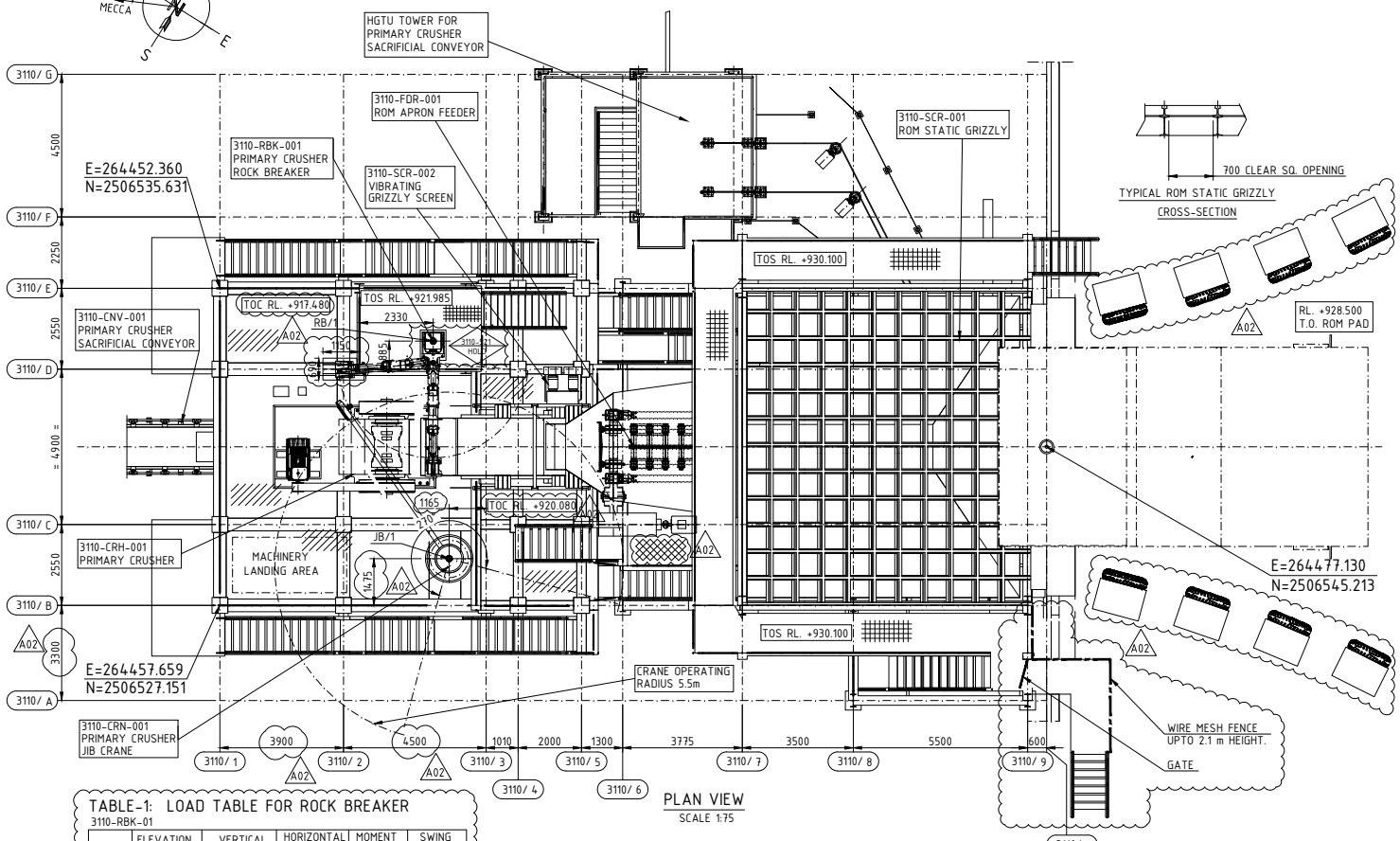
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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16123_005		REVISION 1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31110	DOCUMENT ID OU100666774	REVISION 1
DRAWN	SKH	29/11/2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	30/11/2019	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	02/12/2019	CRUSHING
APPROVED	BGR	02/12/2019	PRIMARY CRUSHER HOUSE - SECTION
EMPLOYER			
SCALE	1:75, 1:50	A1	DRAWING NUMBER MD-216-3110-EG-ME-ELV-1009 REV A02

DRAWING NUMBER MD-216-3110-EG-ME-LPX-1001 REV A02





PLAN VIEW
SCALE 1:75

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

KEY PLAN

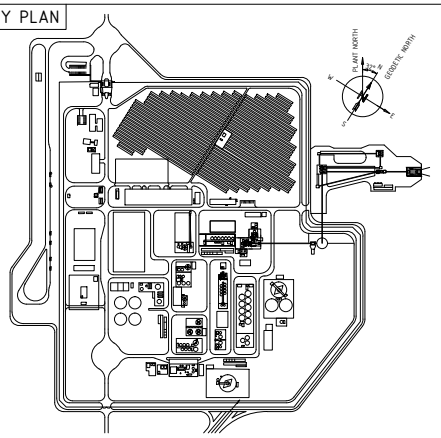


TABLE-1: LOAD TABLE FOR ROCK BREAKER
3110-RBK-01

POINT	ELEVATION IN "m"	VERTICAL LOAD (V) IN "t"	HORIZONTAL TOWARDS JAW CRUSHER (H1) IN "t"	MOMENT ABOUT CENTER (M) IN "t-m"	SWING TORQUE (MK) IN "t-m"
RB/1	(+)921.985	(+)3.5	7.5	20.0	3.0

LOAD COMBINATION FOR ROCK BREAKER : 1. V+H1+M
2. V+H1+MK
3. V+MK

TABLE-2: LOAD TABLE FOR JIB CRANE
3110-CRN-01

POINT	ELEVATION IN "m"	VERTICAL LOAD (V) IN "t"	MOMENT ABOUT CENTER (M) IN "t-m"
JB/1	(+)916.985	(+)15.5	(±)37.0

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS HEAD PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1009	SECTION								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

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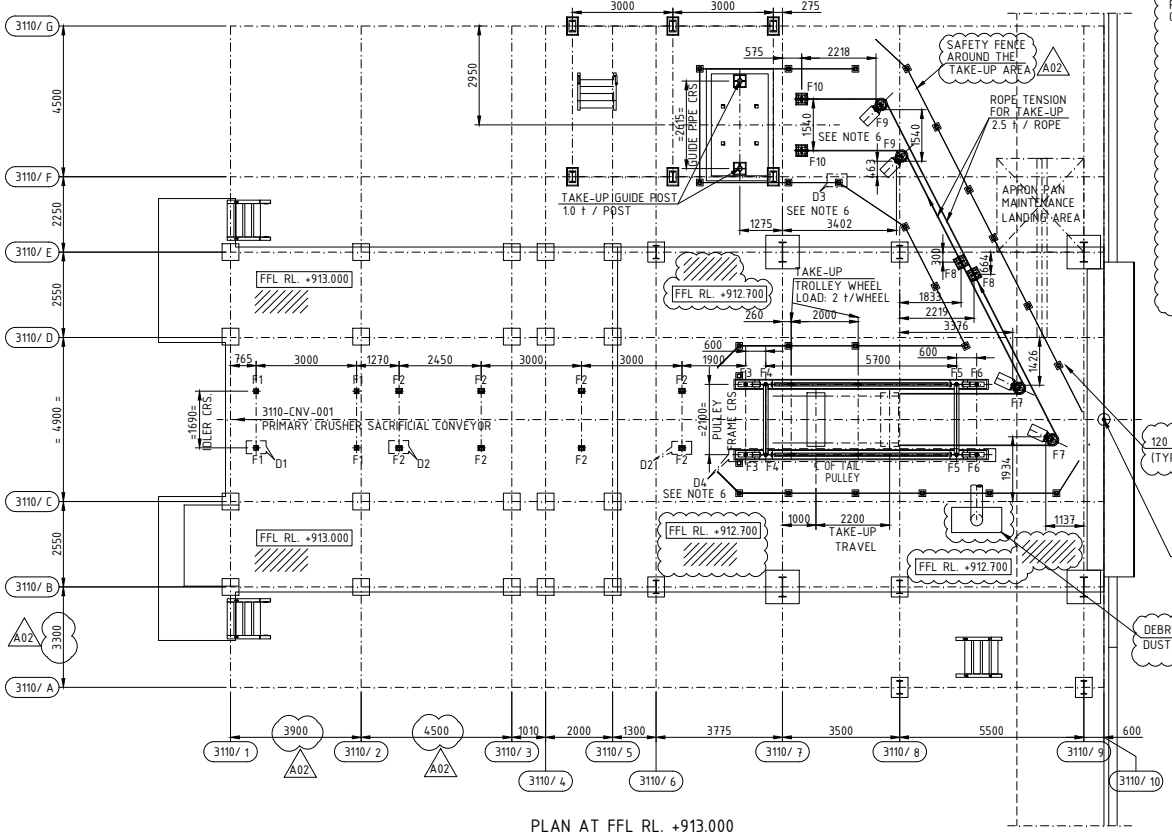
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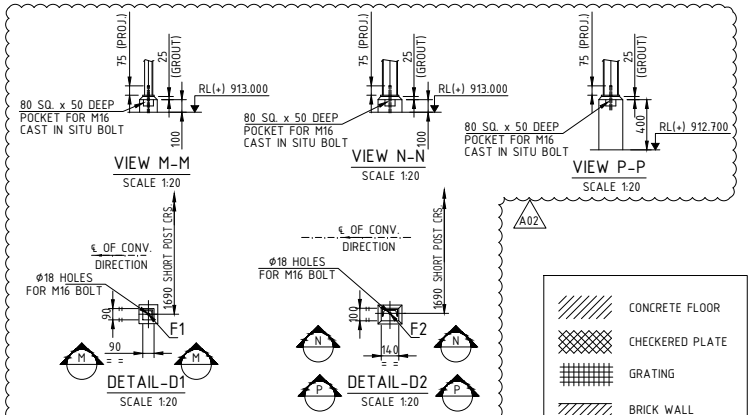
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31107	DOCUMENT ID OU100666771	REVISION 1
DRAWN	SKH	29/11/2019	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	30/11/2019	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	02/12/2019	CRUSHING
APPROVED	BGR	02/12/2019	PRIMARY CRUSHER HOUSE - PLAN VIEW
EMPLOYER			
SCALE	1:75	A1	DRAWING NUMBER MD-216-3110-EG-ME-LPX-1001 REV A02



PLAN AT FFL RL. +913.000

SCALE 1:75



NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.
6. FOR DETAIL D3, D4, F7, F8, F9, F10 REEFER DRAWING 1008

KEY PLAN

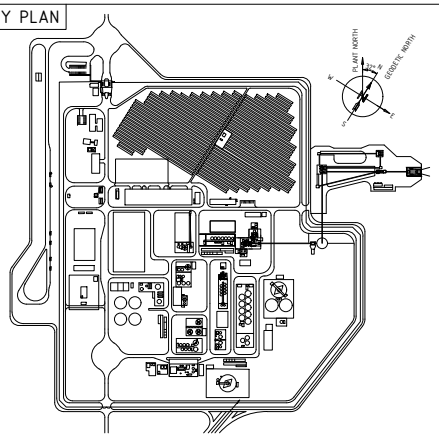


TABLE-3 : LOAD TABLE FOR SHORT POST

FOR CONV. 3110-CNV-001

	POINT	ELEVATION IN "m"	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
NORMAL ZONE	F1	(+)913.000	(+)0.65	(+)0.35	(±)0.11	(±)0.06
IMPACT ZONE	F2	(+)912.500 (+)913.000	(+)0.98	(+)0.55	(±)0.17	(±)0.09

TABLE-4 : LOAD TABLE FOR BUFFER POST

FOR CONV. 3110-CNV-001

POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F3	(+)912.500	(+)5.6	(+)3.6	(±)0.56
F4		(-)5.6	(+)1.0	(±)0.56
F5		(-)5.6	(-)1.0	(±)0.56
F6		(+)5.6	(-)3.6	(±)0.56

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS HEAD PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1009	SECTION								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW	A02	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	02/12/2019	ISSUED FOR APPROVAL	SKH	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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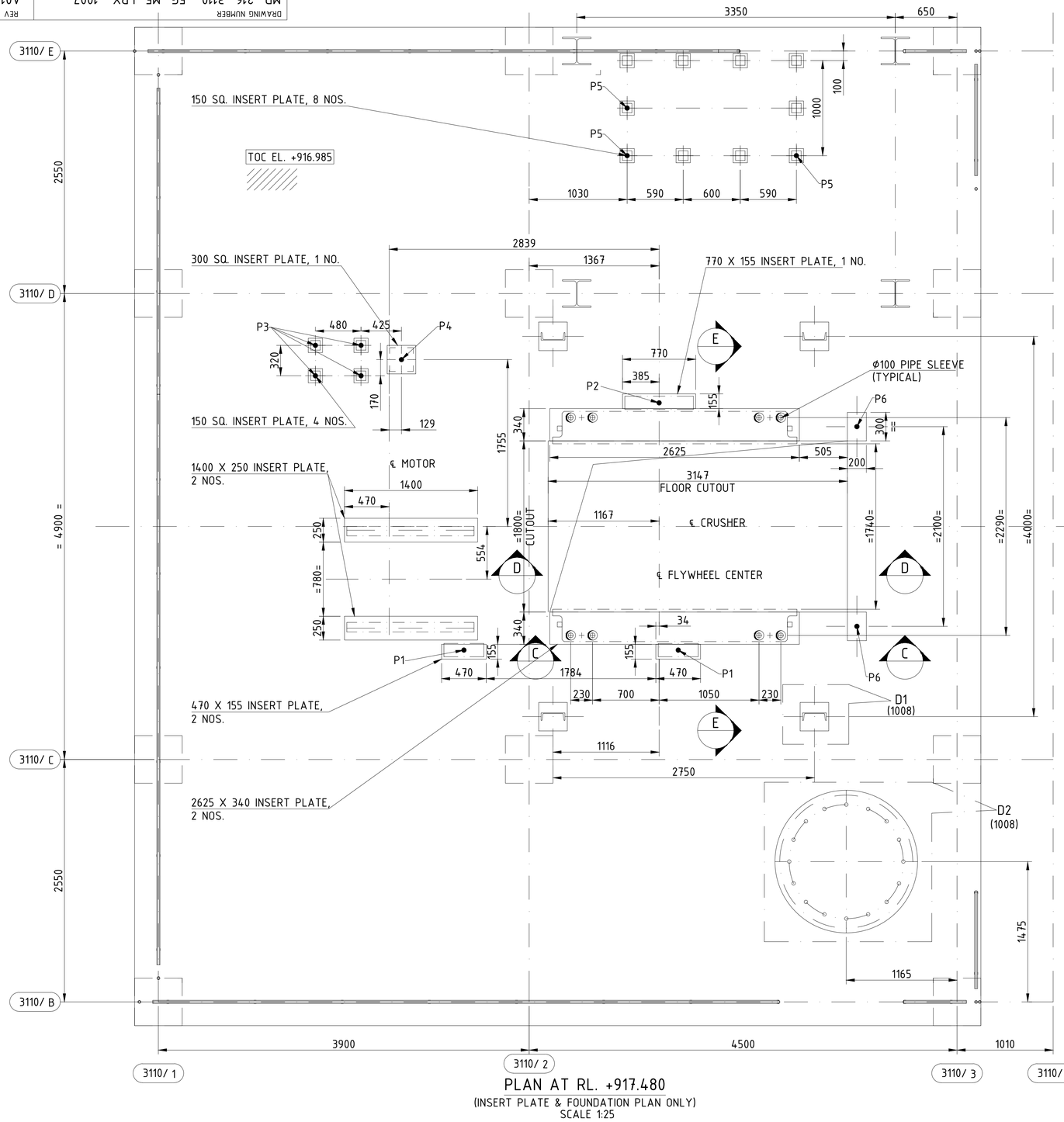
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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16123_003
REVISION	1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31108
DOCUMENT ID	OU100666772
REVISION	1
DRAWN	SKH 29/11/2019
CHECKED	BRD 30/11/2019
ENGINEER	BD 02/12/2019
APPROVED	BGR 02/12/2019
EMPLOYER	
SCALE	1:75
	A1

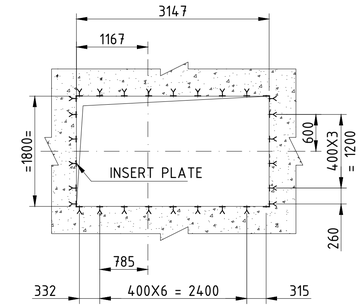
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CRUSHING
PRIMARY CRUSHER HOUSE - PLAN VIEW

DRAWING NUMBER: MD-216-3110-EG-ME-LPX-1002
REV: A02

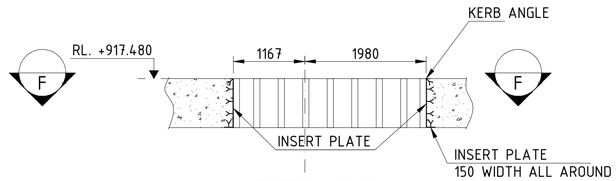
DRAWING NUMBER MD-216-3110-EG-ME-LPX-1007 REV A01



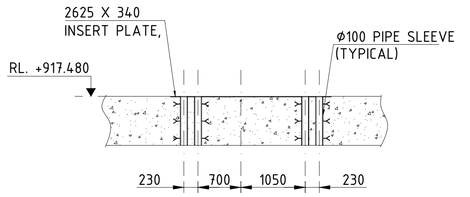
PLAN AT RL. +917.480
(INSERT PLATE & FOUNDATION PLAN ONLY)
SCALE 1:25



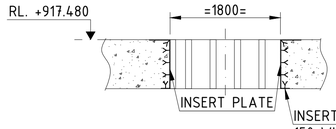
SECTION F-F
SCALE 1:50



SECTION D-D
SCALE 1:50



SECTION C-C
SCALE 1:50



SECTION E-E
SCALE 1:50

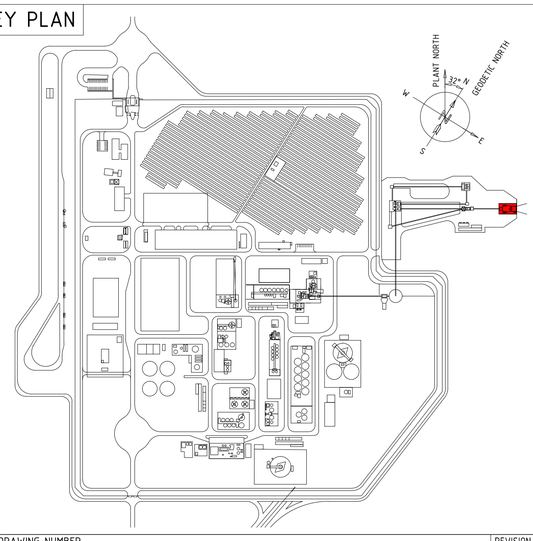
TABLE-7: LOAD FOR INSERTS

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")
P1	(+)916.985	(+)1
P2		(+)1
P3		(+)0.5
P4		(+)0.5
P5		(+)0.5
P6		(+)2.5

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

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5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

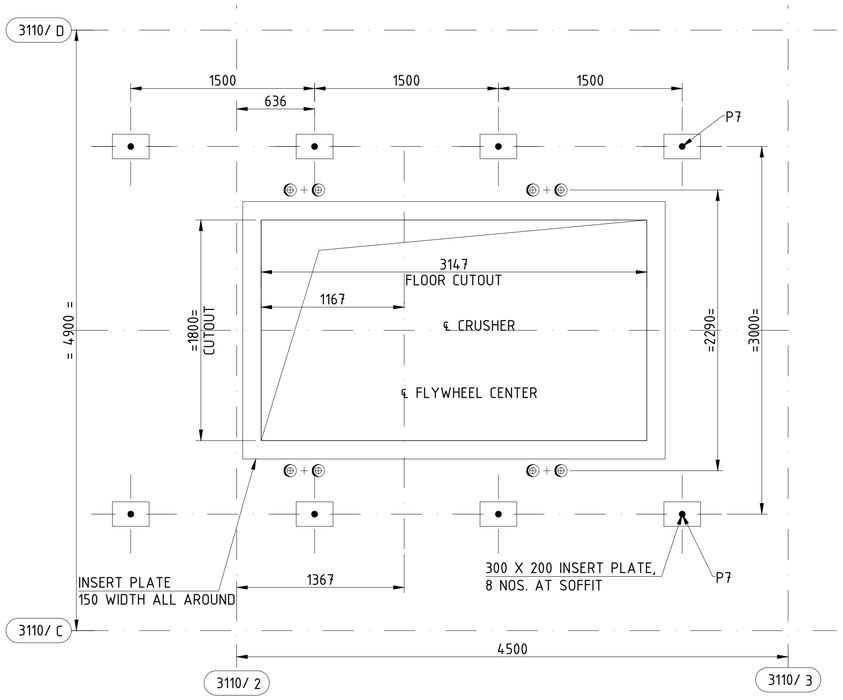
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16123_006 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31156 — DOCUMENT ID: OU100672982 — REVISION: 0

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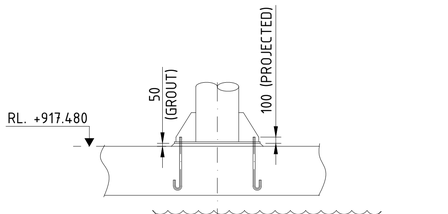
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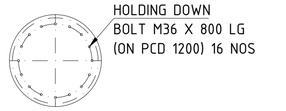
DRAWN	ANN	02/03/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	03/03/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	04/03/2020	CRUSHING
APPROVED	BGR	05/03/2020	PRIMARY CRUSHER HOUSE - PLAN VIEW
EMPLOYER			
SCALE	1:75, 1:50	A1	DRAWING NUMBER: MD-216-3110-EG-ME-LPX-1007 — REV: A01



SOFFIT PLAN AT RL. +917.480
SCALE 1:25



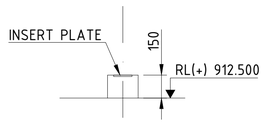
SECTION H-H
SCALE 1:50
3110-522 HOLD



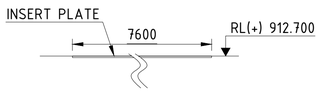
DETAIL D2
SCALE 1:50
3110-522 HOLD

TABLE-8: LOAD FOR CHUTE SUPPORT

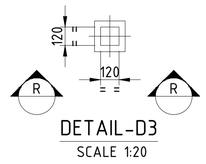
POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P7	(+)917.480 (AT SOFFIT)	(+)2.0	(+)3.5



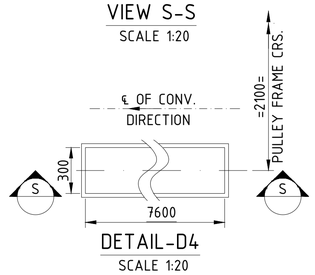
VIEW R-R
SCALE 1:20



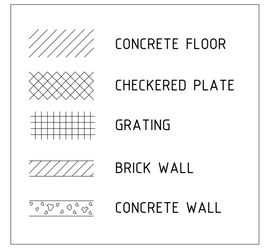
VIEW S-S
SCALE 1:20



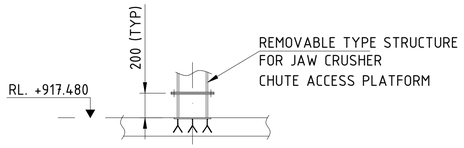
DETAIL-D3
SCALE 1:20



DETAIL-D4
SCALE 1:20



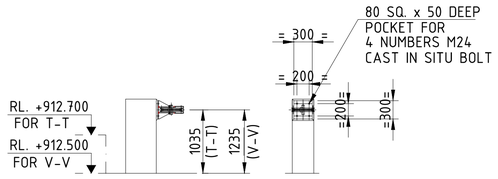
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



SECTION G-G
SCALE 1:25

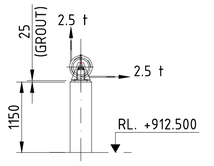


DETAIL D1
SCALE 1:25



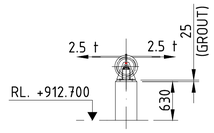
VIEW T-T
VIEW V-V
SCALE 1:50

VIEW X-X

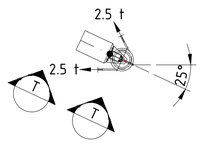


VIEW U-U
SCALE 1:50

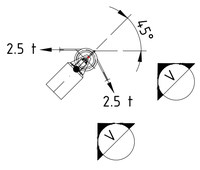
VIEW Y-Y



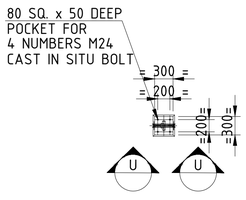
VIEW W-W
SCALE 1:50



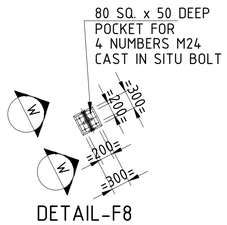
DETAIL-F7
SCALE 1:50



DETAIL-F9
SCALE 1:50

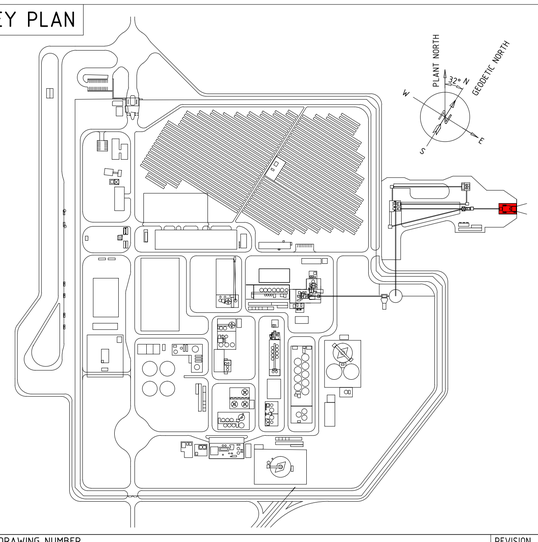


DETAIL-F10
SCALE 1:50



DETAIL-F8
SCALE 1:50

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1009	PLAN VIEW								
MD-216-3110-EG-ME-LPX-10087	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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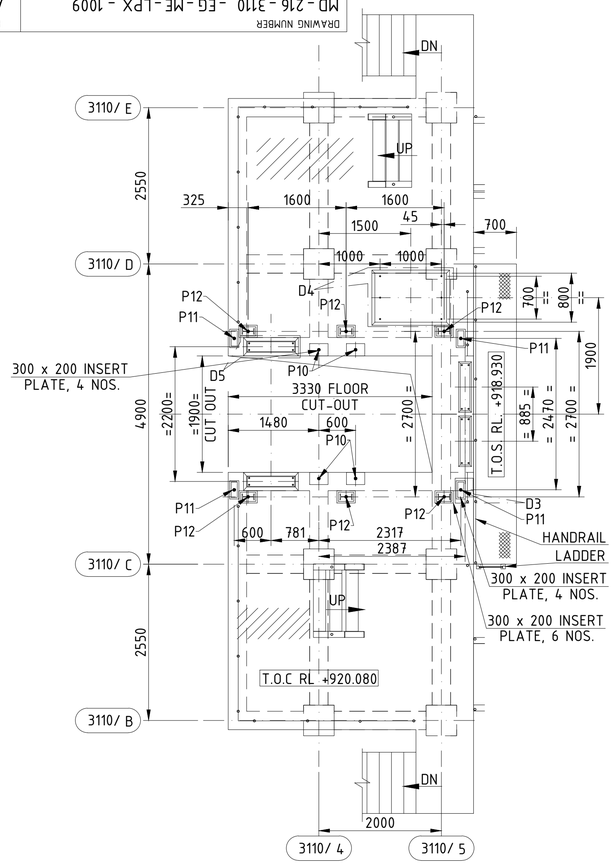
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OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31157	DOCUMENT ID OU100672983 / REVISION 0

DRAWN	ANN	02/03/2020
CHECKED	BRD	03/03/2020
ENGINEER	BD	04/03/2020
APPROVED	BGR	05/03/2020
EMPLOYER		
SCALE	1:75, 1:50	A1

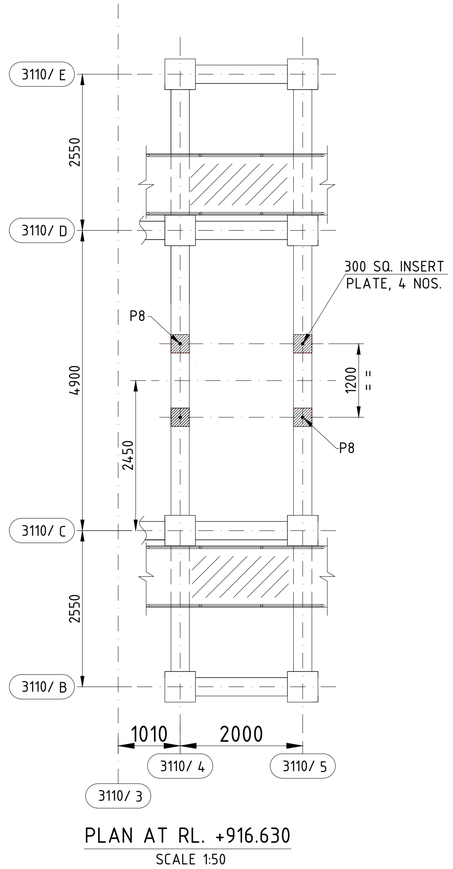
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CRUSHING
PRIMARY CRUSHER HOUSE - PLAN VIEW

DRAWING NUMBER MD-216-3110-EG-ME-LPX-1008 REV A01

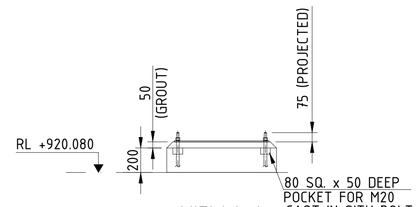
DRAWING NUMBER MD-216-3110-EG-ME-LPX-1009 REV A01



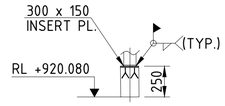
PLAN AT RL. +920.080
(INSERT PLATE & FOUNDATION PLAN ONLY)
SCALE 1:50



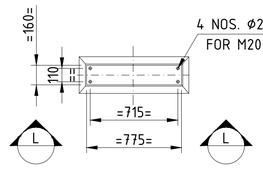
PLAN AT RL. +916.630
SCALE 1:50



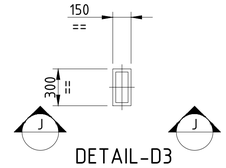
VIEW L-L
SCALE 1:25



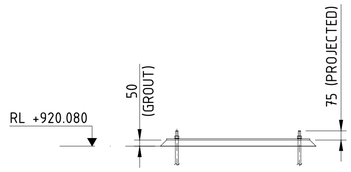
VIEW J-J
SCALE 1:25



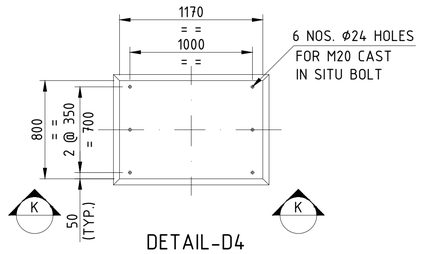
DETAIL-D5
SCALE 1:25



DETAIL-D3
SCALE 1:25

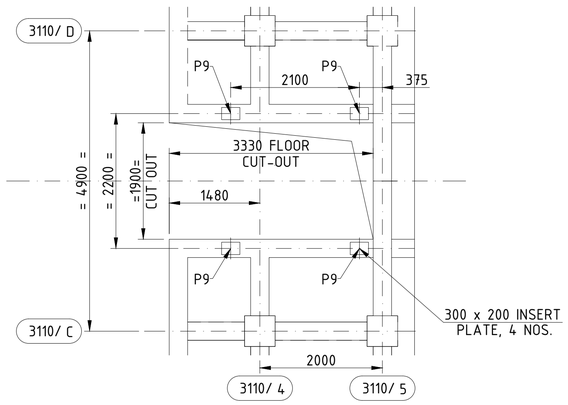


VIEW K-K
SCALE 1:25



DETAIL-D4
SCALE 1:25

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
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3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



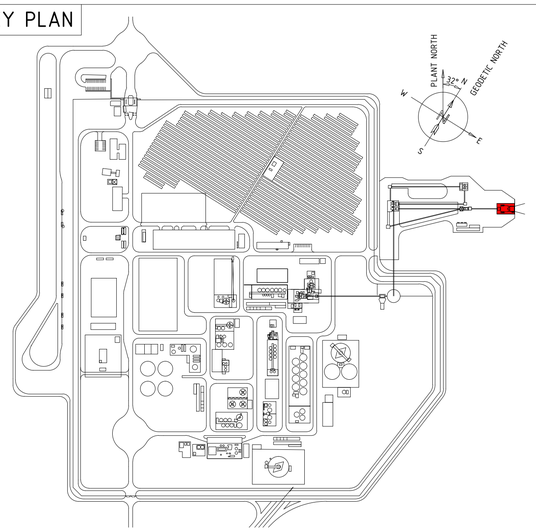
PLAN AT SOFFIT OF RL. +920.080
SCALE 1:50

TABLE-9: LOAD FOR CHUTE SUPPORT

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P8	(+)916.630	(+)1.0	(+)3.5
P9	(+)920.080 (AT SOFFIT)	(+)1.0	(+)3.5
P10	(+)920.080	(+)1.5	(+)2.5
P11		(+)1.5	(+)2.5
P12		(+)1.5	(+)2.5

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1008	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1007	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1008	SECTION								
MD-216-3110-EG-ME-LPX-1002	PLAN VIEW								
MD-216-3110-EG-ME-LPX-1001	PLAN VIEW								
MD-216-3110-EG-ME-ELV-1007	ELEVATION	A01	05/03/2020	ISSUED FOR APPROVAL	ANN	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16123_008	REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31158	DOCUMENT ID OU100672984 / REVISION 0

DRAWN	ANN	02/03/2020
CHECKED	BRD	03/03/2020
ENGINEER	BD	04/03/2020
APPROVED	BGR	05/03/2020
EMPLOYER		
SCALE	1:75, 1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CRUSHING
PRIMARY CRUSHER HOUSE - PLAN VIEW

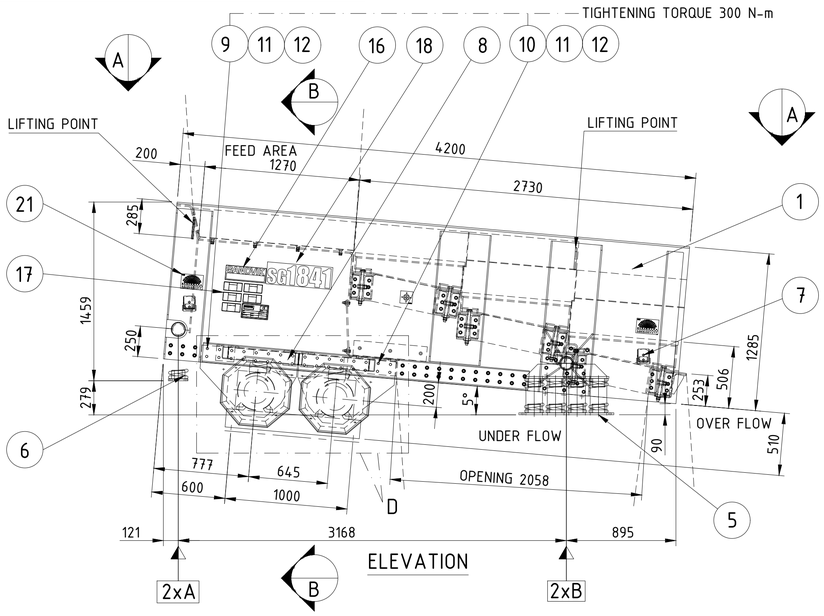
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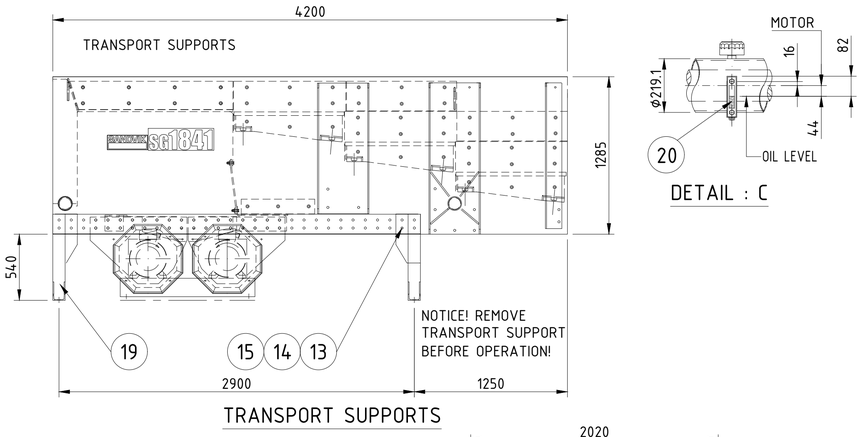
ATTACHMENT 5 L&T INPUT DRAWINGS(GA OF SCREEN)

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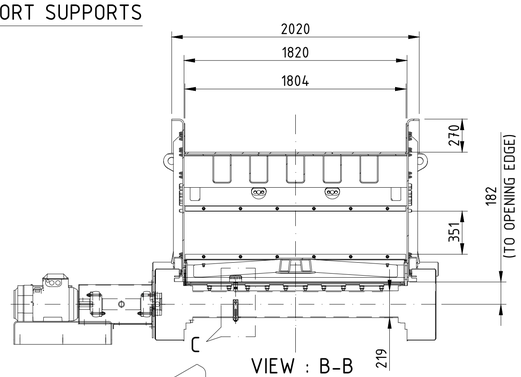
DRAWING NUMBER MD-216-3110-EG-ME-GAD-1001 REV A01



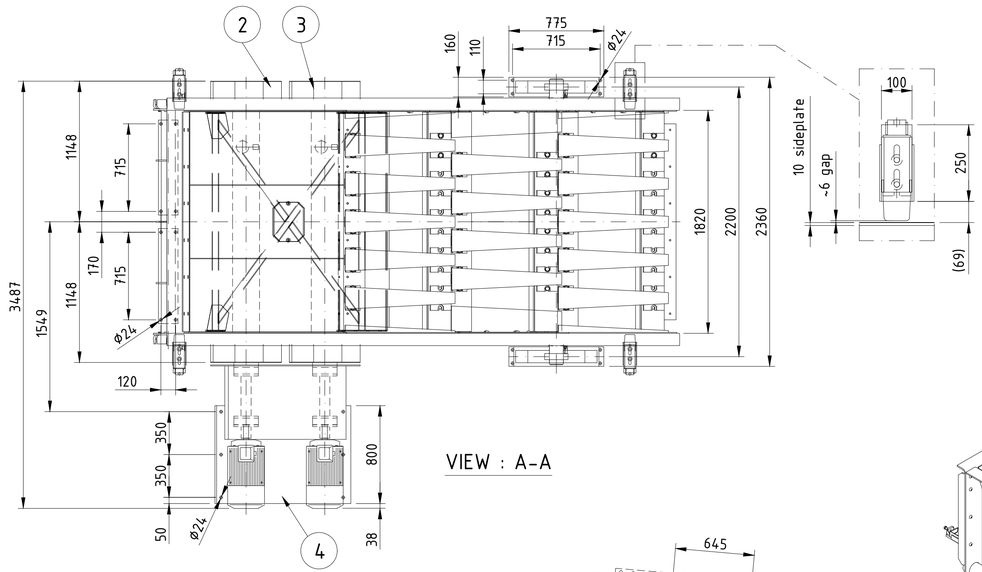


TRANSPORT SUPPORTS

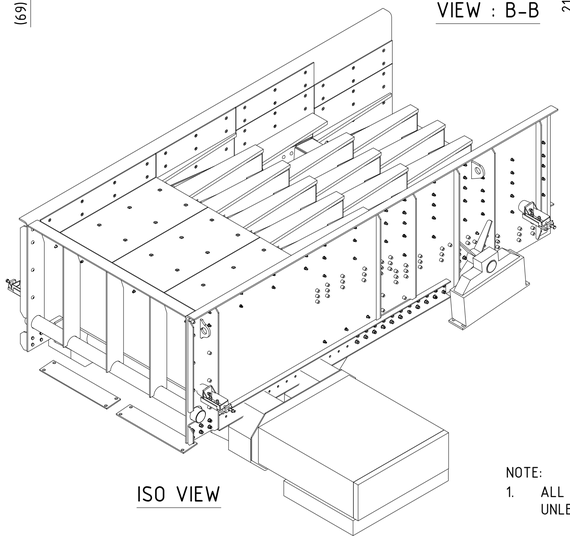
DETAIL : C



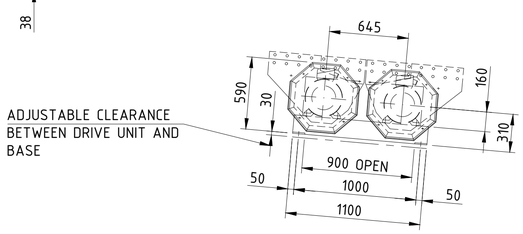
VIEW : B-B



VIEW : A-A



ISO VIEW



DETAIL : D

MECHANISM ROTATIONAL DIRECTION
GRIZZLY SCREEN

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE STATED.

TECHNICAL SPECIFICATION OF VIBRATING GRIZZLY SCREEN

EQUIPMENT TAG NO.	: 3110-SCR-002
EQUIPMENT DESCRIPTION	: VIBRATING GRIZZLY SCREEN
CAPACITY (RATED/DESIGN)	: 702/772 t/h
MATERIAL HANDLED	: FRESH ORE / OXIDE ORE
INPUT PSD	: F100 - 700 mm : F90 - 600 mm : F80 - 480 mm : F50 - 227 mm
MODEL TYPE	: SG 1841
RUNNING SPEED	: 900 rpm
STROKE LENGTH	: 6 - 10 mm
STROKE ANGLE	: 45 DEGREE
INSTALLATION ANGLE	: 5 DEGREE
INSTALLED MOTOR POWER	: INPUT : 2x 11 kW / 400V / 900 rpm @60 Hz
SUPPORT SPRING	: HF 654
TOTAL WEIGHT	: 8,000 kg
MOVING MASS	: 7,250 kg

LOADS WITHOUT MATERIAL

A static	18 200 N	B static	18 200 N
Dynamic load, running (+ / -)			
A dyn.vertical	8000 N	B dyn.vertical	8000 N
A dyn.hor/lat	4000N	B dyn.hor/lat	4000 N
Dynamic load, start and stop (+ / -)			
A dyn.vertical	20 000 N	B dyn.vertical	20 000 N
A dyn.hor/lat	10 000 N	B dyn.hor/lat	10 000 N

LOADS WITH MATERIAL

A static 23 000 N B static 23 000 N

(Dynamic loads are same as without material)

DRIVE UNIT LOADS
Static
Drive unit mass 430 kg / 4120 N

Dynamic load, start and stop (+ / -)
Dyn.vertical 5000 N
Dyn.hor/lat 5000 N

CLEARANCE TO FIXED STEELWORK:
Min. Lateral Clearance, across flow direction - 30 mm
Min. Horizontal Clearance, flow direction - 50 mm
Min. Vertical Clearance - 80 mm
Lifting Height to Remove Springs - 150 mm

ITEM	NAME	SPECIFICATION / DRAWING NO.	QTY.
21	Stainless Steel Serial Plate	726.4606-002	1
20	Oil for shaft	ISO VG 220+EP 5.5 L	2
19	Transport support	723.8843-901	2
18	SG1841 Product Name	726.4592-005	2
17	Safety Decal Kit	726.4620-001	2
16	SG Decal Instruction	726.4611-901	1
15	Washer	847.0016-00	16
14	Lock nut	845.0215-00	16
13	Hexagon head bolt	840.0070-00	16
12	Washer	17	128
11	Hexagon nut	M16	64
10	Hexagon head bolt	M16x70	56
9	Hexagon head bolt	M16x60	8
8	Support plate	BS021-B20-OP17	2
7	Mounting of buffers	BSV-P	1
6	Spring support	MSV10-P4	2
5	Spring support	MS020-P4	2
4	Drive equipment	MSV10-C20	1
3	Vibrating unit	B20	1
2	Vibrating unit	B20	1
1	Screen Body Assembly	SG1841	1

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	14/01/2020	ISSUED FOR REFERENCE	--	BRD	--	BD	

SANDVIK

DRAWING : BG-0134464 Rev A

SANDVIK PROJECT DRAWING NUMBER					REVISION
190300-3110-SCR-002					0
DATE	23/12/2019	DATE	25/12/2019	DATE	11/01/2020
DRAWN	V.S.	CHECKED	A.P.	APPROVED	M.V.

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_H_V_16501_001		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 __GAU01 __ZC01 __MTL01 __31116	0U100668097	0

DRAWN	--	--
CHECKED	BRD	14/01/2020
ENGINEER	--	--
APPROVED	BD	14/01/2020
EMPLOYER		
SCALE	1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
VIBRATING GRIZZLY SCREEN
3110-SCR-002

DRAWING NUMBER MD-216-3110-EG-ME-GAD-1001 REV A01

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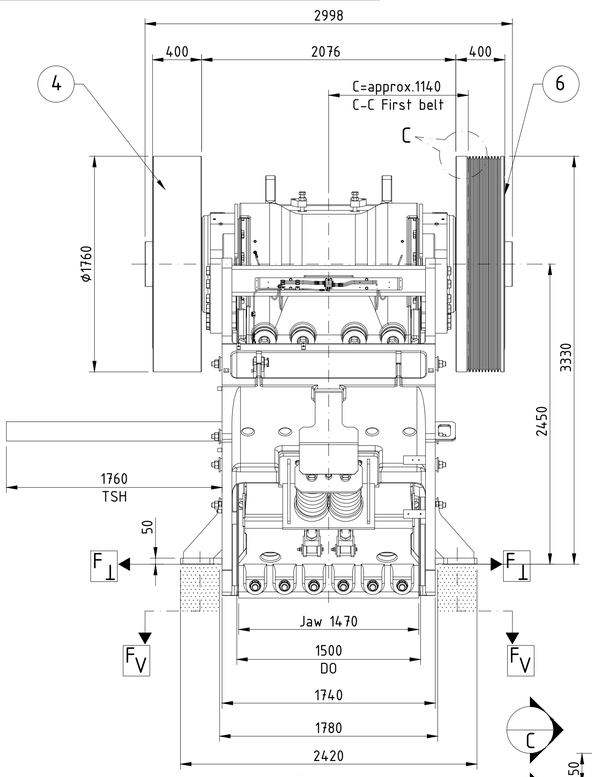
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ATTACHMENT 6 L&T INPUT DRAWINGS(GA OF JAW CRUSHER)

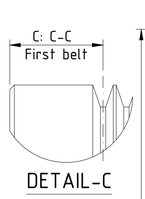
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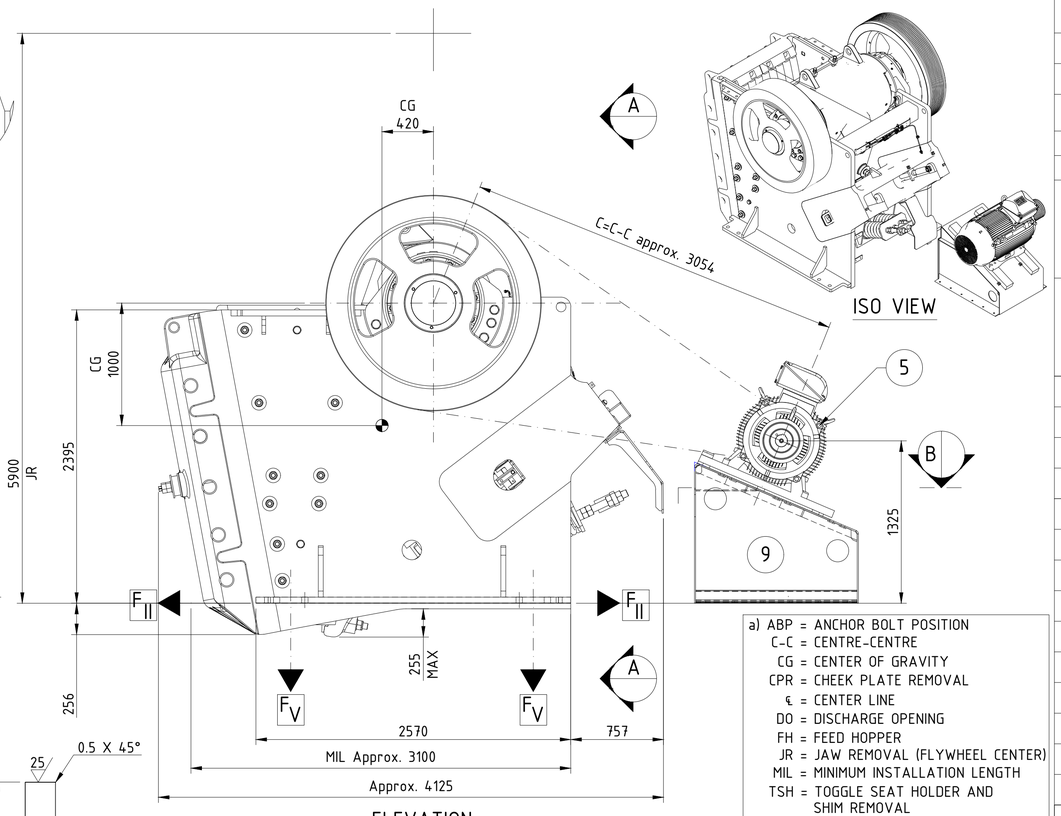
DRAWING NUMBER MD-216-3110-EG-ME-GAD-1007 | REV A01



VIEW : A-A



DETAIL-C

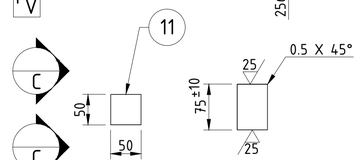


ELEVATION

ISO VIEW

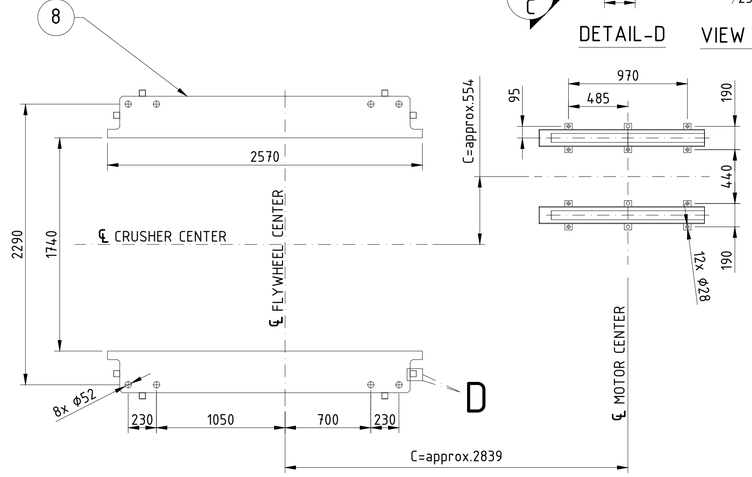
a) ABP = ANCHOR BOLT POSITION
C-C = CENTRE-CENTRE
CG = CENTER OF GRAVITY
CPR = CHEEK PLATE REMOVAL
℄ = CENTER LINE
DO = DISCHARGE OPENING
FH = FEED HOPPER
JR = JAW REMOVAL (FLYWHEEL CENTER)
MIL = MINIMUM INSTALLATION LENGTH
TSH = TOGGLE SEAT HOLDER AND SHIM REMOVAL

b) C= 60 Hz, 200 kW, 6 pole
300 MGT100; 11xSPC/9200; 202 rpm

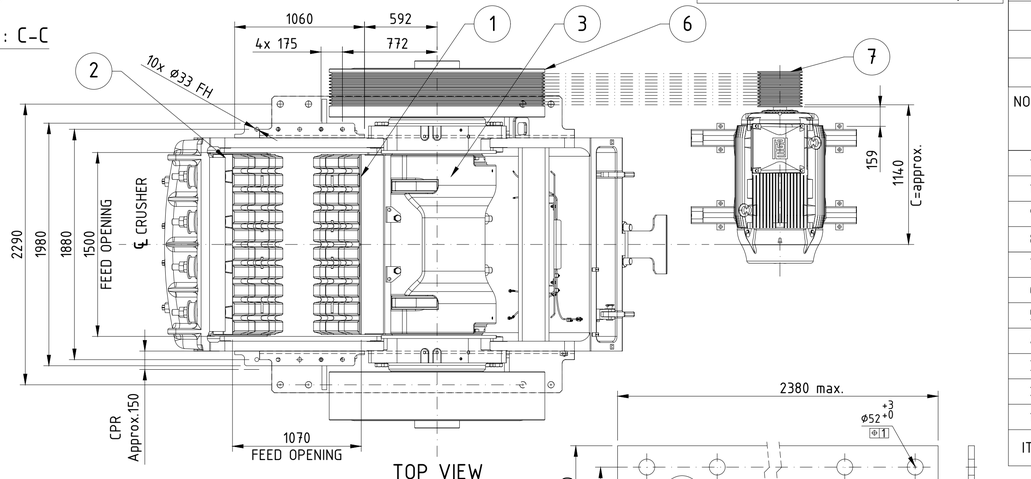


DETAIL-D

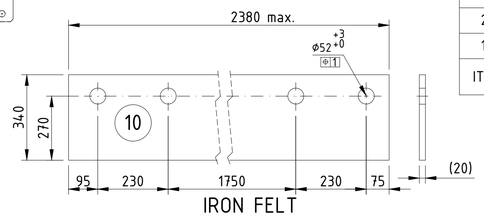
VIEW : C-C



VIEW : B-B



TOP VIEW



IRON FELT
(FOR JAW CRUSHER FOUNDATION)

TECHNICAL SPECIFICATION OF JAW CRUSHER	
EQUIPMENT TAG NO.	: 3110-CRH-001
EQUIPMENT DESCRIPTION	: PRIMARY CRUSHER
EQUIPMENT MODEL	: CJ615
CAPACITY	: 353 / 424 t/h
MATERIAL HANDLED	: FRESH ORE / OXIDE ORE
INPUT PSD	: F100 - 700 mm : F90 - 600 mm : F80 - 480 mm : F50 - 227 mm
OUTPUT PSD	: P98 - 301 / 275 mm : P90 - 247 / 214 mm : P80 - 207 / 181 mm : P50 - 136 / 121 mm
FEED OPENING	: 1500 X 1070 mm
CLOSE SIDE SETTING (css)	: 170 mm
INSTALLED MOTOR POWER	: 200 kW WITH VFD
DRIVE SIDE	: LEFT HAND SIDE
TOTAL WEIGHT	: 52,100 kg
LENGTH	: 4,125 mm
WIDTH	: 3,000 mm
HEIGHT	: 3,330 mm
MOTOR FREQUENCY	: 60 Hz
MOTOR PULLEY DIAMETER	: 300 mm
MOTOR FRAME SIZE	: 355 M/L, 6 Poles
MOVABLE JAW PLATE WEIGHT	: 3,800 kg
FIXED JAW PLATE WEIGHT	: 3,555 kg
HEAVIEST COMPONENT WEIGHT	: 26,269 kg (MOVING JAW WITH FLY WHEEL)

	VERTICAL FORCE	HORIZONTAL LONGITUDINAL FORCE	HORIZONTAL TRANSVERSAL FORCE
	F_V	F_{II}	$F_{\perp}$
STATIC LOAD 1	200 kN		
DYNAMIC LOAD 2	± 31 kN	± 13 kN	± 3 kN

NOTE: THE DYNAMIC LOAD FREQUENCY IS 3.3 HZ FOR THE VERTICAL AND LONGITUDINAL LOAD USING THE 90 DEGREE METHOD SETTINGS. THE FORCES ARE DERIVED BASED ON A RIGID FOUNDATION ASSUMPTION.

ITEM	NAME	SPECIFICATION PART/DRAWING NO.
11	STOPPER	400.3093-001
10	IRON FELT	400.3101-001
9	MOTOR FRAME	SPC U 615-04-160
8	BASE FRAME	400.0814-901
7	DRIVE PULLEY	879.0705-00
6	DRIVEN PULLEY	400.0312-003
5	MOTOR	972.1897-00
4	FLY WHEEL	400.0312-001
3	ECCENTRIC SHAFT	402.4330-02
2	FIXED JAW	400.0488-001
1	MOVABLE JAW	400.0485-001

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE STATED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	15/01/2020	ISSUED FOR REFERENCE	--	BRD	--	BD	

SANDVIK

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DRAWING : 400-0909-001 Rev 06	SANDVIK PROJECT DRAWING NUMBER					REVISION
	190300-3110-CRH-001, SHEET 1 OF 2					0
DRAWING : 400-0810- Rev 3	DATE	23/12/2019	DATE	25/12/2019	DATE	14/01/2020
	DRAWN	V.S.	CHECKED	A.P.	APPROVED	M.P.

LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_H_V_16503_002			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31122		OU100668105	0

DRAWN	--	--
CHECKED	BRD	15/01/2020
ENGINEER	--	--
APPROVED	BD	15/01/2020
EMPLOYER		
SCALE	1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
JAW CRUSHER
3110-CRH-001

DRAWING NUMBER: MD-216-3110-EG-ME-GAD-1007 | REV: A01